



Modeling nuclear-effects in neutrino-nucleus interactions

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Agenda

Did you know? NuWro was once called WroNG!

- ♦ Part I

- ♦ Implementation of the 2020 Valencia model

- ♦ Fine-tuning NuWro's nucleon-FSI module

- ♦ Part II

- ♦ AI integration in eWro

Multinucleon knockout model in NuWro

Implementation of MEC models

Within the context of Nieves *et al.* (Valencia 2012) model

Theoretical MEC models only predicts : $\frac{d^2\sigma}{dE'_l d\Omega(\hat{\mathbf{k}})}$

What we need is : $\frac{d^8\sigma}{dE'_l d\Omega(\hat{\mathbf{k}}) d\mathbf{p}_1 d\mathbf{p}_2}$

So it is then necessary to develop tools to produce complete events for given values for specific energy-momentum transfer by **assigning outgoing nucleons isospin and momentum.**

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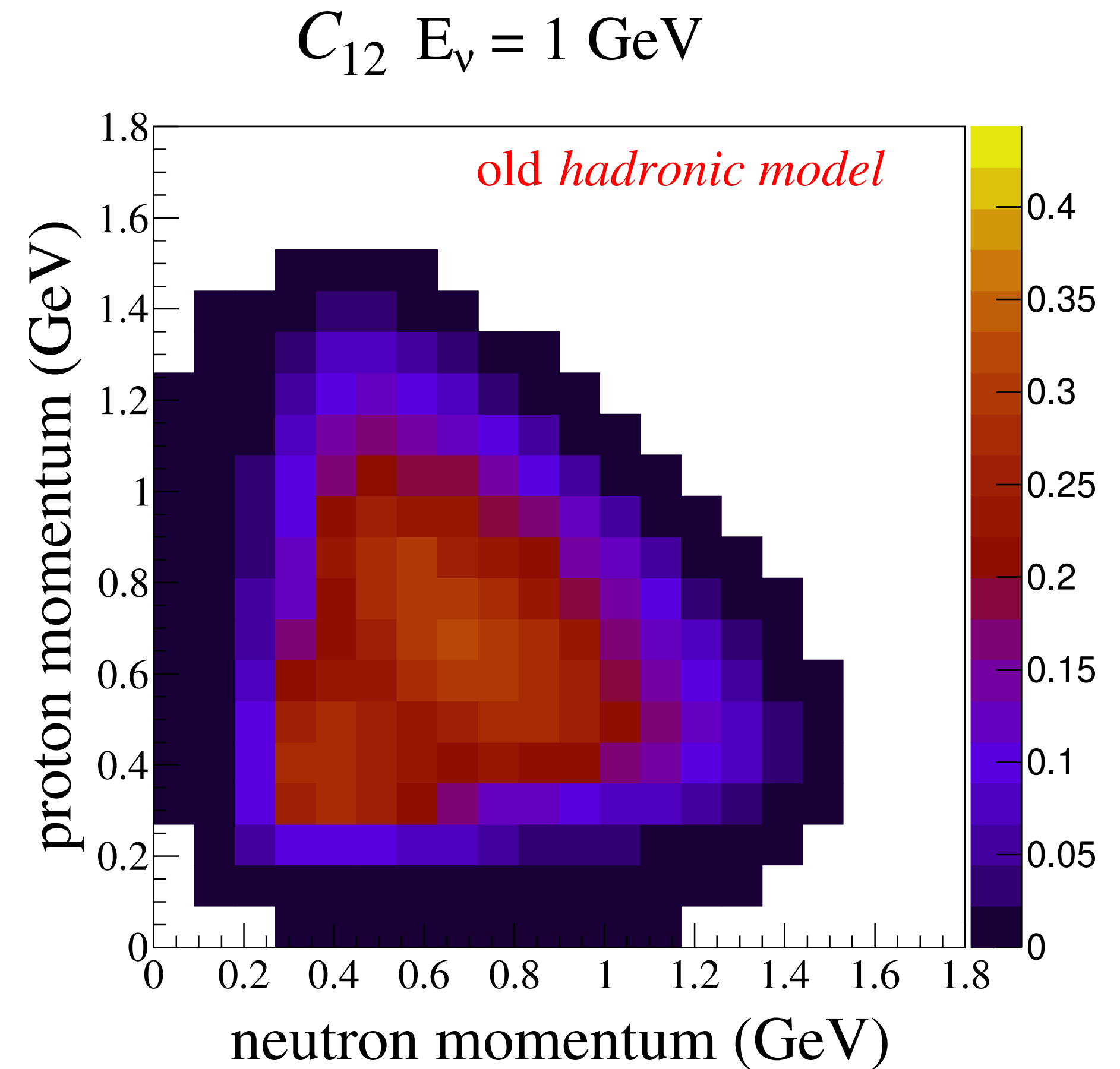
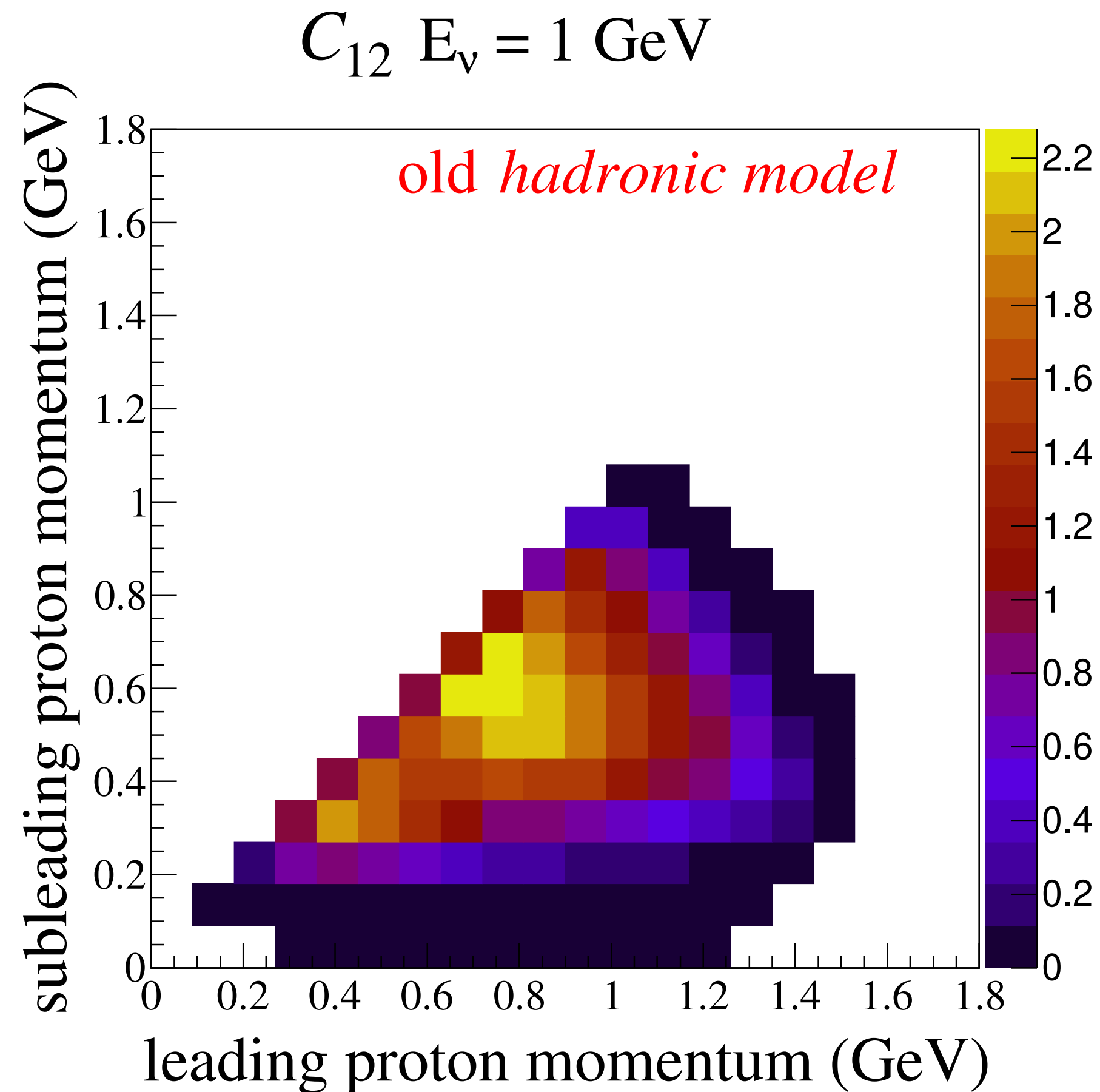
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Phenomenological treatment of $\mathbf{p}_1, \mathbf{p}_2$

Modeling outgoing nucleon kinematics

Implementation of Nieves *et al.* (Valencia 2012) model in NuWro



The 2020 Valencia model

The 2020 Valencia model

J. E. Sobczyk *et al.* Phys. Rev. C. 102 (2020) 024601

- Separation of 2p2h and 3p3h contribution from the total MEC cross section.

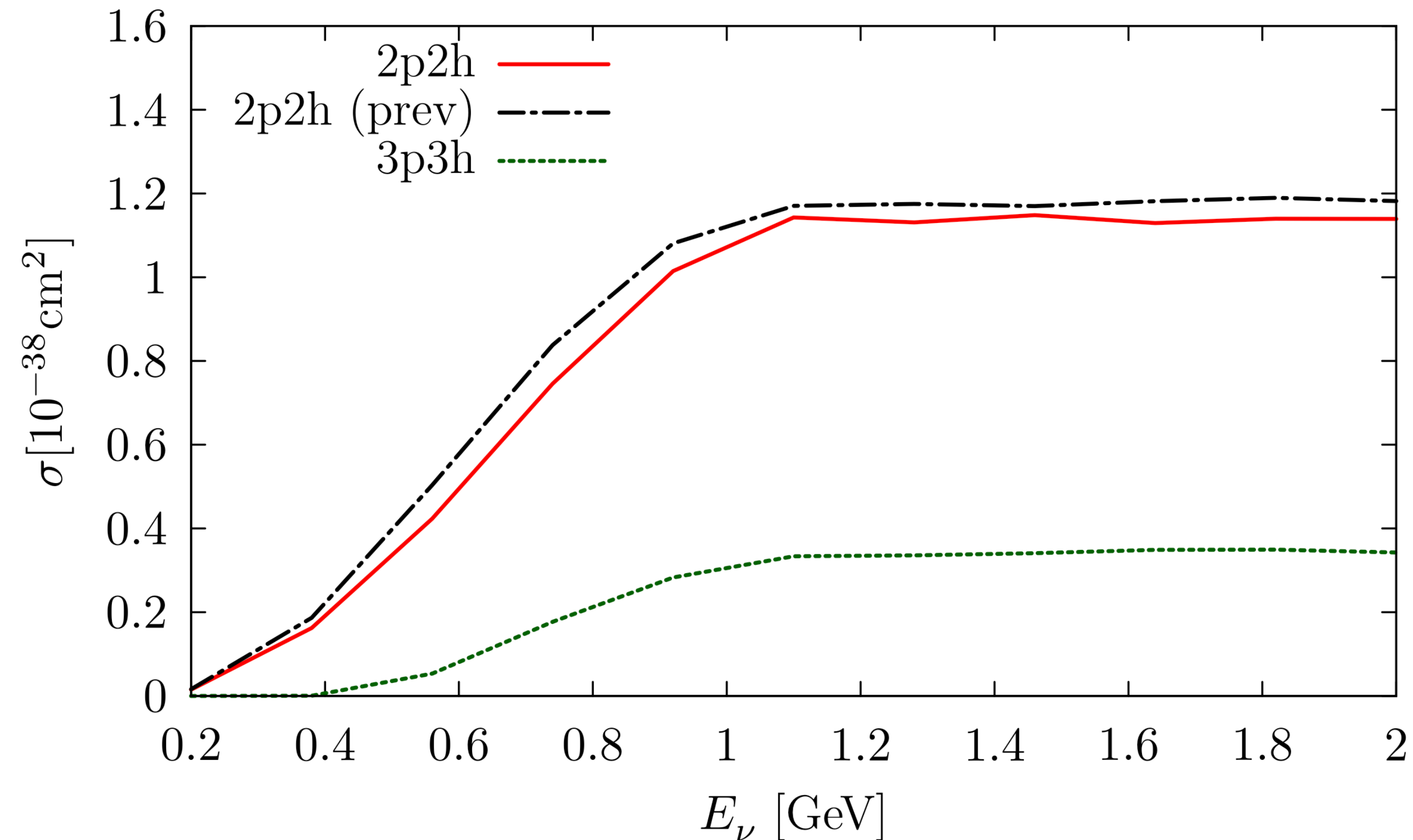


Figure from J.E. Sobczyk *et al.* Phys. Rev. C. 102 (2020) 024601

The 2020 Valencia model

J. E. Sobczyk *et al.* Phys. Rev. C. 102 (2020) 024601

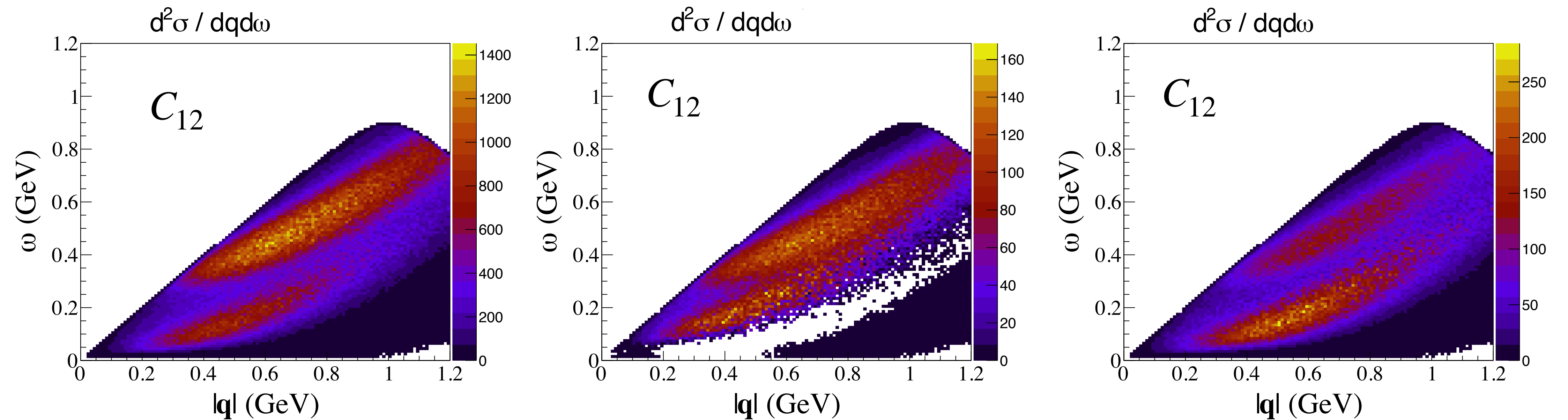


Figure from code by J.E. Sobczyk

The 2020 Valencia model

J. E. Sobczyk *et al.* Phys. Rev. C. 102 (2020) 024601

- Correlation strength **differs** based on different outgoing nucleon pair.
- Different outgoing pair **must be treated differently** in Monte Carlo generators !

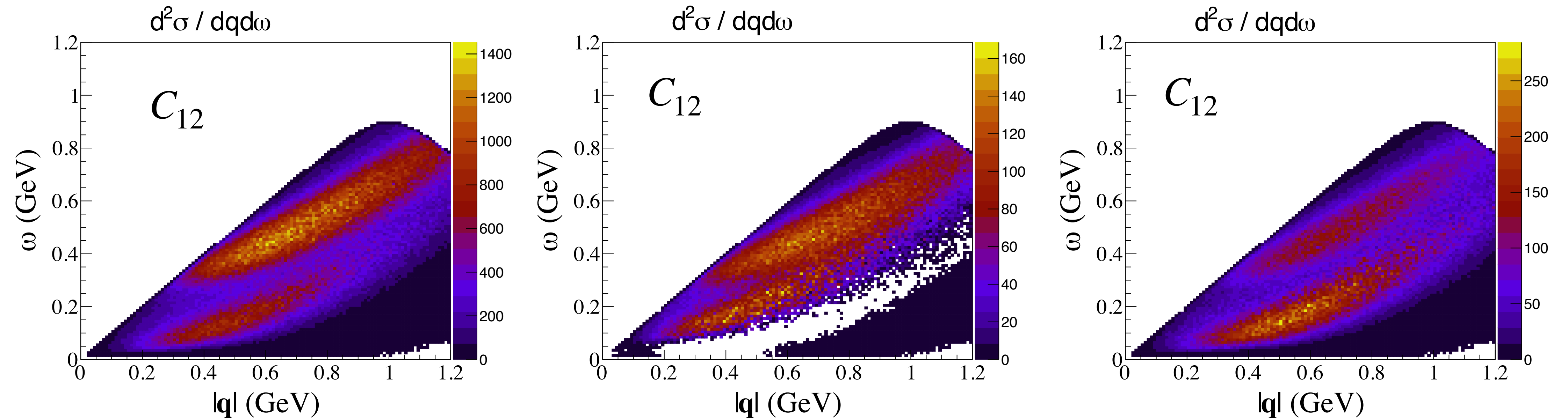


Figure from code by J.E. Sobczyk

The 2020 Valencia Model

J. E. Sobczyk *et al.* Phys. Rev. C. 102 (2020) 024601

- Predictions on momenta of outgoing nucleons.

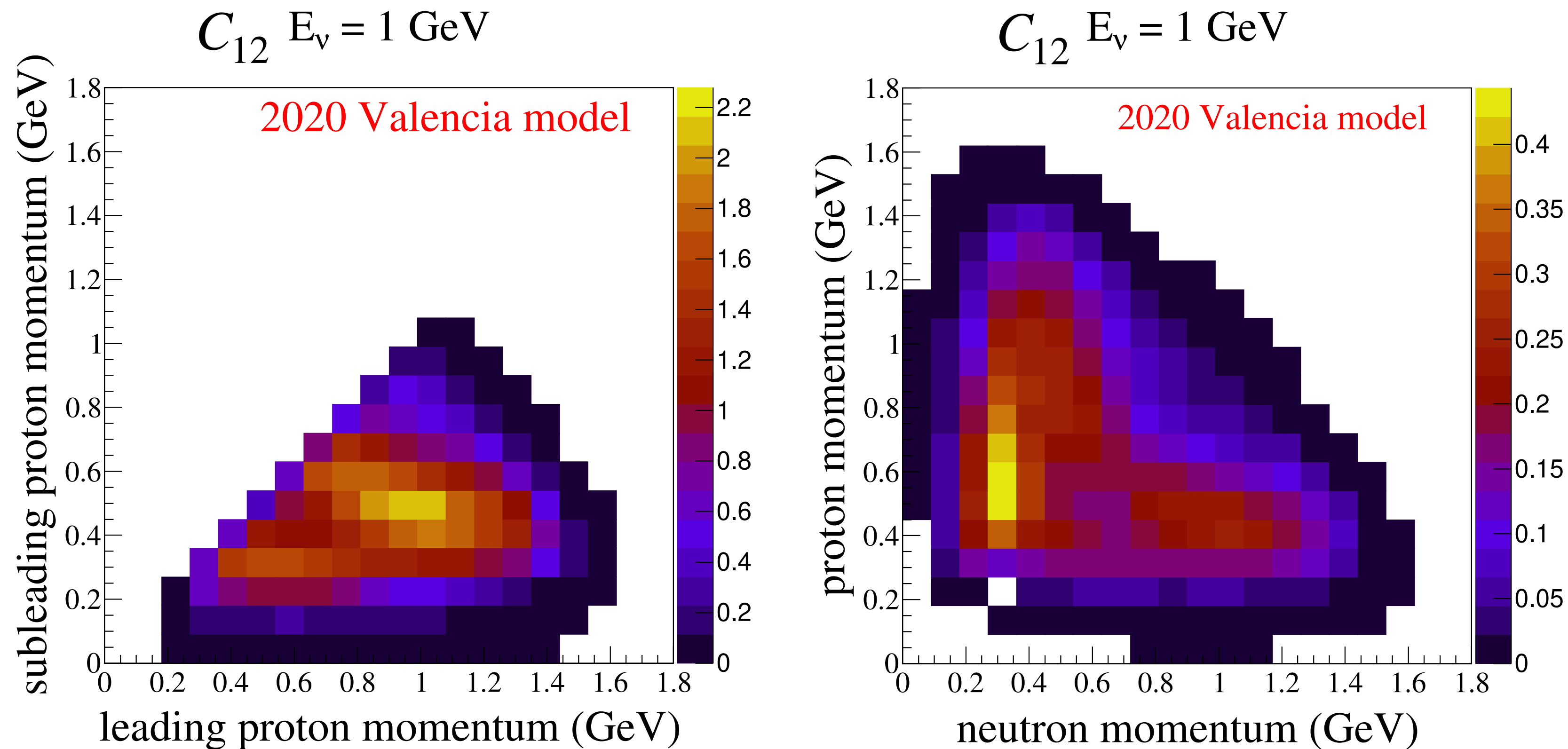
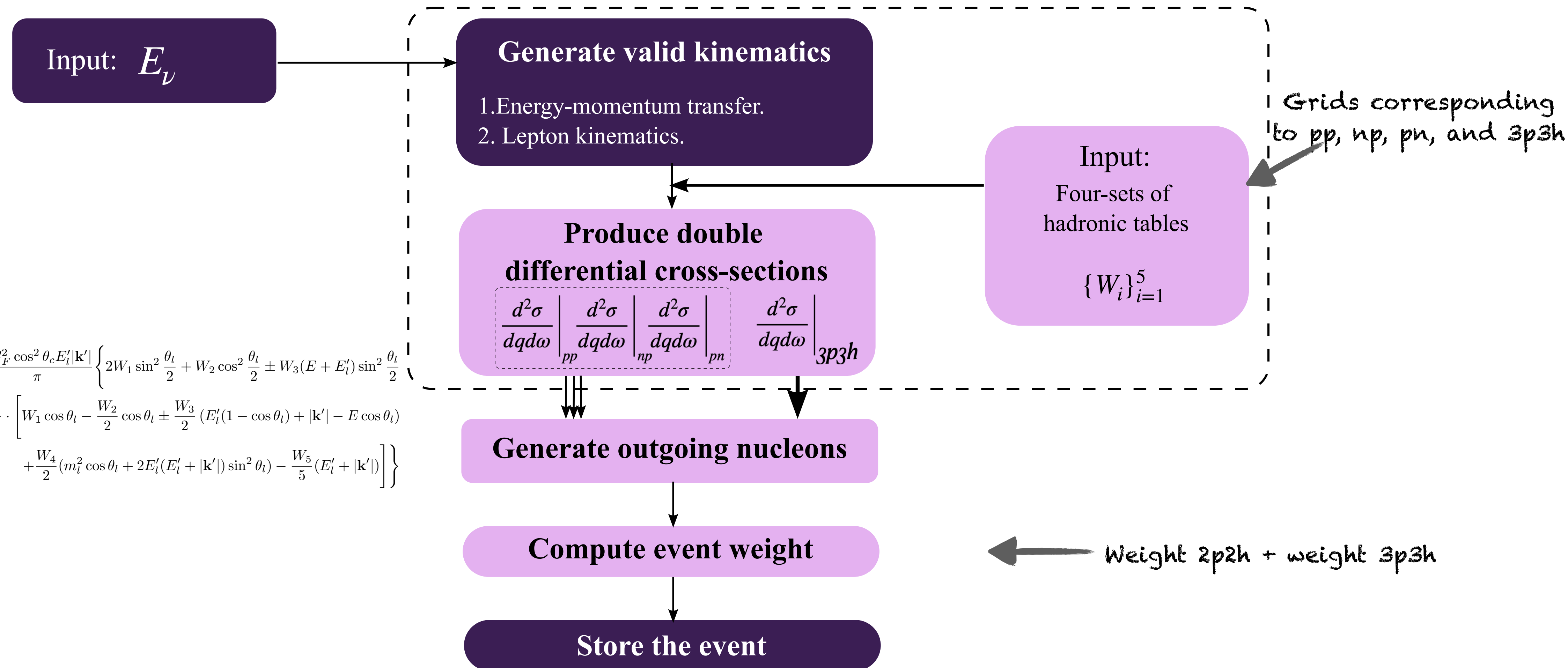


Figure from code by J.E. Sobczyk

NuWro implementation of the 2020 Valencia model

New implementation of MEC models

Within the context of the 2020 Valencia model

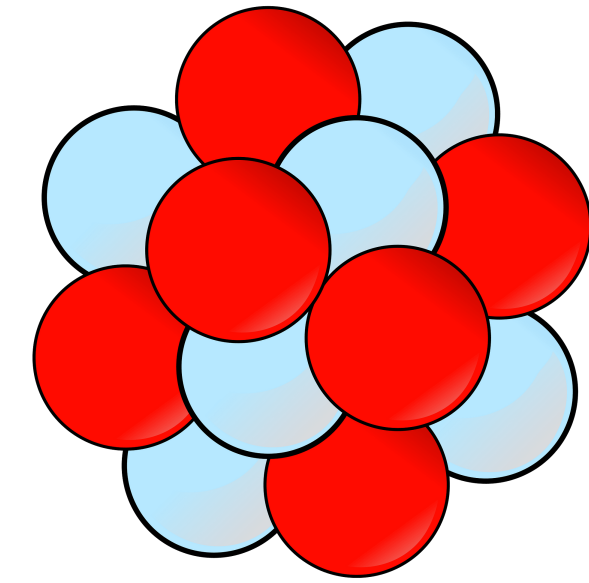


$$\frac{d^2\sigma}{dE'_l d\cos\theta_l} = \frac{2G_F^2 \cos^2\theta_c E'_l |\mathbf{k}'|}{\pi} \left\{ 2W_1 \sin^2 \frac{\theta_l}{2} + W_2 \cos^2 \frac{\theta_l}{2} \pm W_3 (E + E'_l) \sin^2 \frac{\theta_l}{2} \right. \\ \left. + \frac{m_l^2}{(E'_l + |\mathbf{k}'|) E'_l} \cdot \left[W_1 \cos\theta_l - \frac{W_2}{2} \cos\theta_l \pm \frac{W_3}{2} (E'_l (1 - \cos\theta_l) + |\mathbf{k}'| - E \cos\theta_l) \right. \right. \\ \left. \left. + \frac{W_4}{2} (m_l^2 \cos\theta_l + 2E'_l (E'_l + |\mathbf{k}'|) \sin^2 \theta_l) - \frac{W_5}{5} (E'_l + |\mathbf{k}'|) \right] \right\}$$

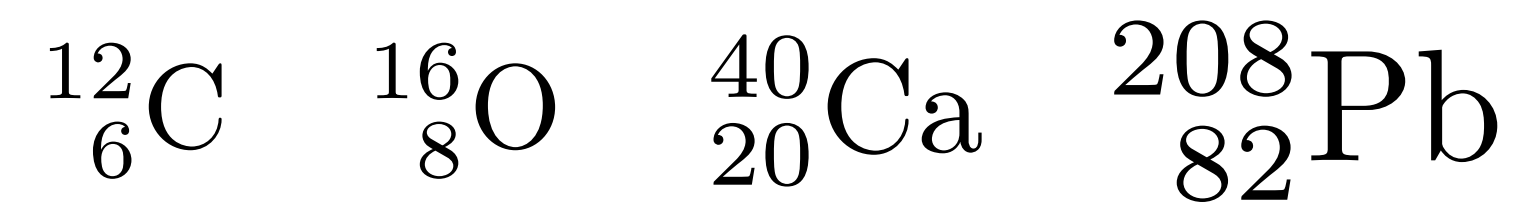
H. Prasad *et al.* arXiv:2501.11470v1

Hadronic tables

Grids available in NuWro 25.01



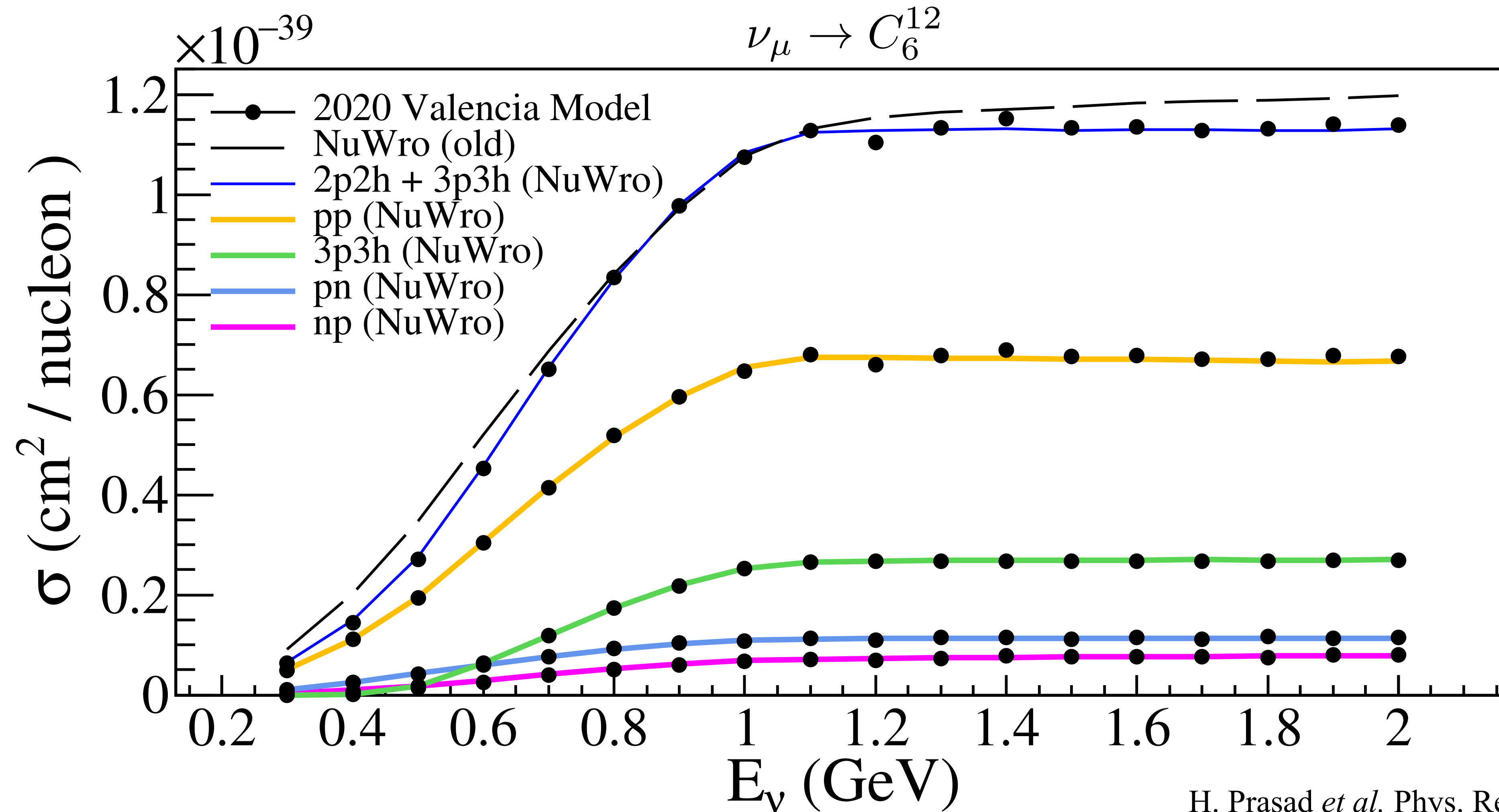
- Available hadronic grids for **mec_kind=6**



- **For the rest of the slides, I will show results for carbon only**

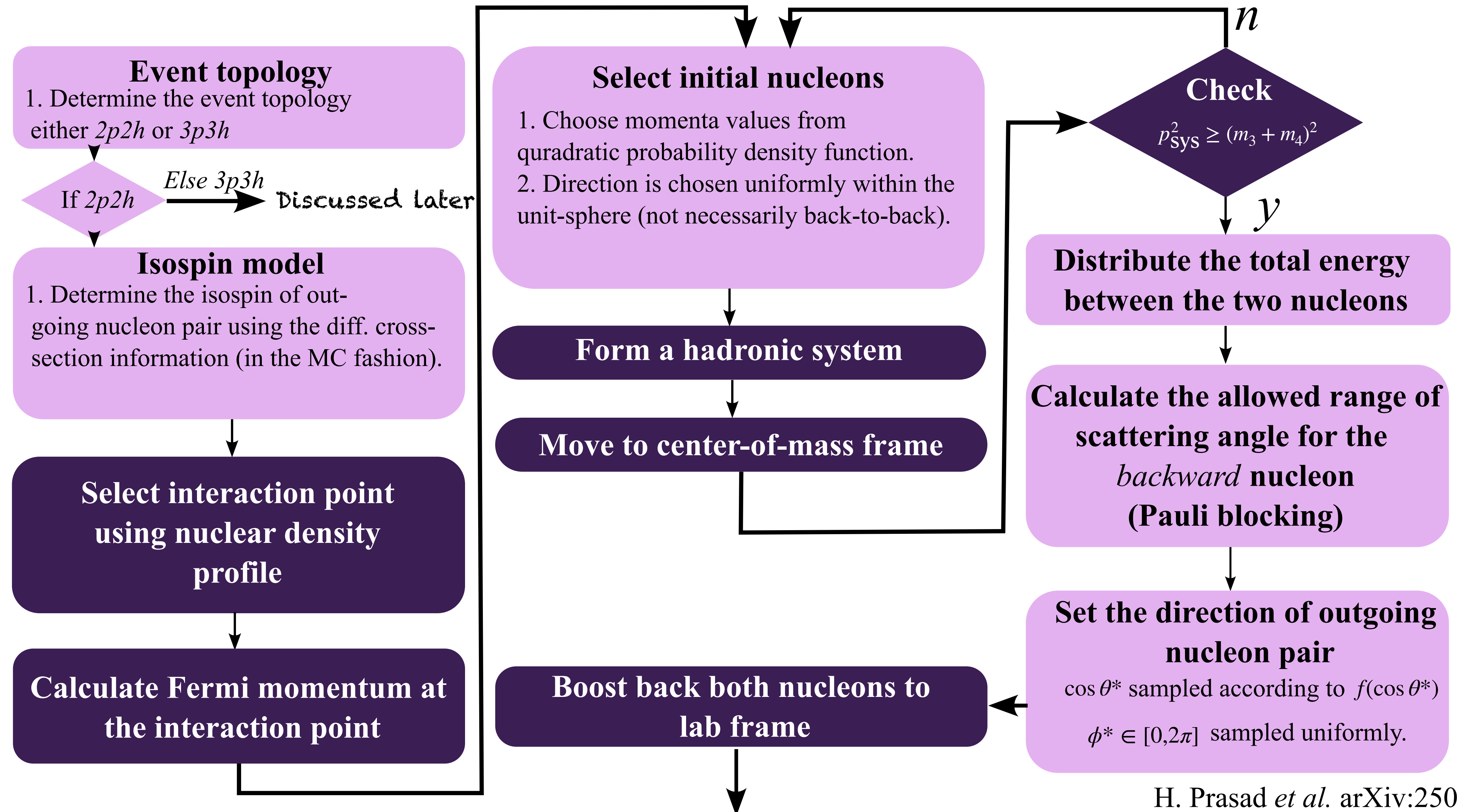
Inclusive part

Nuwro implementation of the 2020 Valencia model



H. Prasad *et al.* Phys. Rev. D 111 (2025) 036032

New implementation of MEC models

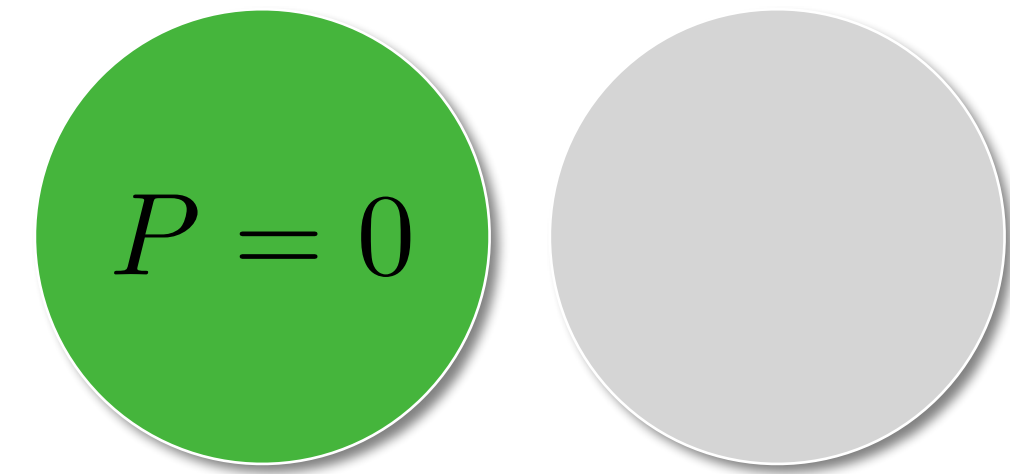
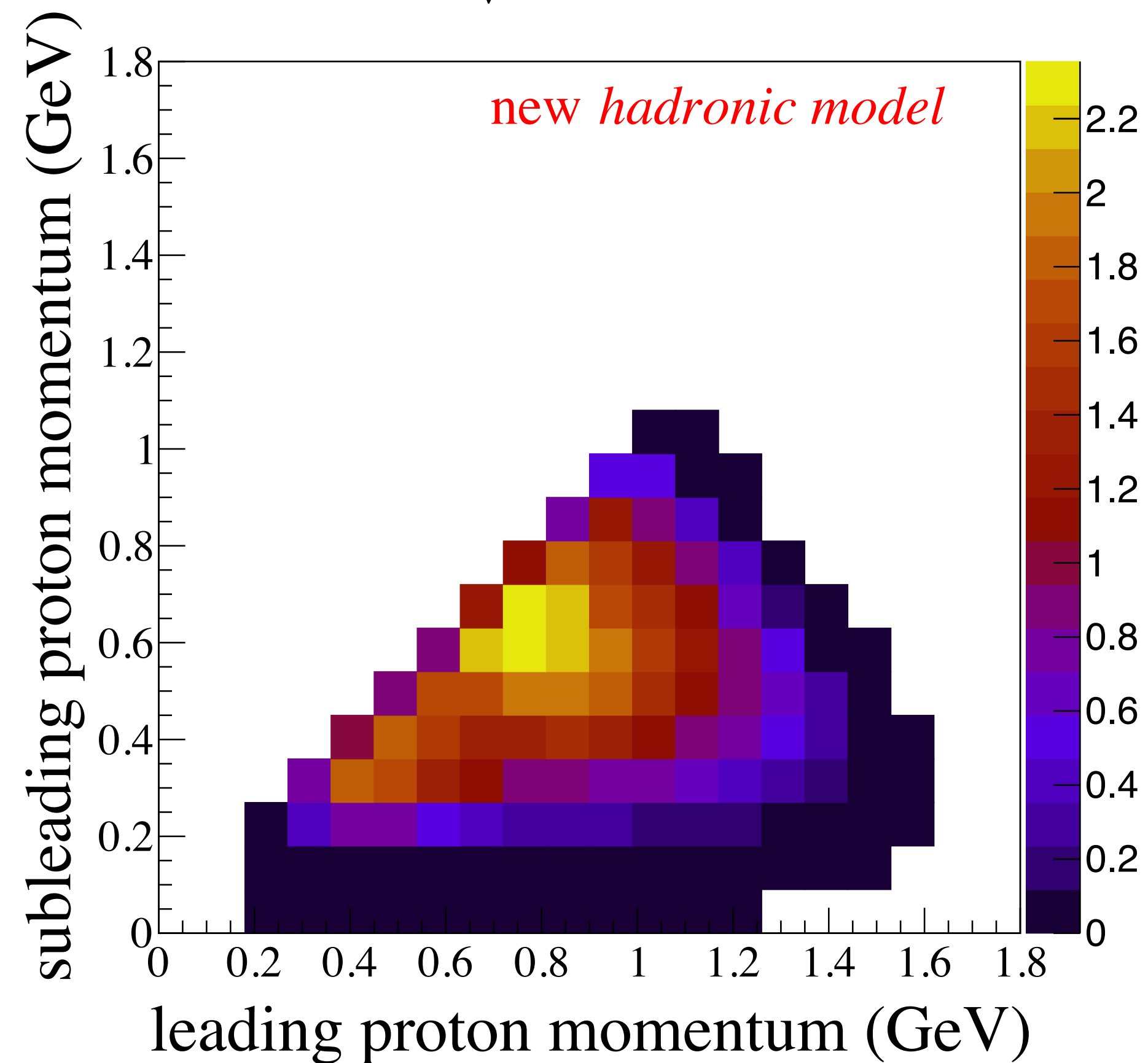


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Nucleon sampling function

Caught in action

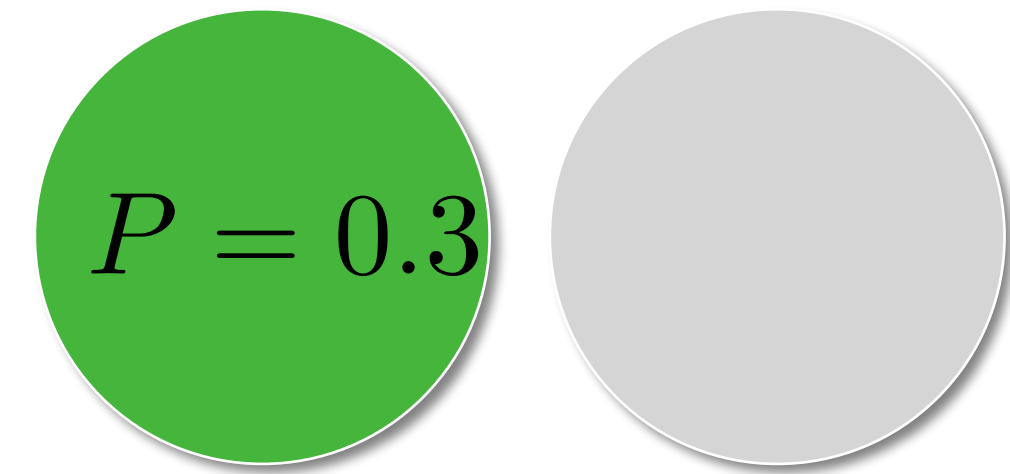
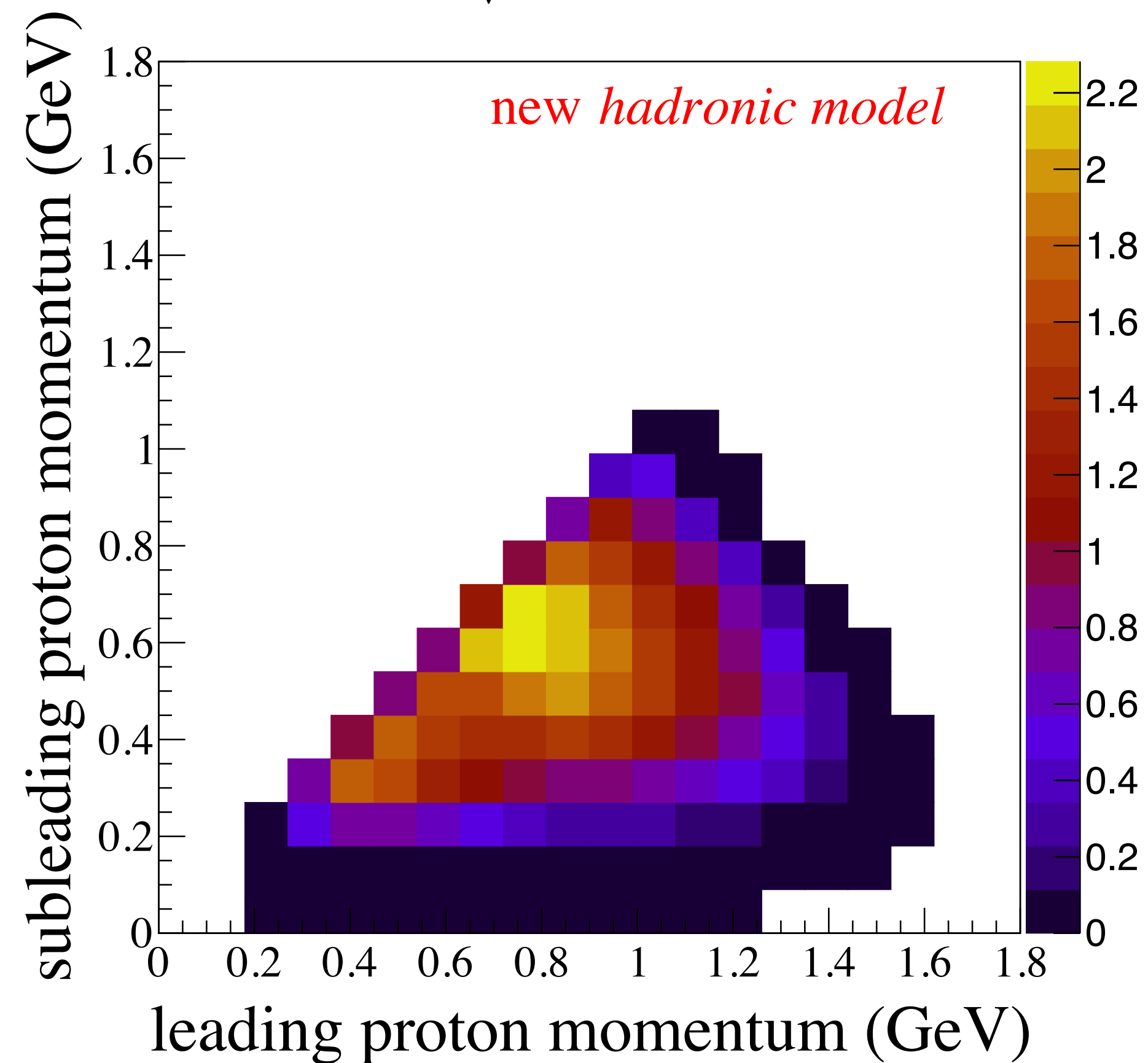
$$E_\nu = 1 \text{ GeV}$$



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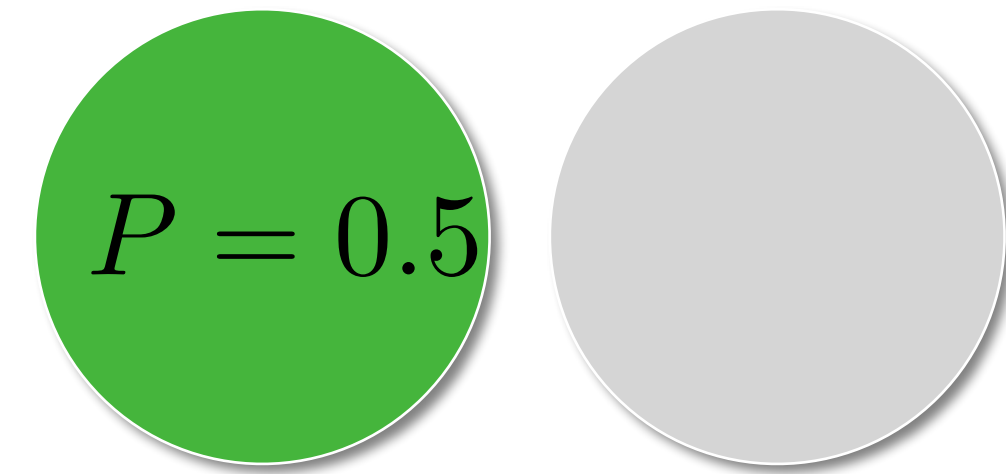
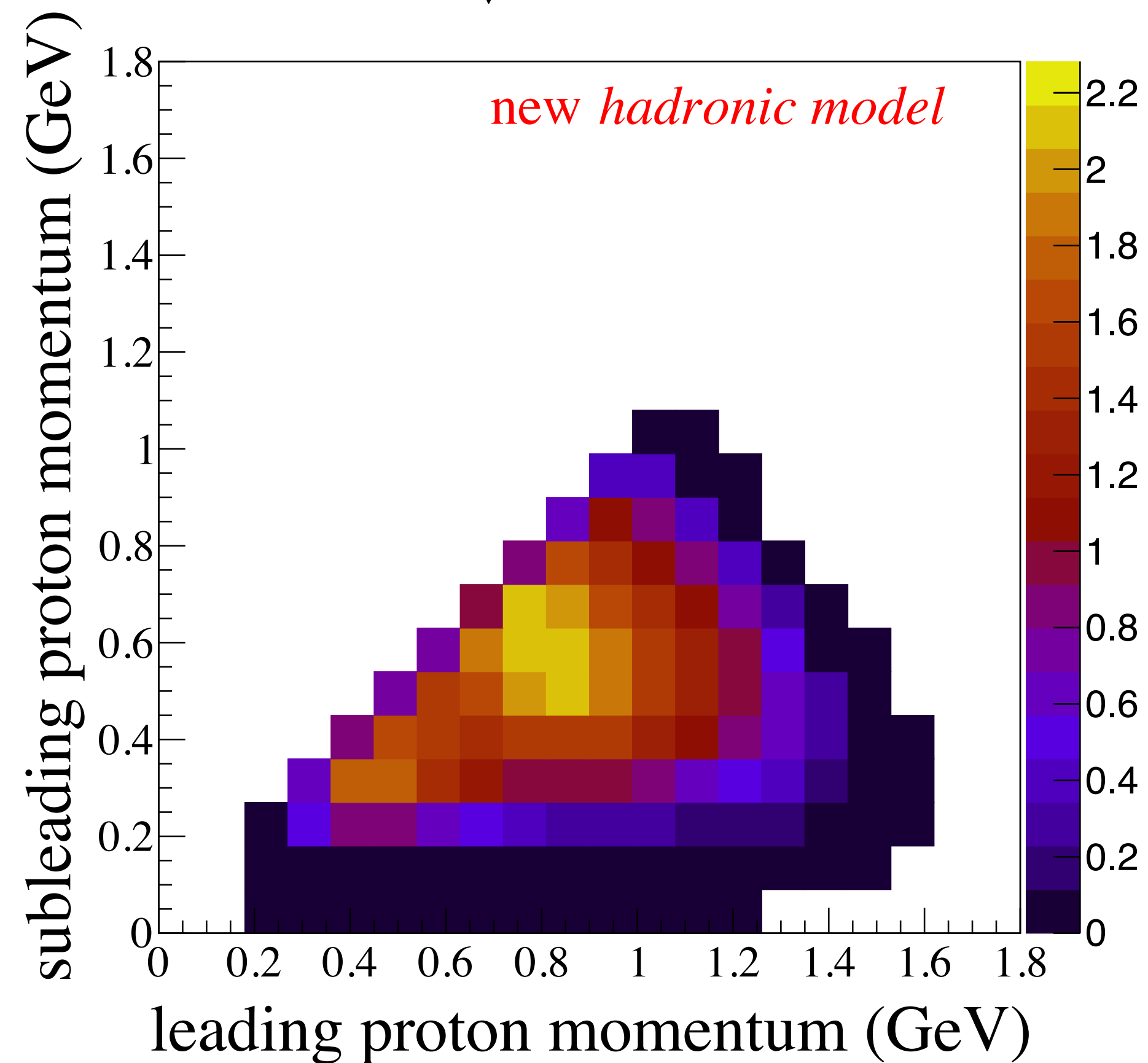
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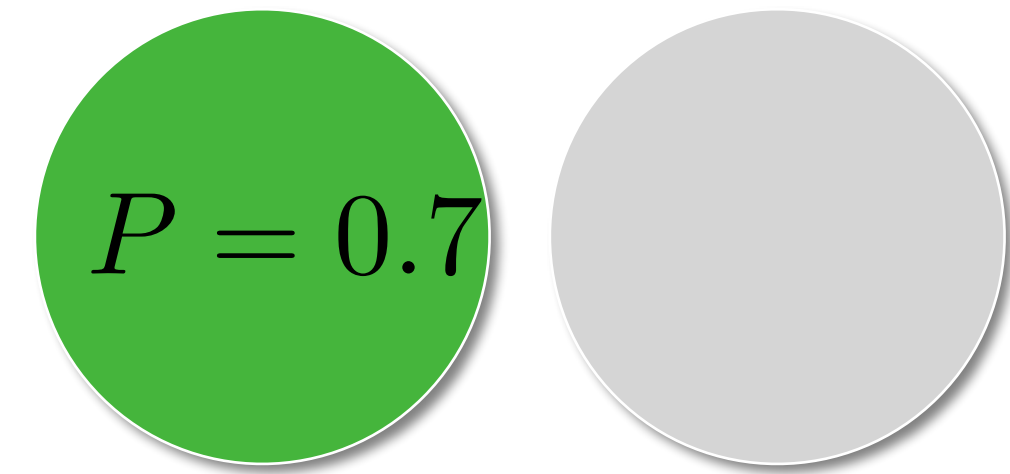
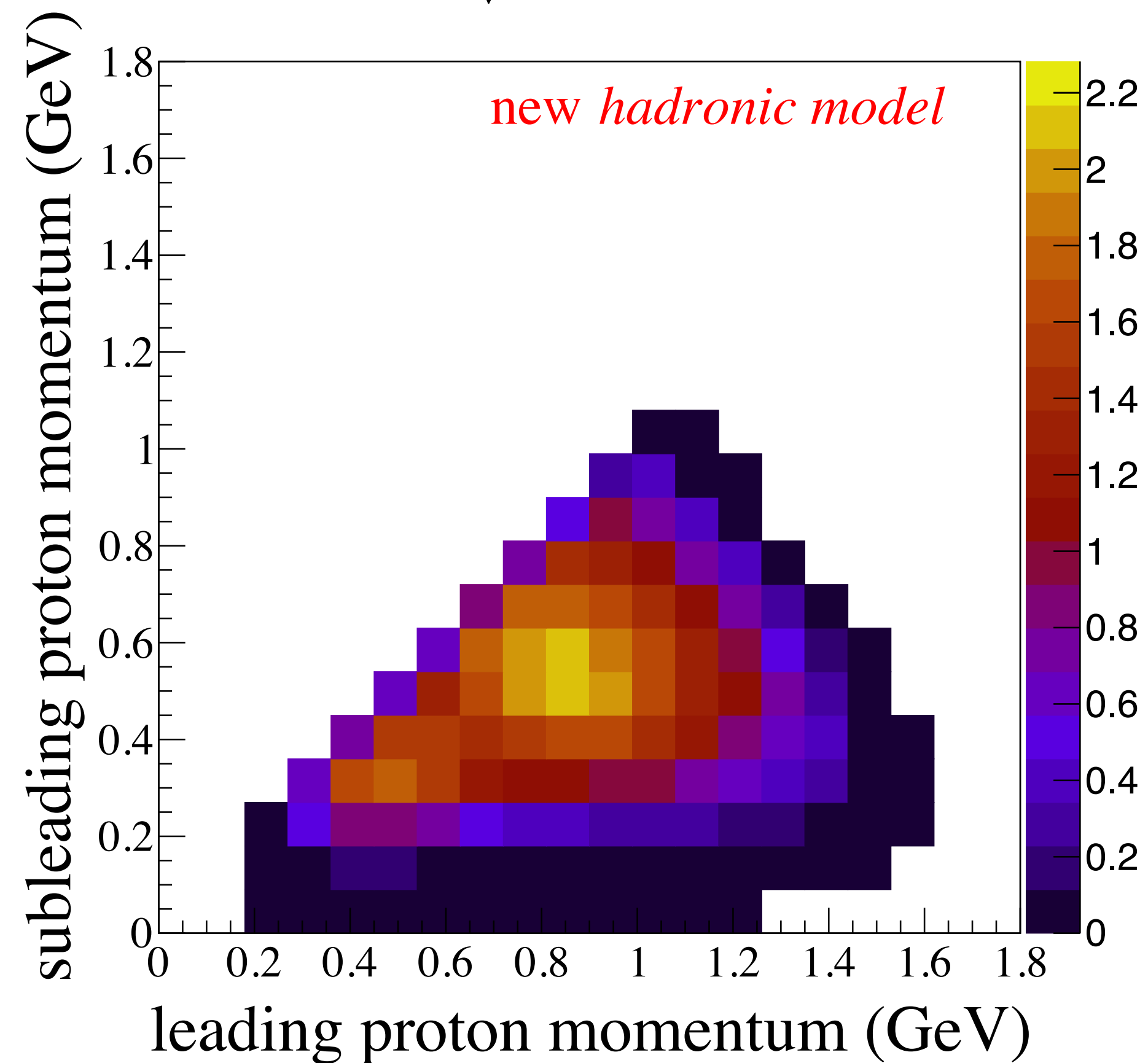
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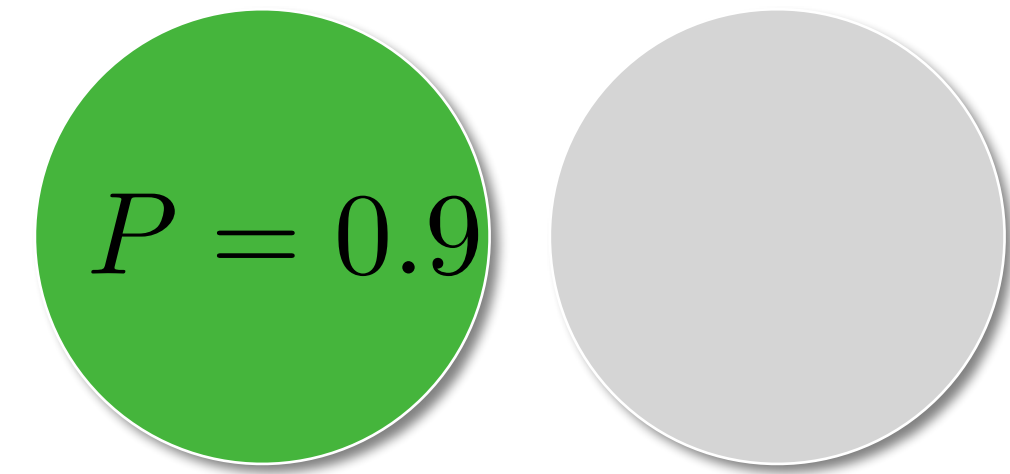
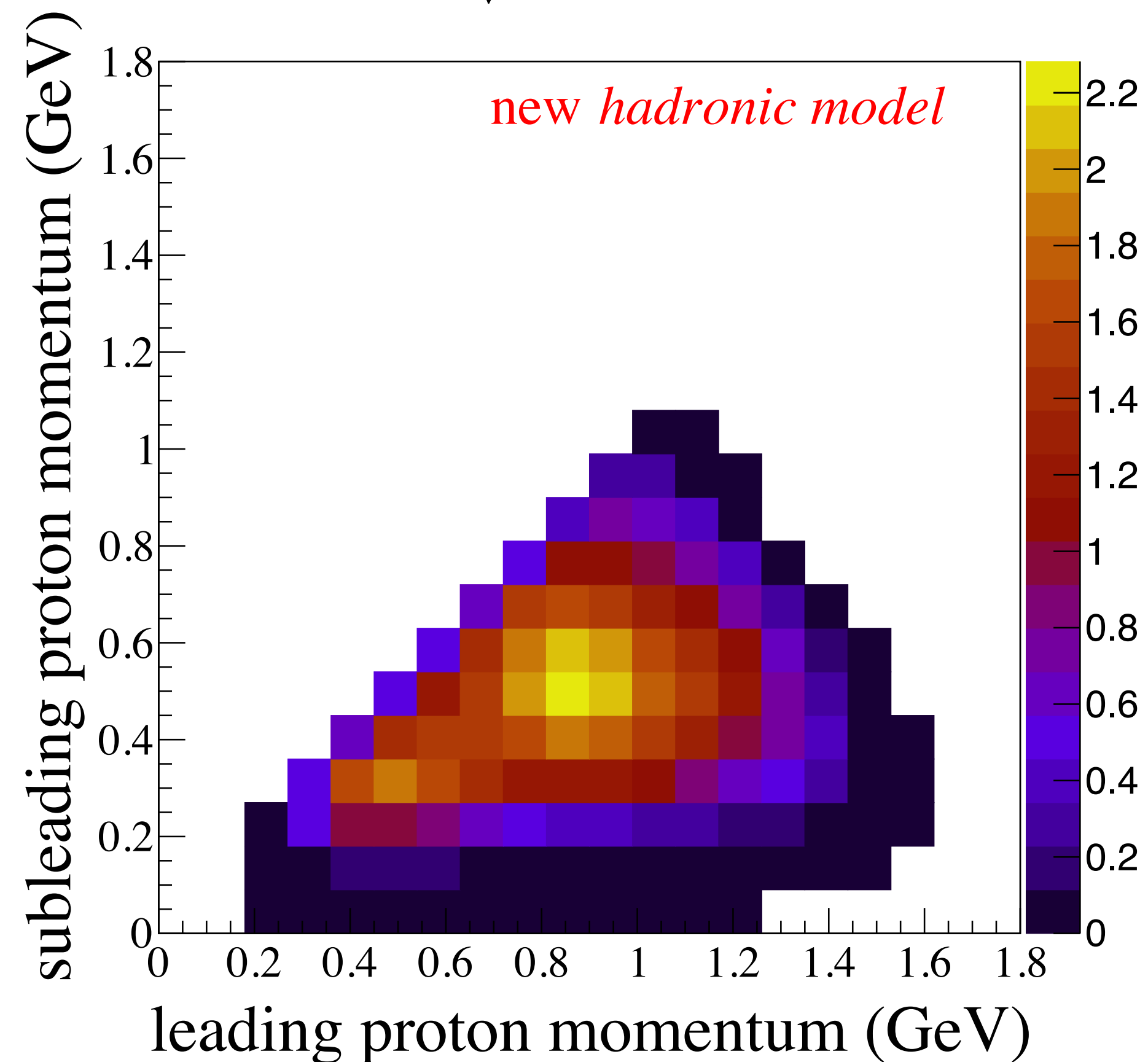
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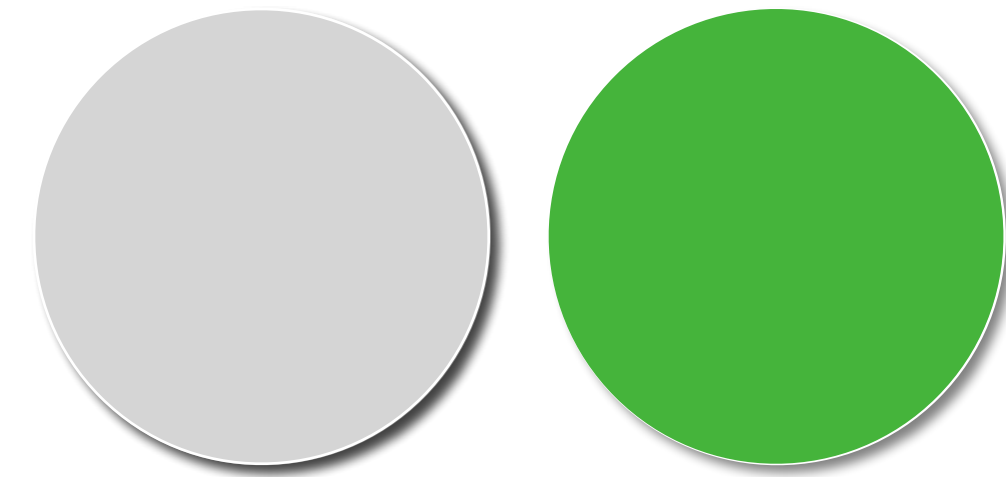
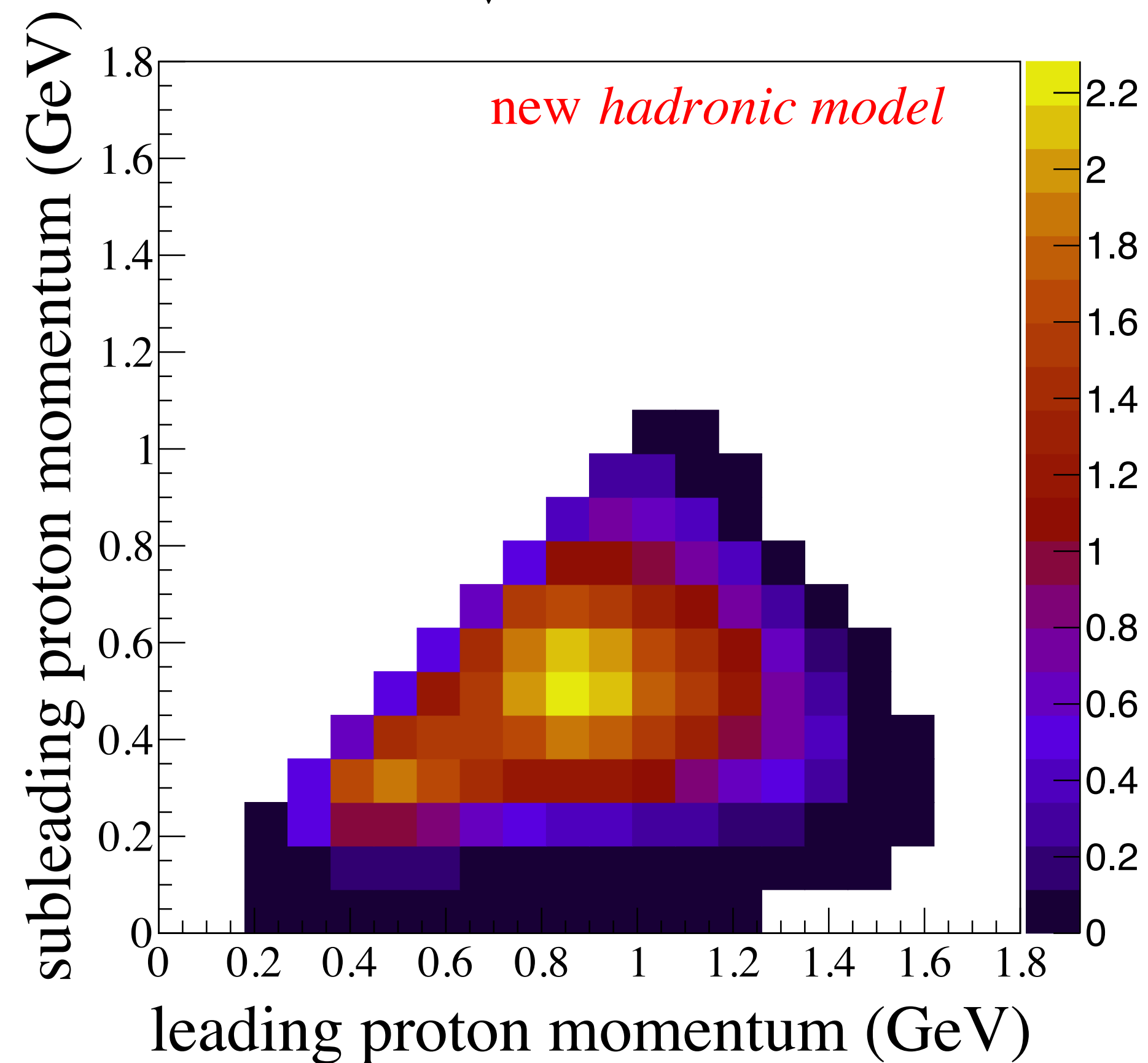
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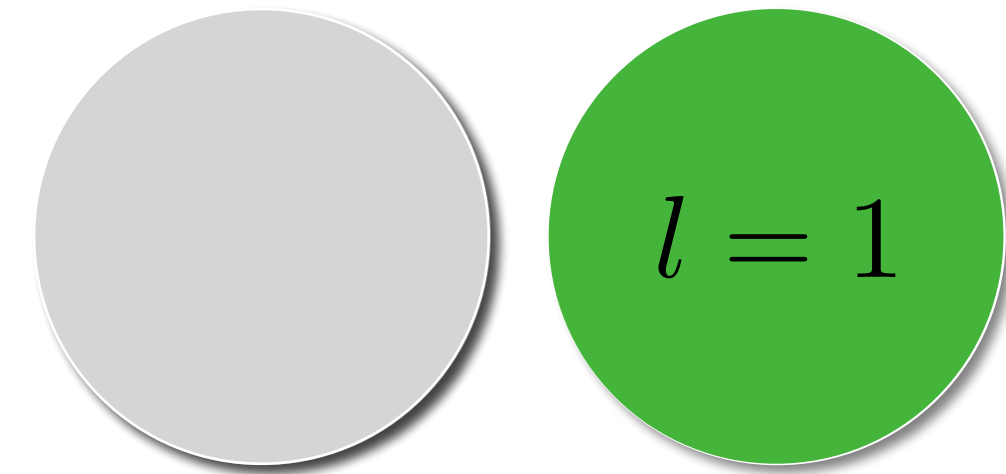
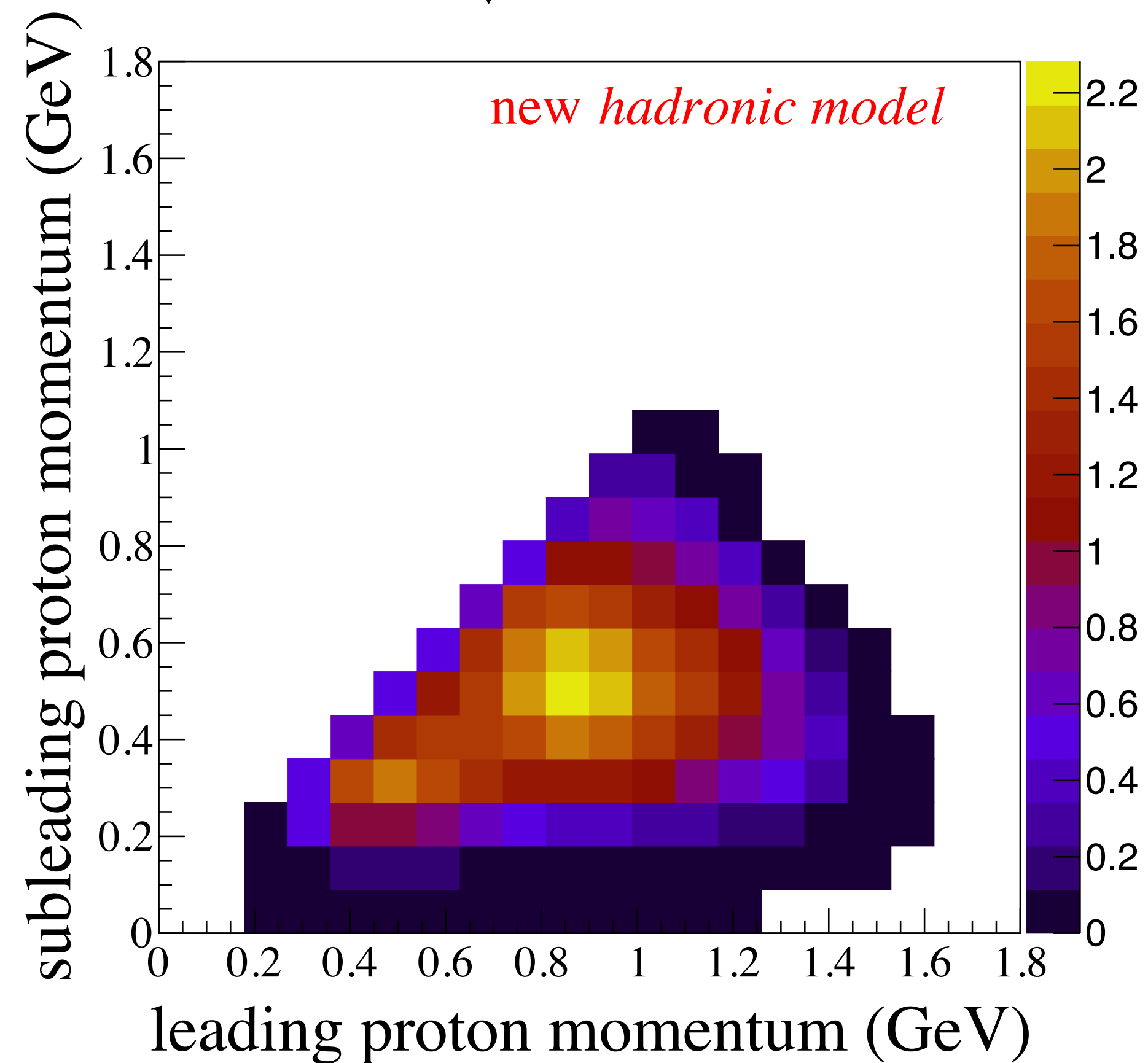
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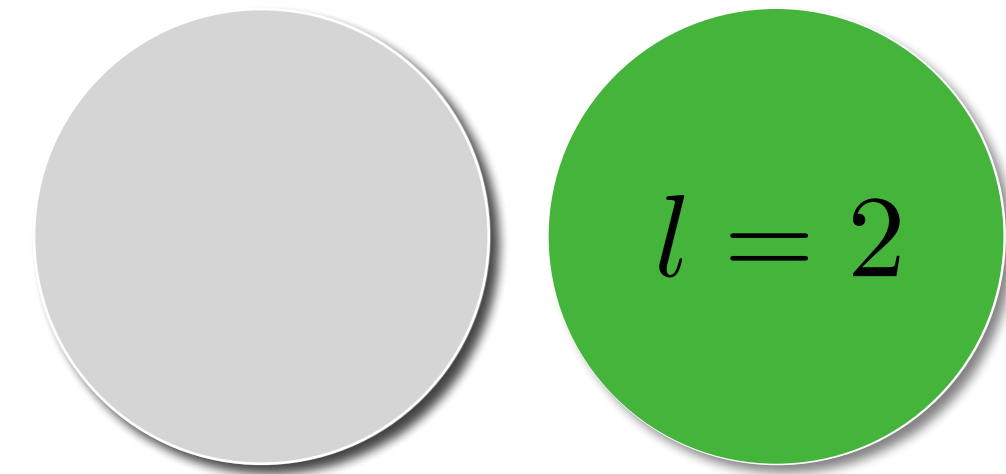
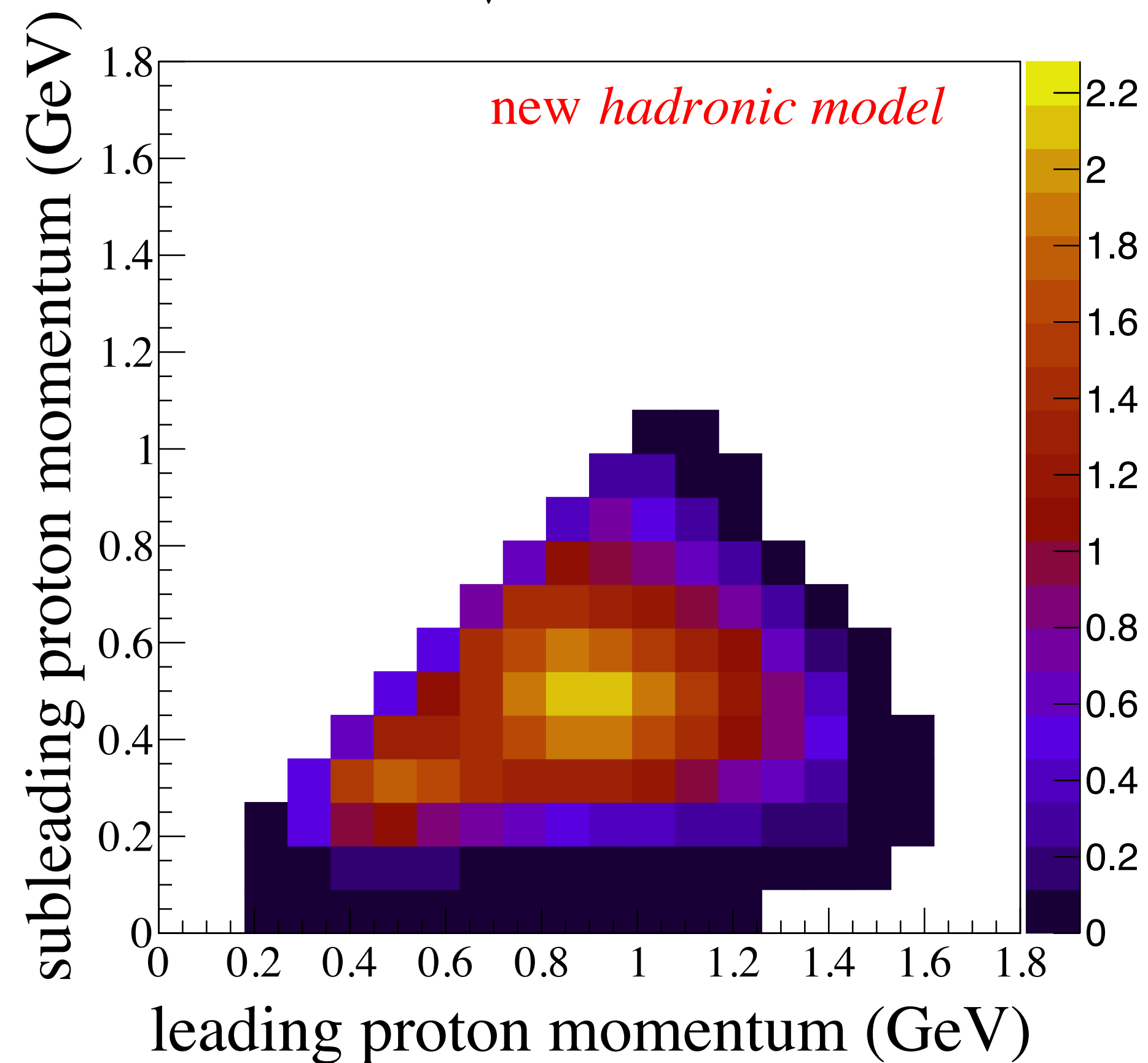
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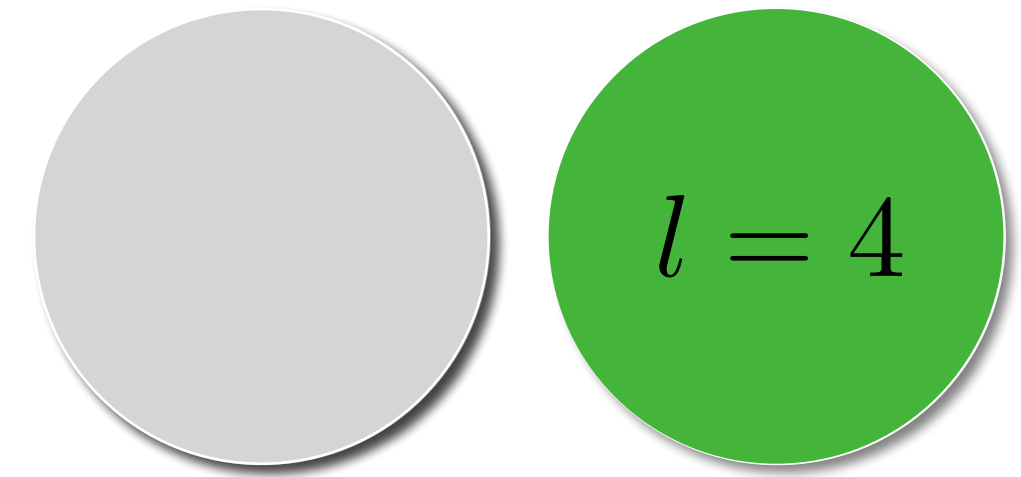
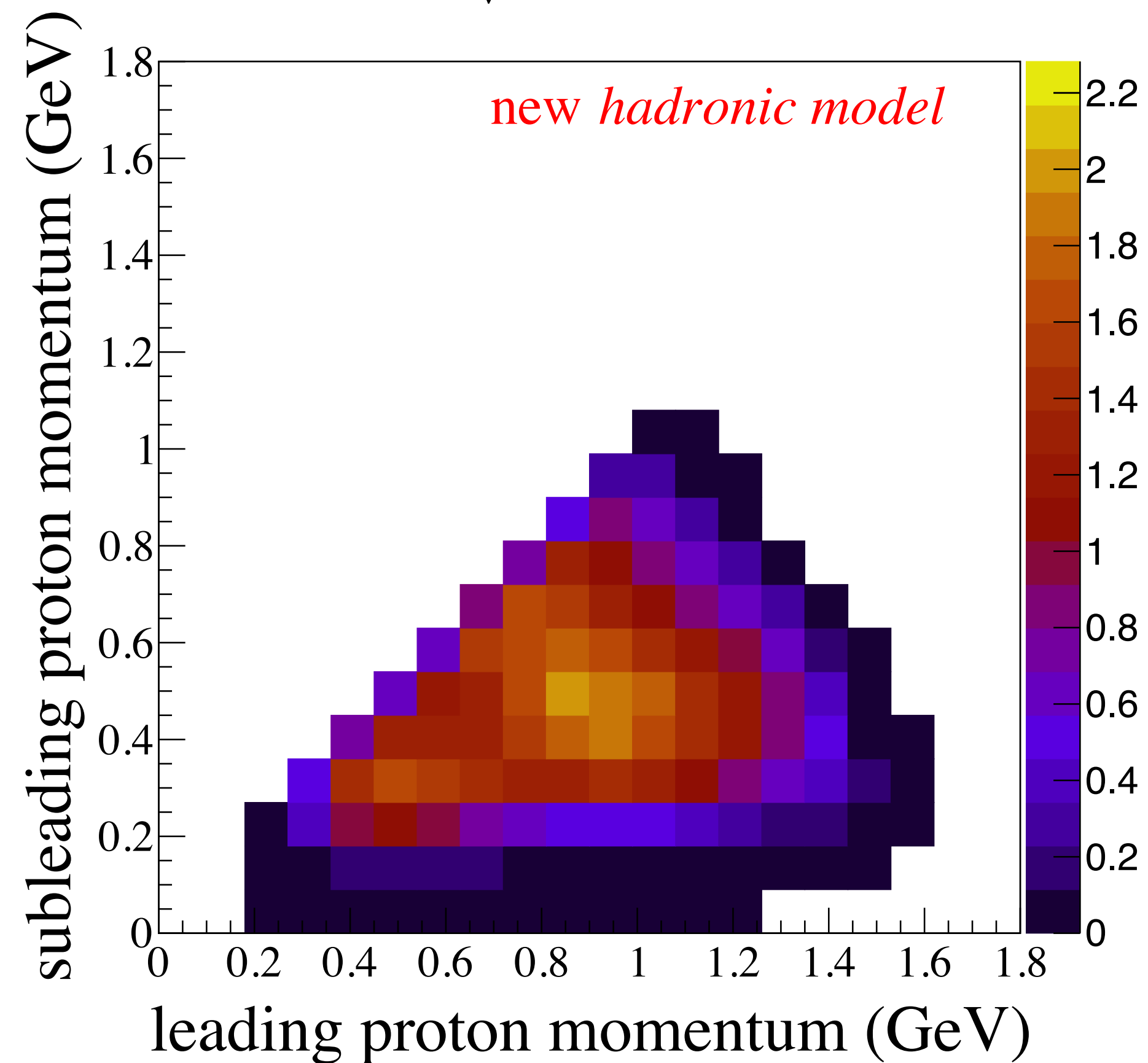
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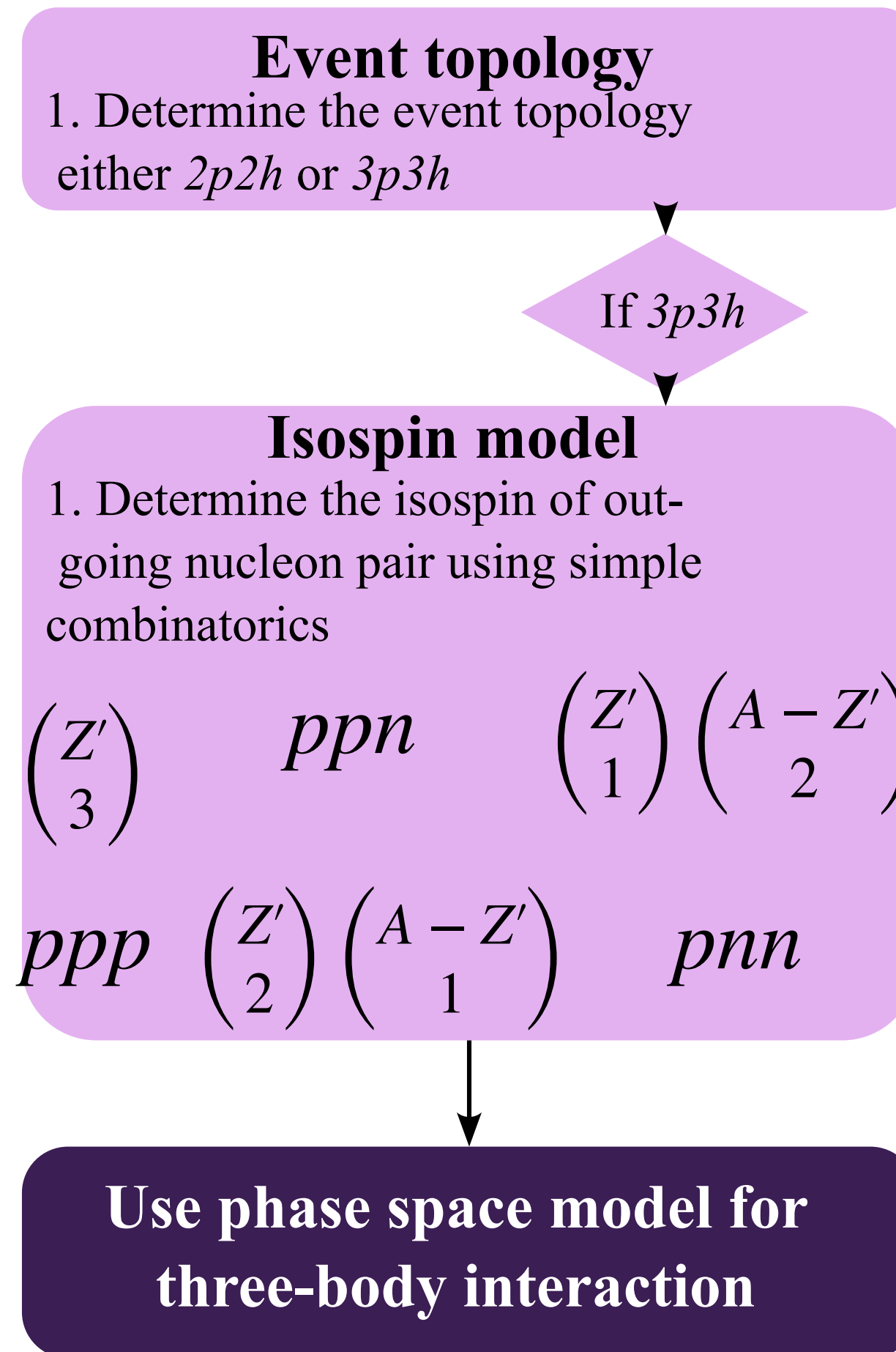
Nucleon sampling function

Caught in action

$$E_\nu = 1 \text{ GeV}$$



New implementation of MEC models



$$Z' = Z \pm 1$$

Depending on (anti) neutrino - nucleus interaction

Theory gives no hints so we choose combinatorics
for isospin and phase space for momenta !!

Implementation results

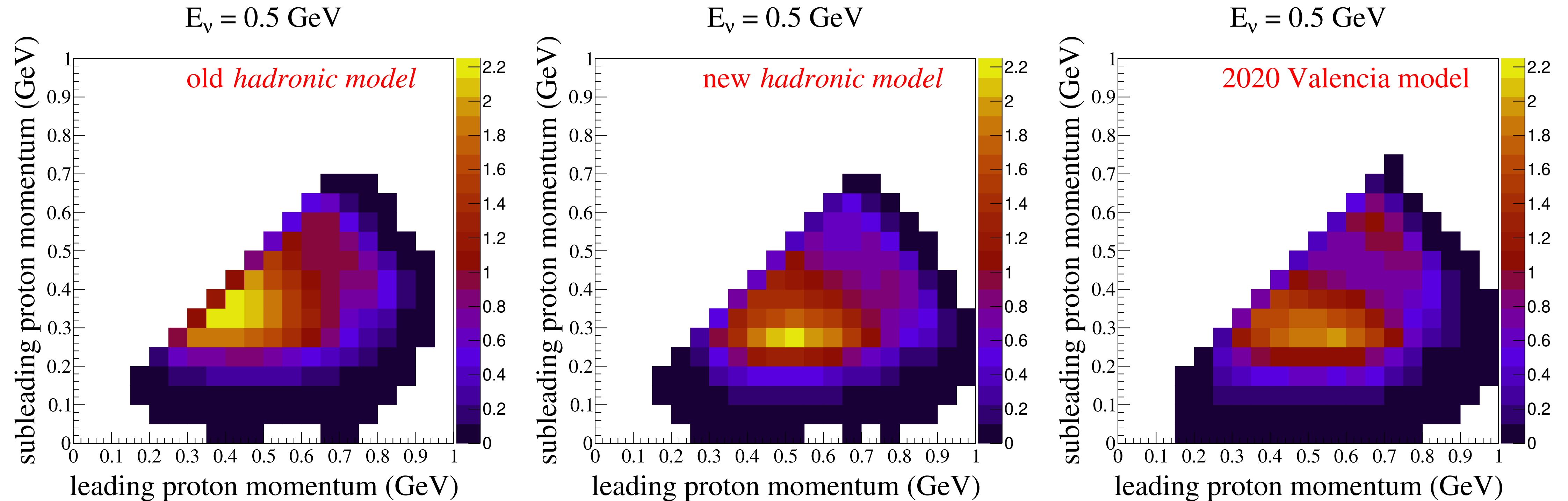
Momentum distribution of outgoing nucleons

Two proton in final state (primary vertex)

H. Prasad *et al.* Phys. Rev. D 111 (2025) 036032

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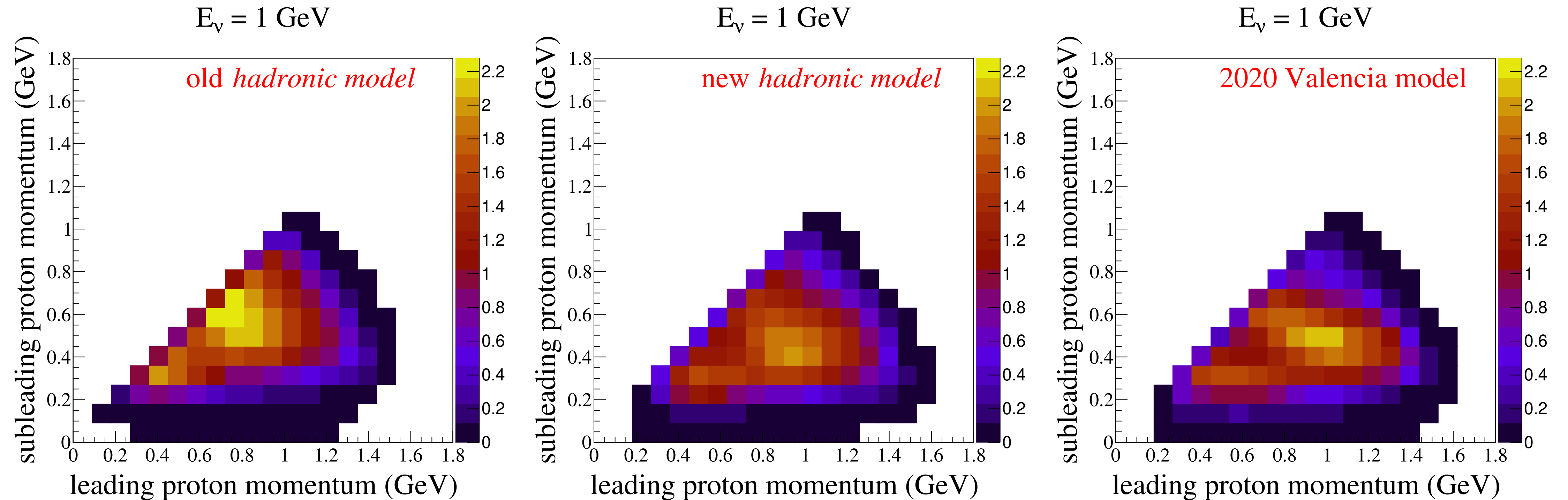
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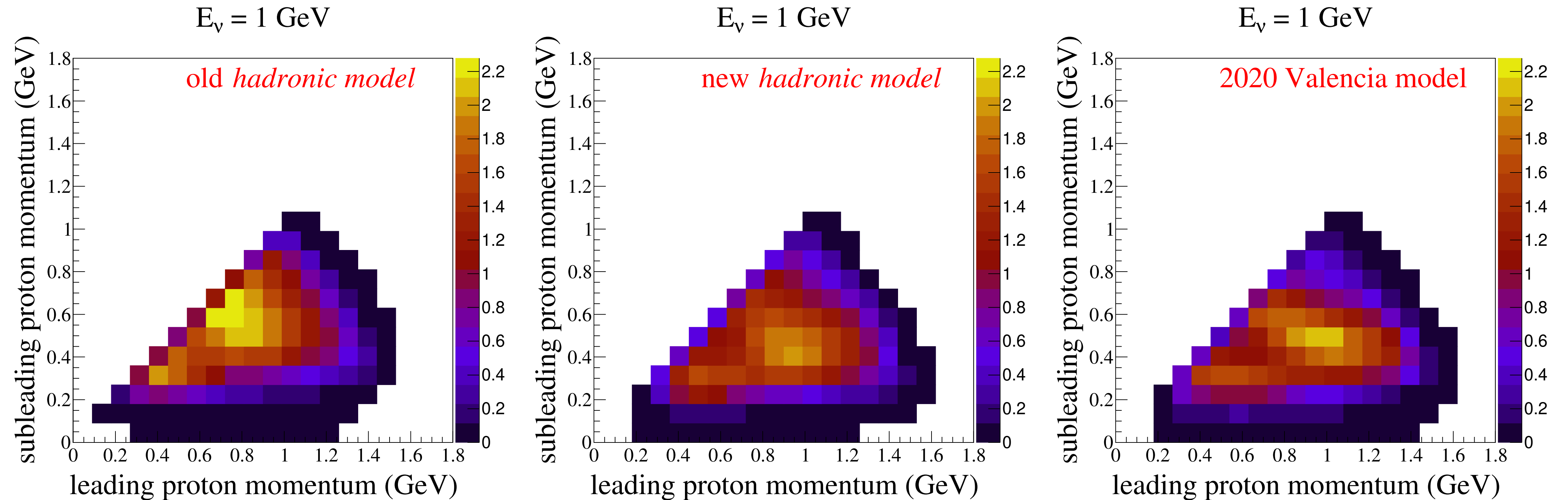
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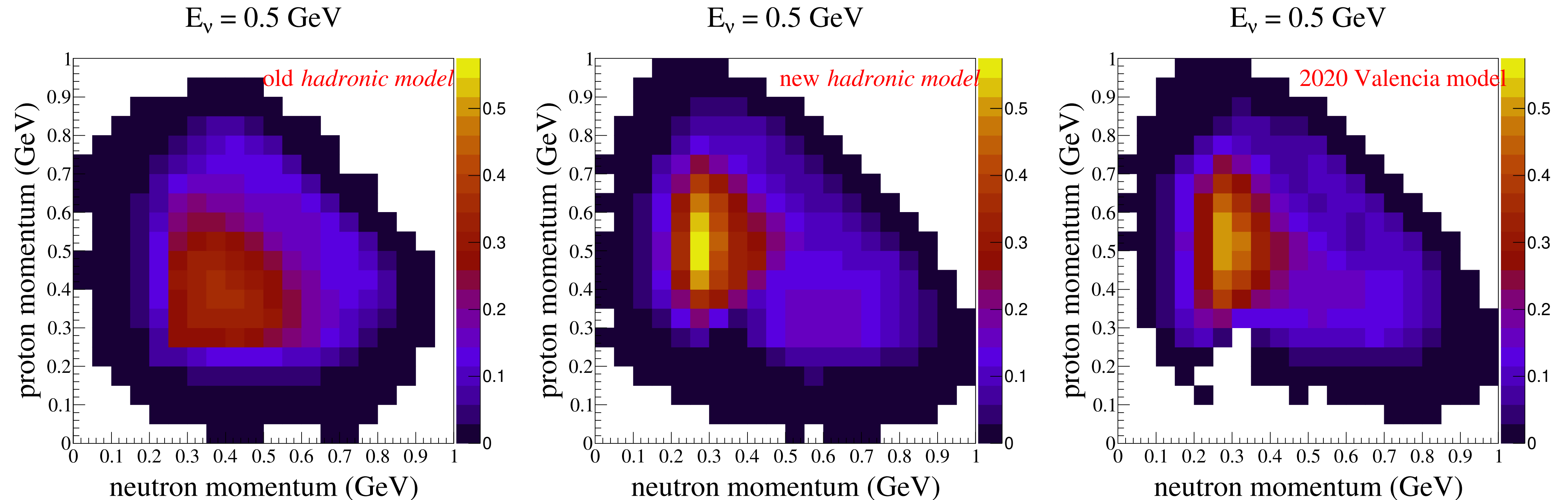
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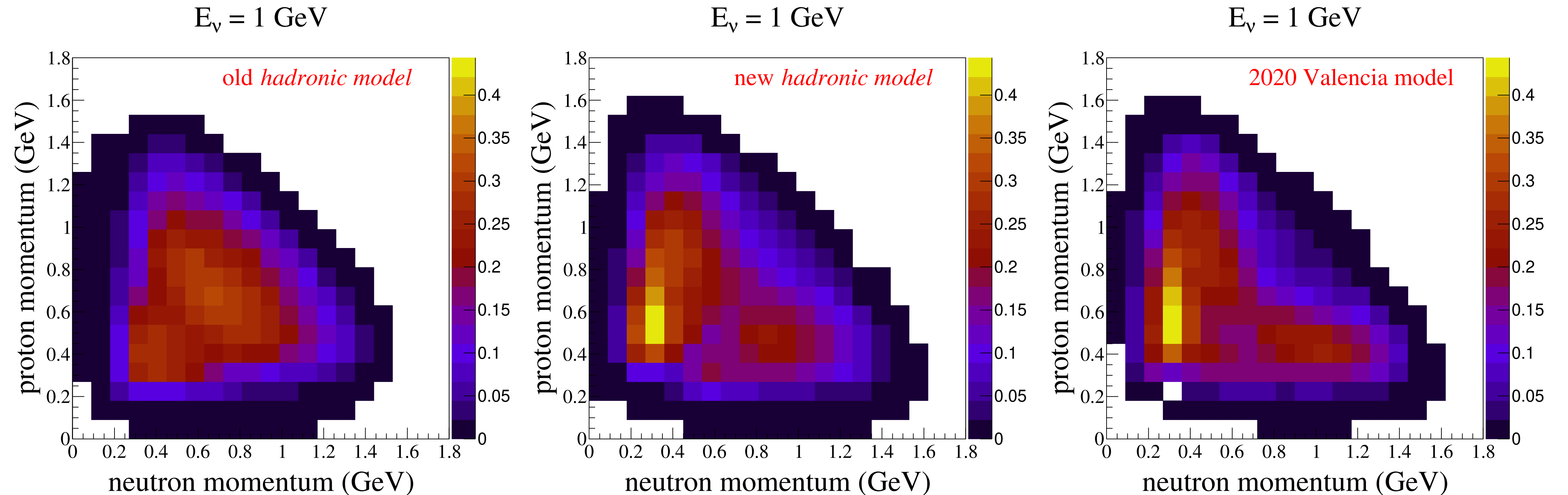
A neutron and a proton in the final state (primary vertex)



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Momentum distribution of outgoing nucleons

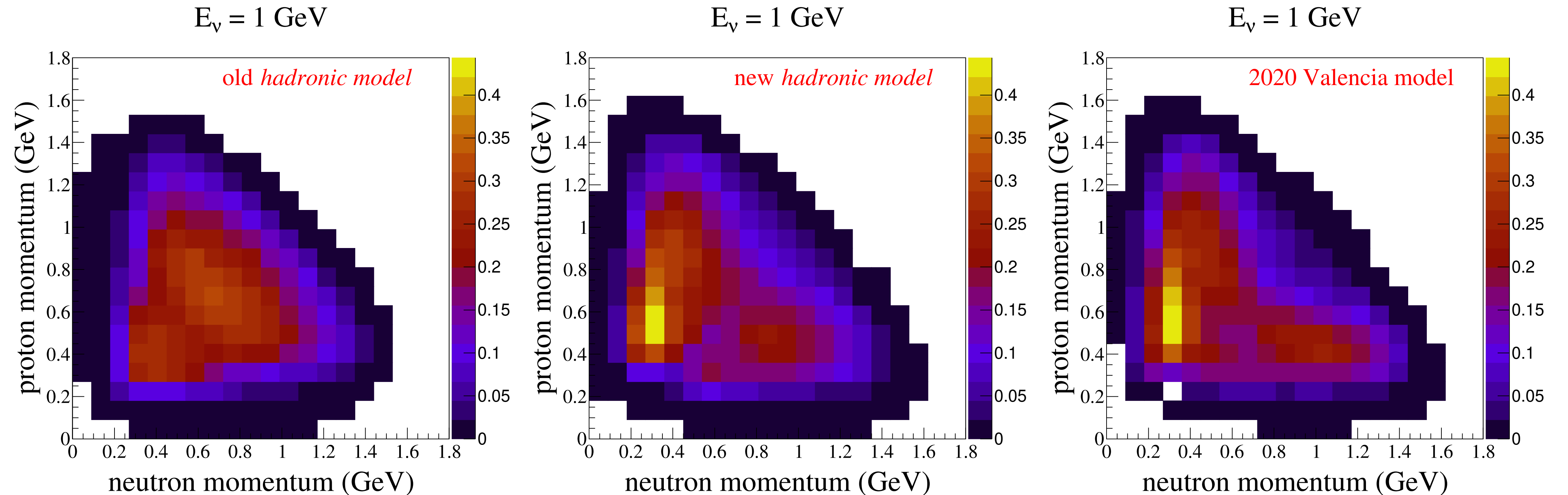
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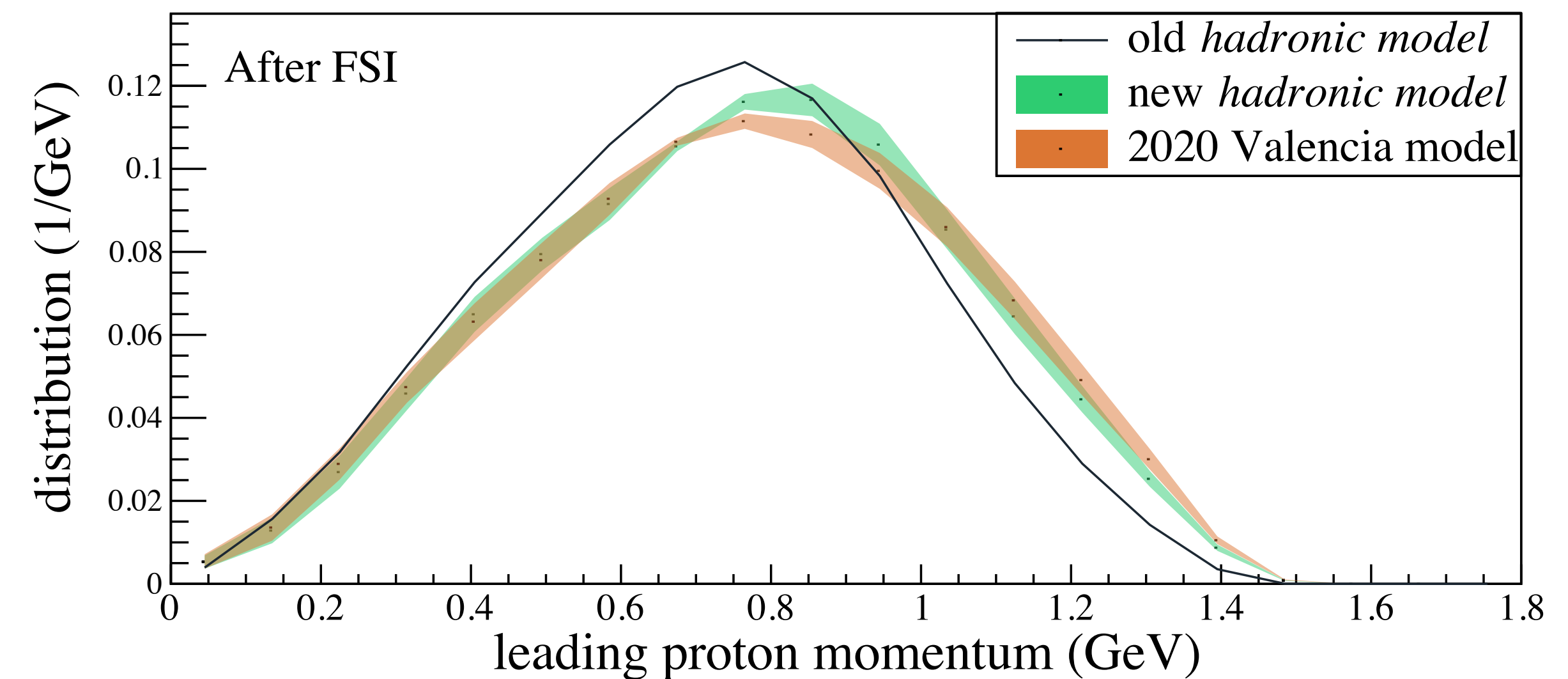
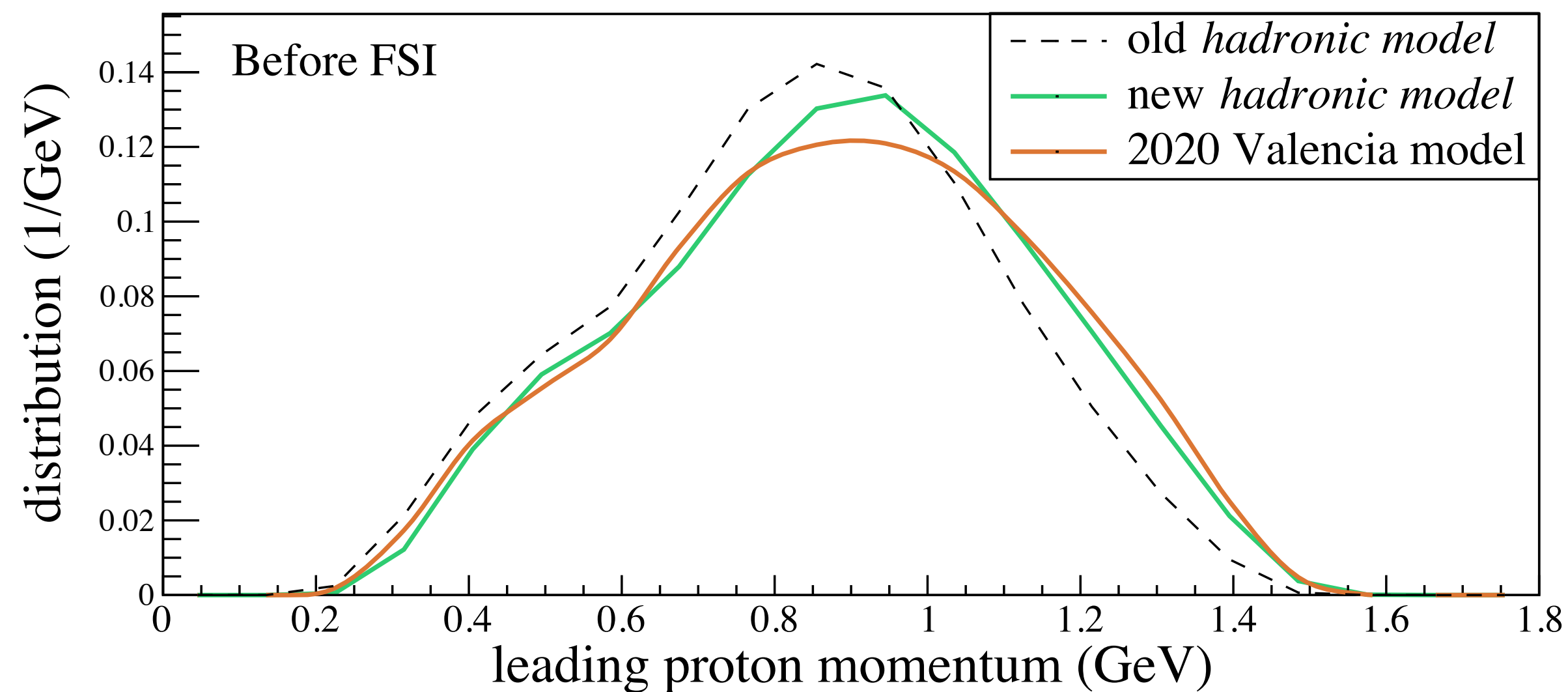
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Impact of FSI

A distribution of momentum of the maximal proton

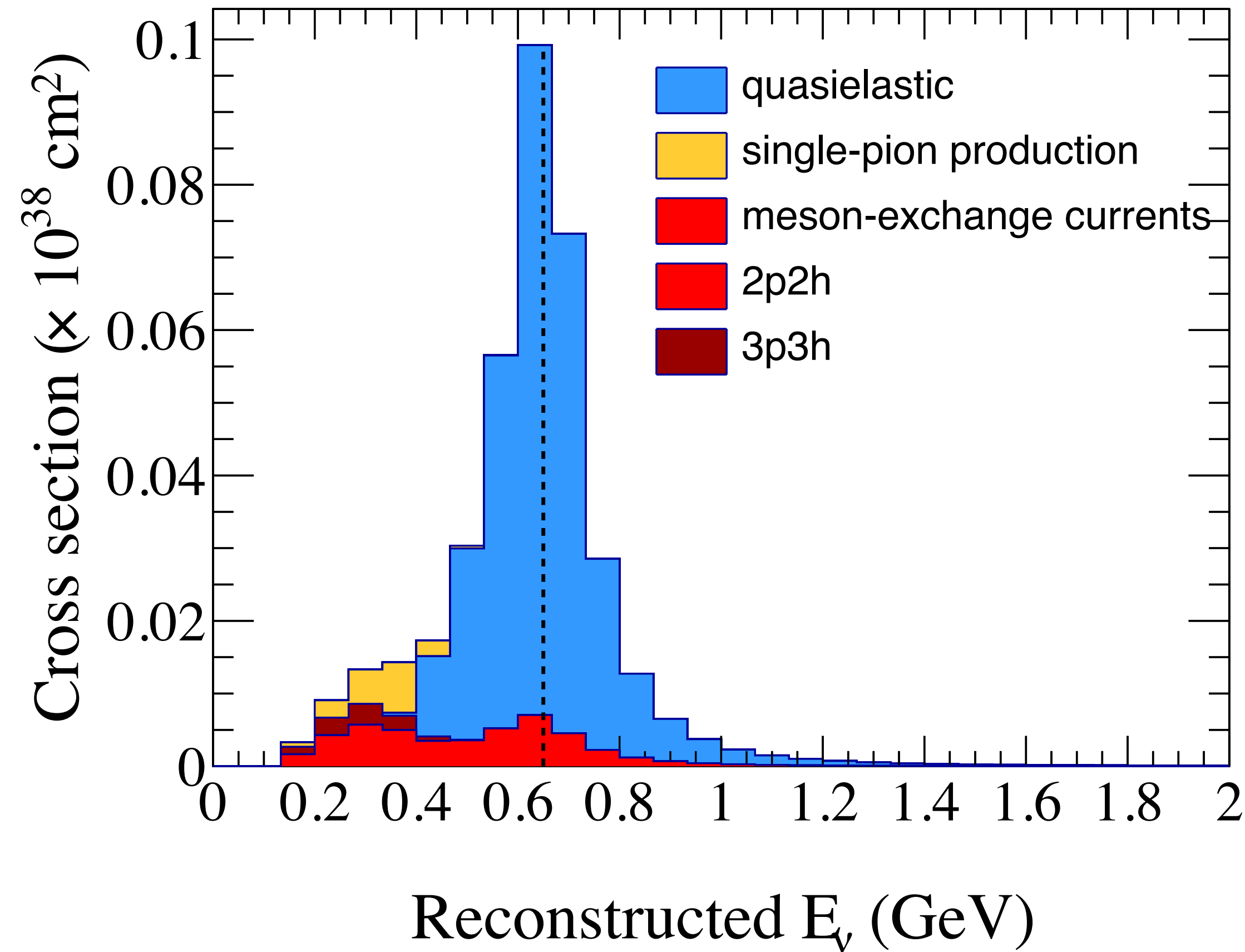


We claim that the difference is smaller than uncertainty of FSI !!

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Impact on reconstructed neutrino energy

ν_μ (E=650 MeV) at carbon



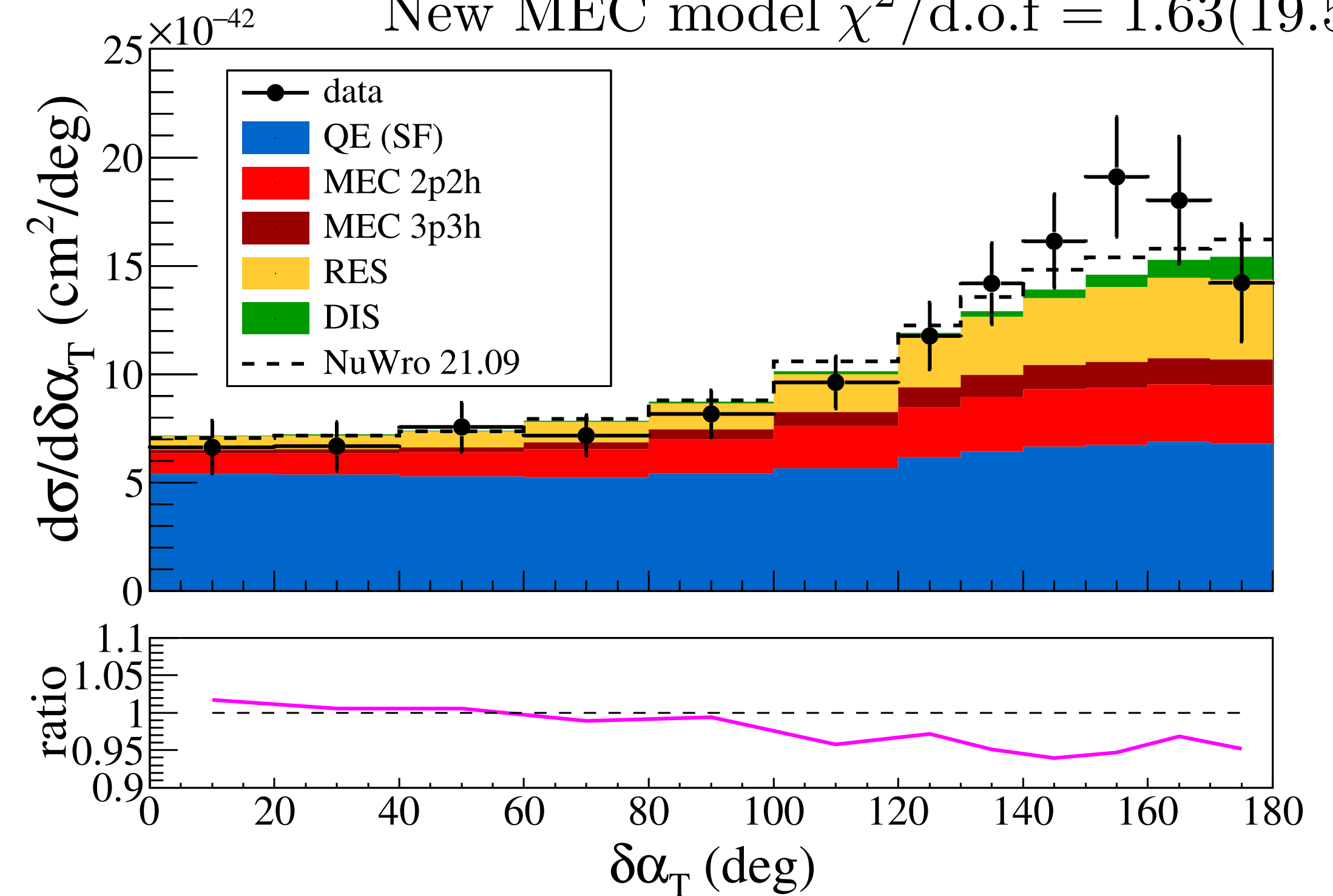
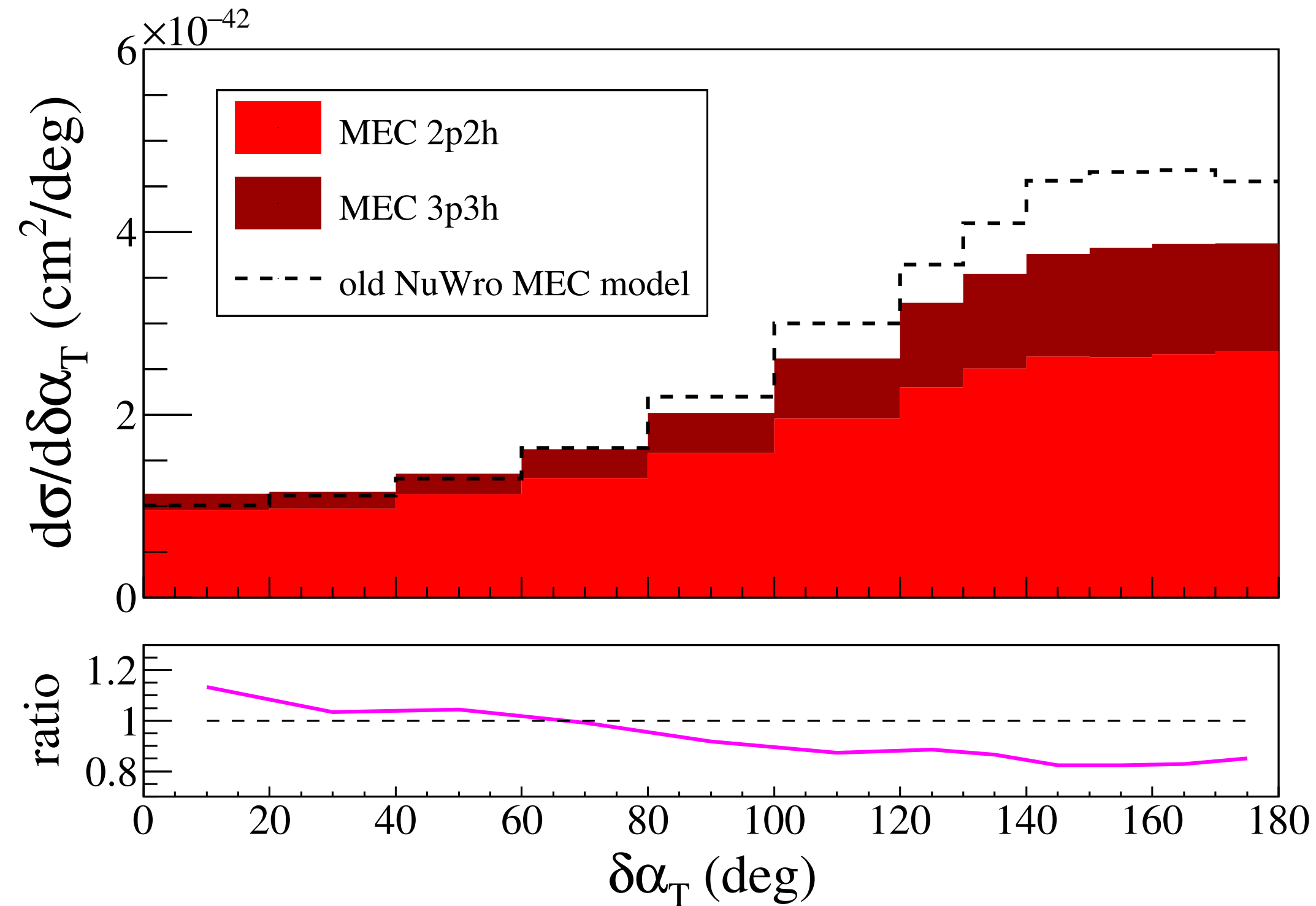
$$E_\nu = \frac{2(m_n - E_B)E_l - [(m_n - E_B)^2 + m_l^2 - m_p^2]}{2[(m_n - E_B) - E_l + k_l \cos \theta_l]}$$

MINER ν A CC1p0 π data*

Comparison : old vs new MEC model

Old MEC model $\chi^2/\text{d.o.f} = 1.76(21.32/12)$

New MEC model $\chi^2/\text{d.o.f} = 1.63(19.52/12)$



Further improvements

Improvements in missing energy treatment in both QE and MEC

- Bug removed from both QE and MEC
 - In MEC, proper treatment of the binding energy before and after nucleons pass through NuWro's cascade.
 - This led to some unphysical events
 - Reported by the JSNS² collaboration (by Eric)
- The latest version of NuWro is free from this bug. The MEC code will be public soon.

Further improvements

Comparison with JSNS² data Phys. Rev. Lett. 134 (2025) 081801

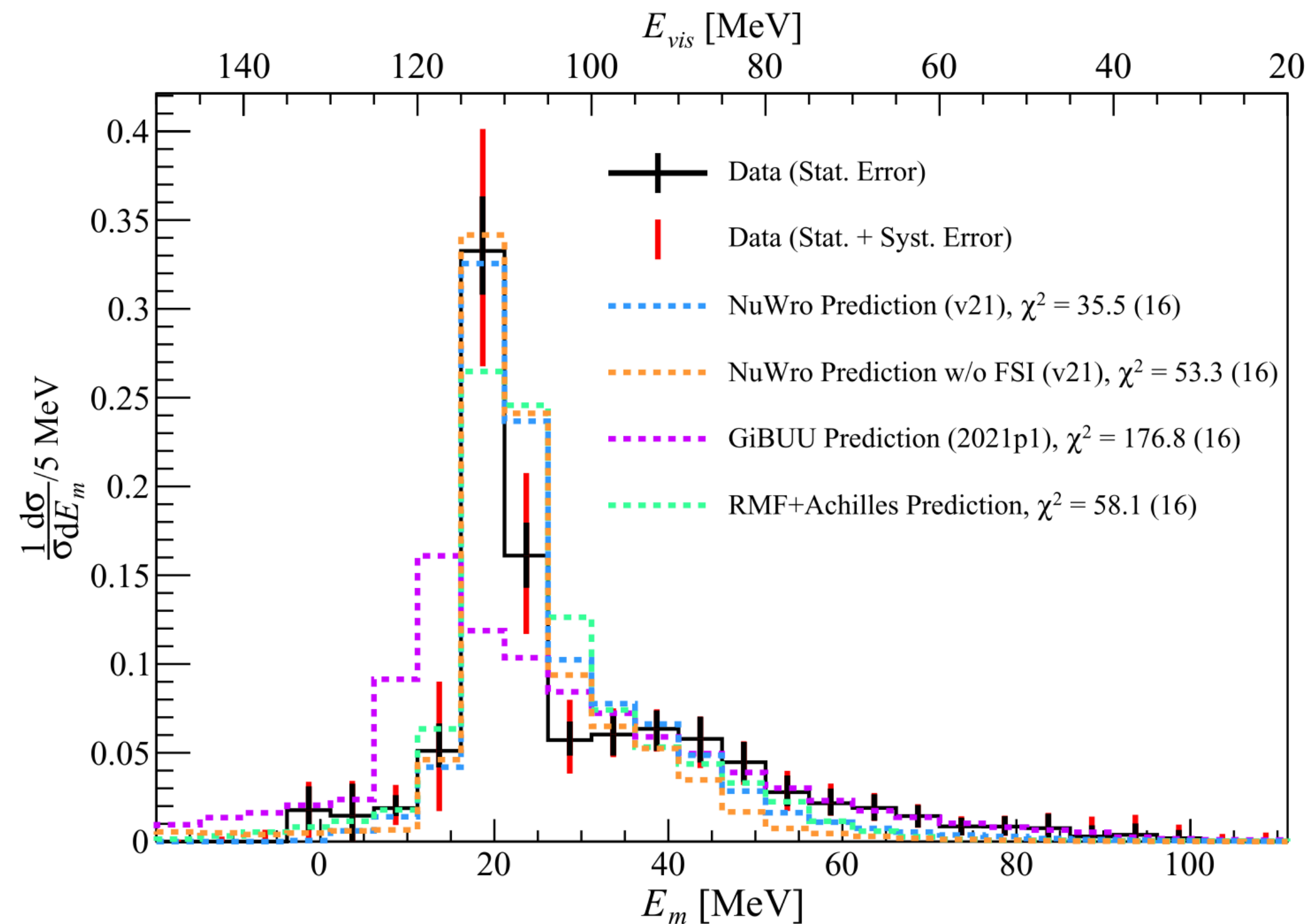
Further improvements

Comparison with JSNS² data Phys. Rev. Lett. 134 (2025) 081801

We present the first measurement of the missing energy due to nuclear effects in monoenergetic, muon neutrino charged-current interactions on carbon, originating from $K^+ \rightarrow \mu^+ \nu_\mu$ decay at rest ($E_{\nu_\mu} = 235.5$ MeV), performed with the J-PARC Sterile Neutrino Search at the J-PARC Spallation Neutron Source liquid scintillator based experiment. Toward characterizing the neutrino interaction, ostensibly $\nu_\mu n \rightarrow \mu^- p$ or $\nu_\mu {}^{12}\text{C} \rightarrow \mu^- {}^{12}\text{N}$, we define the *missing* energy as the energy transferred to the nucleus (ω) minus the kinetic energy of the outgoing proton(s), $E_m \equiv \omega - \sum T_p$, and relate this to visible energy in the detector, $E_m = E_{\nu_\mu}(235.5 \text{ MeV}) - m_\mu(105.7 \text{ MeV}) + [m_n - m_p(1.3 \text{ MeV})] - E_{\text{vis}}$. The missing energy, which is naively expected to be zero in the absence of nuclear effects (e.g., nucleon separation energy, Fermi momenta, and final-state interactions), is uniquely sensitive to many aspects of the interaction, and has previously been inaccessible with neutrinos. The shape-only, differential cross section measurement reported, based on a $(77 \pm 3)\%$ pure double-coincidence kaon decay-at-rest signal (621 total events), provides detailed insight into neutrino-nucleus interactions, allowing even the nuclear orbital shell of the struck nucleon to be inferred. The measurement provides an important benchmark for models and event generators at hundreds of MeV neutrino energies, characterized by the difficult-to-model transition region between neutrino-nucleus and neutrino-nucleon scattering, and relevant for applications in nuclear physics, neutrino oscillation measurements, and Type-II supernova studies.

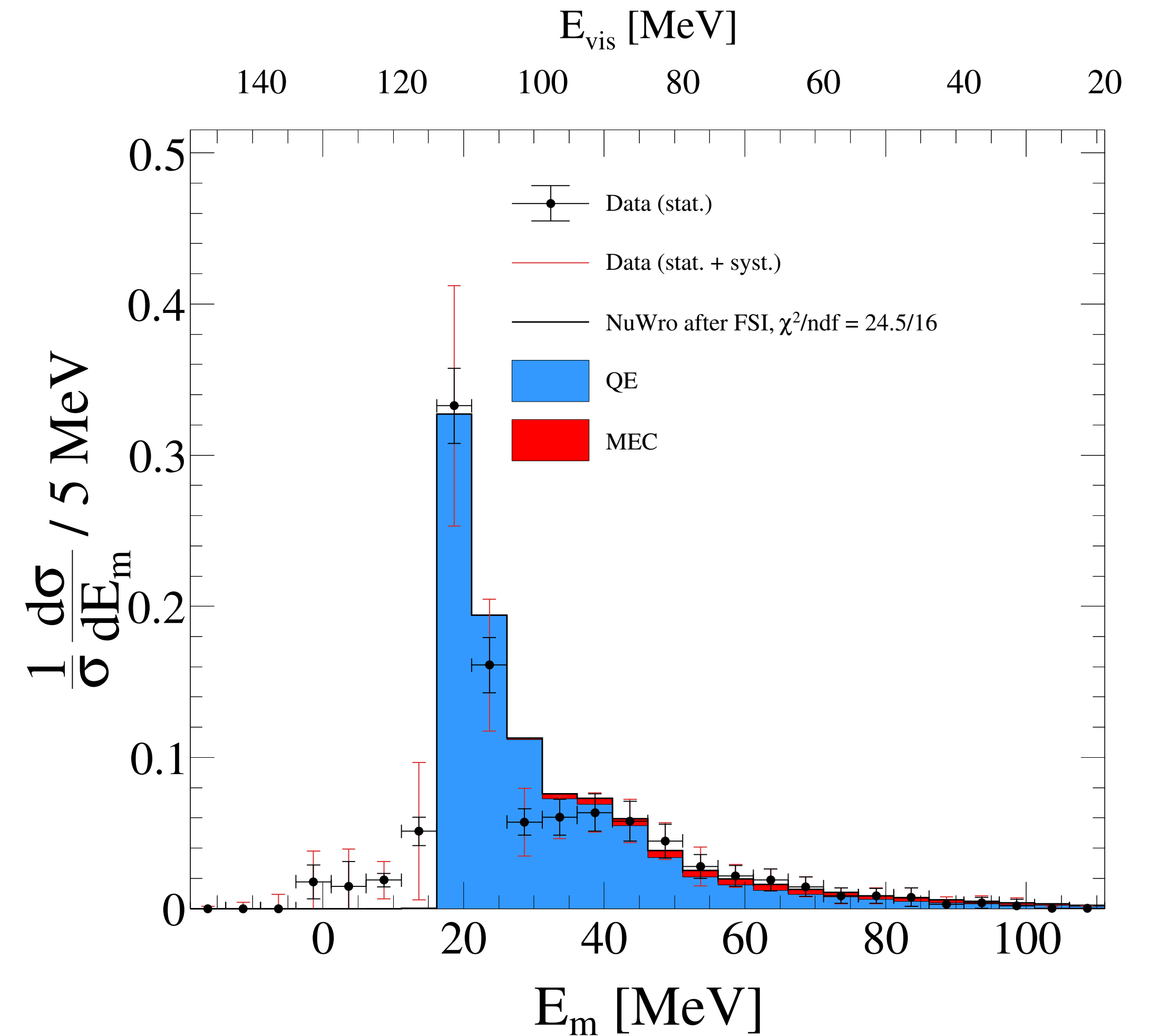
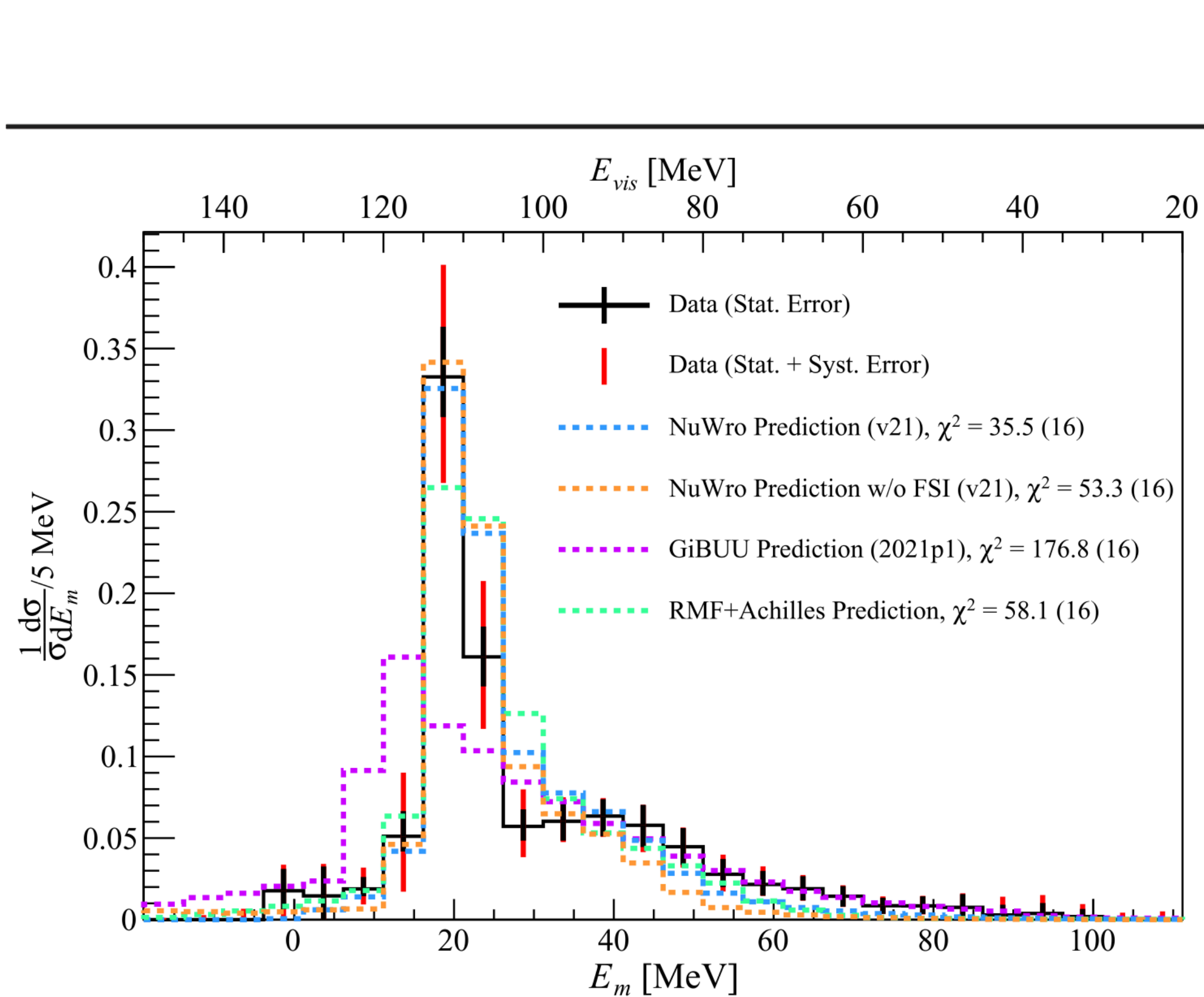
Further improvements

Comparison with JSNS² data Phys. Rev. Lett. 134 (2025) 081801



Further improvements

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Fine-tuning NuWro's nucleon-FSI module

Role of final-state interactions (FSI)

Motivation

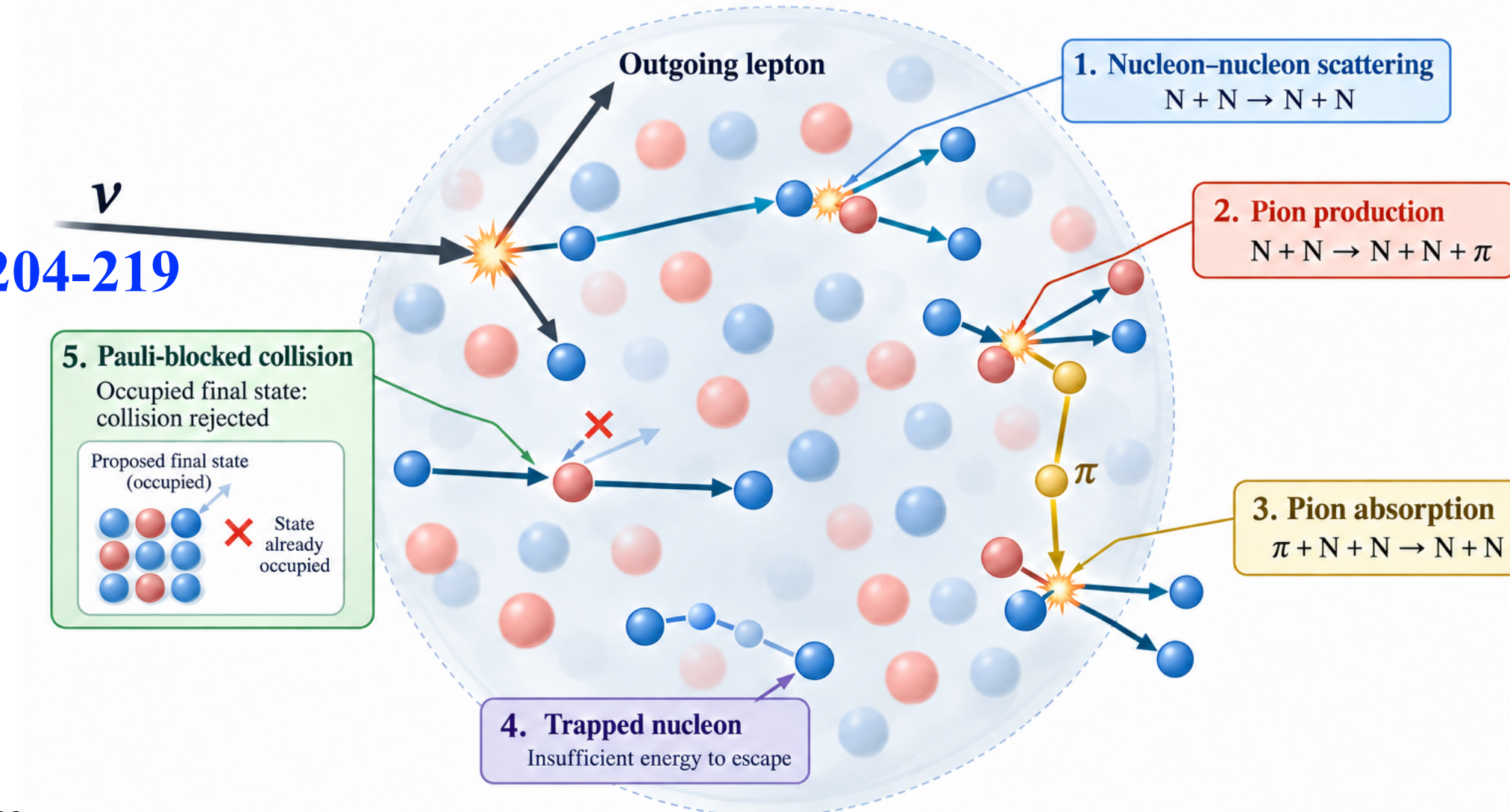
- FSIs are interactions between particles produced at the primary vertex as they pass through the nucleus.
- **Why FSI Matters?**
 - FSI can alter interaction topology
 - FSI can affect reconstructed neutrino energy
 - Oscillation parameters are sensitive to reconstructed neutrino energy (*see a recent [talk](#) from Laura Munteanu*)
 - Quality of reconstruction depends on control over FSI
- **Two-fold objective**
 - Fine-tune NuWro's nucleon-FSI module over neutrino-scattering data
 - Quantify uncertainty from nucleon-FSI modeling! (**Previously: $\pm 30\%$ from transparency data**)

NuWro's FSI Module

Semi-classical cascade description

Final-state interactions Semi-classical intranuclear cascade

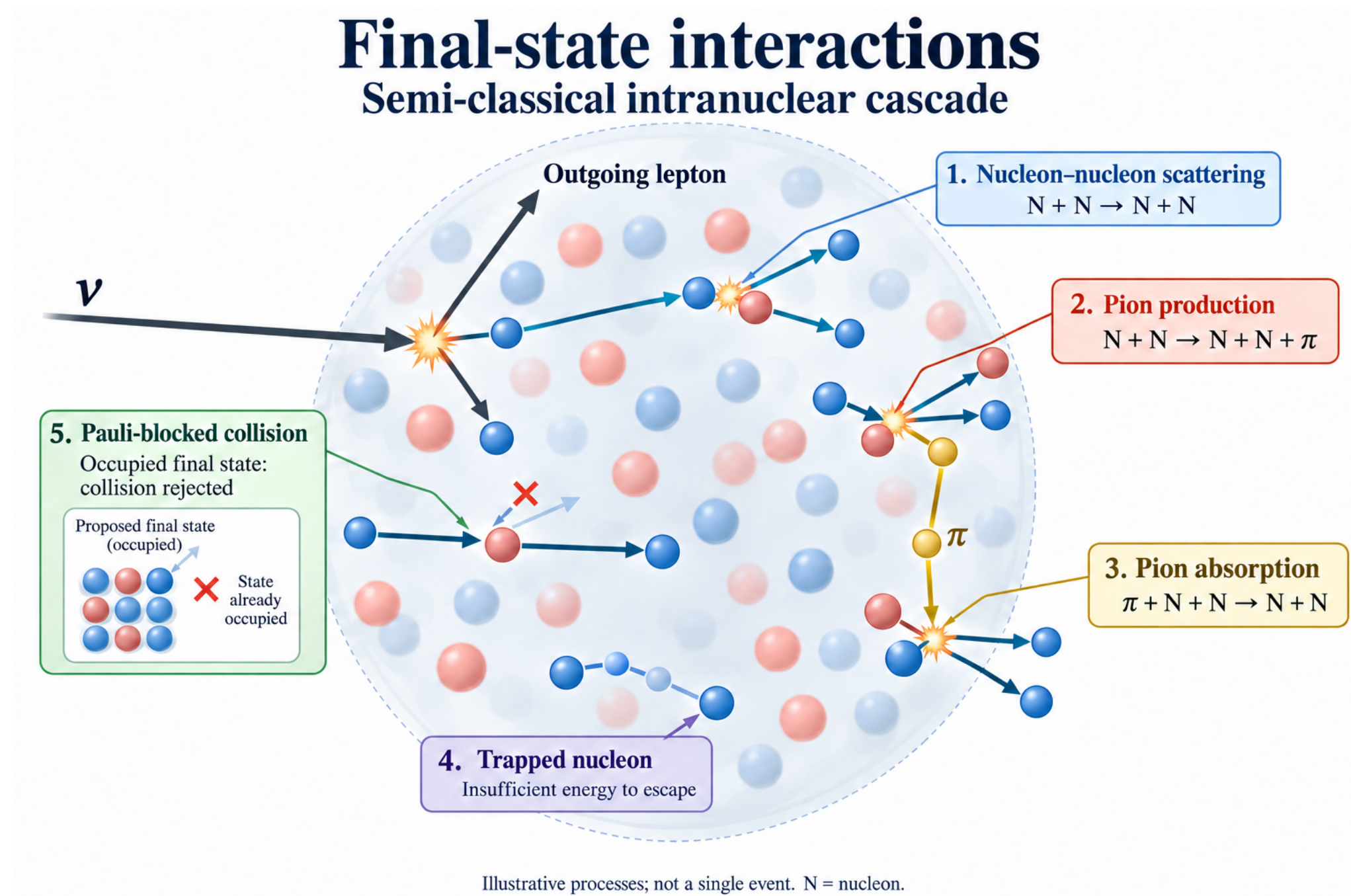
- Space like
- Based on the algorithm from Metropolis *et al.* **Phys. Rev. 110 (1958) 185-203 and 204-219**
- Energies transferred in collisions are large compared to binding energies.
- De-broglie wavelength is smaller than the mean free path
- Nucleon reinteractions are independent of each other



- Particles formed at the primary vertex are propagated by a constant step $\Delta x = 0.2$ fm
- At each step, the fate of the particle is decided $\tilde{p}(\Delta x) = \exp(-\Delta x/\lambda)$ $\lambda = (\rho\sigma)^{-1}$
- Pauli blocking and nucleon-nucleon (NN) correlation effects are included $p = 1 - \tilde{p}$
- Total and elastic free NN cross sections are from: **M. Tanabashi *et al.* Phys. Rev. D 98 (2018) 030001**
- Single-pion production in the overall cross-section is from: **J. Bystricky *et al.* J. Physique 48 (1987) 1901**
- Recent improvements in nuclear effects (*Covered by Rwik*)

Fine-tuning NuWro's nucleon-FSI module

The Idea behind it



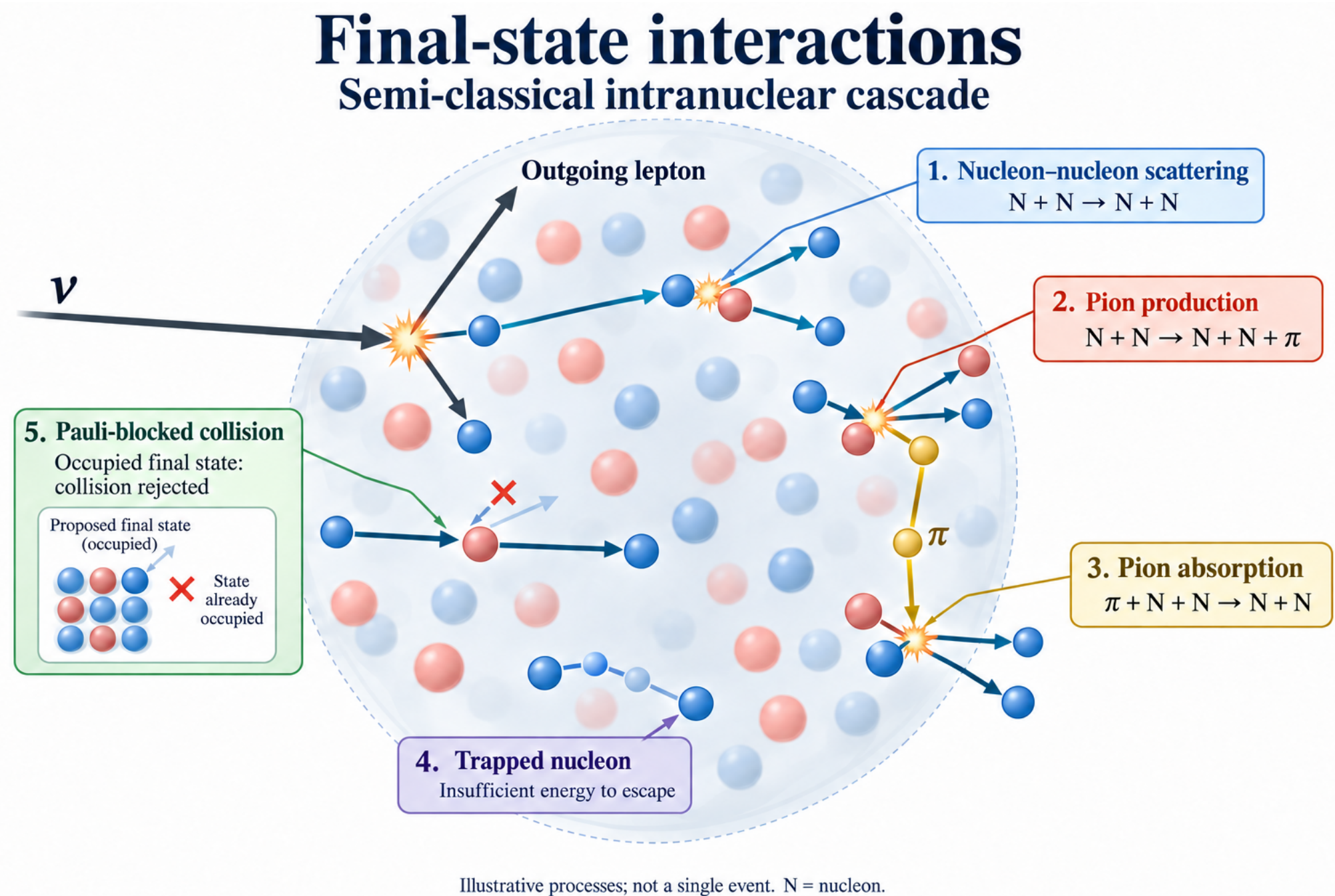
$$\tilde{p}(\Delta x) = \exp(-\Delta x/\lambda)$$

$$\lambda \rightarrow s\lambda$$

Mean-free-path $\rightarrow$ Interaction-probability $\rightarrow$ Strength of FSI

Fine-tuning NuWro's nucleon-FSI module

The Idea behind it



Full event-resampling is a waste of time!

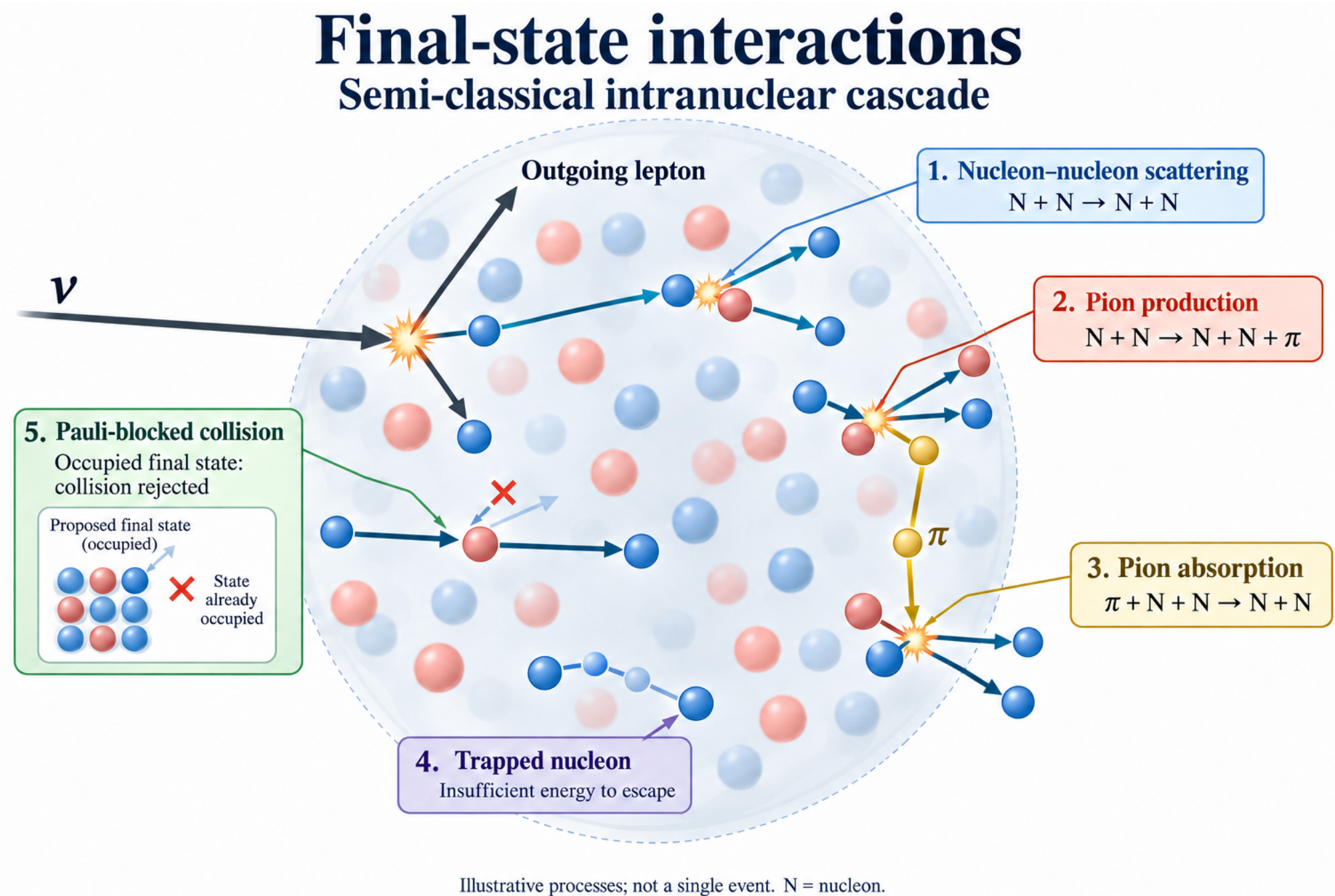
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Fine-tuning NuWro's nucleon-FSI module

The Idea behind it



Full event-resampling is a waste of time!
We need to reweight FSI!

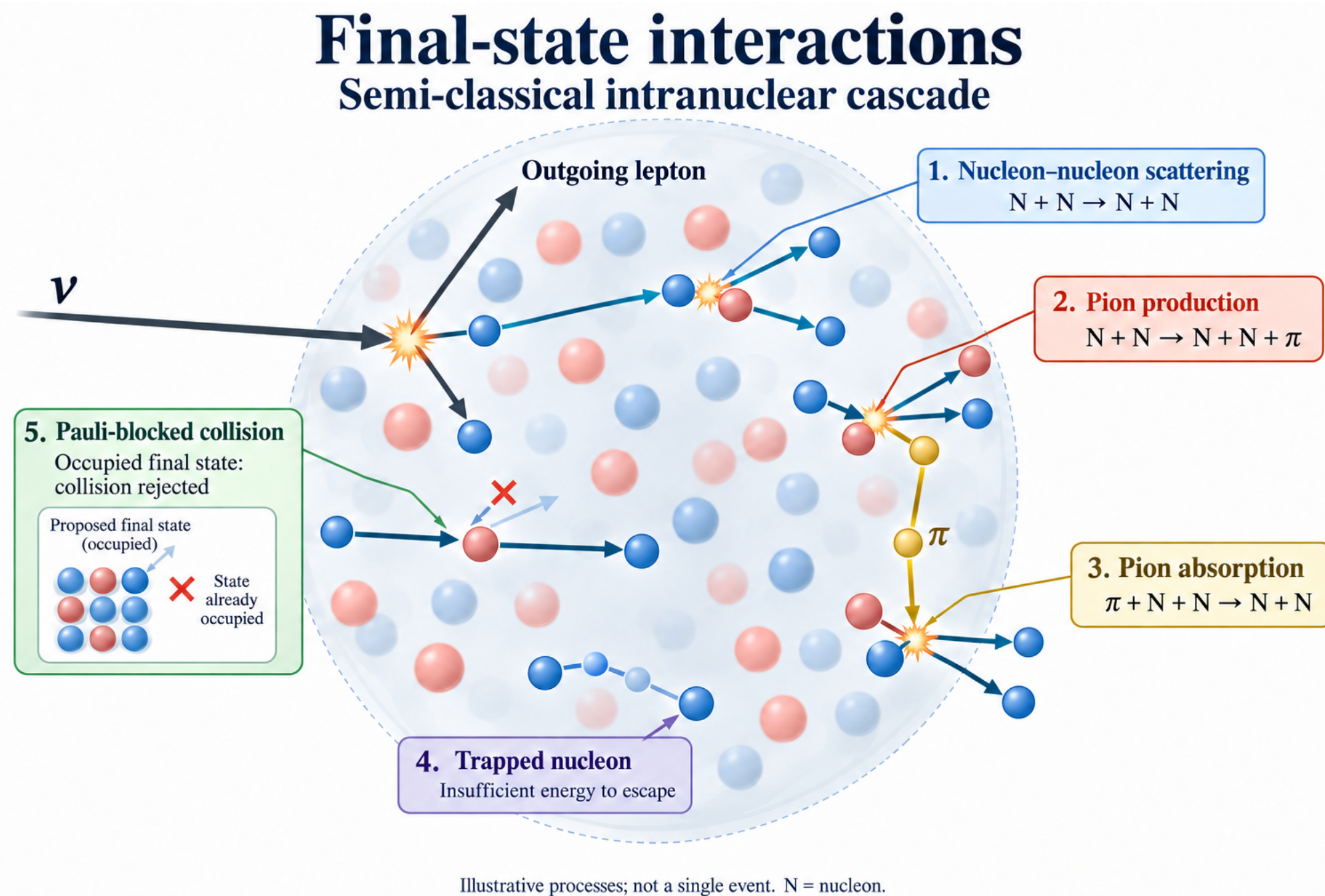
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Mean-free-path $\rightarrow$ Interaction-probability $\rightarrow$ Strength of FSI

Fine-tuning NuWro's nucleon-FSI module

The Idea behind it



Full event-resampling is a waste of time!

We need to reweight FSI!

- Interaction vertex
- Non-interaction vertex
- Pauli-blocked vertex

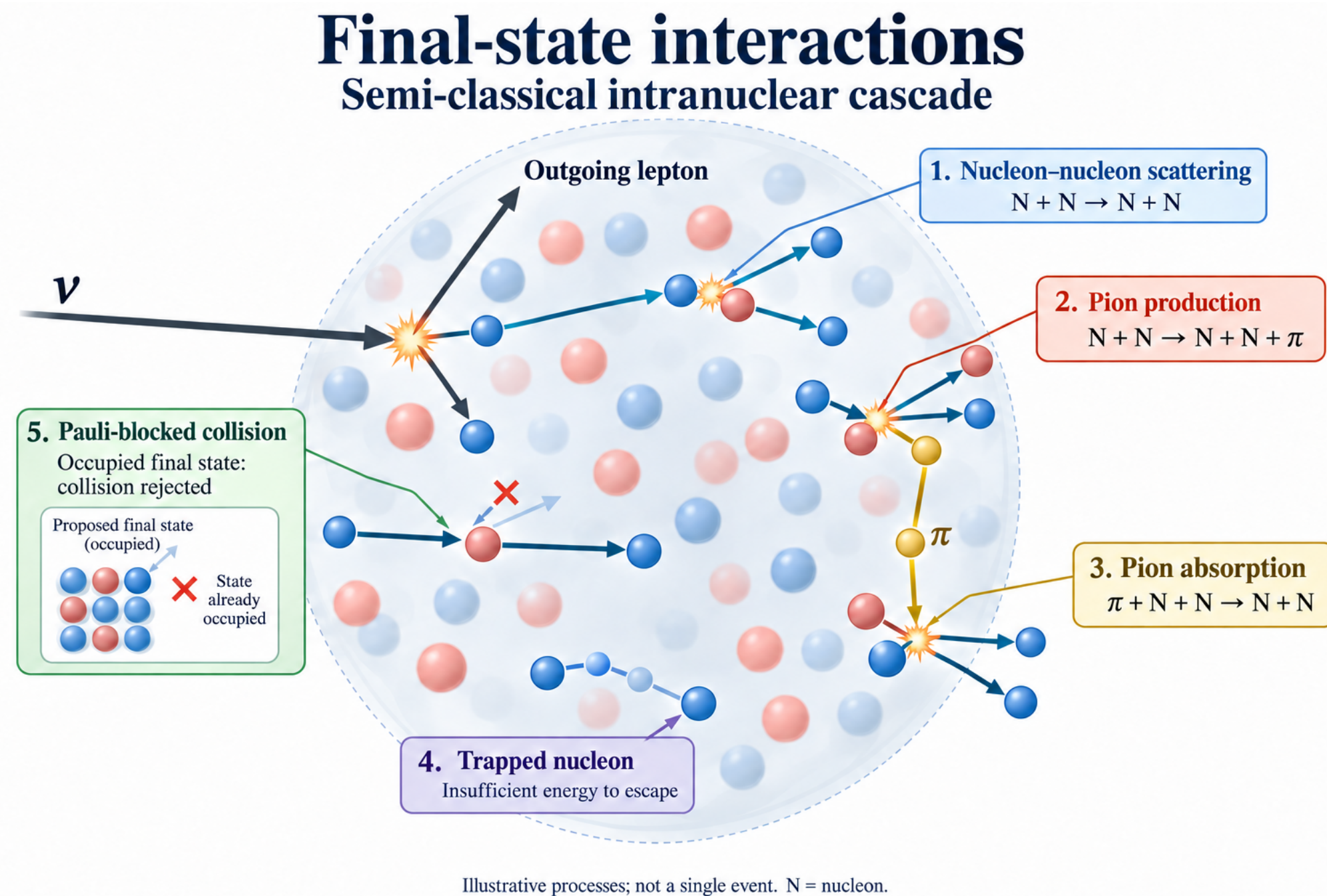
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Fine-tuning NuWro's nucleon-FSI module

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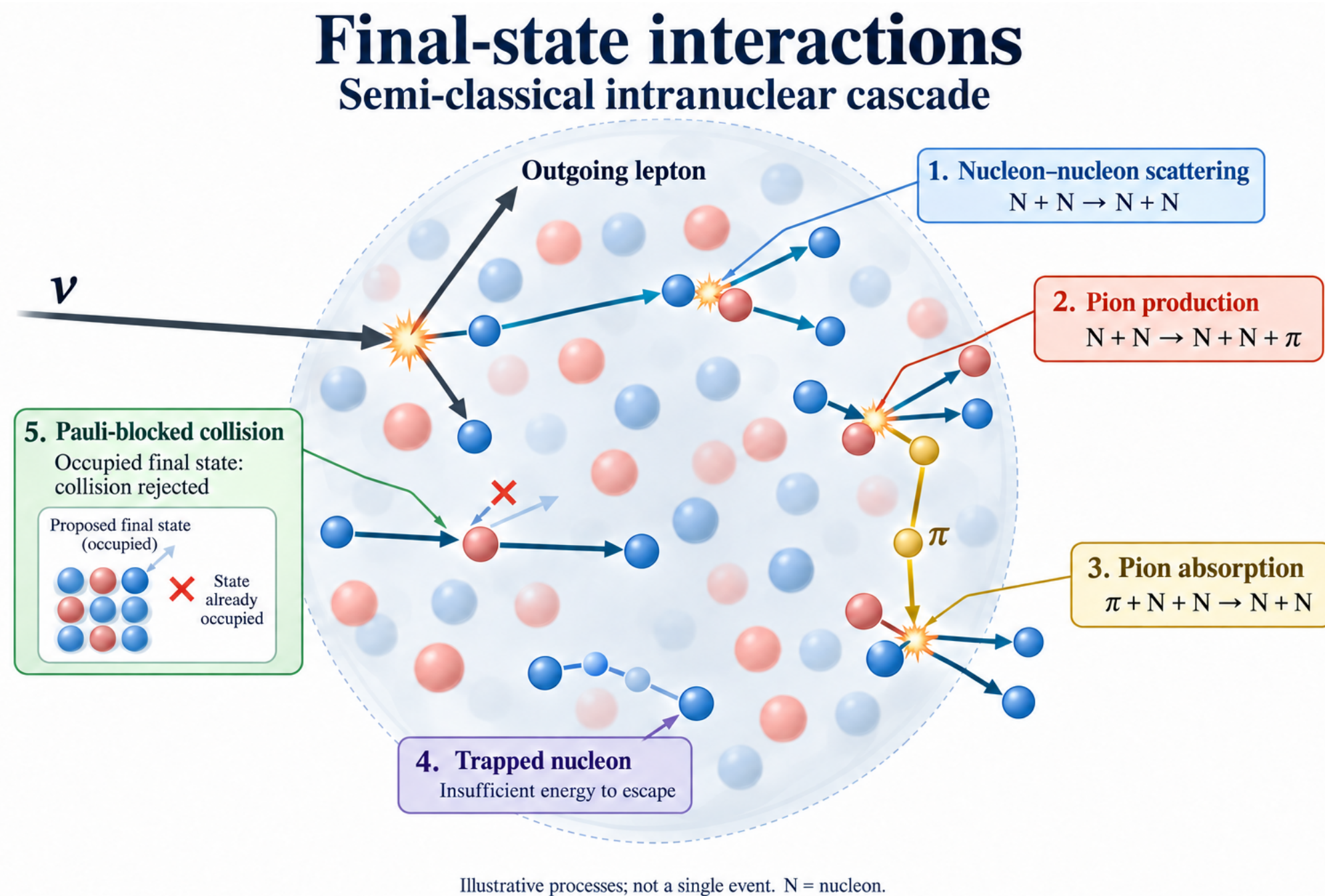
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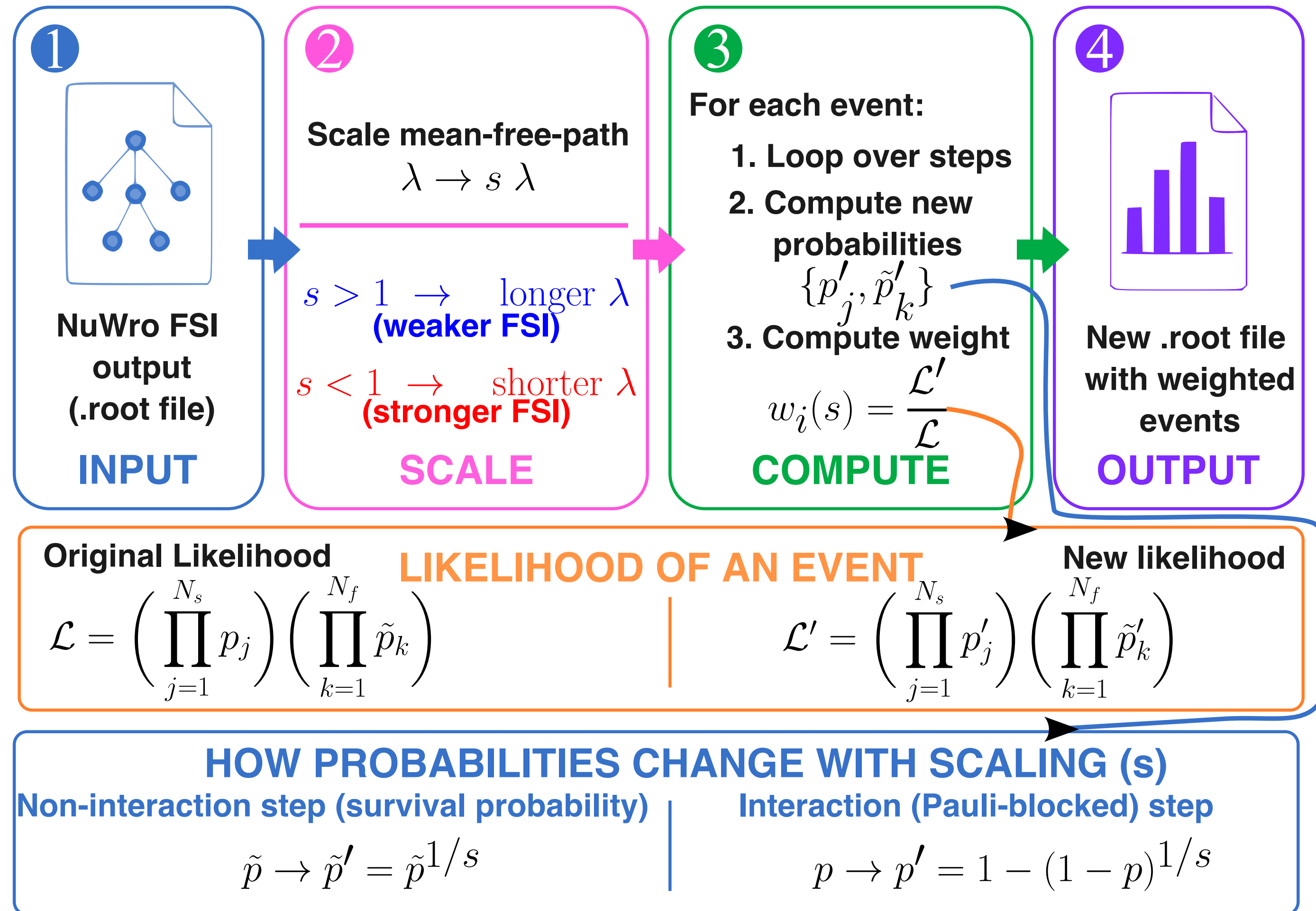
Fine-tuning NuWro's nucleon-FSI module

Exact-reweighting scheme

- We propose a scheme to reweight nucleon-FSI.
- Information like the **total number of (non-)interaction steps** comes in handy!
- Some additional information, such as the **sequence of interaction probabilities for PB, the successful interaction steps,** and the **product of probabilities of non-interaction steps**, needs to be stored.

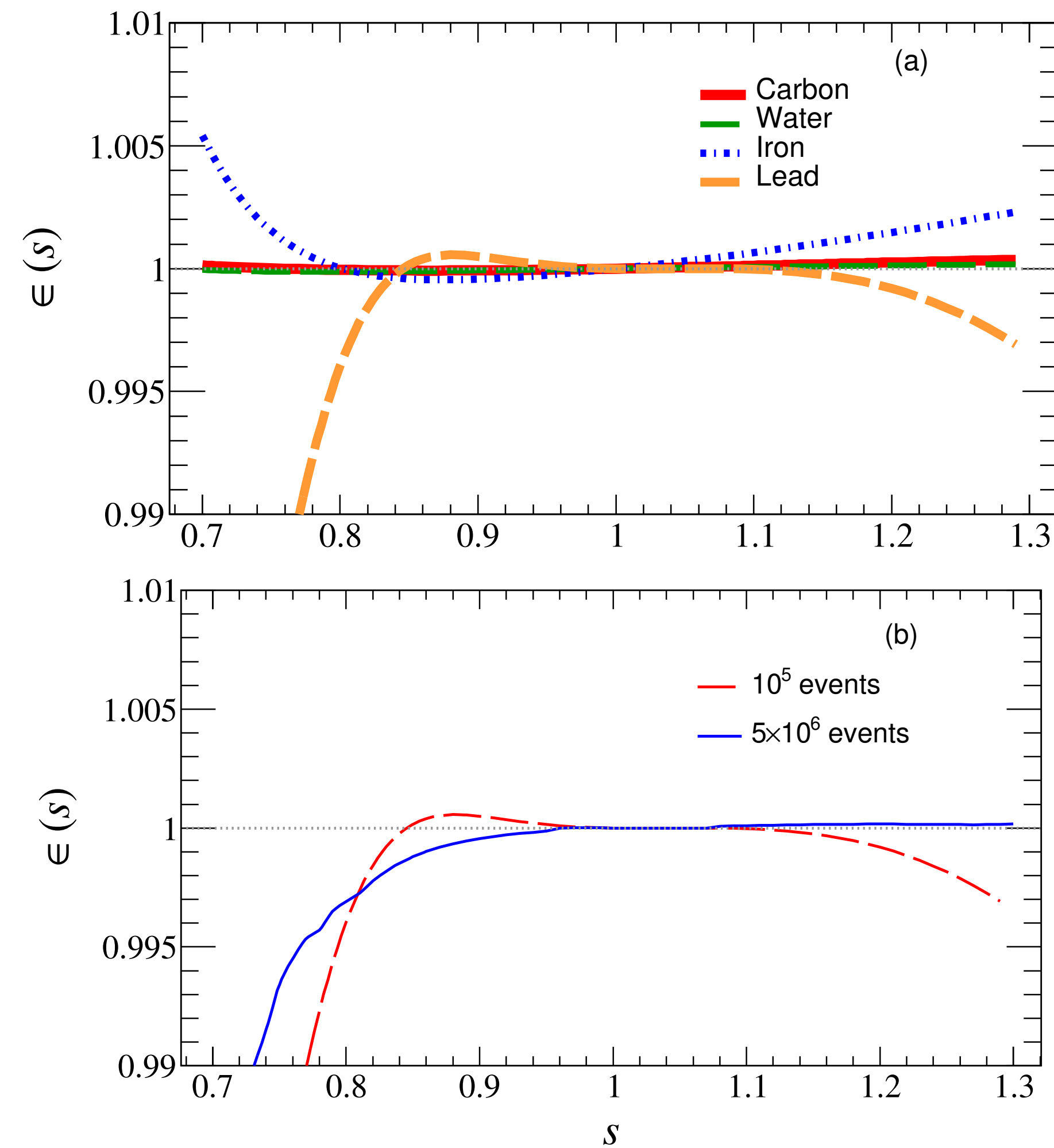
Fine-tuning NuWro's nucleon-FSI module

Exact-reweighting scheme



Exact-reweighting scheme

Performance



Fine-tuning NuWro's nucleon-FSI module

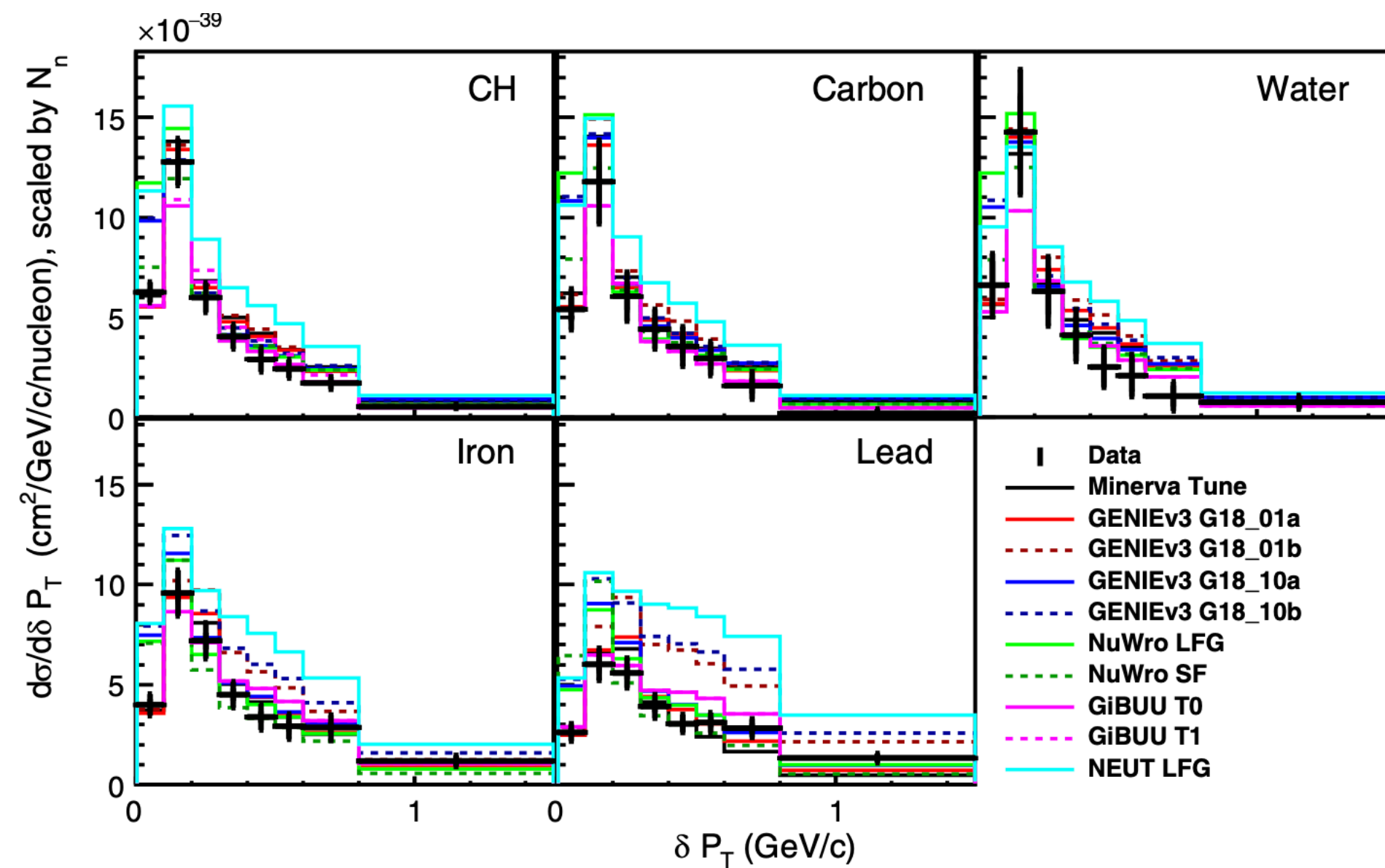
Exact-reweighting scheme

- To fine-tune the nucleon-FSI module, we used the $\text{CC1p0}\pi$ data from the MINERvA experiment (see [Phys. Rev. D. 112 \(2025\) 052005](#)) using NuMI medium energy configuration (~ 6 GeV) performed simultaneously over **carbon, oxygen, iron, and lead**.

Fine-tuning NuWro's nucleon-FSI module

Exact-reweighting scheme

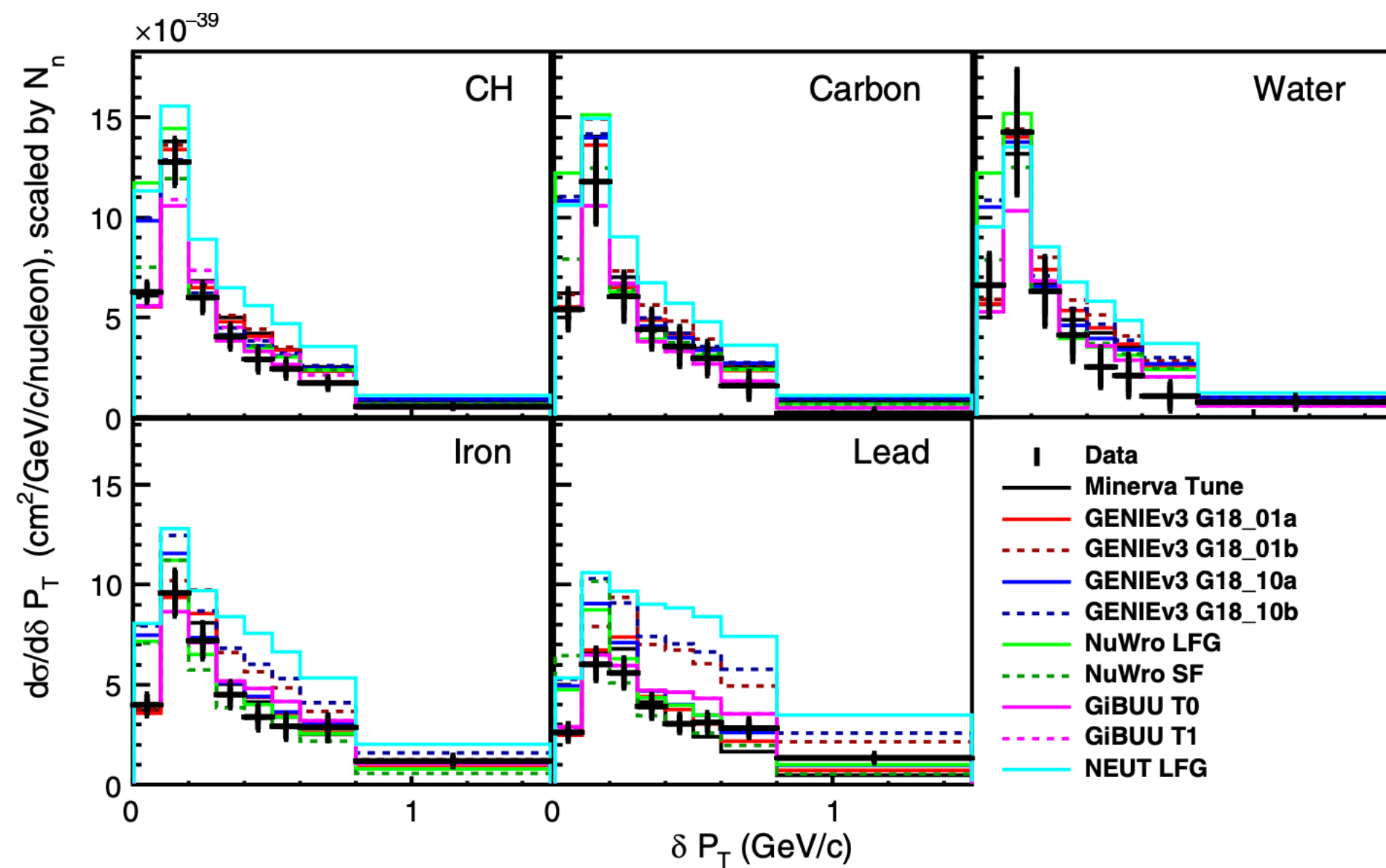
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Fine-tuning NuWro's nucleon-FSI module

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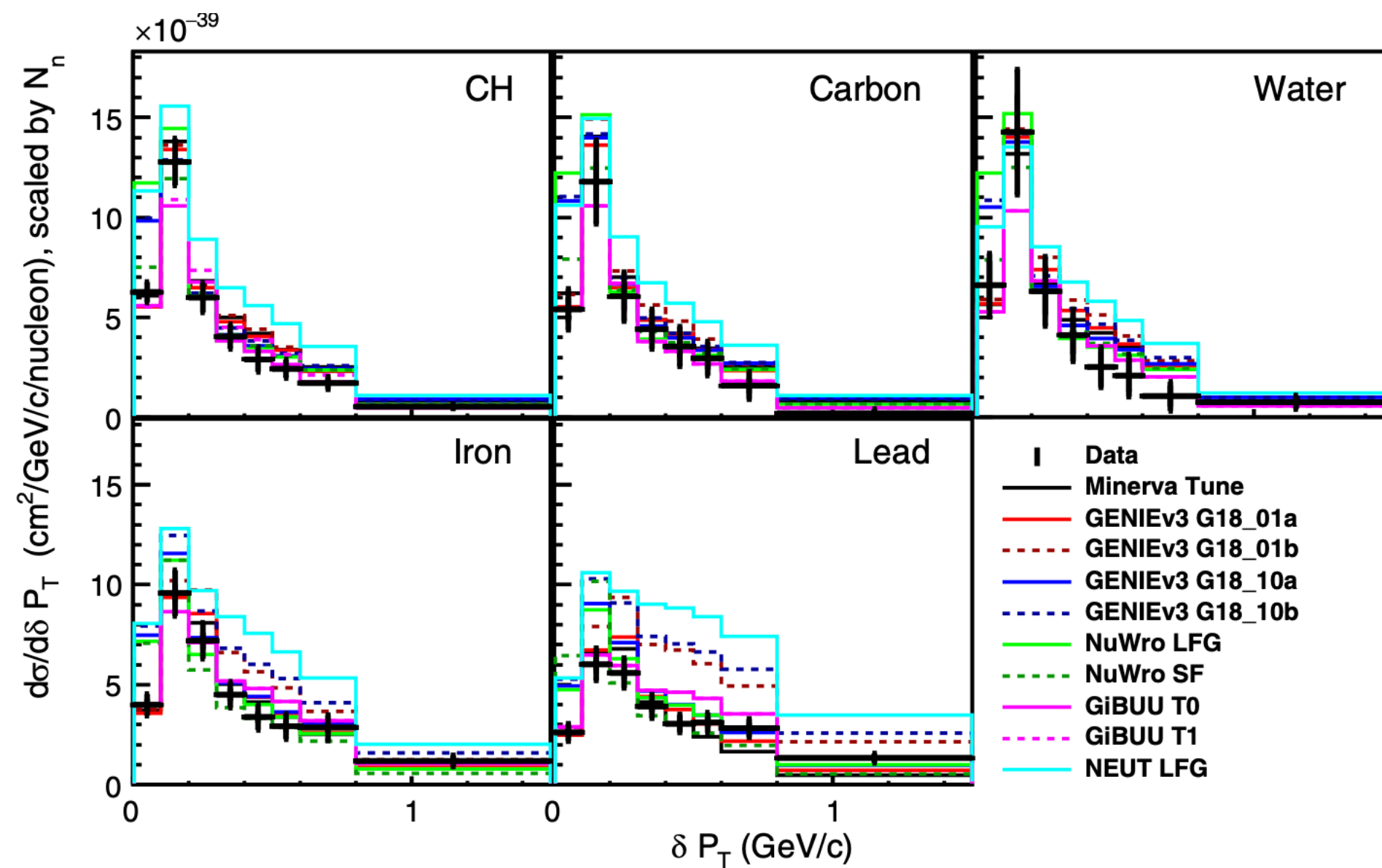
Signal definition:

$$\begin{aligned} 2 \text{ GeV} < |\mathbf{p}_\mu| < 20 \text{ GeV} & \quad \theta_\mu < 17^\circ \\ 0.5 \text{ GeV} < |\mathbf{p}_p| < 1.1 \text{ GeV} & \quad \theta_p < 70^\circ \end{aligned}$$

Fine-tuning NuWro's nucleon-FSI module

Exact-reweighting scheme

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$$0.5 \text{ GeV} < |\mathbf{p}_p| < 1.1 \text{ GeV} \quad \theta_p < 70^\circ$$

A good starting point to fine-tune NuWro nucleon-FSI model !

Fine-tuning NuWro's nucleon-FSI module

Fitting the scaling parameter (s)

Fine-tuning NuWro's nucleon-FSI module

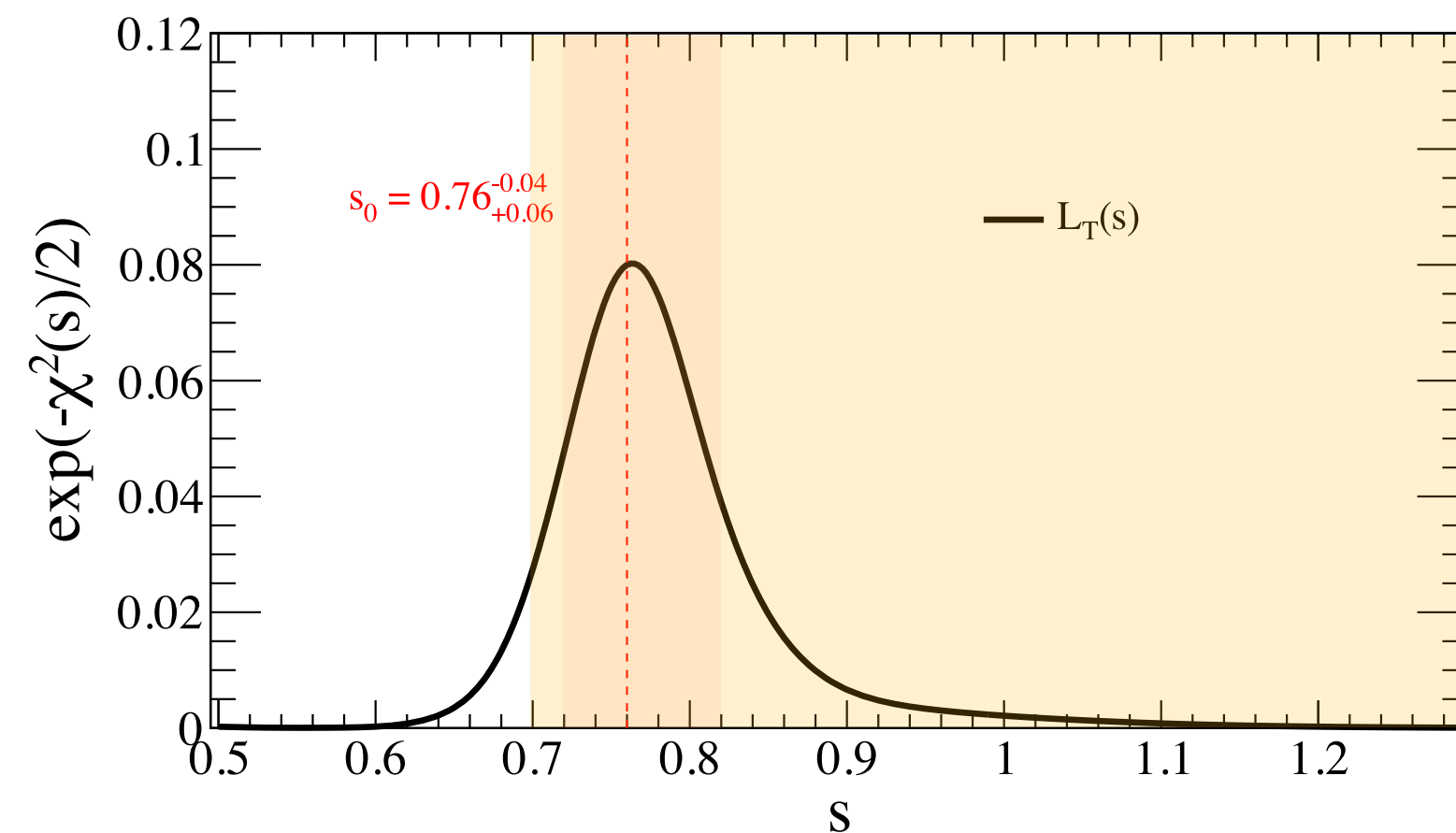
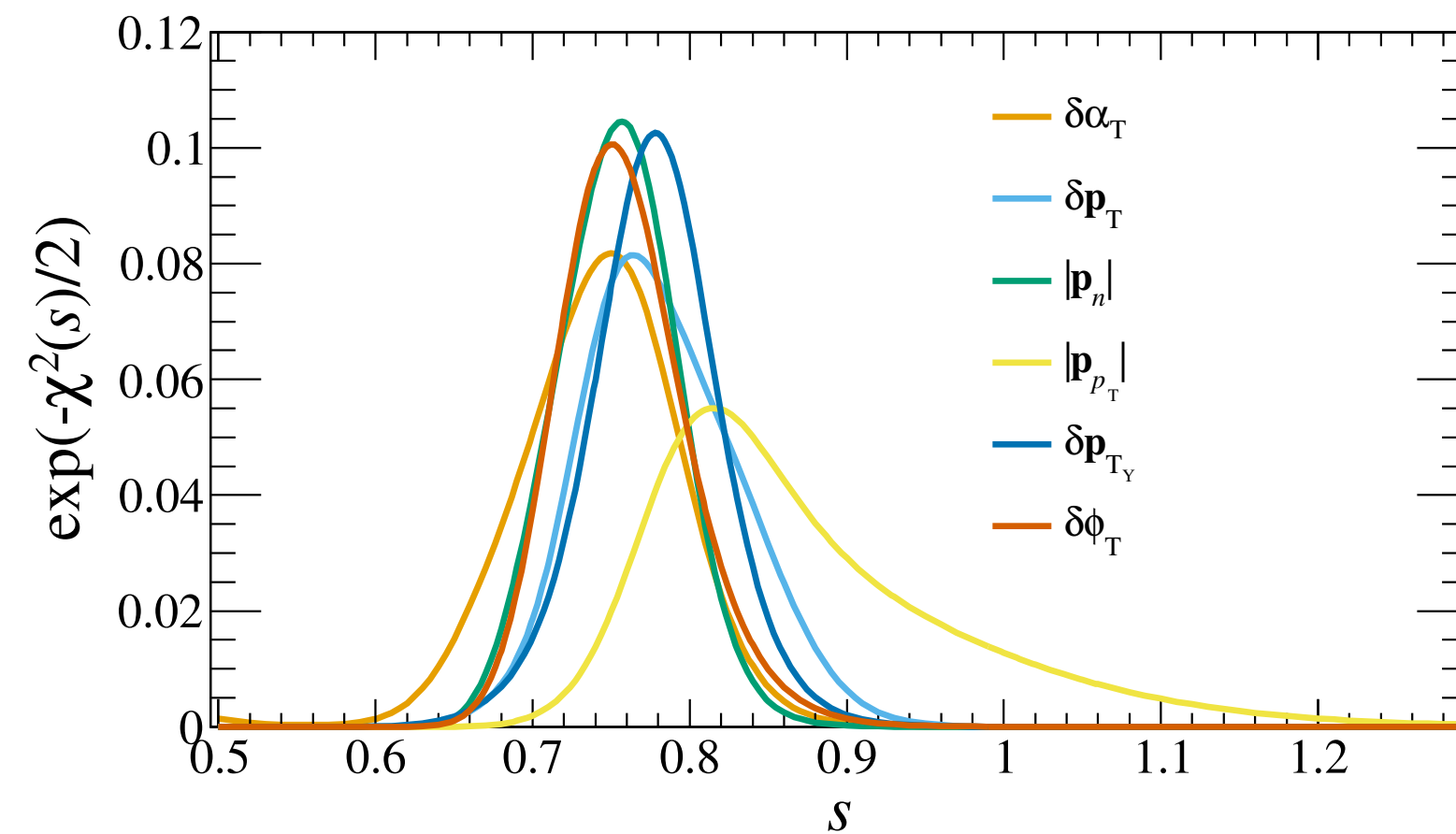
Fitting the scaling parameter (s)

Reweight NuWro events → Apply event selection cuts → Optimize χ^2/dof over exclusive variables

Fine-tuning NuWro's nucleon-FSI module

Fitting the scaling parameter (s)

Reweight NuWro events $\rightarrow$ Apply event selection cuts $\rightarrow$ Optimize χ^2/dof over exclusive variables



$\pm 30\%$

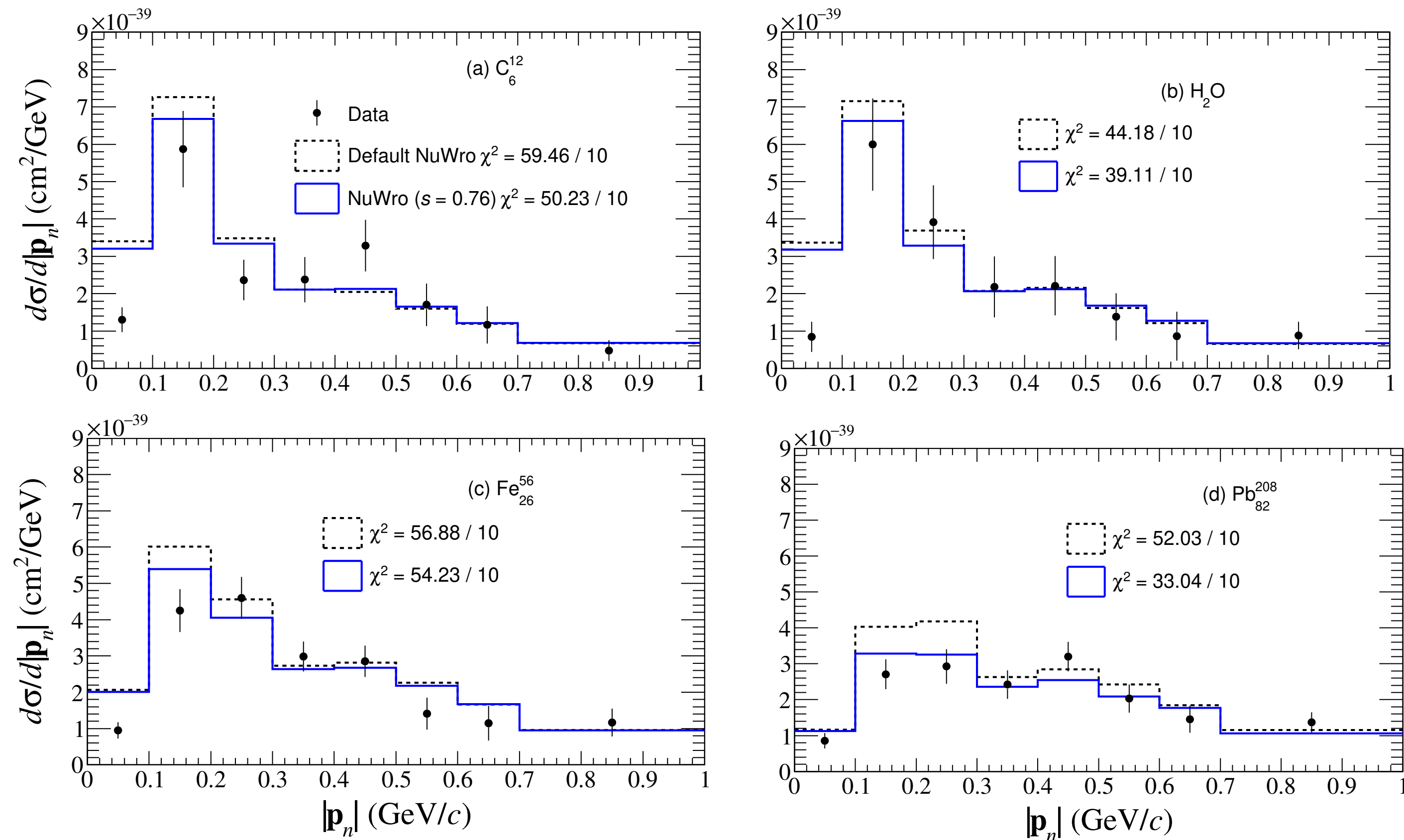
From nuclear transparency studies, Kajetan Niewczas, J. T. Sobczyk Phys. Rev. C 100 025501

$\pm 10\%$

Constraint from neutrino-scattering data (*latest!*) Phys Rev. D. 113 (2026) 096023

Fine-tuning NuWro's nucleon-FSI module

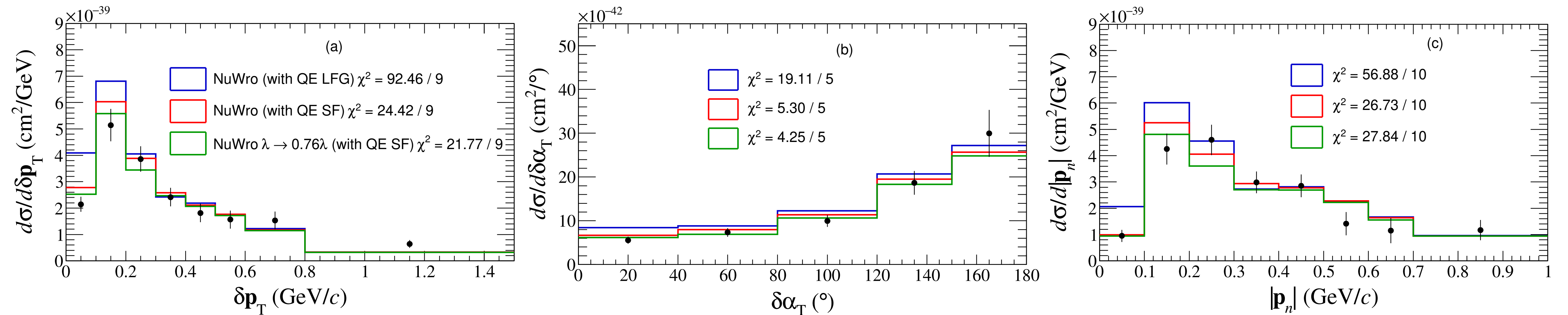
Performance



χ^2/dof improves in all cases!

Fine-tuning NuWro's nucleon-FSI module

Performance over exclusive observables



Data is also sensitive to nuclear modeling

Modeling lepton-nucleus interactions using Deep Neural Networks in eWro

Modeling lepton-nucleus interactions

Motivation

- Electron-nucleus scattering serves as a benchmark for modeling nuclear effects
- Theoretical models are available for nuclear modeling; however
 - They may/maynot agree perfectly with the inclusive cross-section data.
- Precise measurements of systematic uncertainty in the *exclusive part* of theoretical models require precise control over the *inclusive part*
- Deep neural network fits to inclusive electron-nucleus scattering cross-section data are available for precise control over the *inclusive part*
 - Nuclear modeling must come from existing theoretical models for full event generation.

Electron-scattering in NuWro

eWro

- There are switches to run NuWro for electron-nucleus interactions

```
dyn_qel_el      = 1
dyn_res_el      = 1
el_costh_lab    = 0.5
```

- Currently, eWro runs only include quasielastic (QE) and single-pion production (SPP) interactions

eWro interaction scheme



Initial state

SF

Hole spectral function

GFG

Global Fermi gas

LFG

Local Fermi gas



Interaction channels

QE

Quasielastic scattering

SPP

Single pion production



FSI Final-state interactions

Energy transfer modification in the SF model

Quantum mechanical

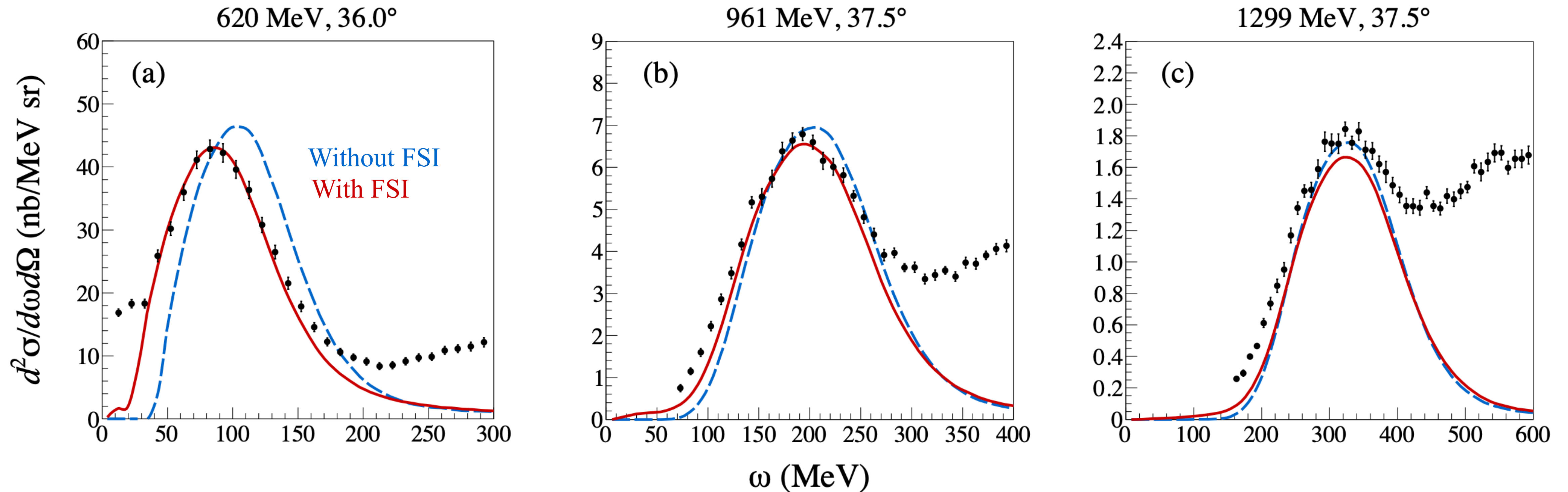
Semi-classical

Cascade

Hadron scattering

Electron-scattering in NuWro

Quasielastic interaction via spectra function (SF) formalism

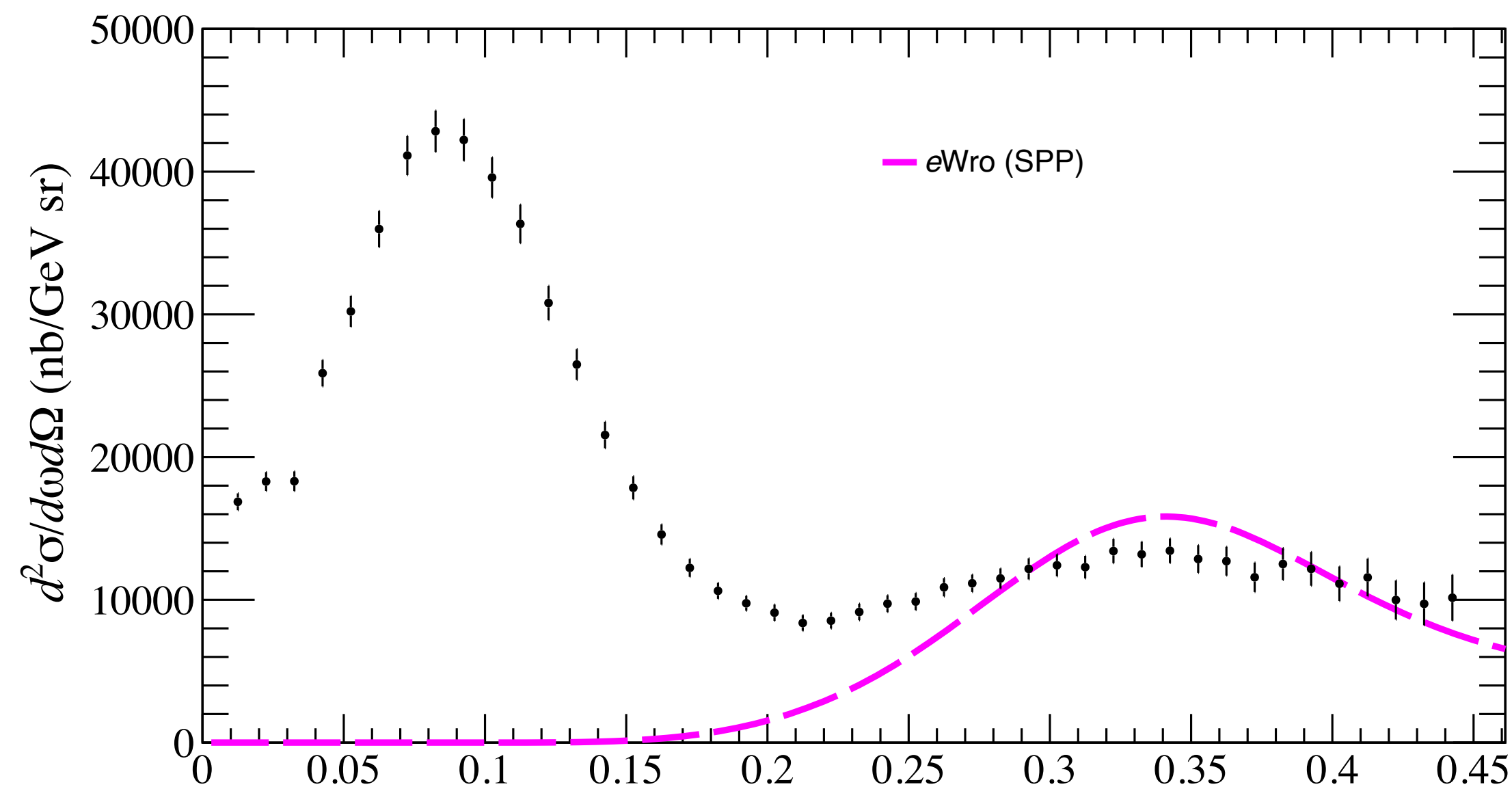


A. M. Ankowski, R. D. Banerjee *et al.* Spectral function approach in NuWro: modeling of multinucleon final states in quasielastic scattering [arXiv.2508.10101](https://arxiv.org/abs/2508.10101)

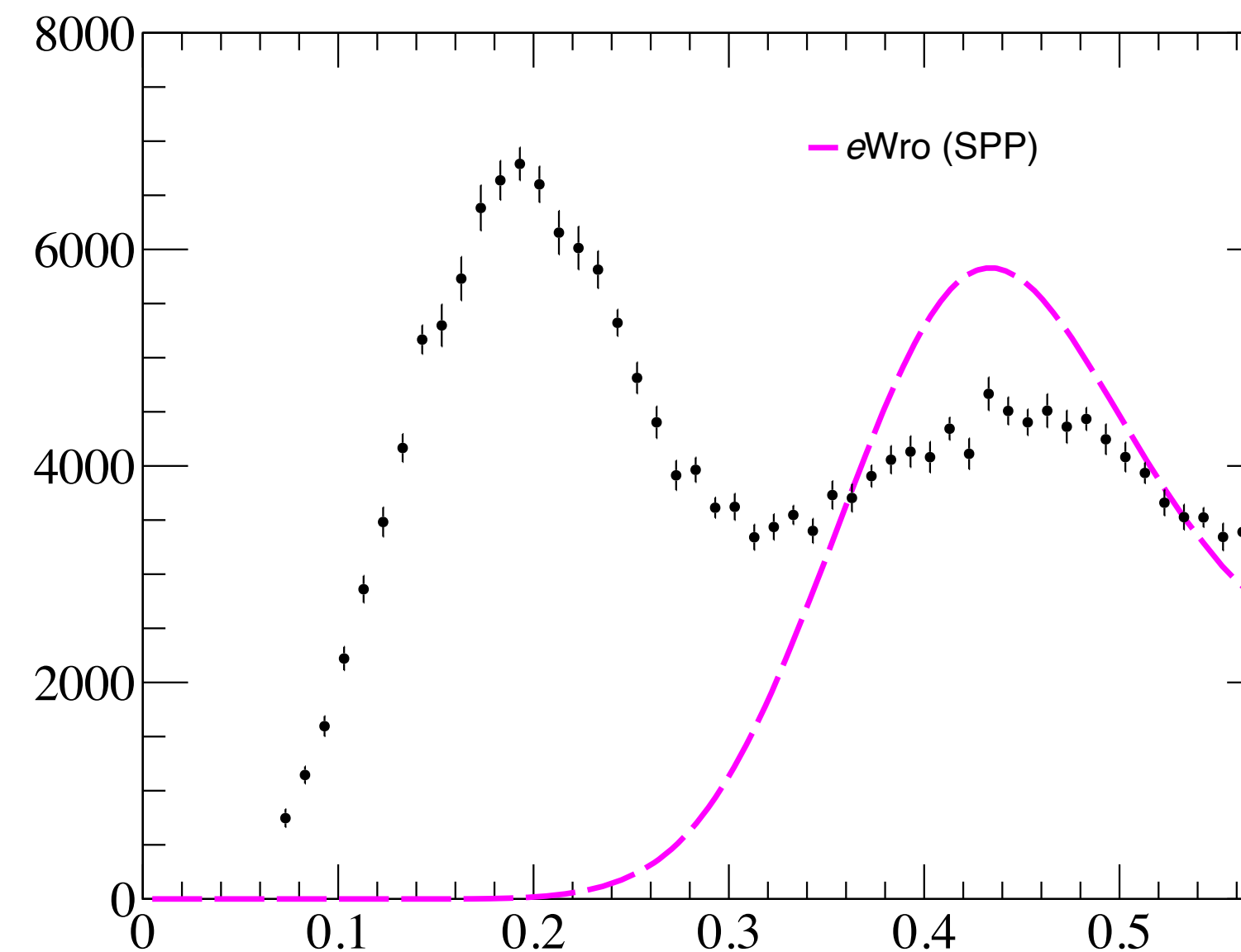
Electron-scattering in NuWro

Quasielastic interaction via spectra function (SF) formalism

0.62 GeV 36 deg



0.961 GeV 37.5 deg



Jakub Żmuda, Krzysztof M. Graczyk, Cezary Juszczak Jan T.
Sobczyk *et al.* Vol. 46 (2015) *ACTA PHYSICA POLONICA B* No
11

eWro + DNN

- Construction of a hybrid model with inputs from a DNN on the inclusive cross section and nuclear modeling using available theoretical models.
- For the inclusive part, the plan is to use inputs from neural-network fits.
- The exclusive part will be modeled based on the available theoretical inputs.

$$\frac{d^3\sigma}{d\omega d\Omega dX}$$

Modeling electron-nucleus scattering

Empirical fits to $C^{12}(e, e'p)$ data using DNN

- Input: E, θ, ω
- Output: $d^2\sigma/d\omega d\Omega \times \left(\frac{10^9}{137^2 E \cos(\theta/2)} \frac{\cos^2(\theta/2)}{4E^2 \sin^4(\theta/2)} \right)^{-1}$

- An ensemble of 50 neural networks
- The mean of the ensemble is taken as the weight
- Uncertainty in prediction is estimated as uncertainty over the ensembles of trained models

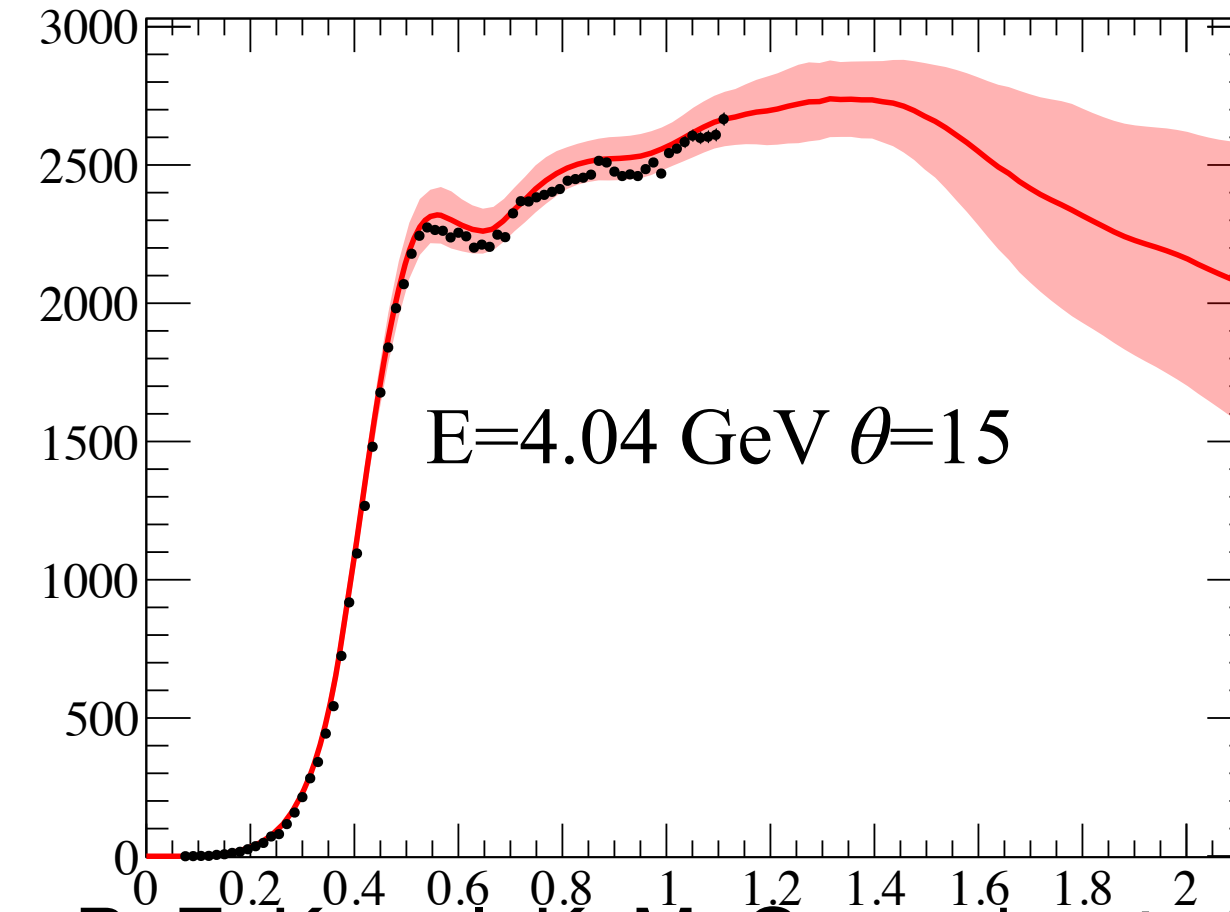
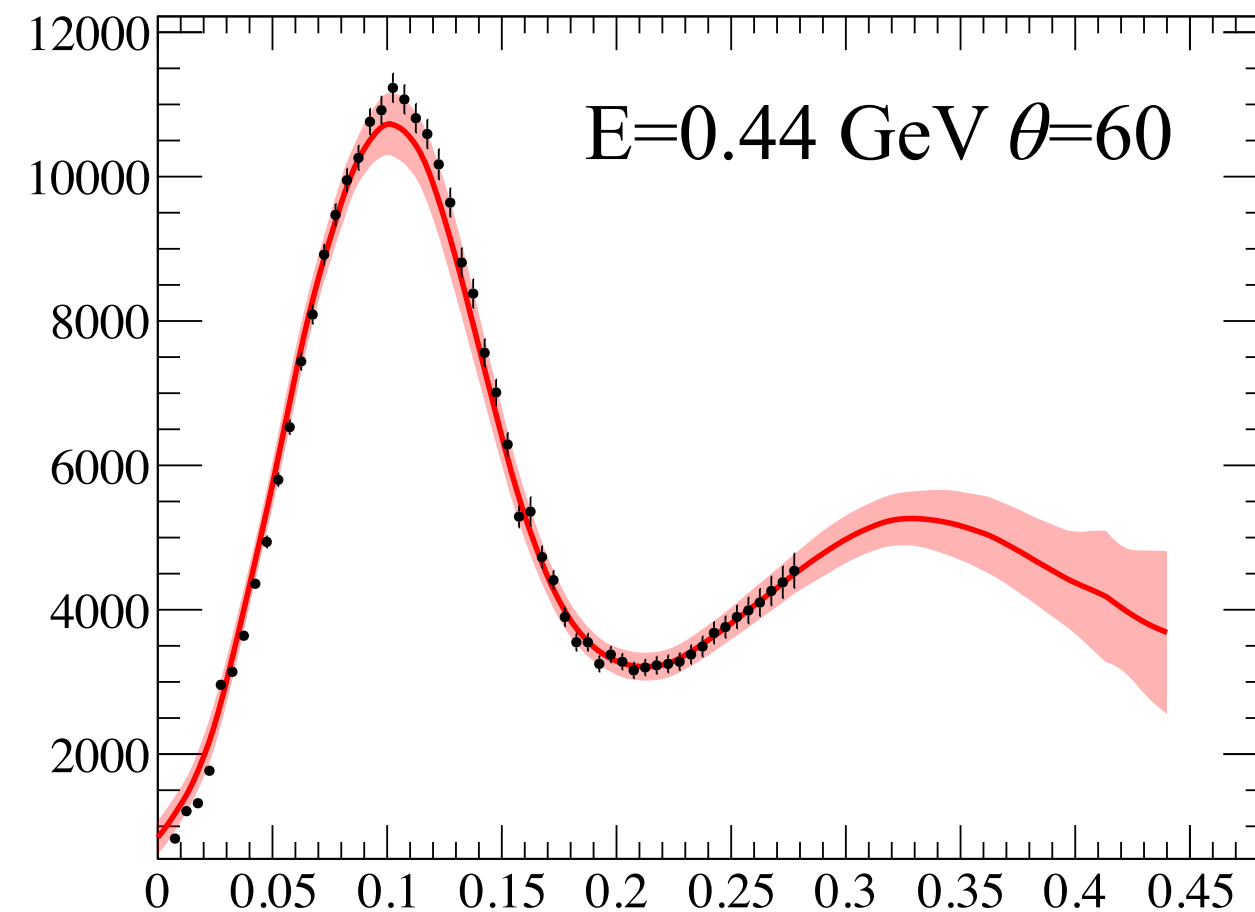
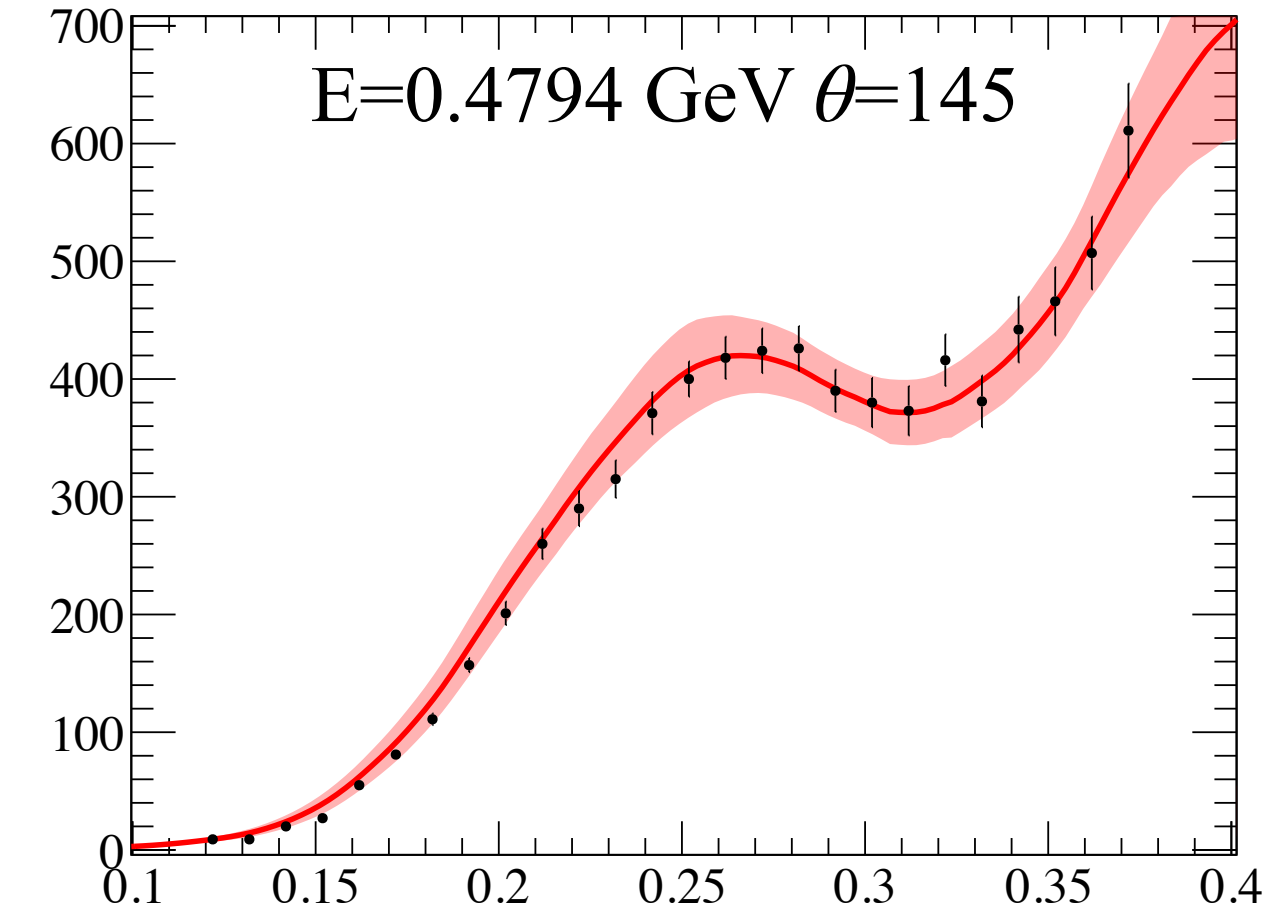
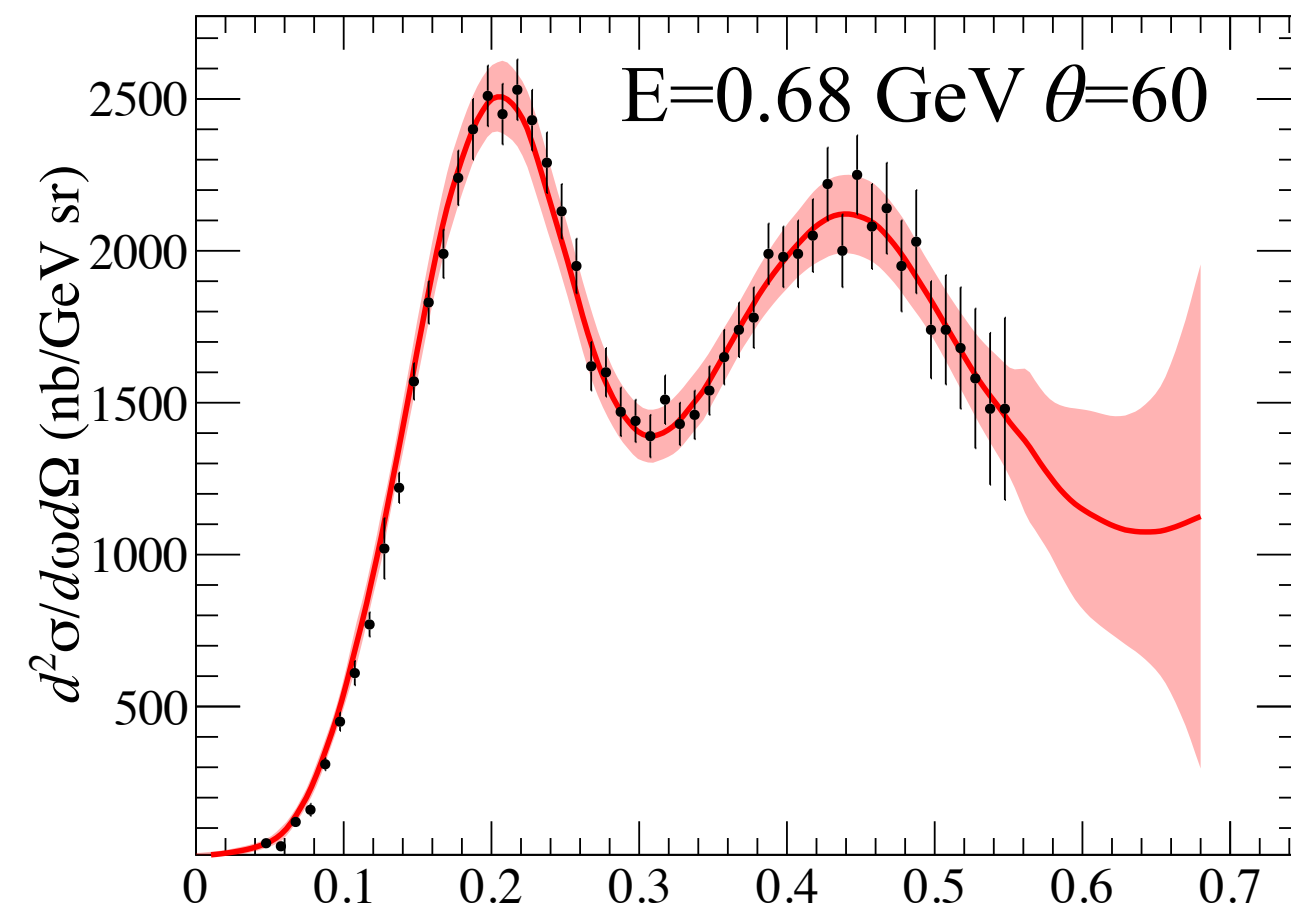


B. E. Kowal, K. M. Graczyk, *et al.*, Empirical fits to electron-carbon scattering data, **Phys. Rev. C. 110 025501 (2024)**

B. E. Kowal, K. M. Graczyk, *et al.*, Re-optimization of deep neural network model for electron-carbon scattering using new experimental data, **Phys. Rev. C 112 (2025) 055504**

Modeling electron-nucleus scattering

Empirical fits to $C^{12}(e, e'p)$ data using DNN



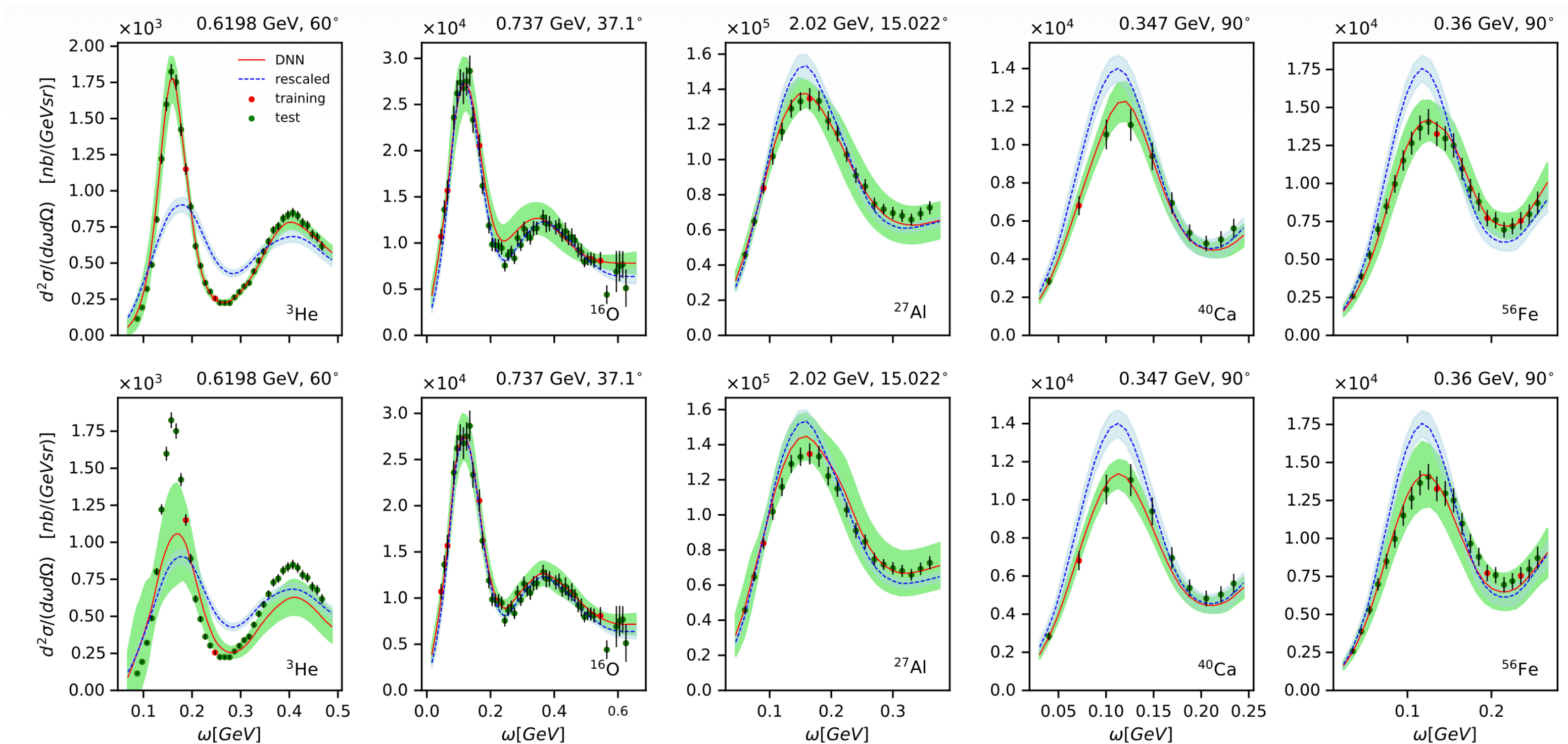
B. E. Kowal, K. M. Graczyk, *et al.*, Empirical fits to electron-carbon scattering data, **Phys. Rev. C. 110 025501 (2024)**

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Modeling electron-nucleus scattering

Extrapolating to other nuclear targets (through Transfer-learning)

- The idea is to extend the model's predictions to other nuclear targets without training from scratch.
- Assumption: The model is sufficiently deep to store general nuclear information.
- Last two layers are fine-tuned to new data, while other layers are fixed



K. M. Graczyk, B. E. Kowal, *et al.*, *Electron-Nucleus Cross Sections from Transfer Learning*, **Phys. Rev. Lett.** **135** 052502 (2025)

Modeling electron-nucleus scattering

Challenges !

- Neural networks predict the inclusive cross section (*decomposition into different interaction mechanisms is required*)
- Much more information (such as the kinematics of the final-state particles) is required to reconstruct a **complete event**.
- **A way out is:**
 - **Decompose the inclusive cross section using existing theoretical models**
 - **Reweight the inclusive responses from the existing theoretical models to match the DNN**
$$f_{QE} \cdot d^2\sigma/d\omega d\Omega|_{\text{inclusive}}^{\text{Neural Network}} + f_{MEC} \cdot d^2\sigma/d\omega d\Omega|_{\text{inclusive}}^{\text{Neural Network}} + f_{SPP} \cdot d^2\sigma/d\omega d\Omega|_{\text{inclusive}}^{\text{Neural Network}} = 1$$

Modeling electron-nucleus scattering

Challenges !

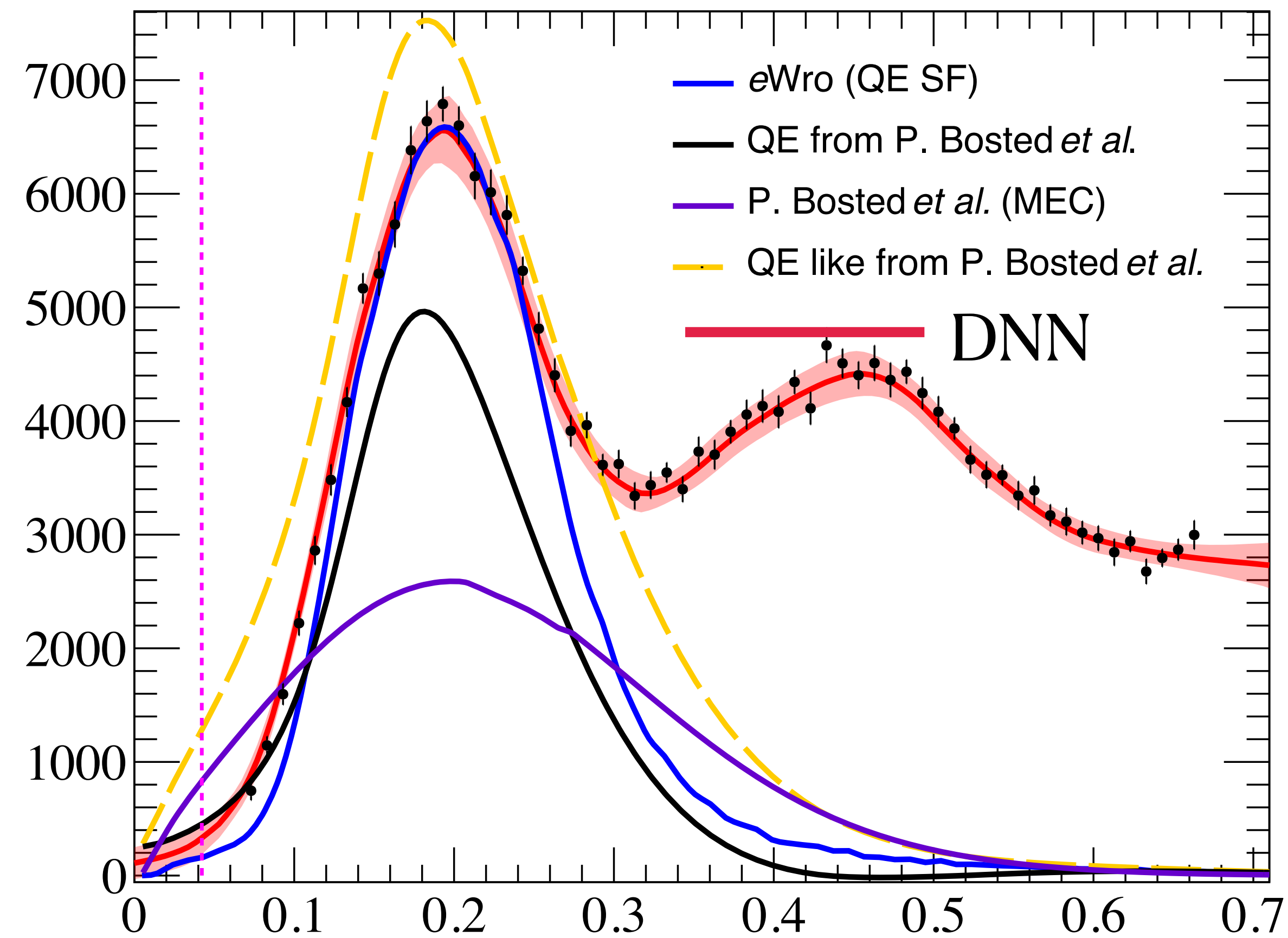
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No MEC model in eWro!

Modeling electron-nucleus scattering

Challenges !

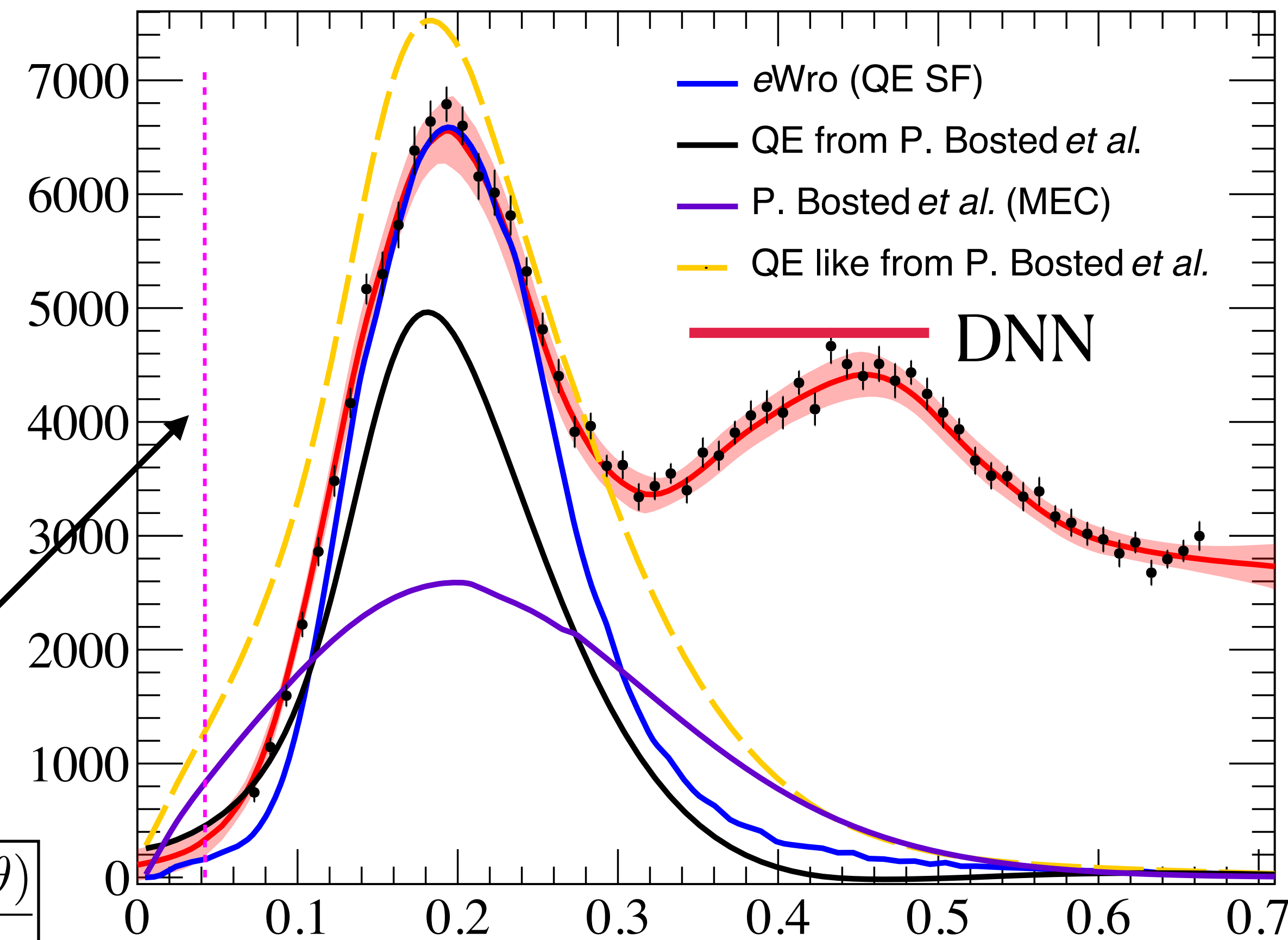


B. E. Kowal, K. M. Graczyk, *et al.*,
Empirical fits to electron-carbon
scattering data, [Phys. Rev. C. 110](#)
[025501 \(2024\)](#)

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Modeling electron-nucleus scattering

Challenges !



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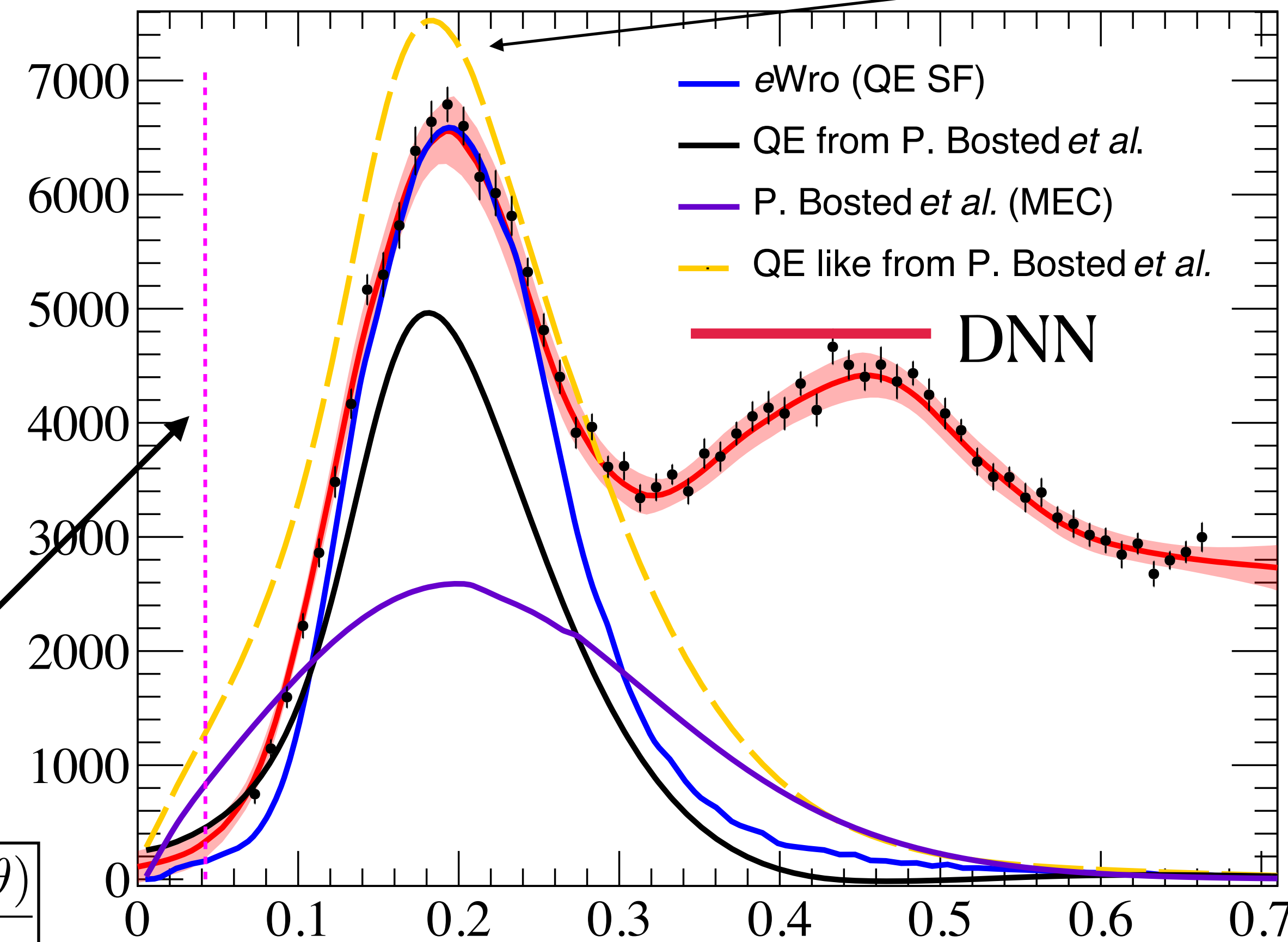
B. E. Kowal, K. M. Graczyk, *et al.*, Re-optimization of deep
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new experimental data, [Phys. Rev. C 112 \(2025\) 055504](#)

$$\omega_{\min} = \frac{M_A E_x + 0.5 E_x^2 + E^2 (1 - \cos \theta)}{M_A + E (1 - \cos \theta)}$$

Modeling electron-nucleus scattering

Challenges !

Acts as an envelope!



B. E. Kowal, K. M. Graczyk, *et al.*,
Empirical fits to electron-carbon
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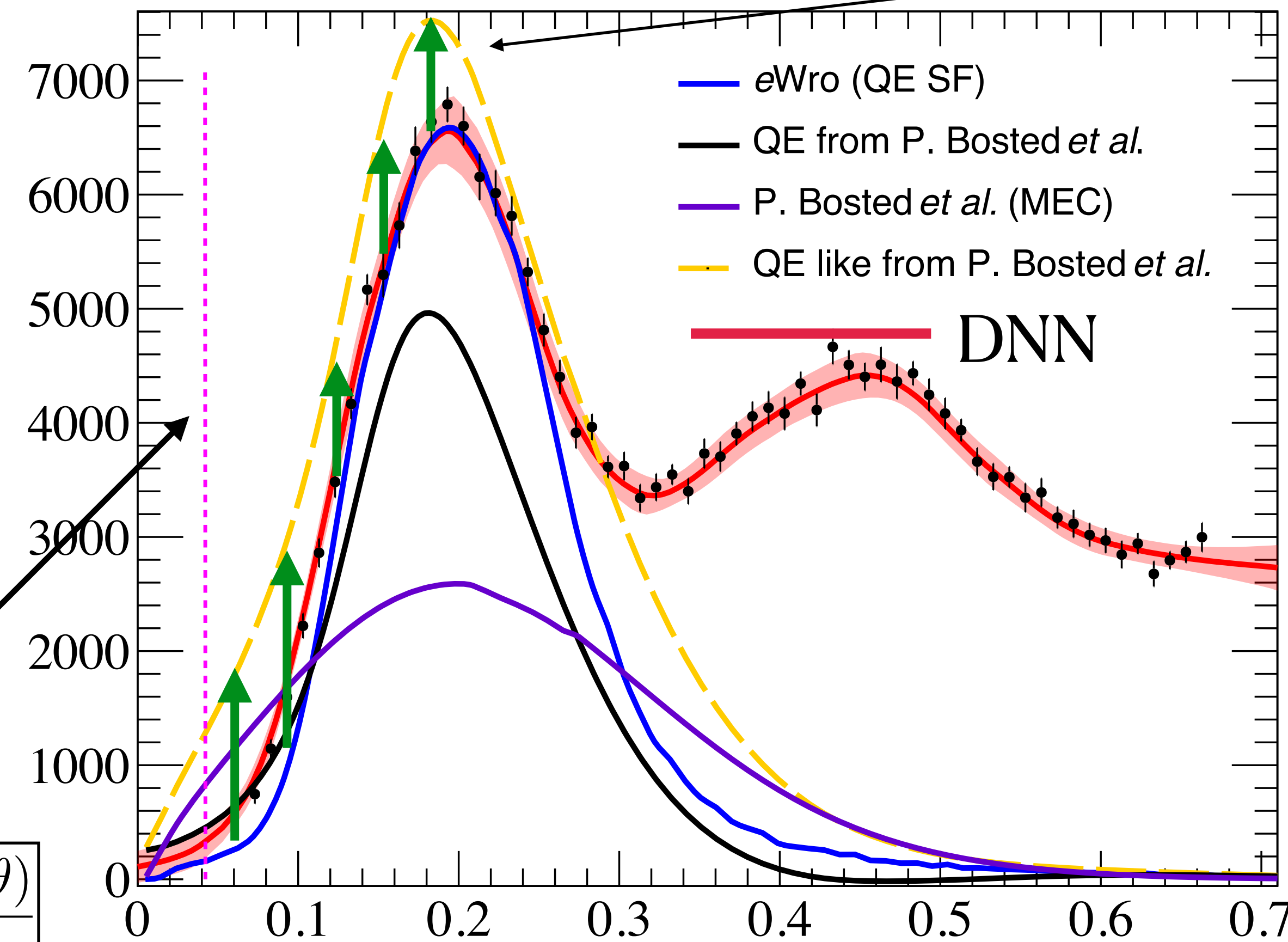
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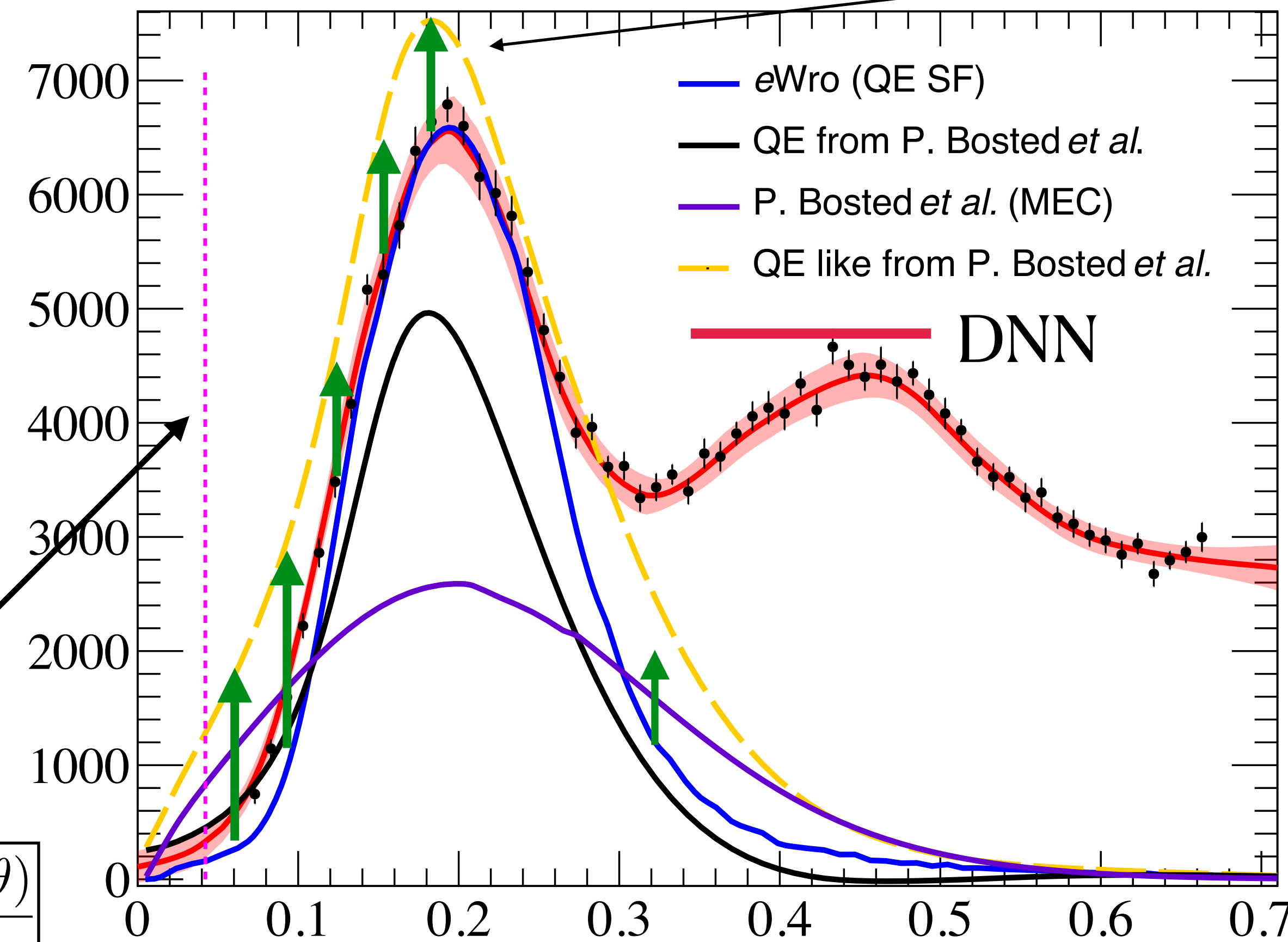
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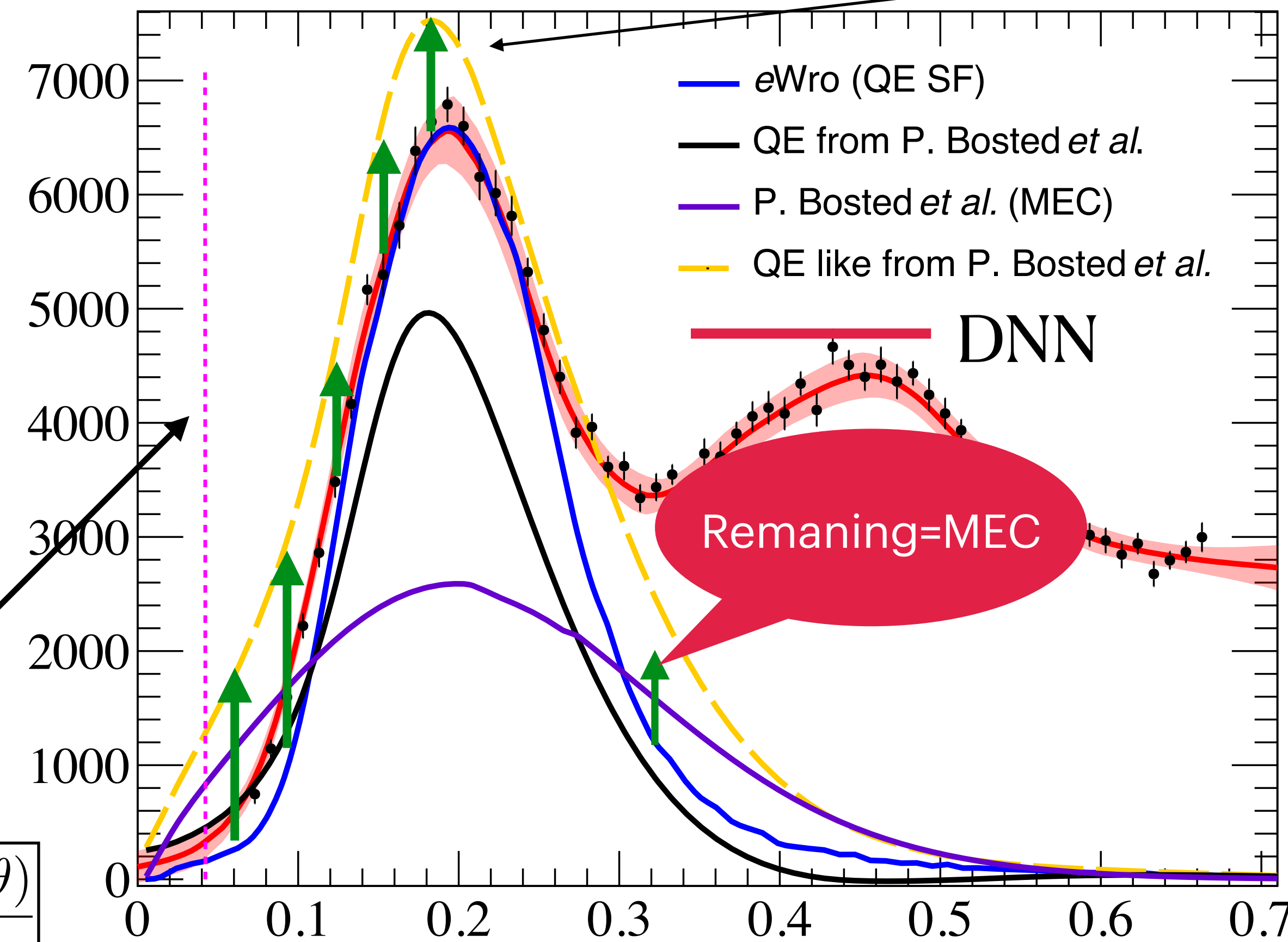
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Modeling electron-nucleus scattering

Challenges !

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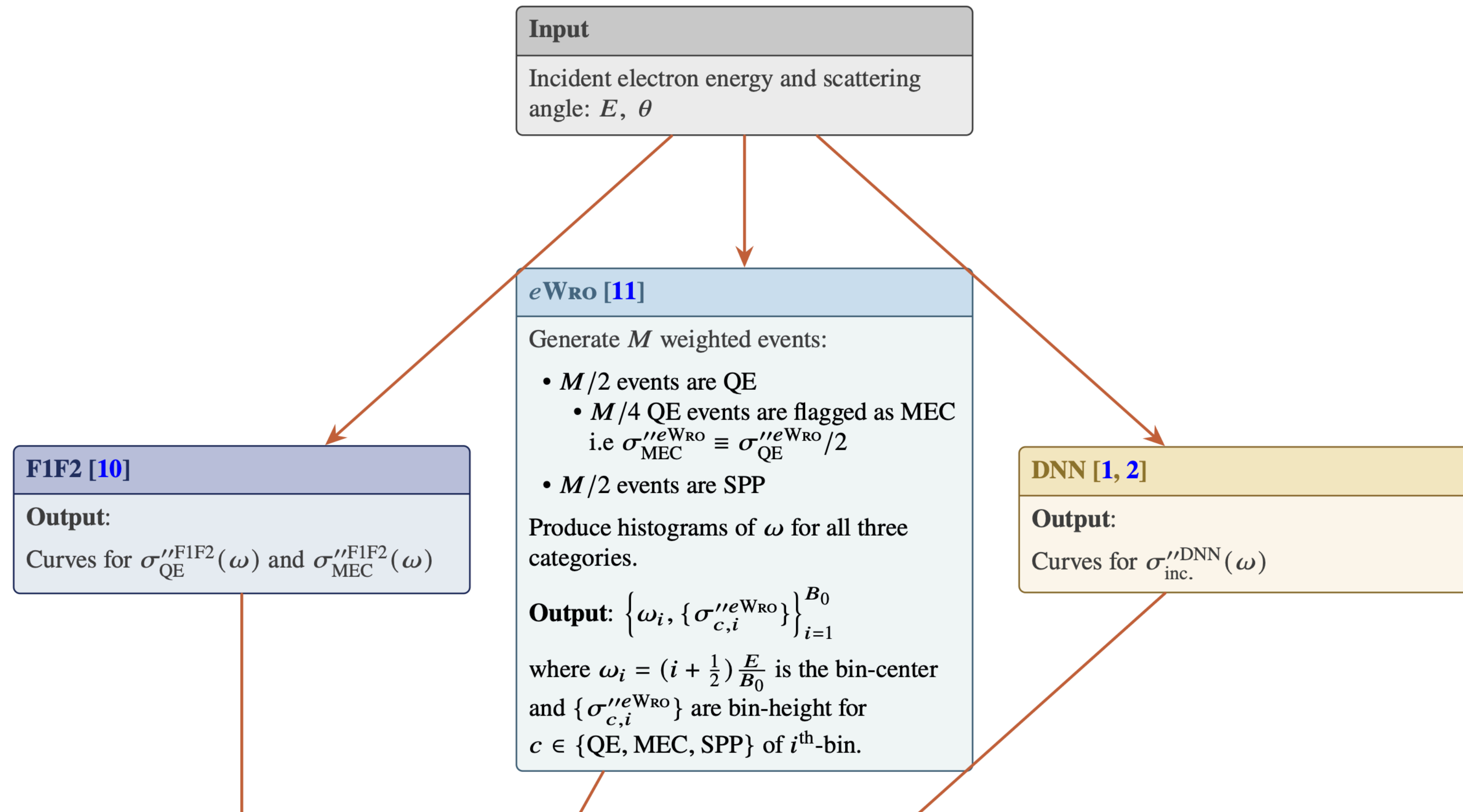
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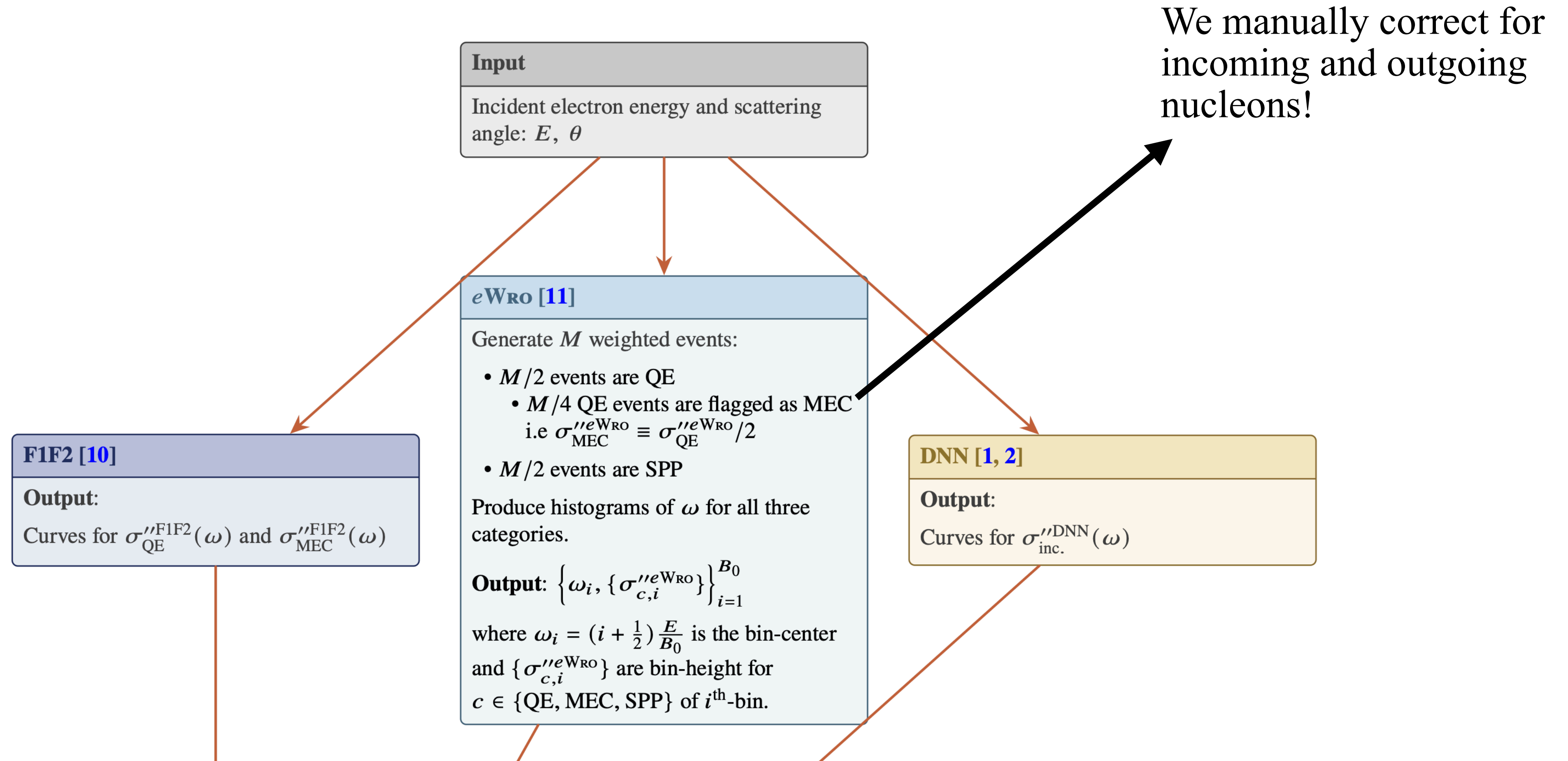
Birth of eWroNet

Schematics



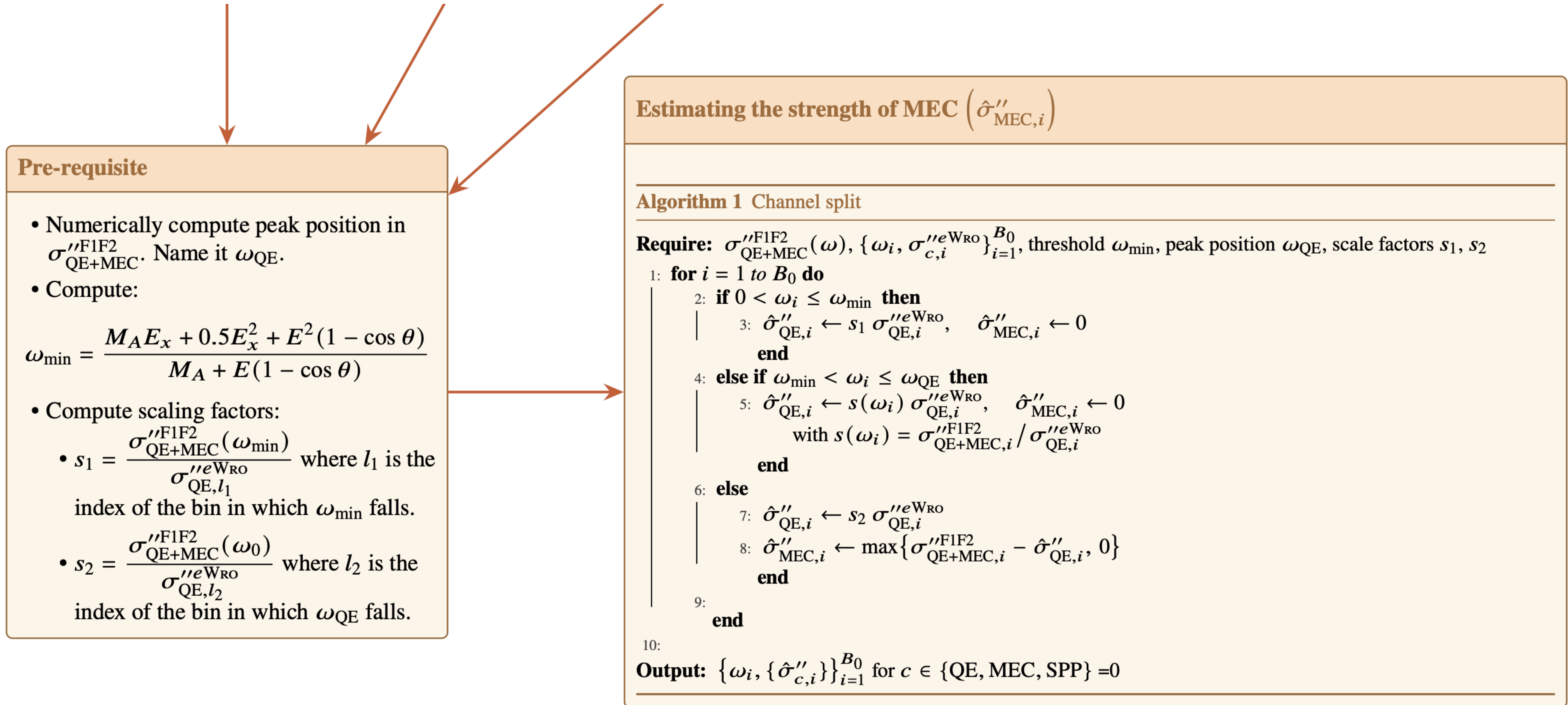
Birth of eWroNet

Schematics



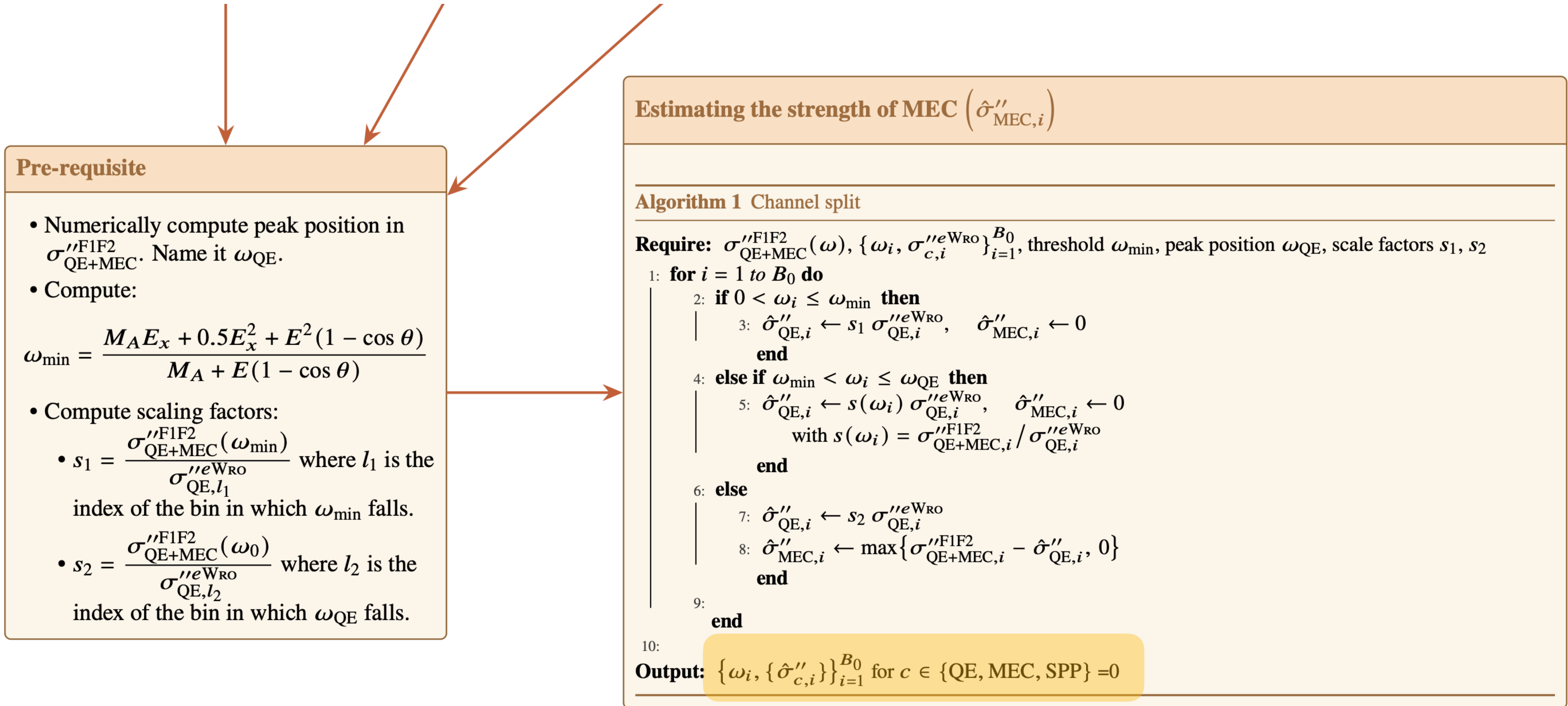
Birth of eWroNet

Schematics



Birth of eWroNet

Schematics



eWroNet

Schematics

eWro interaction scheme

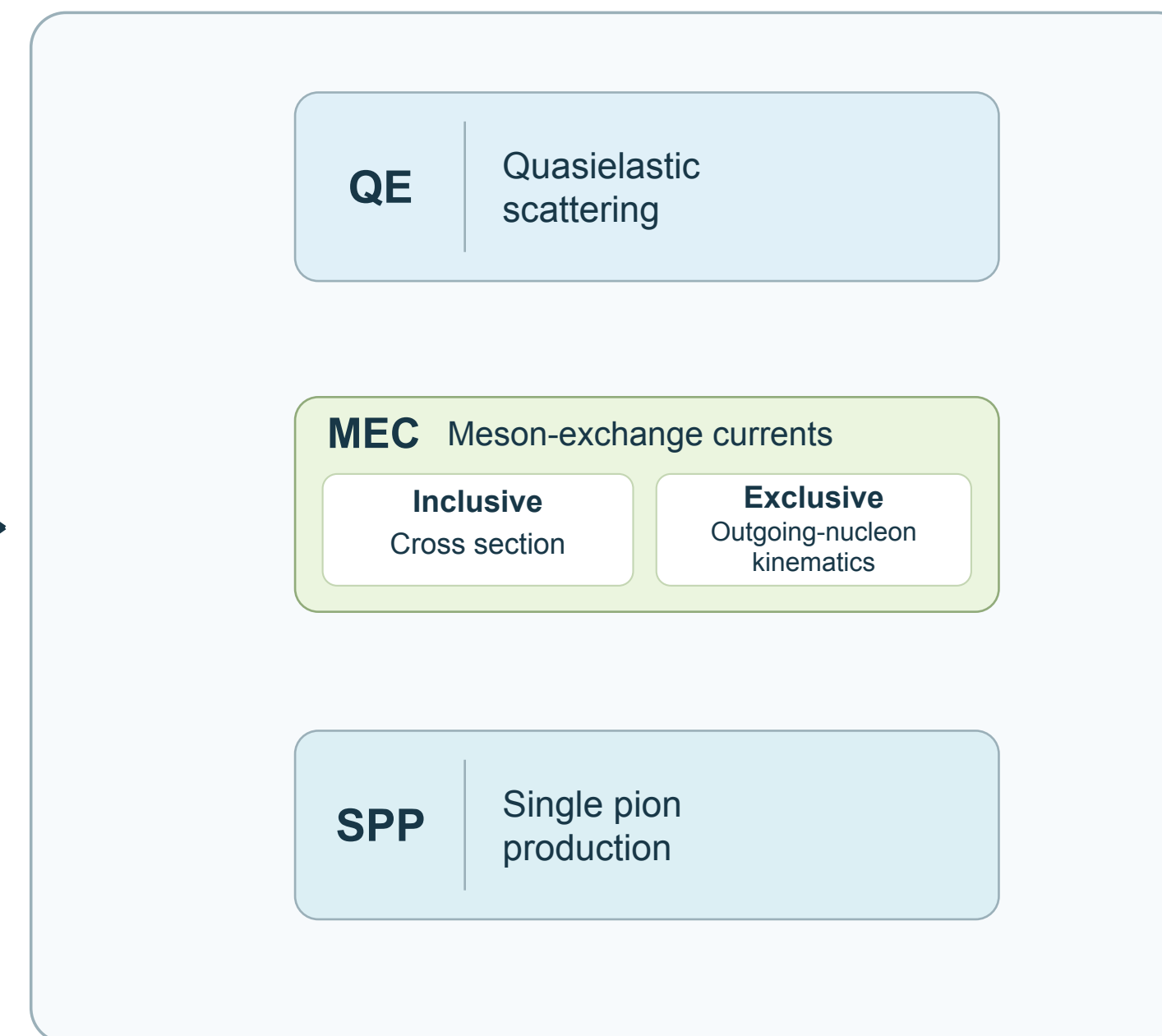


Initial state



DNN & F1F2

Interaction channels

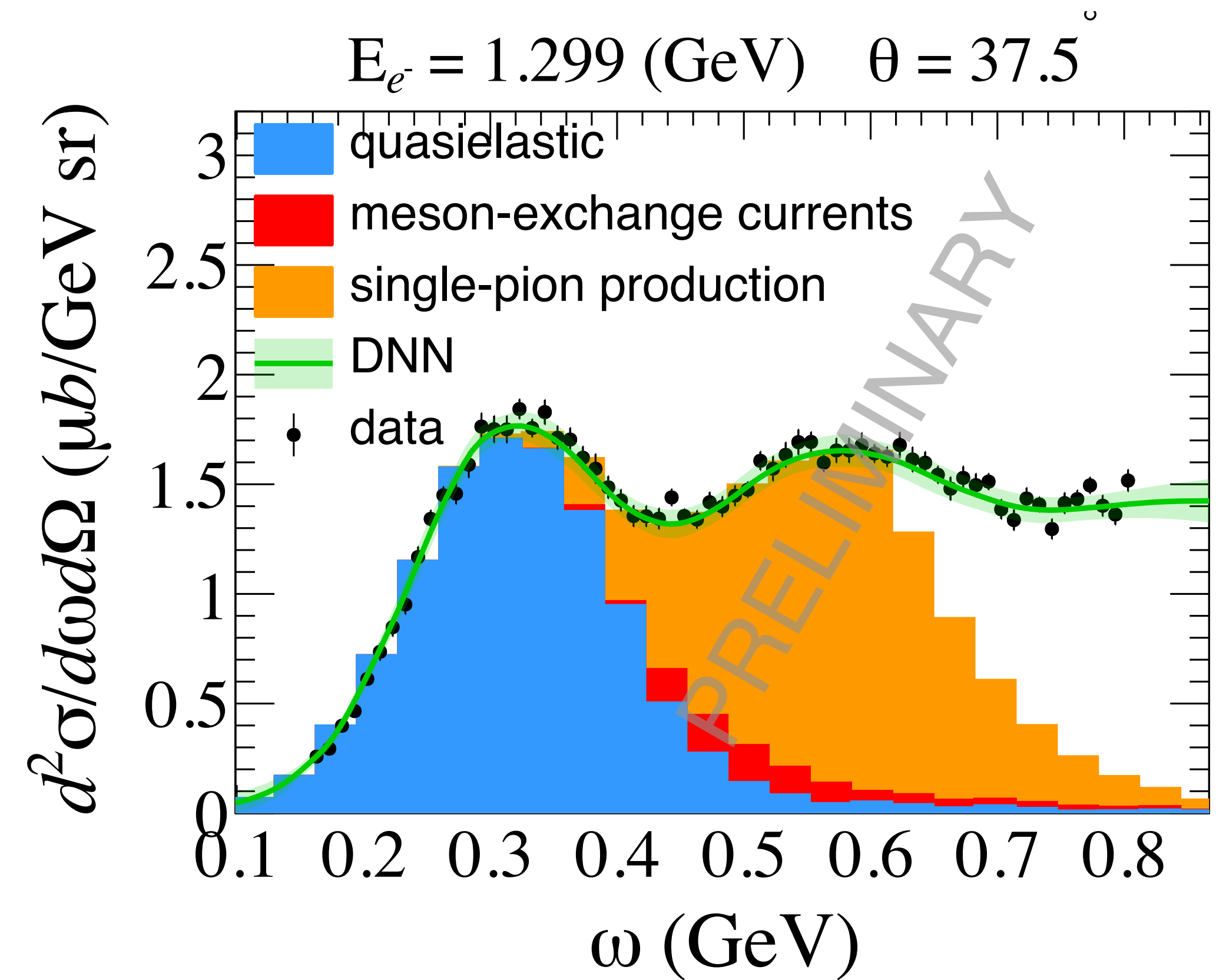
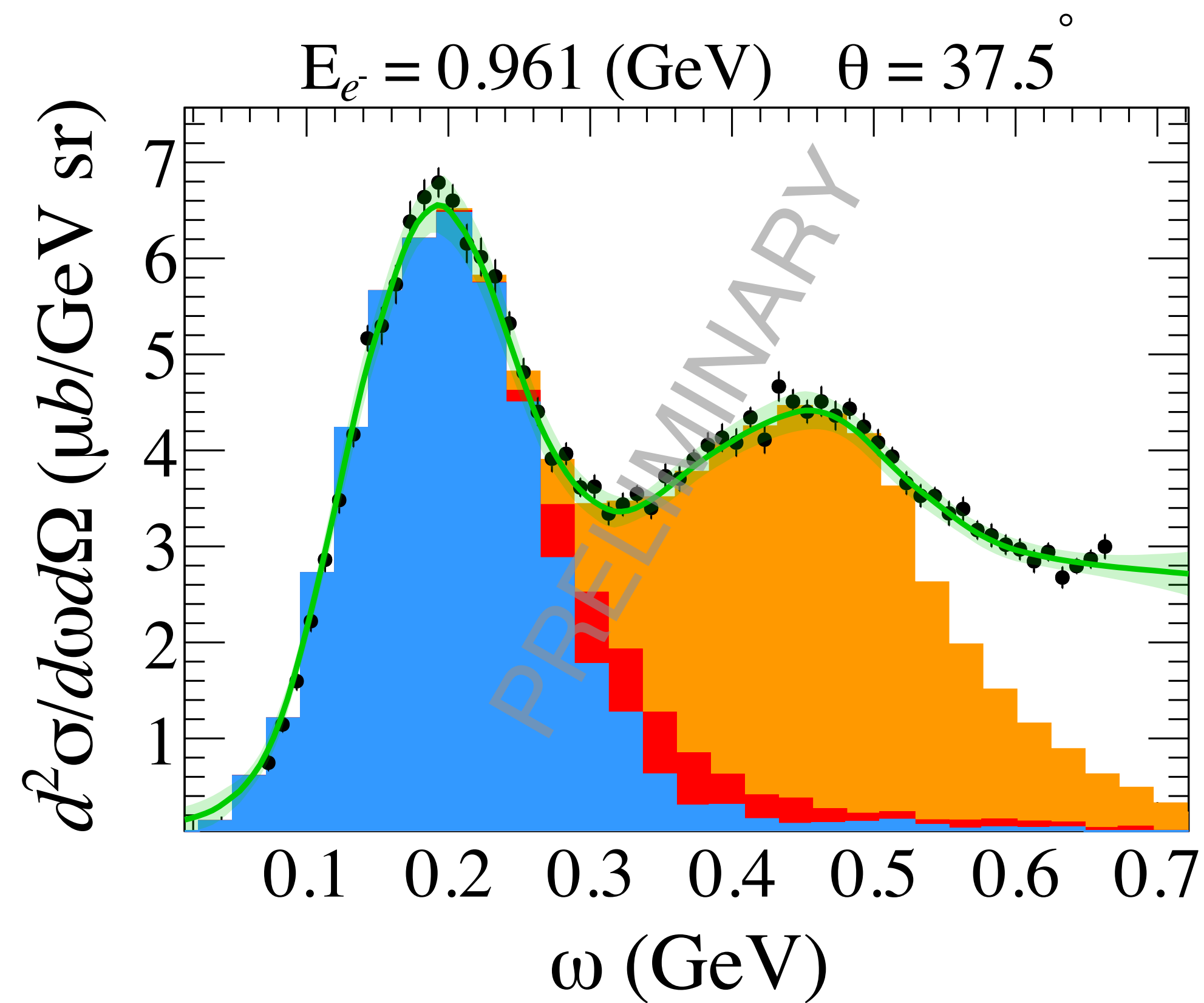


FSI Final-state interactions



eWroNet

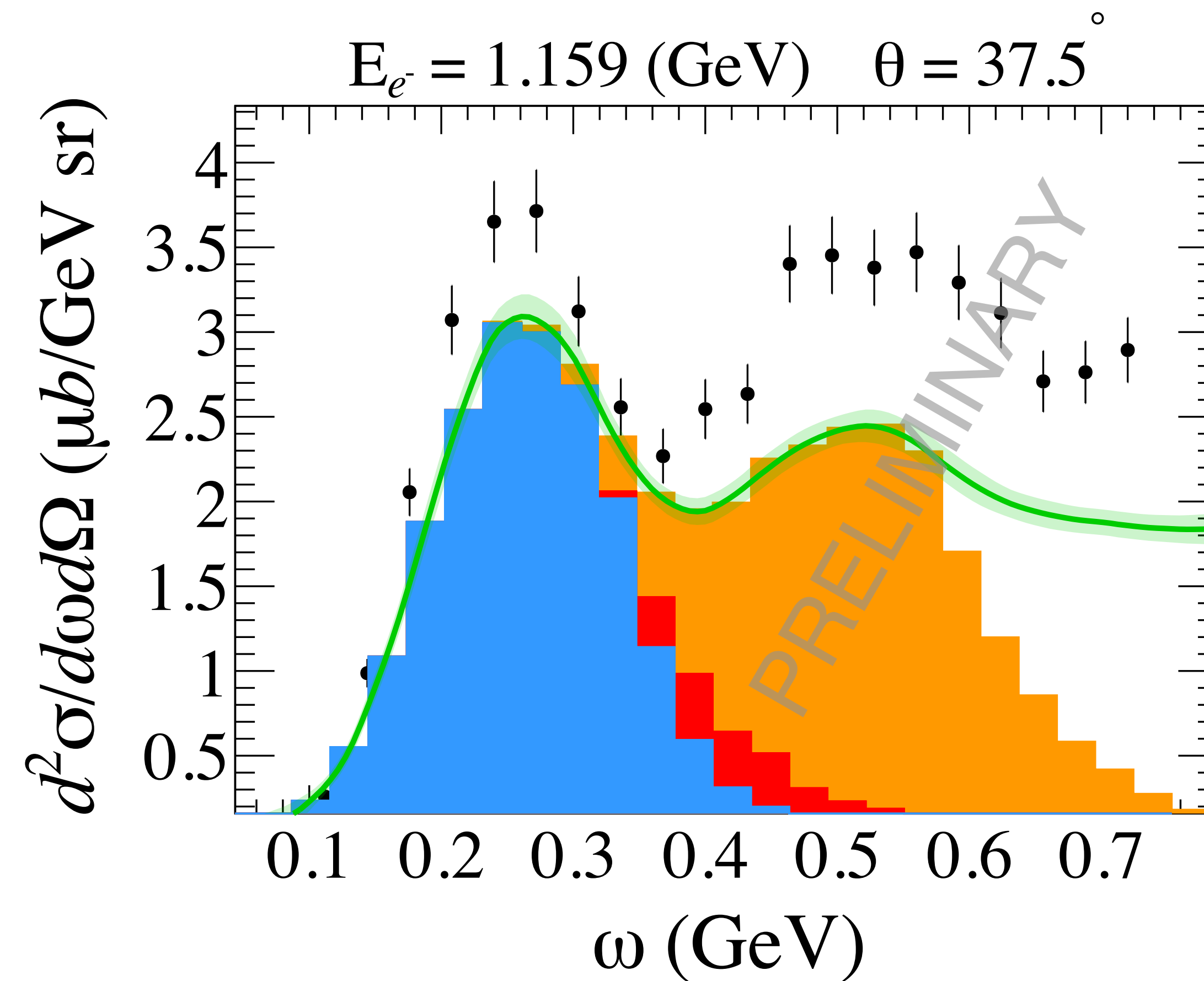
Performance



Target: carbon

eWroNet

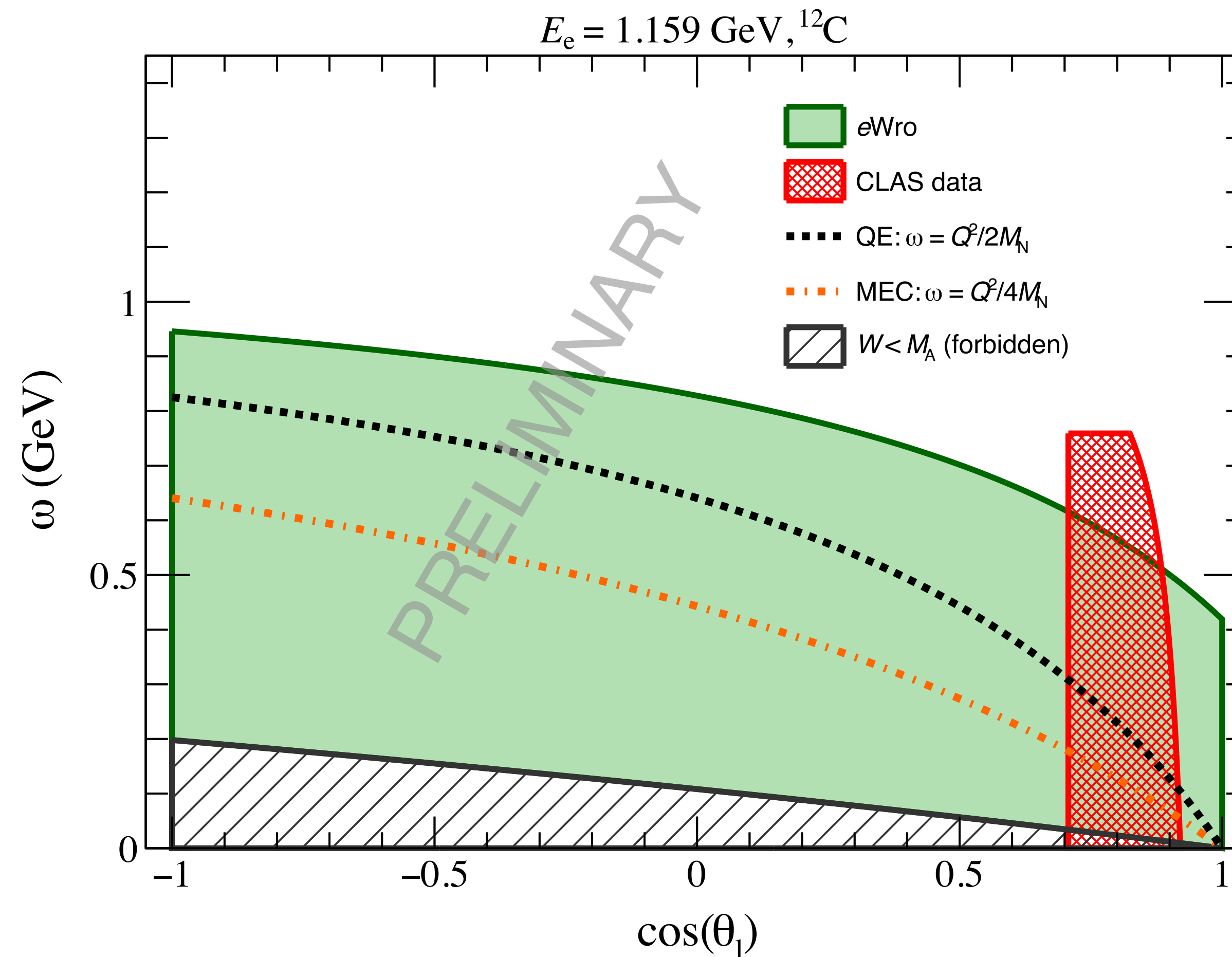
Performance against CLAS data (Nature Vol 599, Page 569)



Target: carbon

CLAS Data (Nature Vol 599, Page 569)

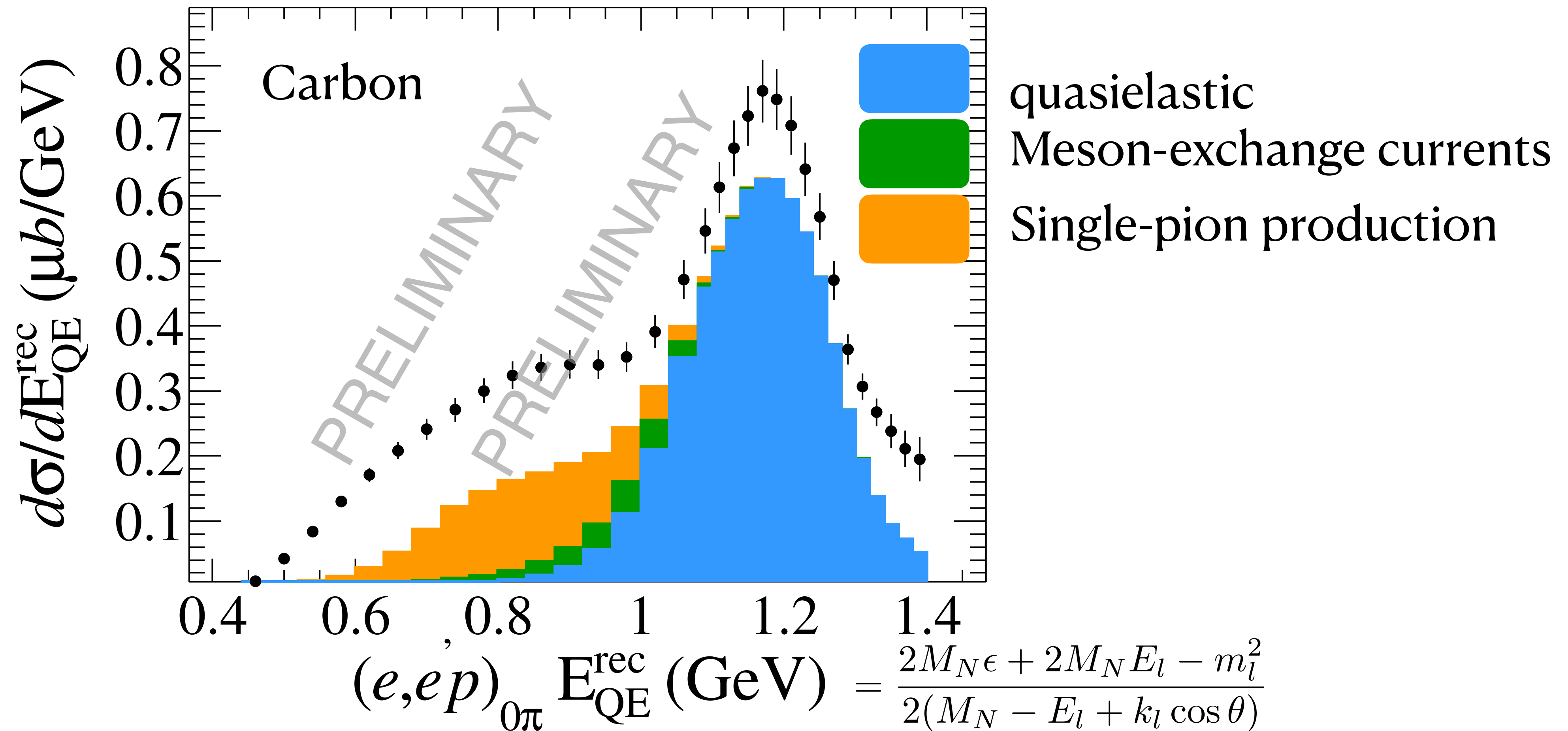
Phasespace covered by CLAS data and eWroNet



CLAS Data (Nature Vol 599, Page 569)

Performance of eWroNet

$15^\circ \leq \theta_l \leq 45^\circ$ $E_e = 1.159$ GeV



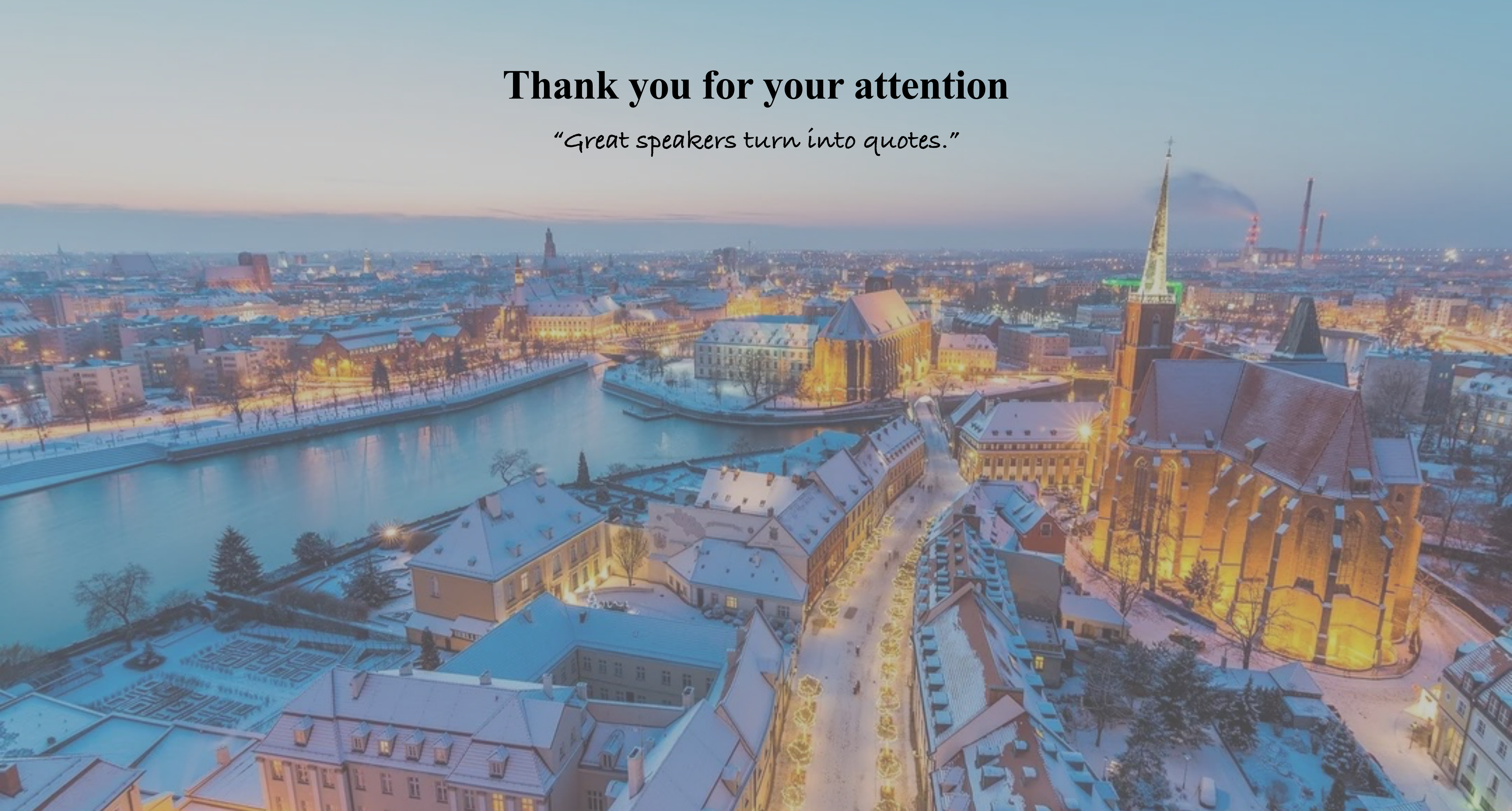
Summary

“And the physics must go on...”

- ♦ Improvements in modeling nuclear effects in the MEC channel
- ♦ A fine-tuned NuWro cascade module with reduced uncertainty
- ♦ Development of a hybrid model (DNN+MC) for future nuclear effects benchmarking
 - ♦ Currently done for carbon; in the future, many other nuclei will also be covered.
- ♦ Users are recommended to use the latest version of NuWro (i.e 25.11.2)
- ♦ Stay tuned to NuWro for upcoming updates

Thank you for your attention

"Great speakers turn into quotes."



Backup slides

Motivation

Meson exchange currents

- Motivated by electron-nucleus scattering data.
- A **multi-nucleon knockout** process.
- Energy exchange between nucleons is through in-medium meson production.
- **Significant contribution around ~ 1 GeV.** An important interaction mode for all upcoming neutrino experiments.
- Dominated by two-body current interactions (2p2h). Small contribution coming from three body current interactions as well (3p3h) before FSI !!
- Highly model-dependent (*will be shown in the upcoming slides !*). Various models are present in the market. Must be implemented in Monte Carlo generators as well.

MEC models in NuWro

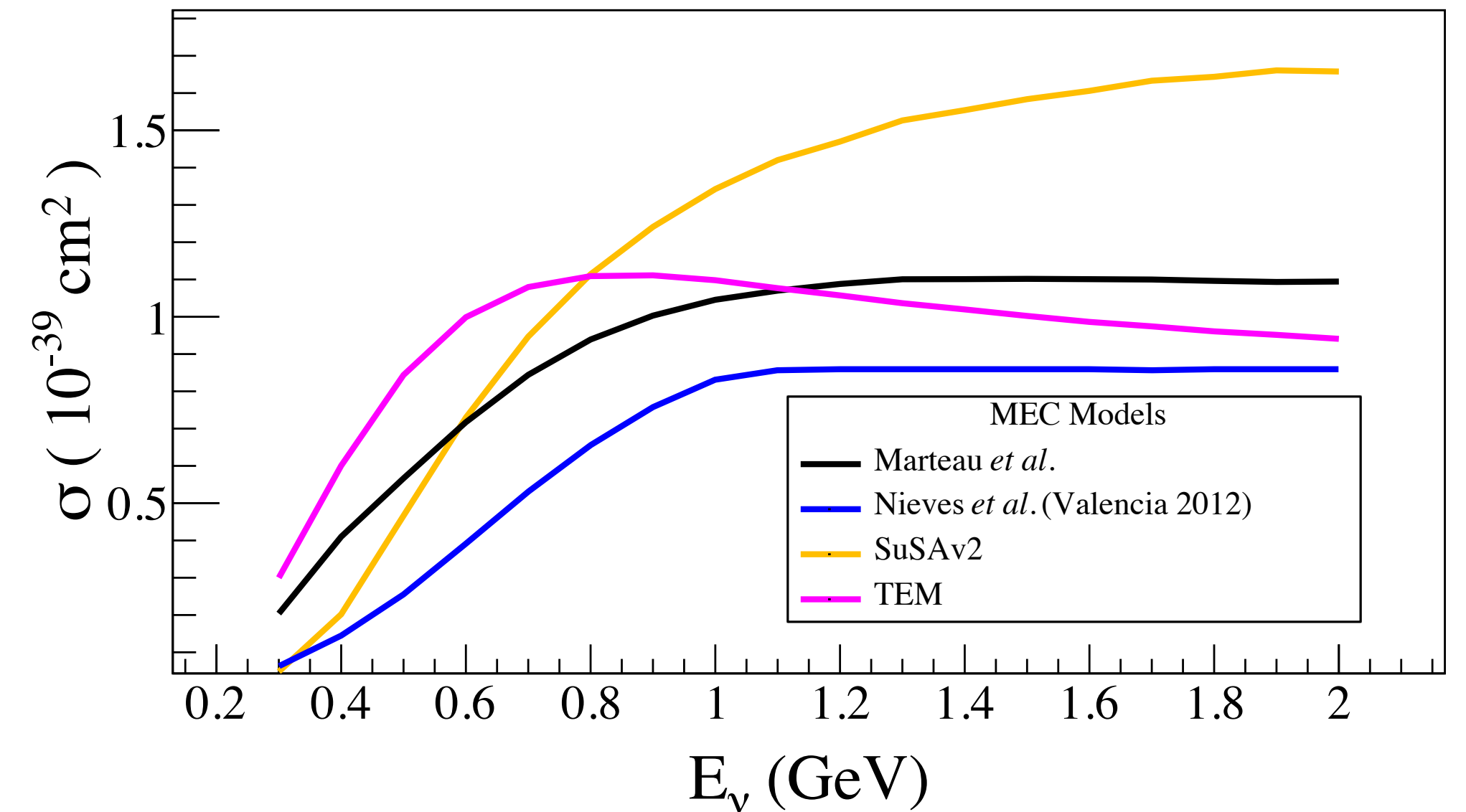
Implementation and coverage

- Four MEC models are featured in NuWro :
 - Nieves *et al.* model (also referred as Valencia 2012 model).
 - Marteau *et al.* model.
 - SuSAv2 model.
 - Transverse enhancement (TE) model.

MEC models in NuWro

Implementation and coverage

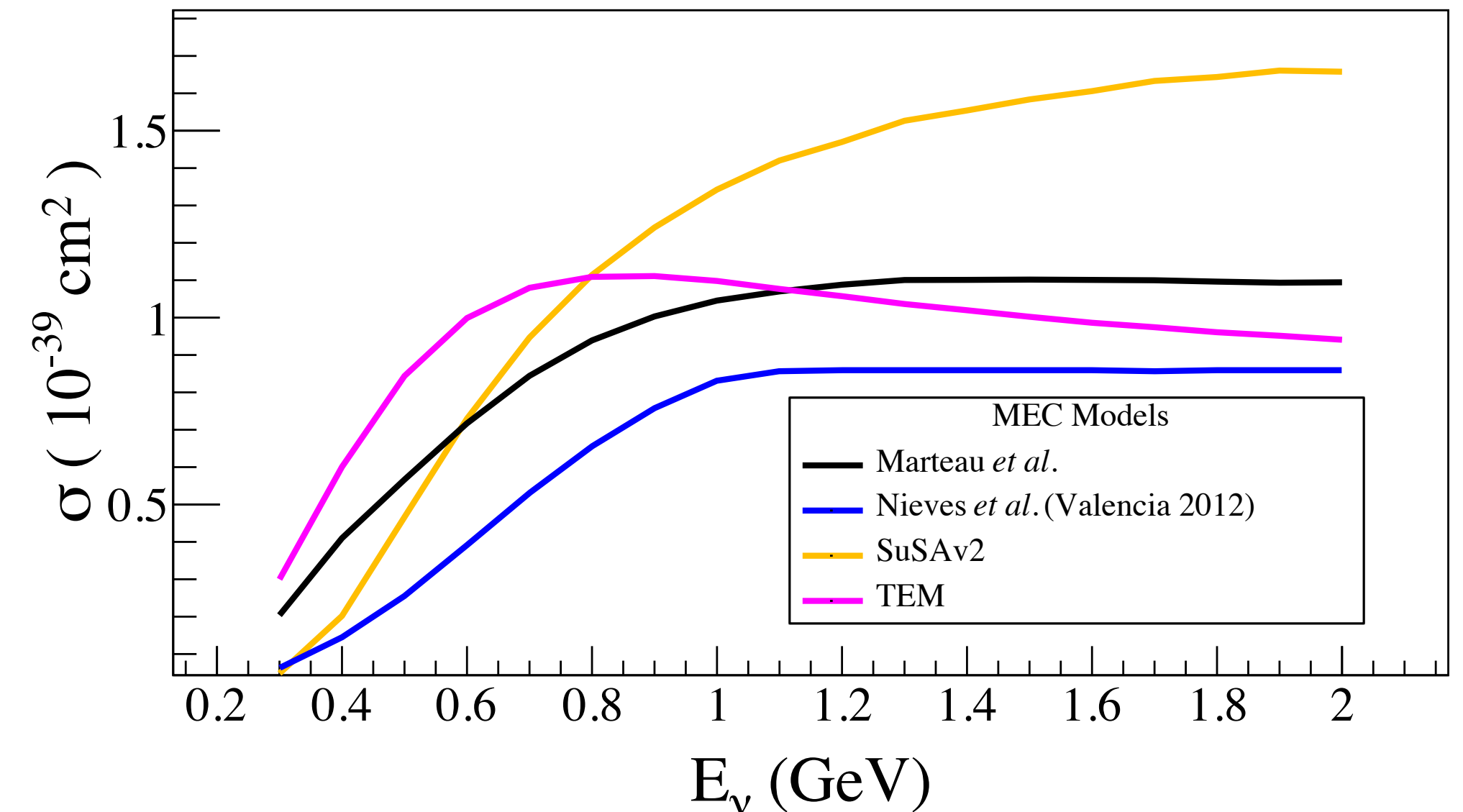
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MEC models in NuWro

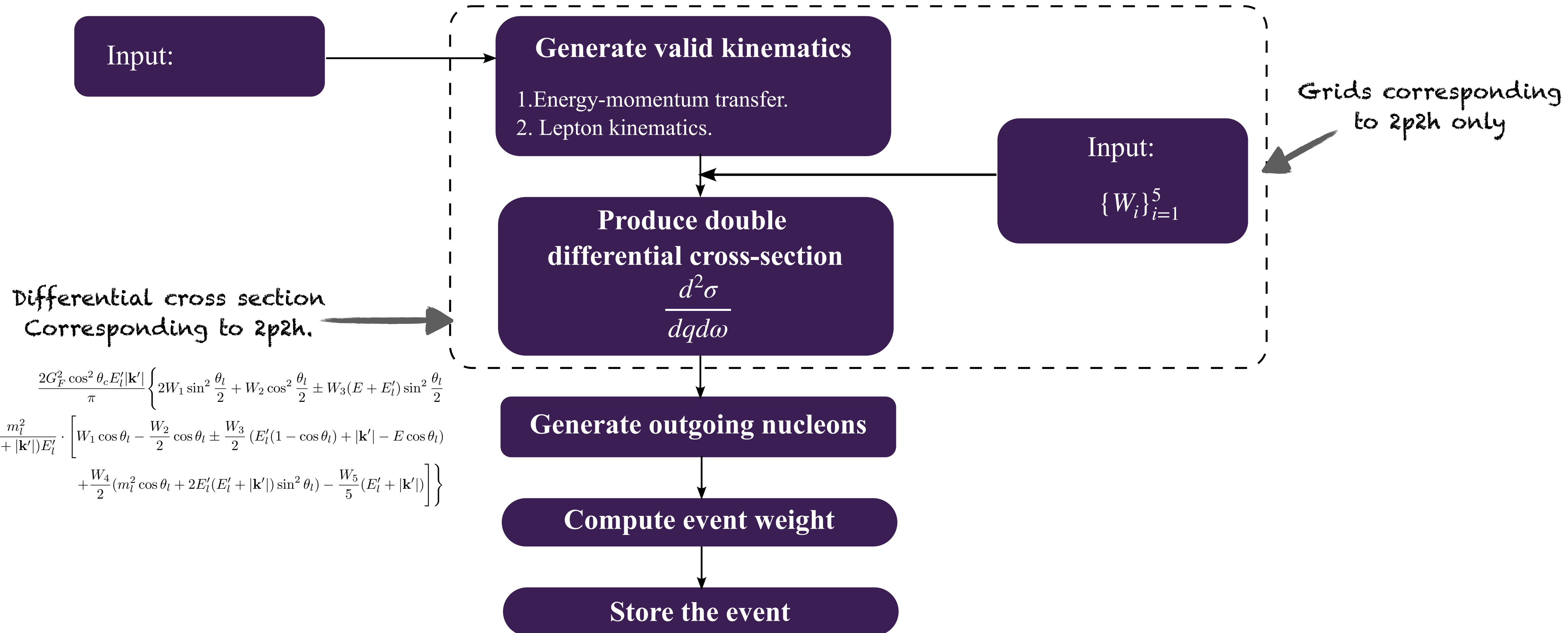
Implementation and coverage

- Four MEC models are featured in NuWro :
 - Nieves *et al.* model (also referred as Valencia 2012 model).
 - Marteau *et al.* model.
 - SuSAv2 model.
 - Transverse enhancement (TE) model.
- Their implementation and coverage differs.
 - TE model is implemented with **analytic formulas**.
 - Implementation of Valencia 2012 and SuSav2 models are done using **tabularization of the five response functions $W^{\mu\nu}(q, \omega)$** .
 - Marteau model is implemented in a **hybrid way**.
- The exclusive parts of different MEC models are modelled identically, referred to as the “**old hadronic model**”.



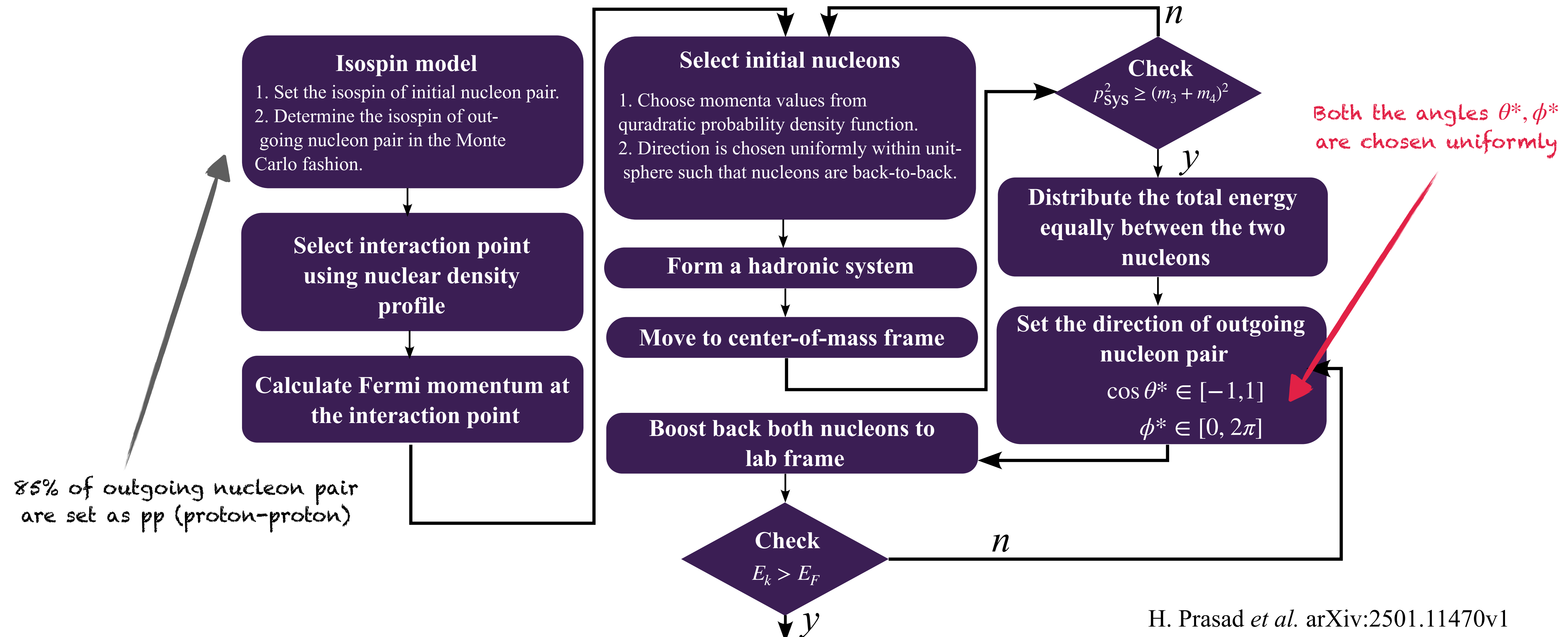
Implementation of MEC models

Within the context of Nieves *et al.* (Valencia 2012) model



Implementation of MEC models

Within the context of Nieves *et al.* (Valencia 2012) model



H. Prasad *et al.* arXiv:2501.11470v1

Minimisation procedure

Optimising the nucleon phase space

- Specify parameter space
- $$P \in [-1,1] \quad n \in \{1,2,\dots,10\}$$
- $$\Delta P = 0.01 \quad \Delta n = 1$$

- Choose neutrino energy values GeV
- For any given (P,n), produce outgoing nucleon distribution
- Choose only those bins in 2020 Valencia model where
- Compare the differences between two distributions using


$$\tilde{\chi}^2 = \frac{1}{2N_{\text{bins}}} \sum_{i,j}^{N_{\text{bins}}} \frac{(N_{ij}^{\text{NuWro}} - N_{ij}^{\text{Valencia}})^2}{(N_{ij}^{\text{NuWro}} + N_{ij}^{\text{Valencia}})}$$

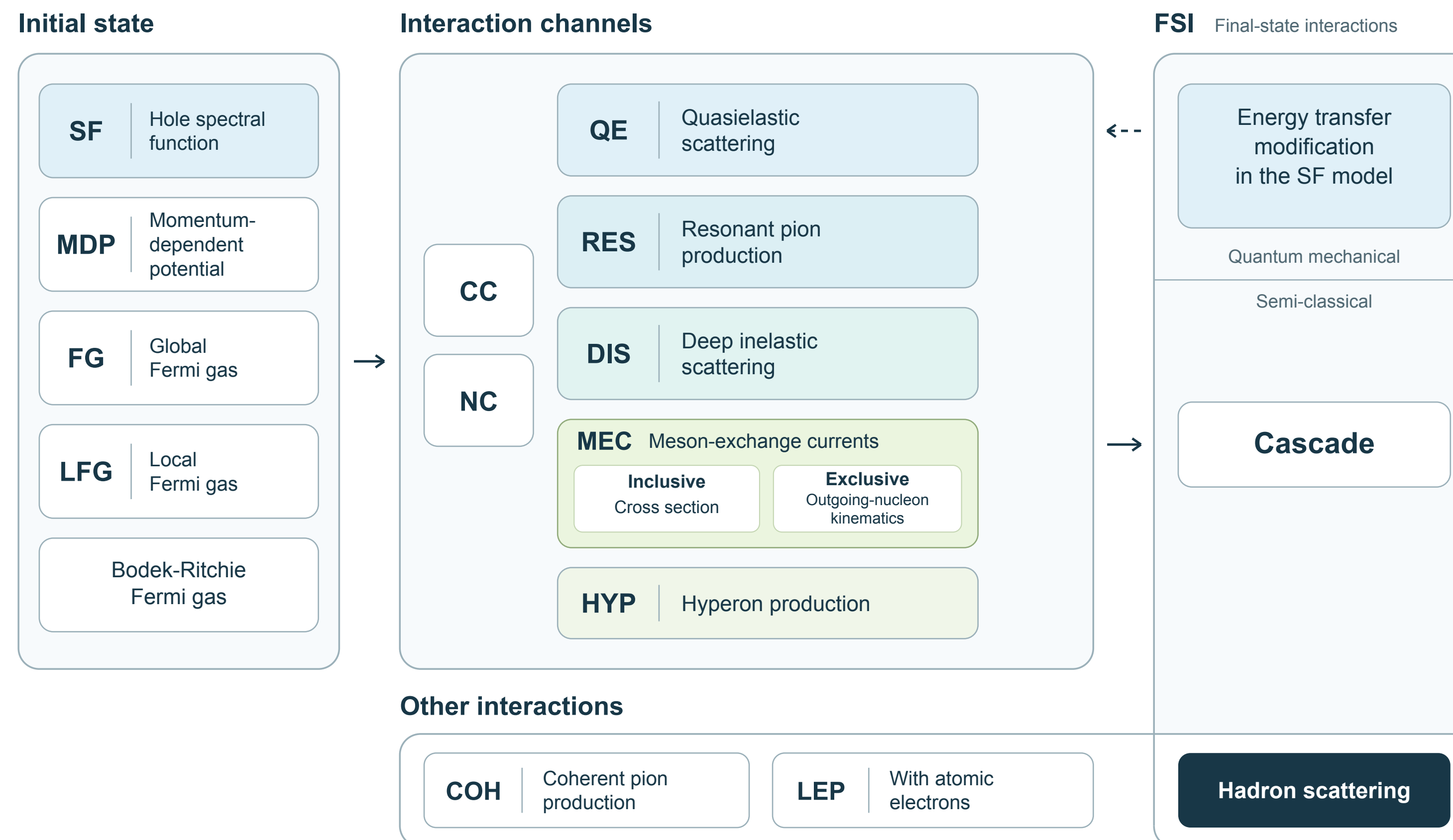
- Find the minimum $\tilde{\chi}^2$ within the parameter space
- Find the global minimum

$$\hat{\tilde{\chi}}_{pp}^2 = \min_{\tilde{\chi}^2} \left\{ \sum_{E_\nu} \tilde{\chi}_{pp}^2(P, l)_{pp} \right\}$$

NuWro

Schematics

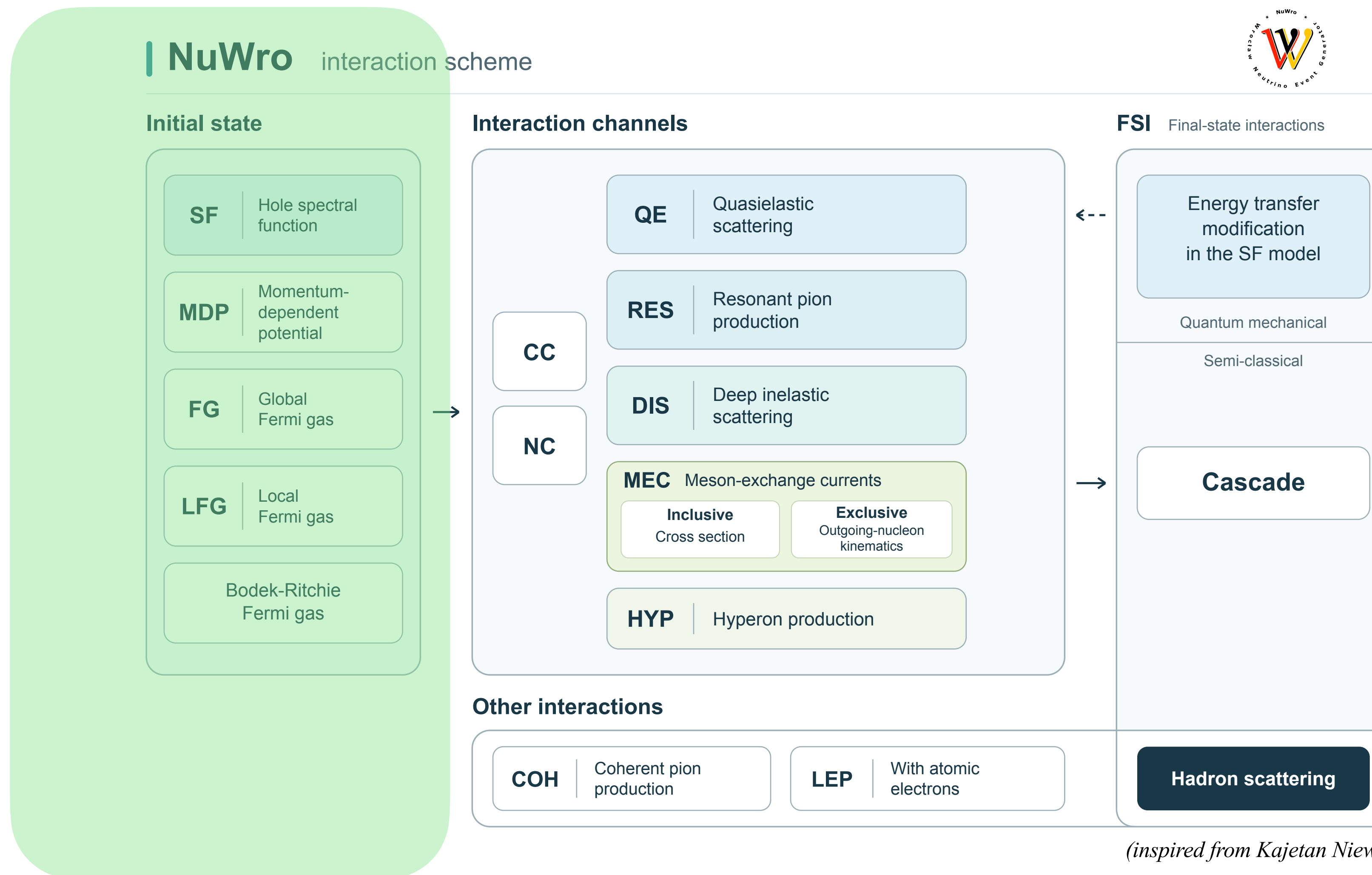
NuWro interaction scheme



(inspired from Kajetan Niewczas's illustrations)

NuWro

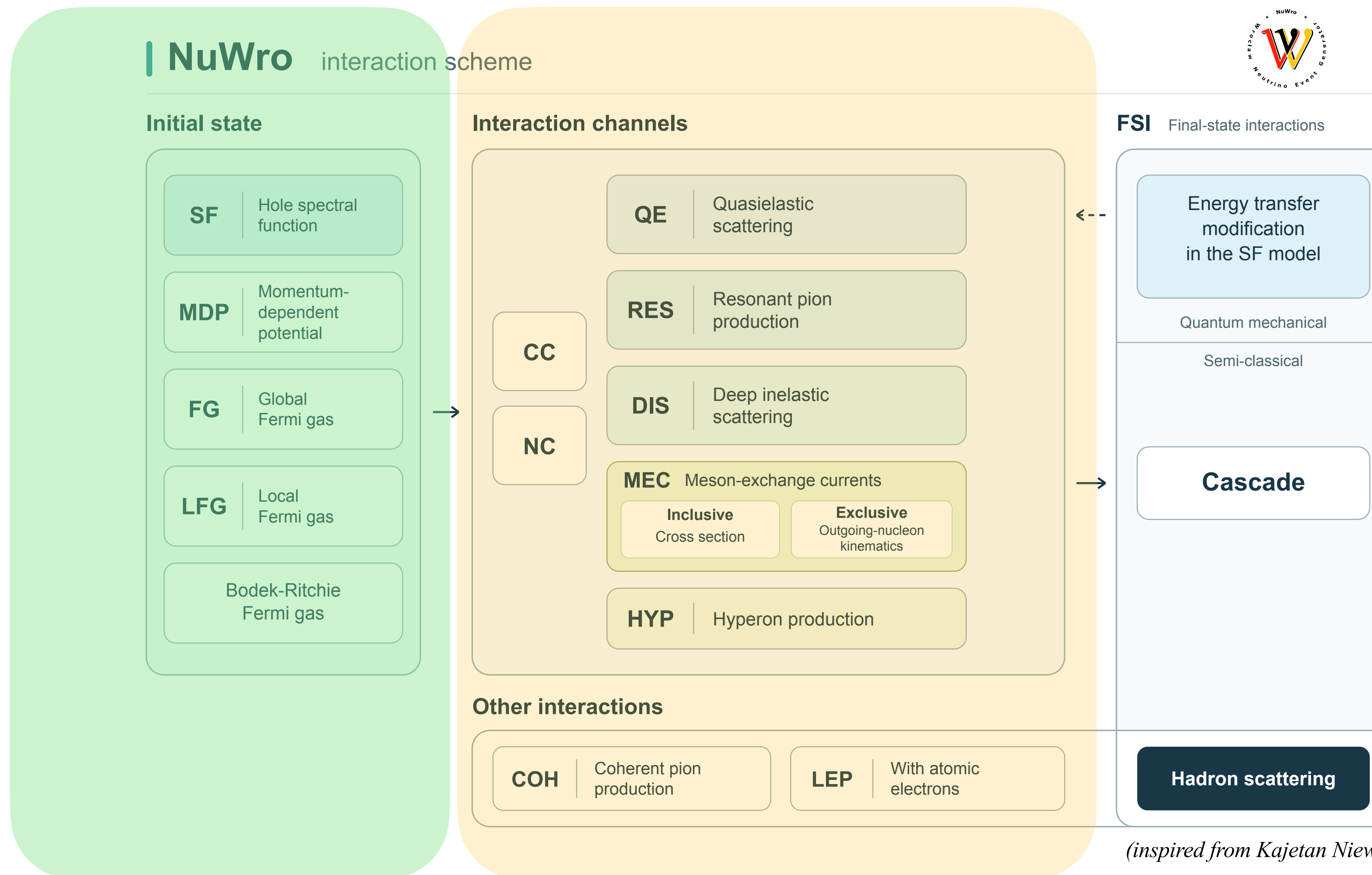
Schematics



(inspired from Kajetan Niewczas's illustrations)

NuWro

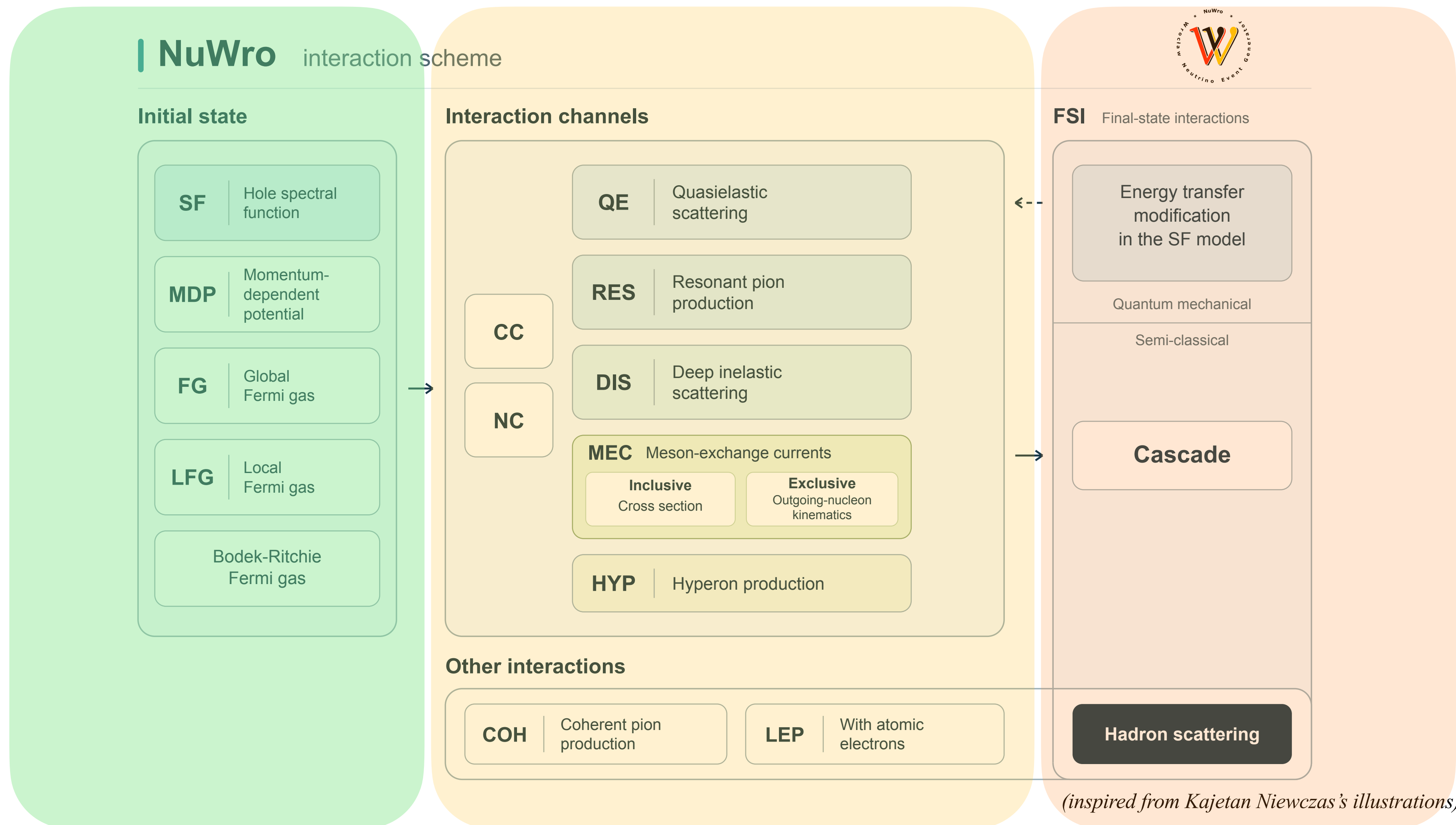
Schematics



(inspired from Kajetan Niewczas's illustrations)

NuWro

Schematics



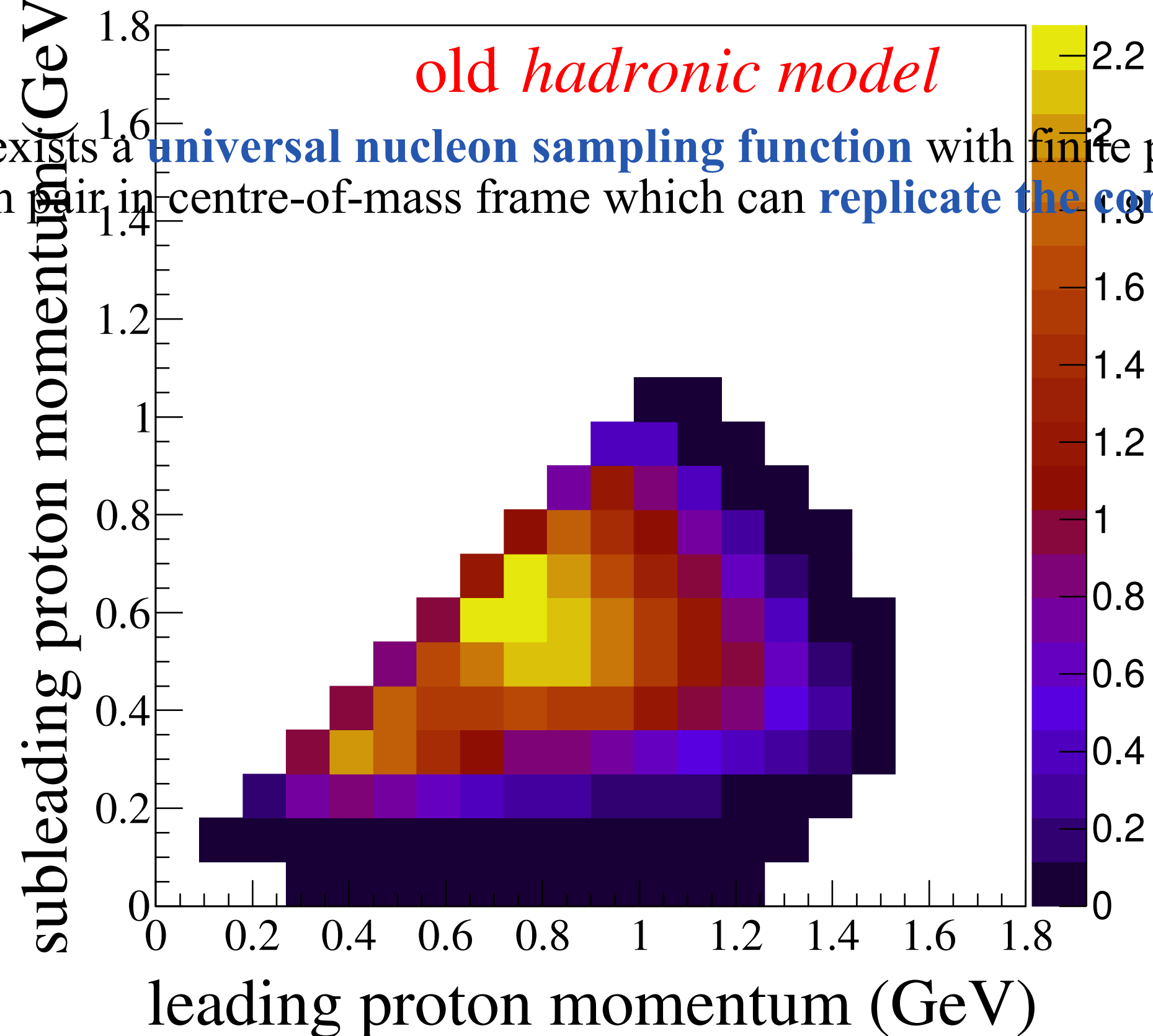
Modelling Correlations Between Nucleons

Two protons in final state

$E_\nu = 1 \text{ GeV}$

Assumption

There exists a **universal nucleon sampling function** with finite parameters which contains information about the direction of outgoing nucleon pair in centre-of-mass frame which can **replicate the correlations** between outgoing nucleons in the lab frame



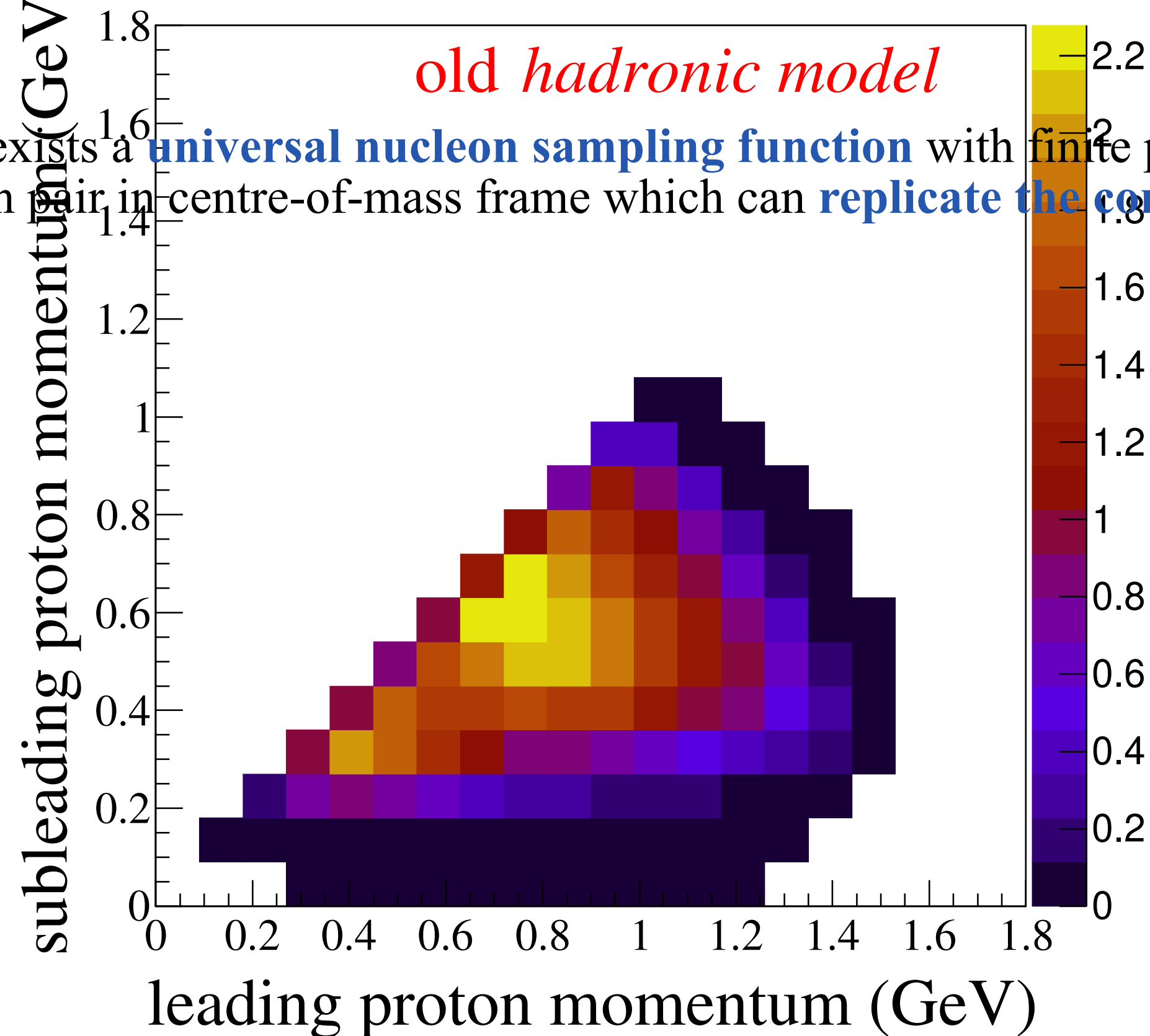
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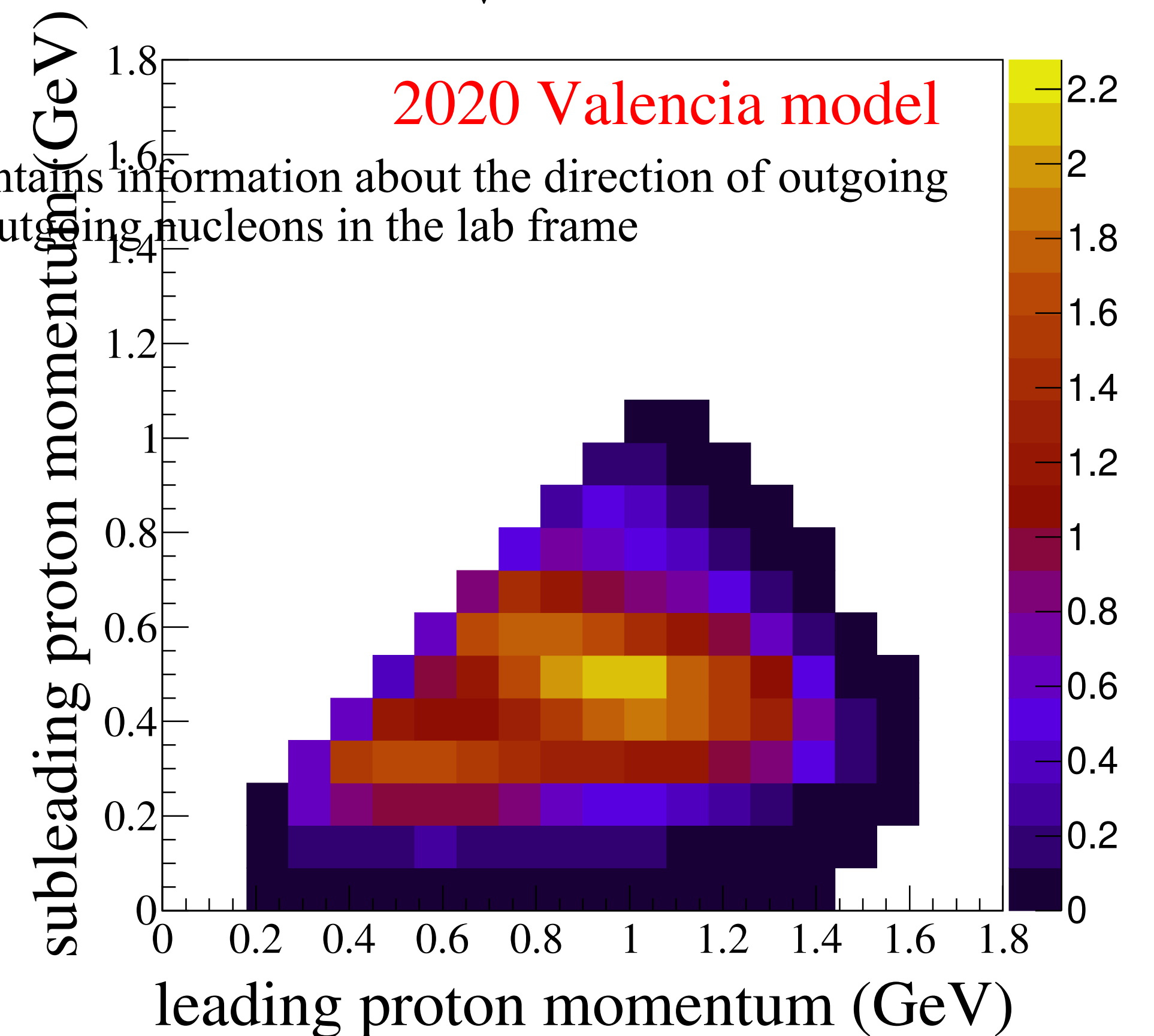
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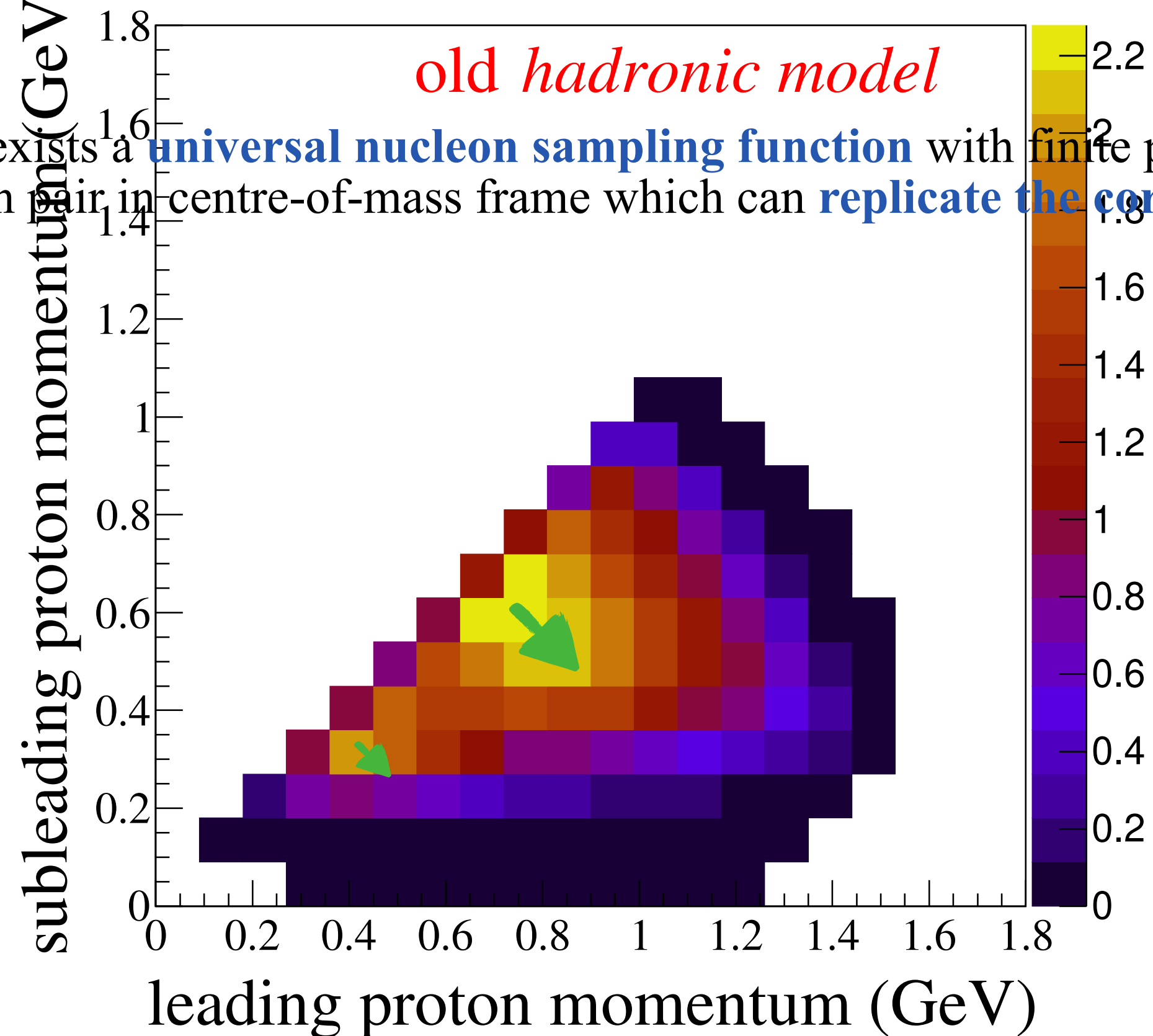
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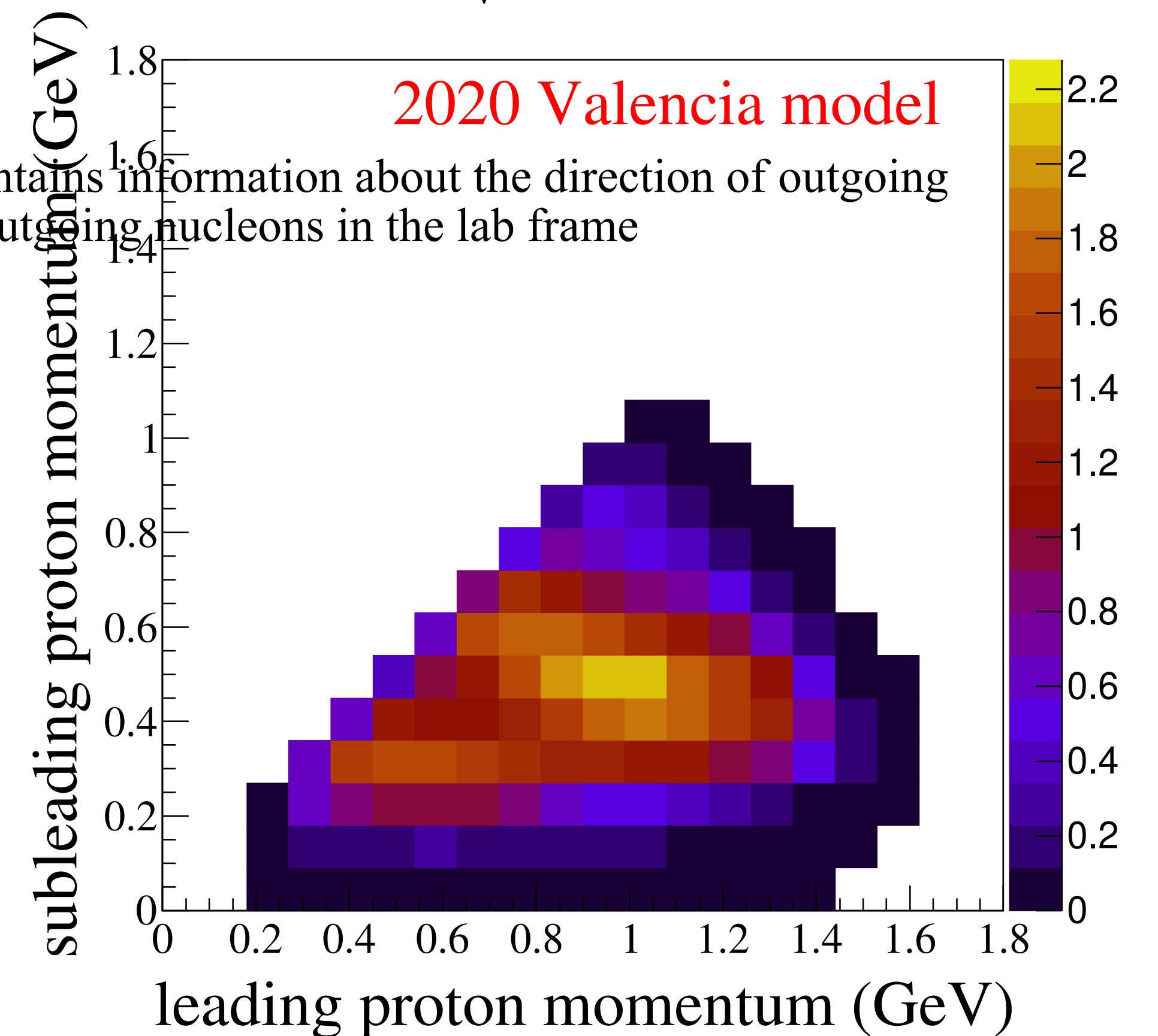
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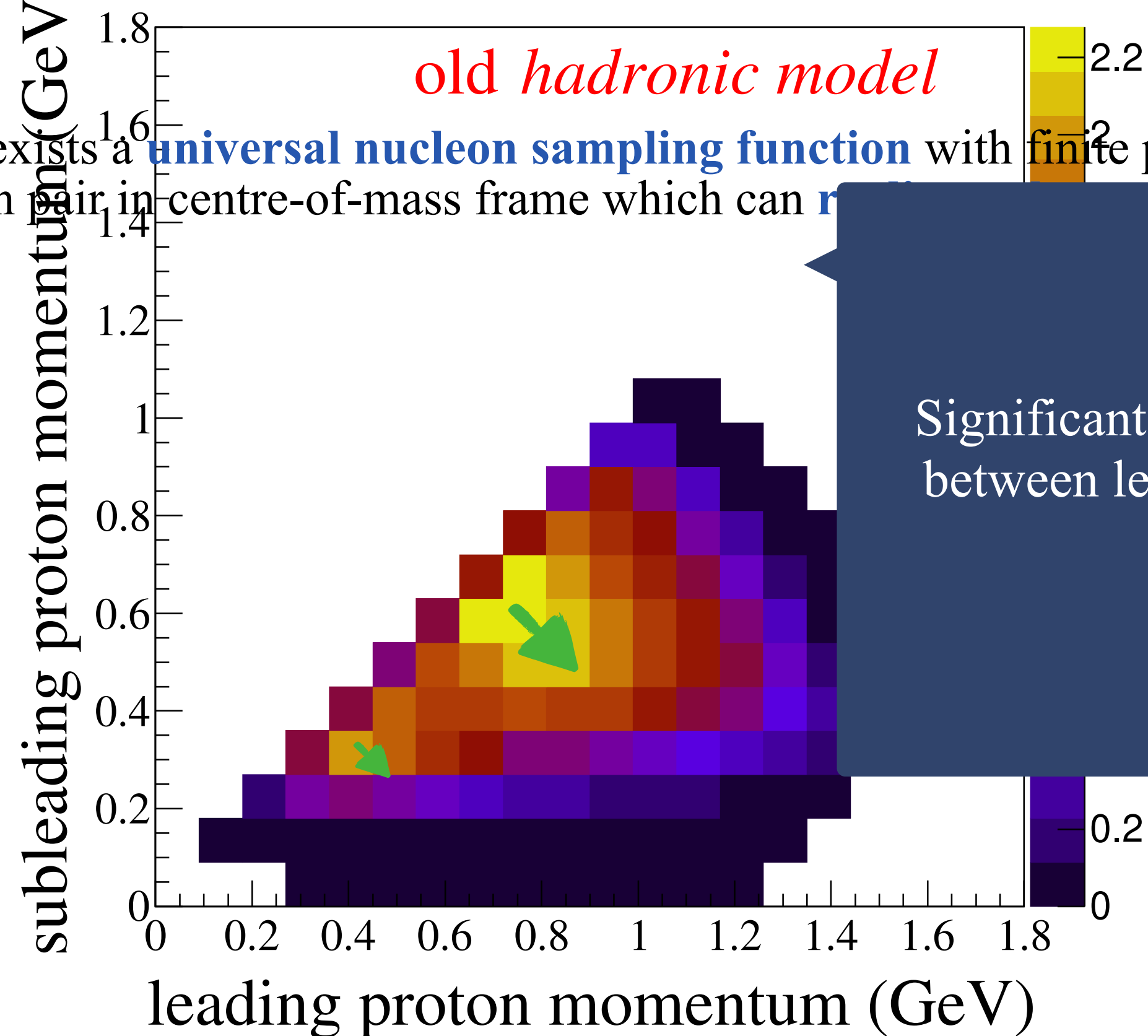
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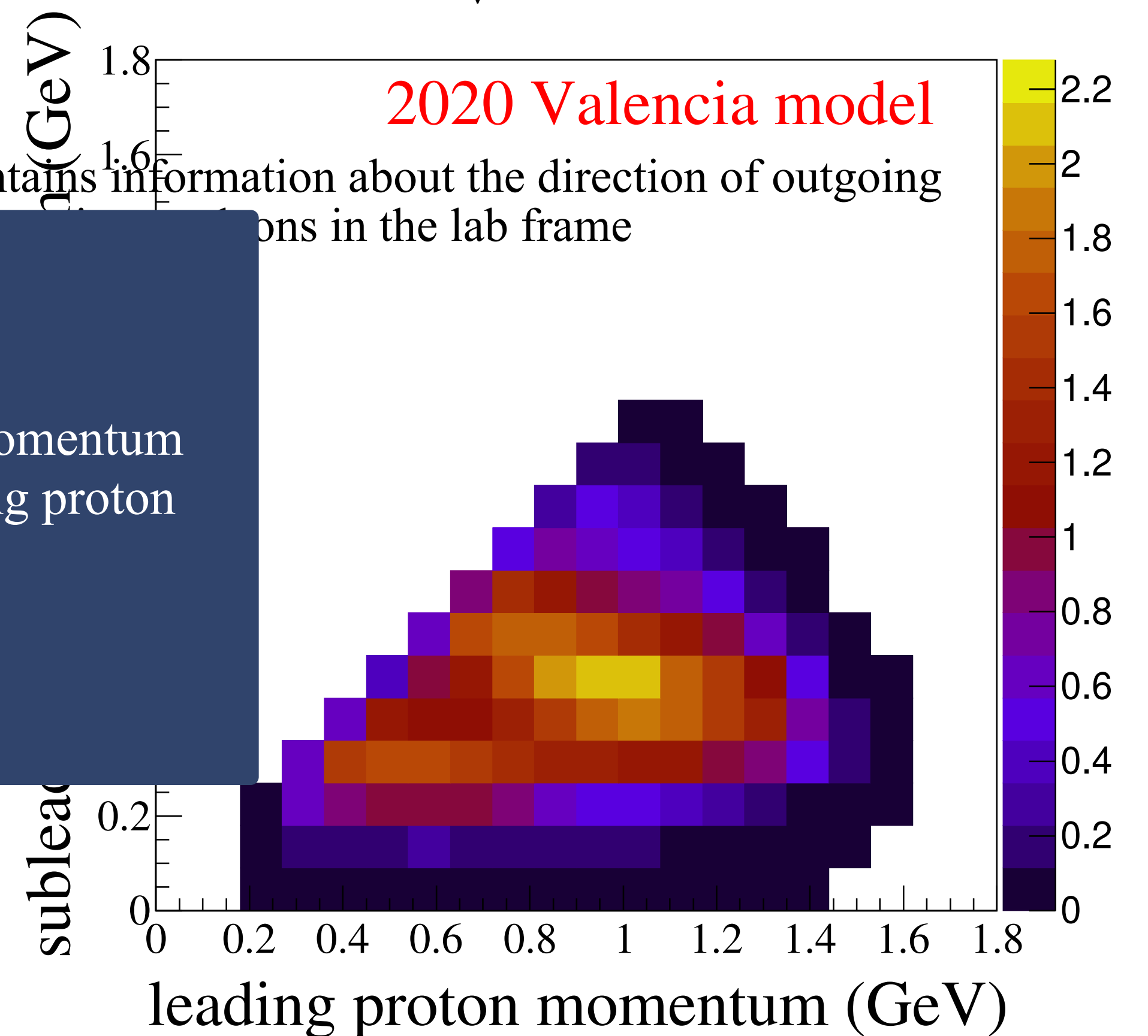
There exists a **universal nucleon sampling function** with finite parameters which contains information about the direction of outgoing nucleon pair in centre-of-mass frame which can be transformed to the lab frame



Significant difference in the momentum between leading and subleading proton momentum

$E_\nu = 1 \text{ GeV}$

2020 Valencia model



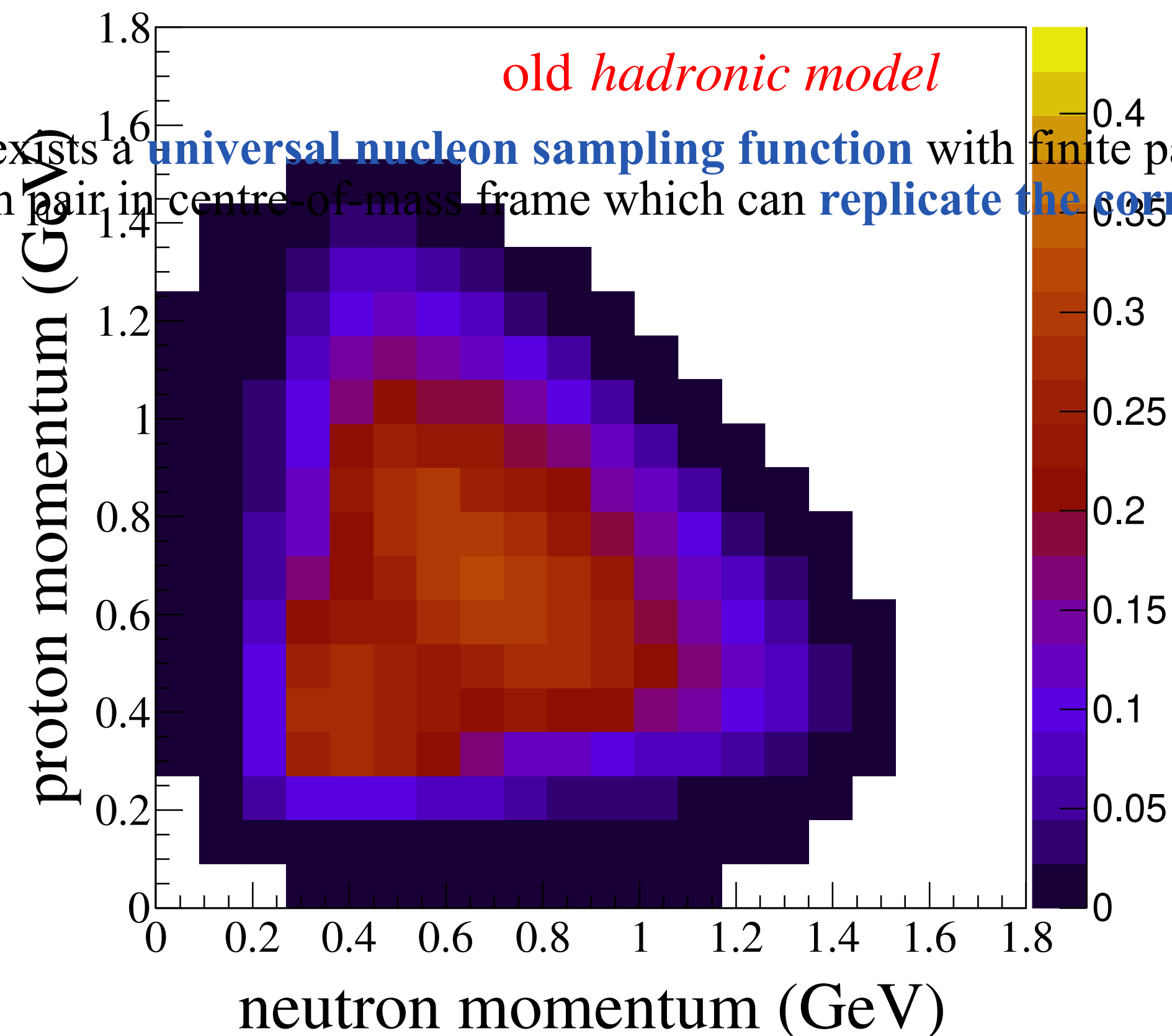
Modelling Correlations Between Nucleons

Neutron and a proton in the final state

$$E_\nu = 1 \text{ GeV}$$

Assumption

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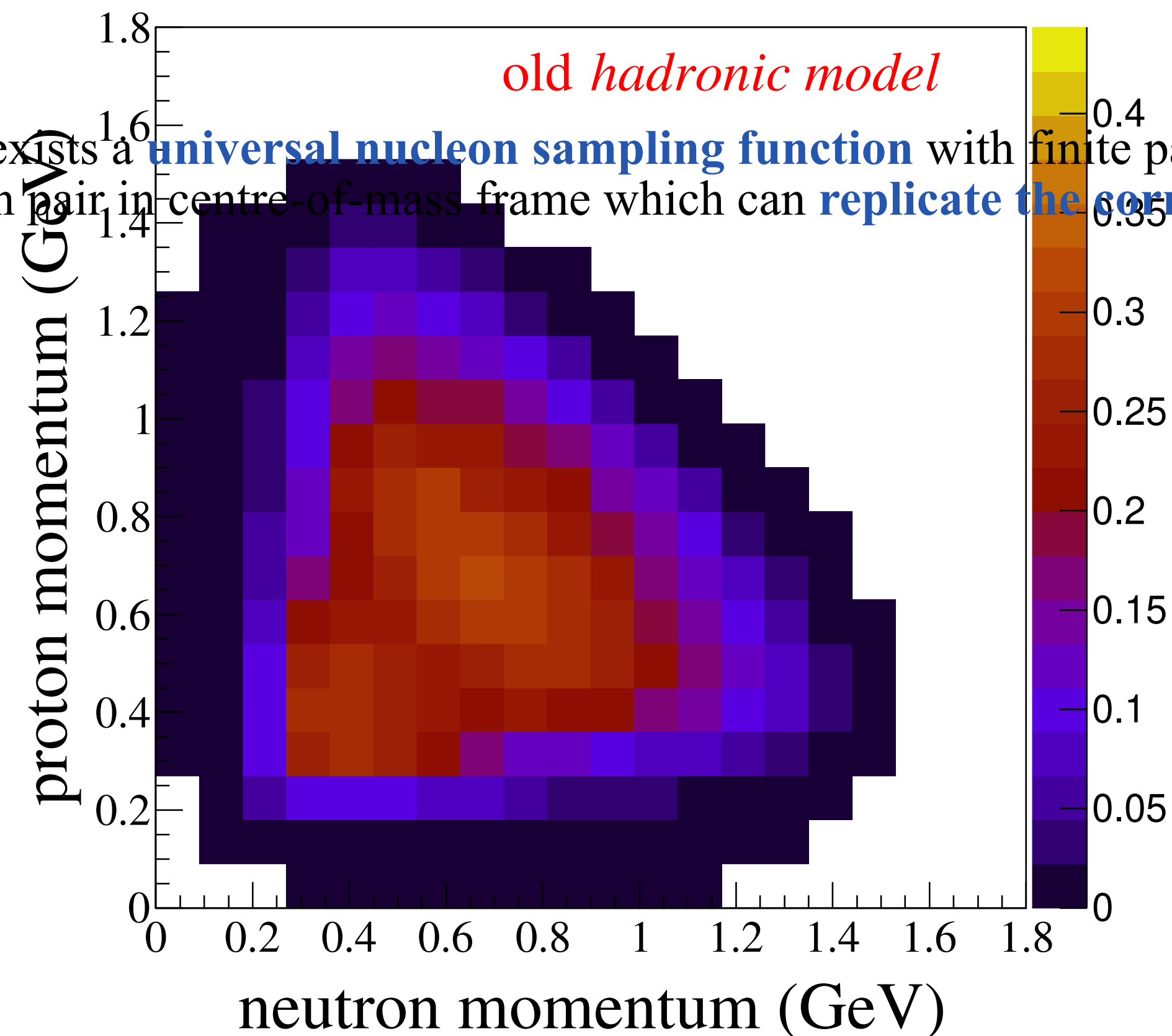
Modelling Correlations Between Nucleons

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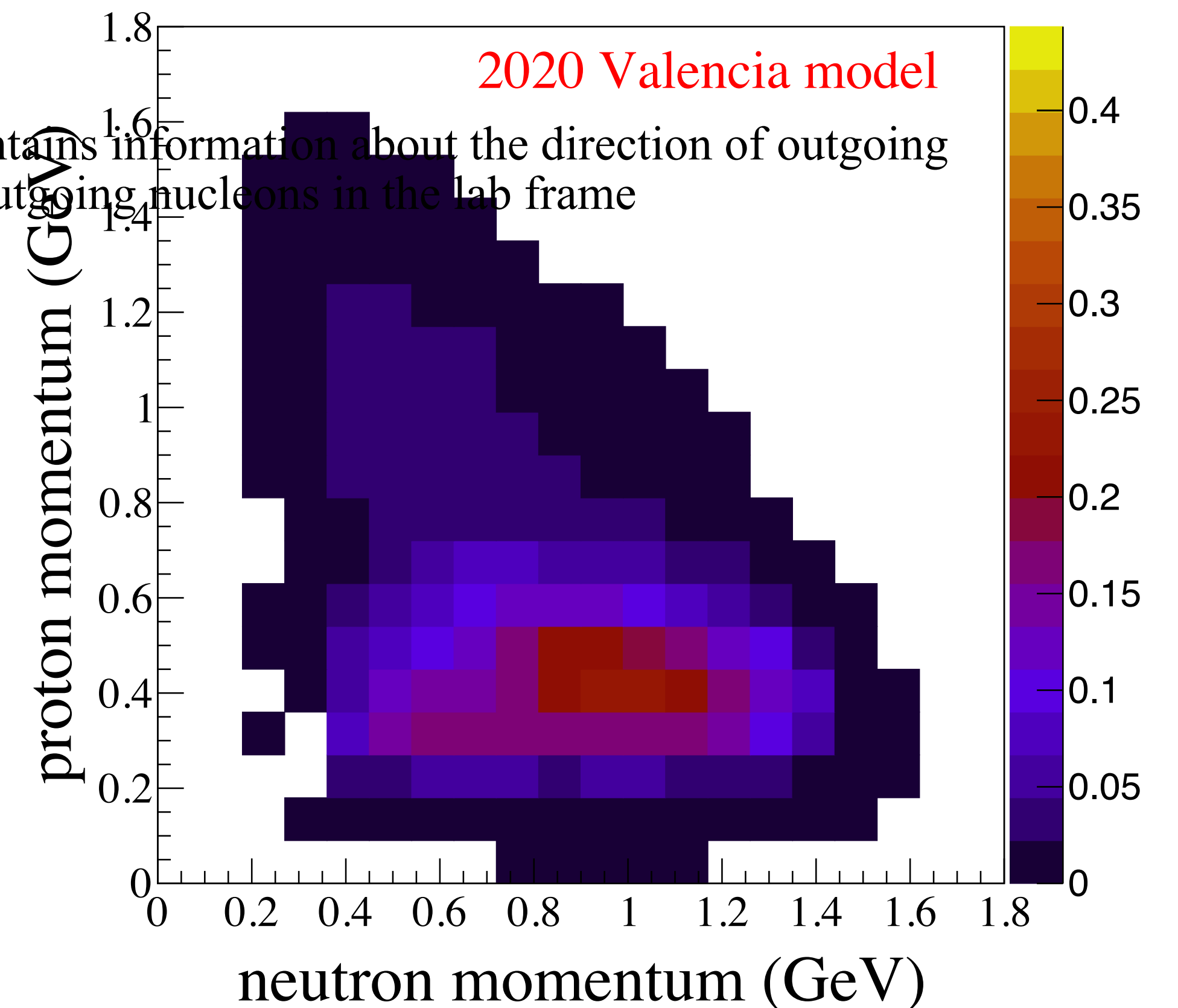
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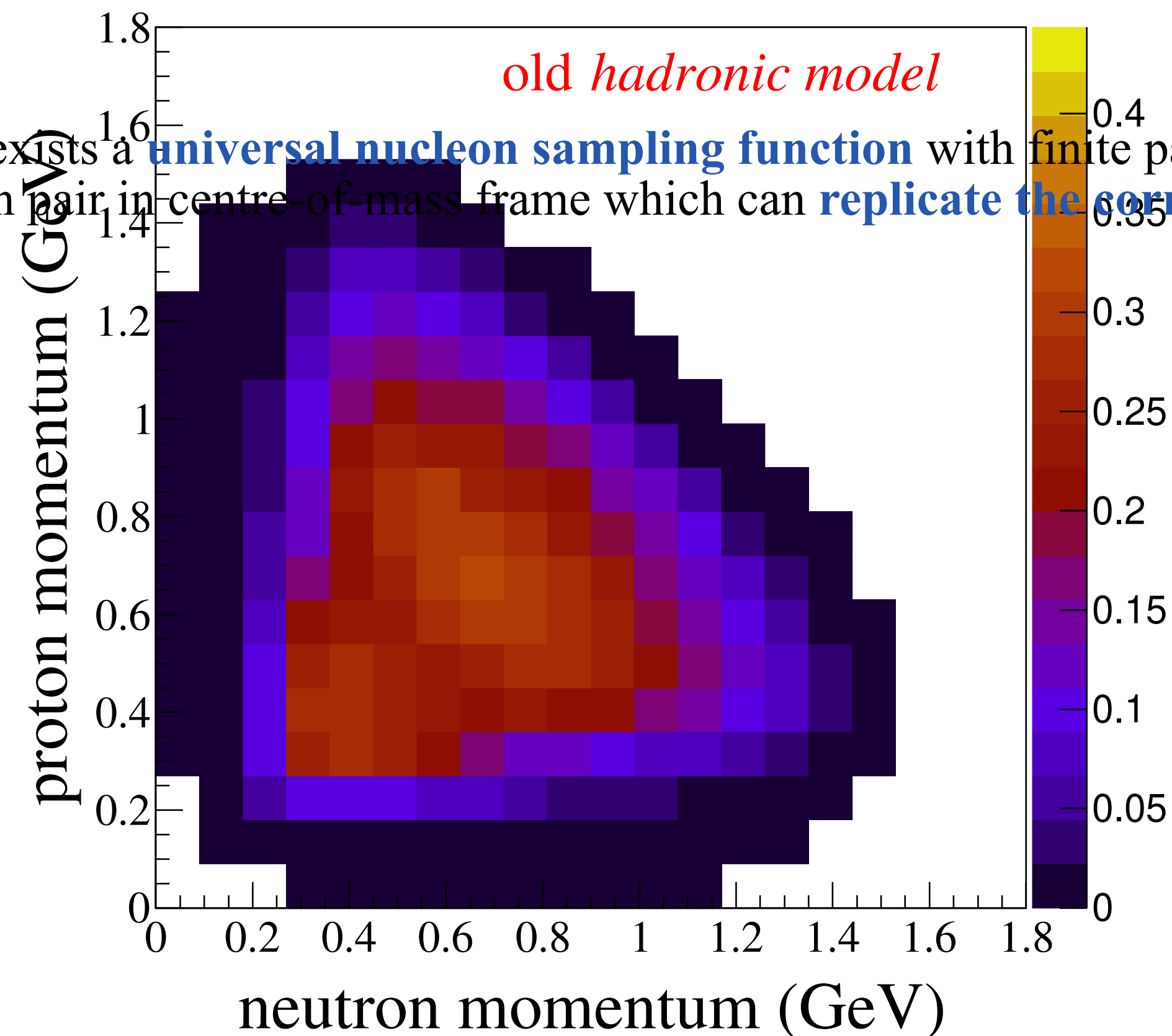
Modelling Correlations Between Nucleons

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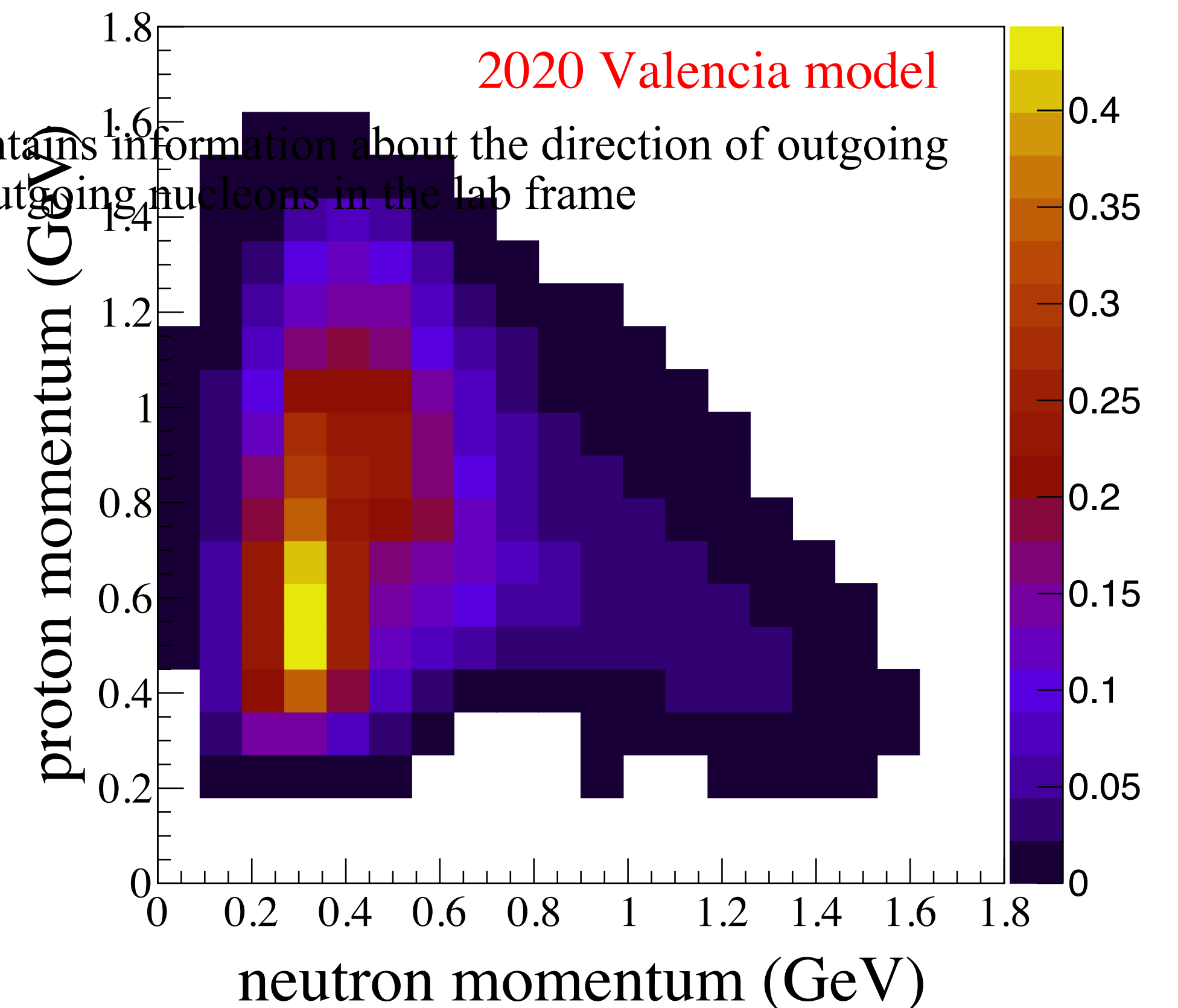
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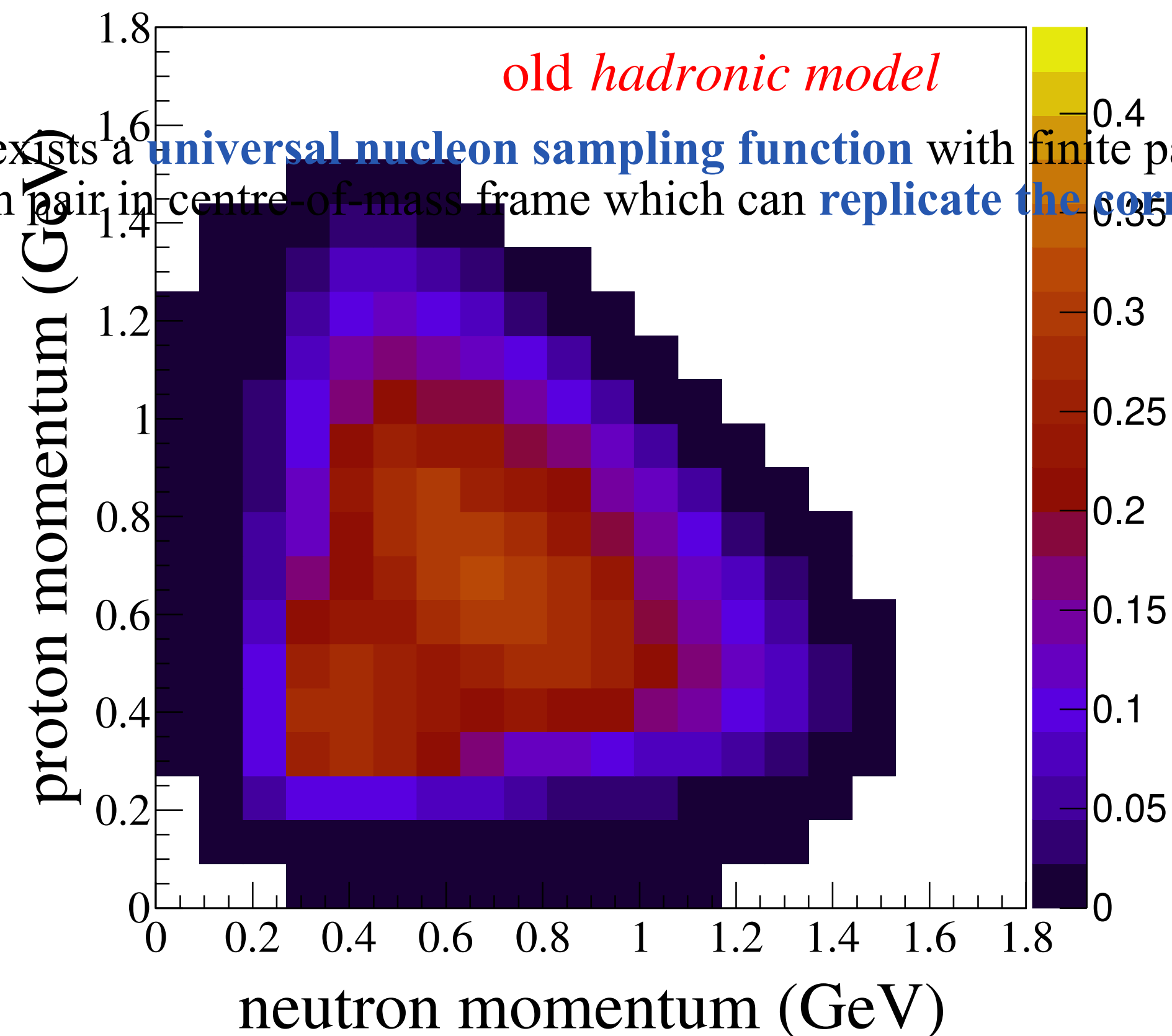
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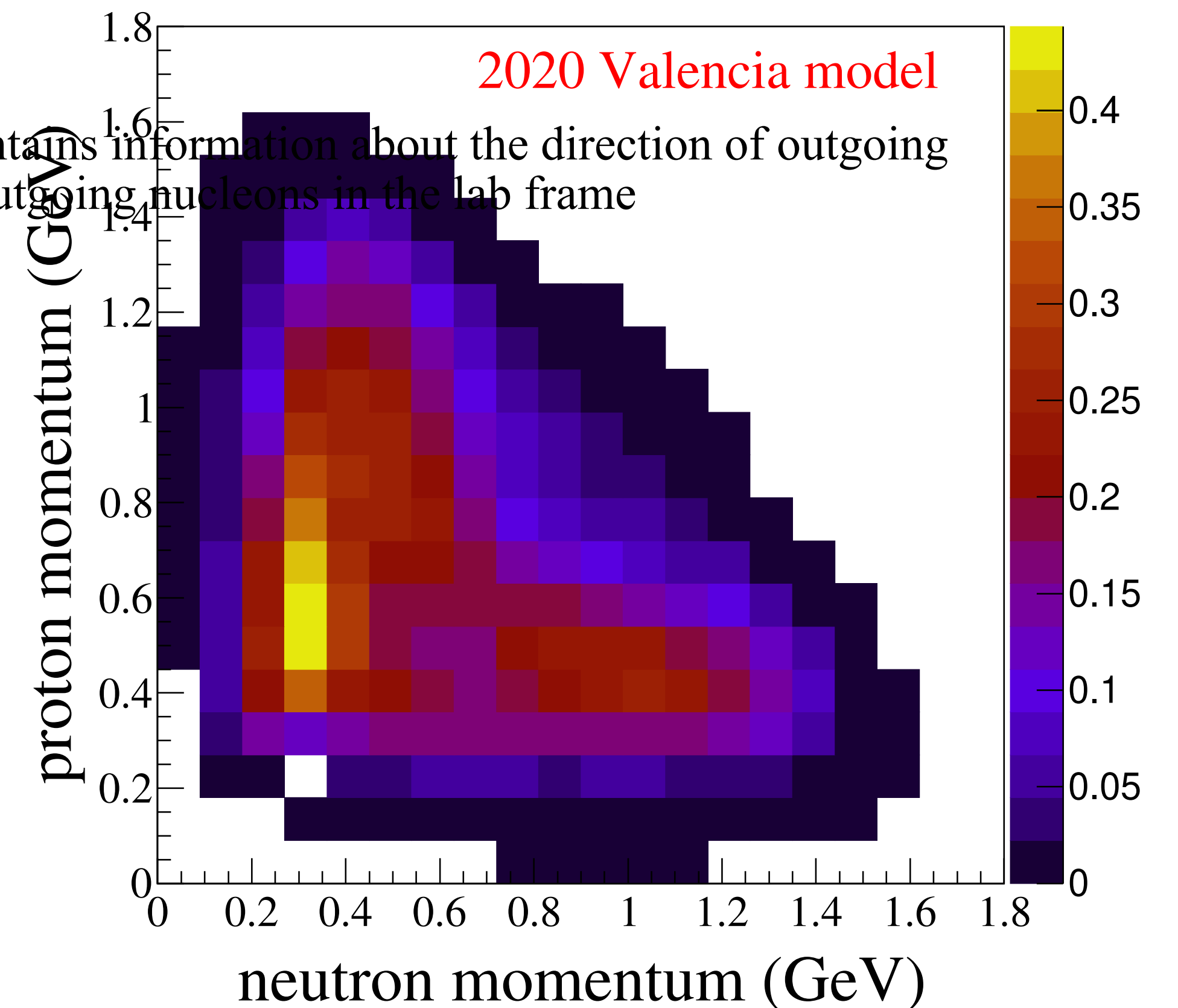
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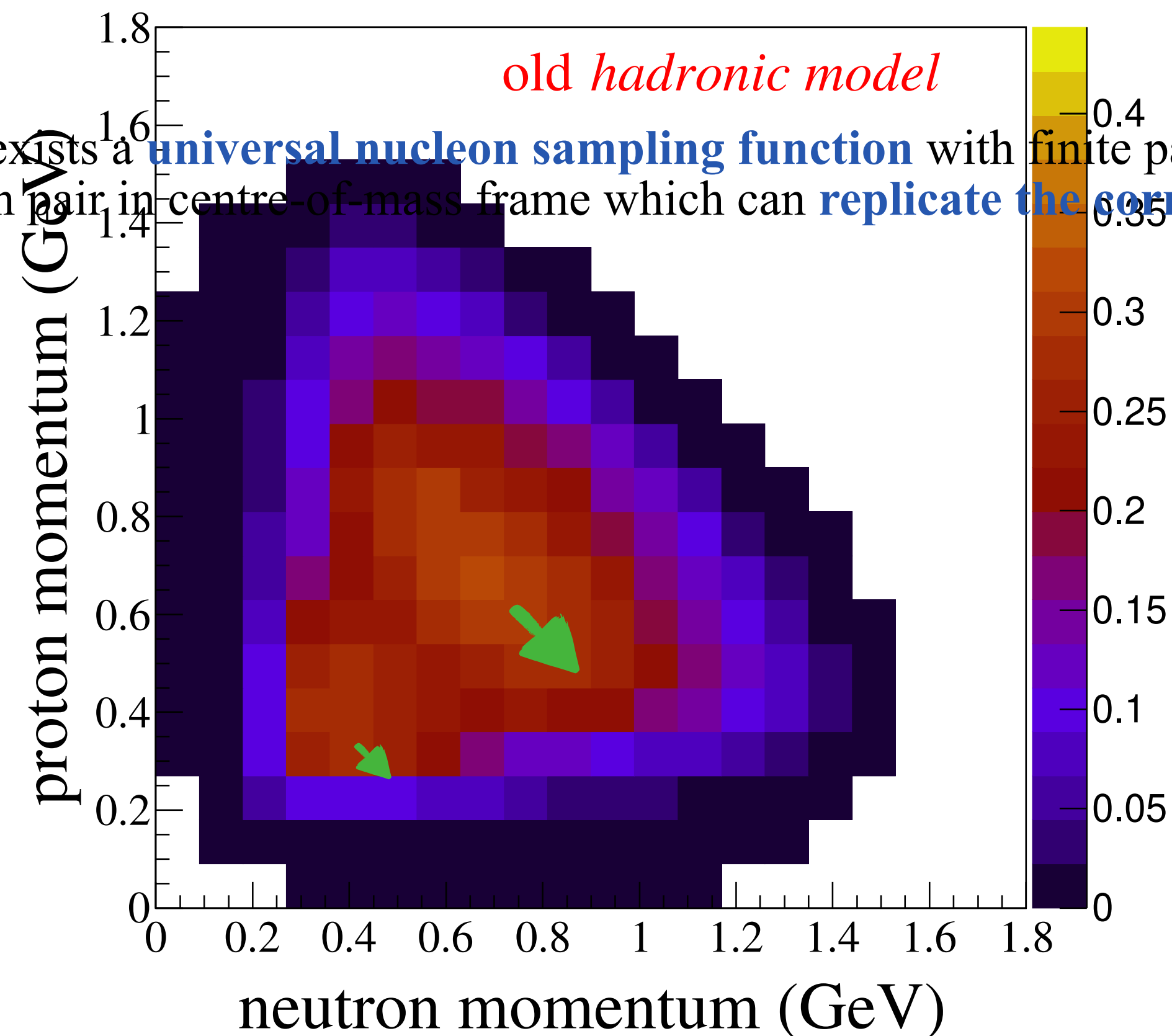
Modelling Correlations Between Nucleons

Neutron and a proton in the final state

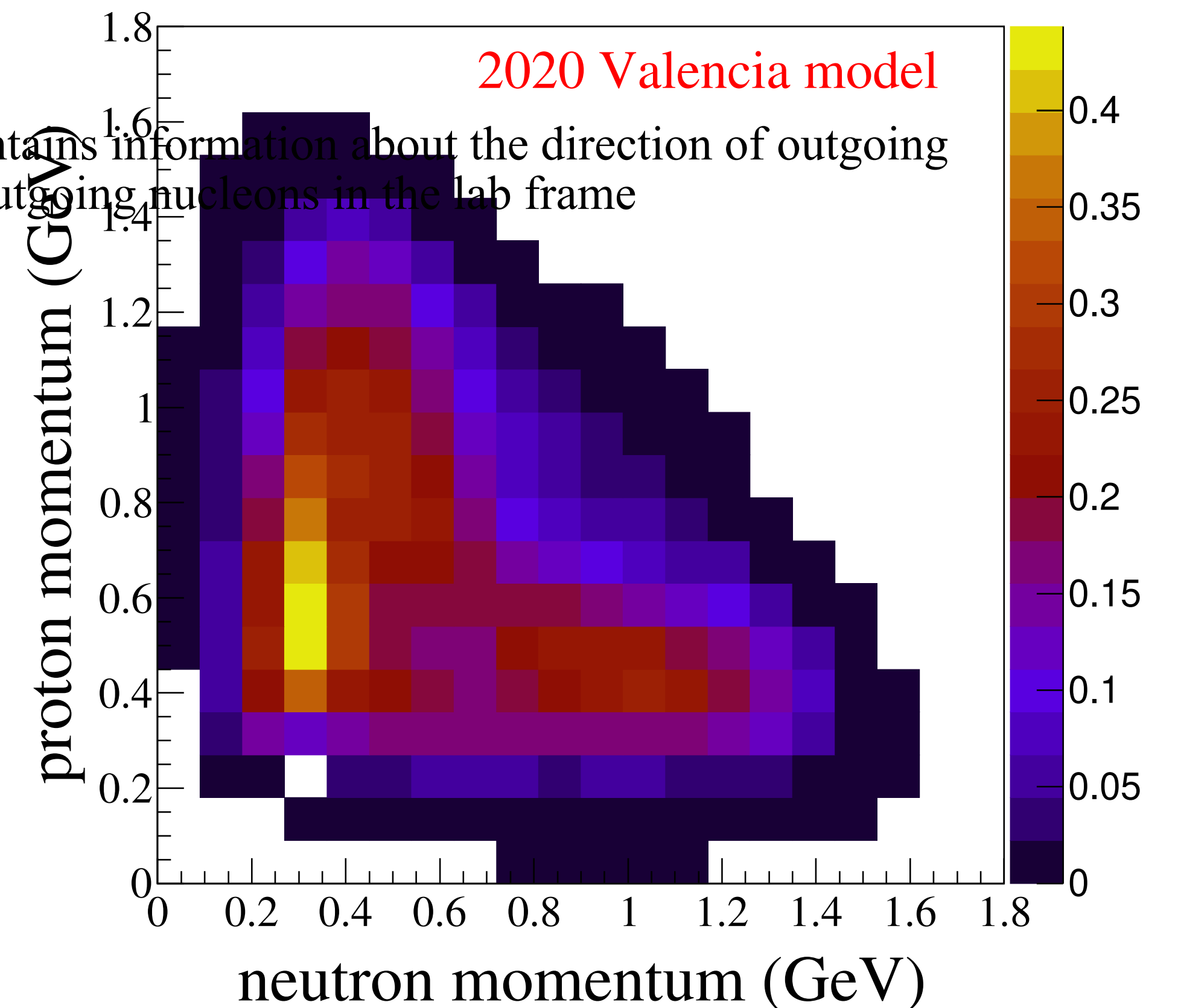
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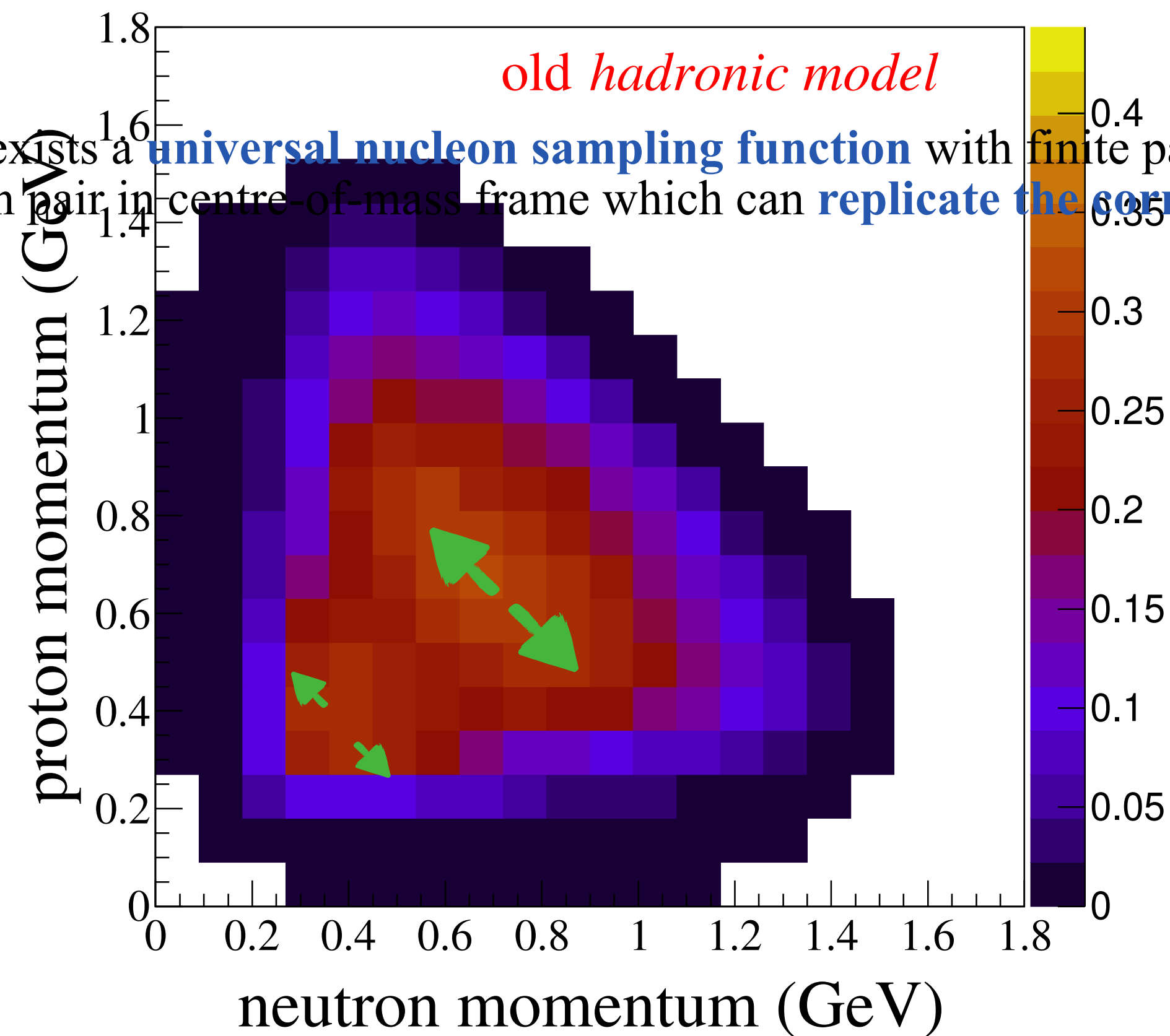
Modelling Correlations Between Nucleons

Neutron and a proton in the final state

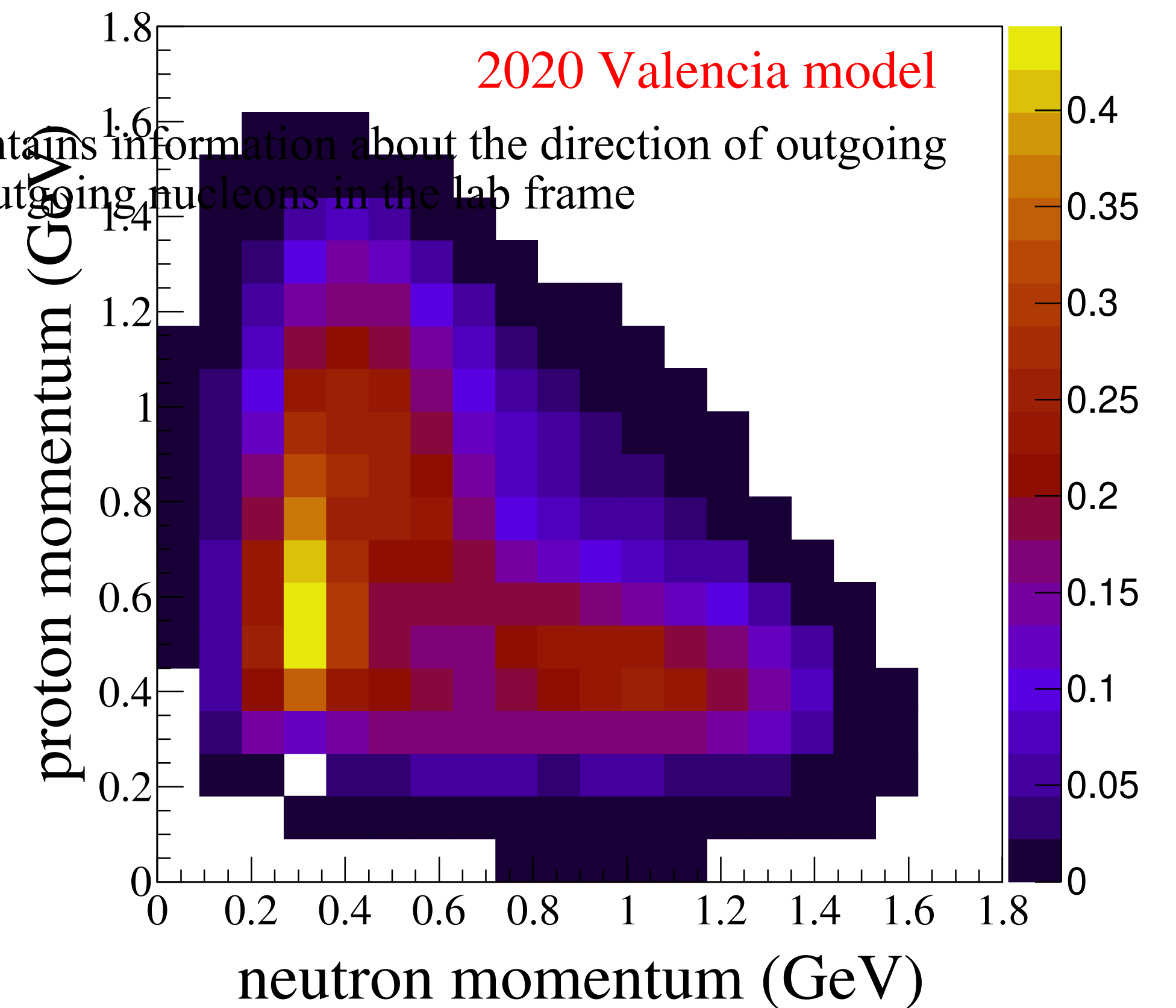
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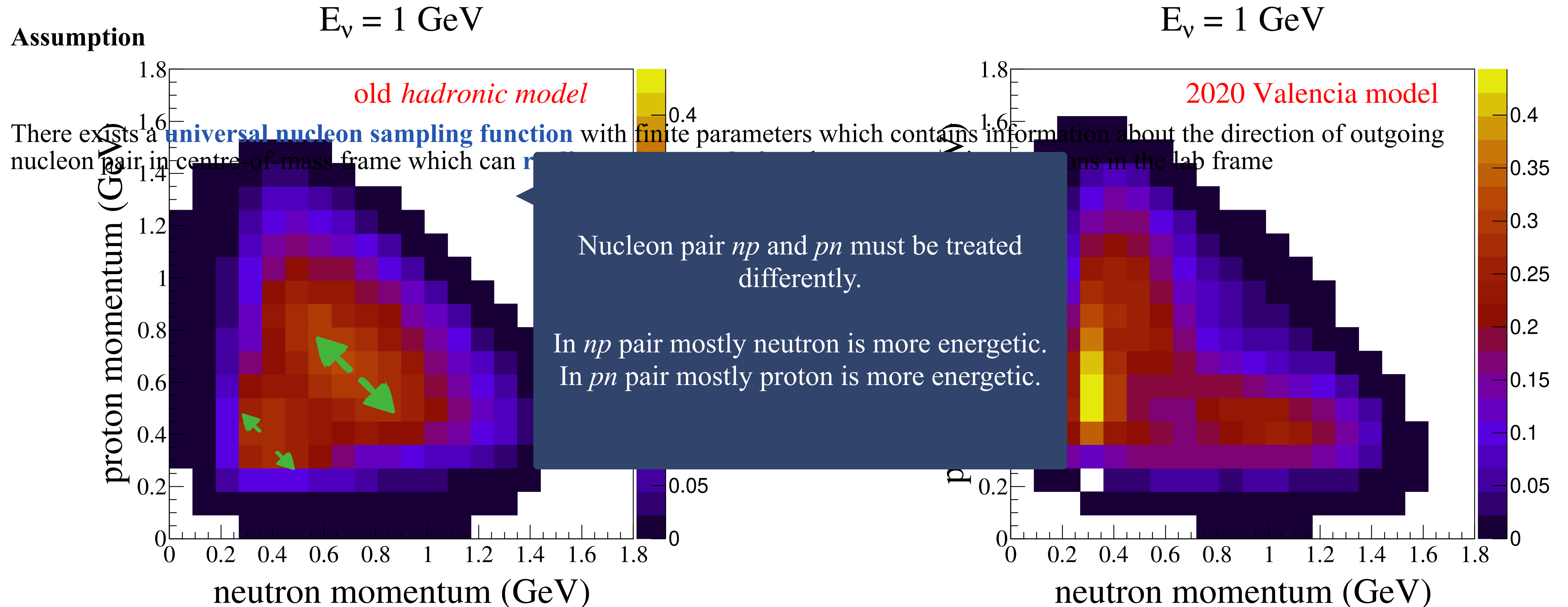


$E_\nu = 1 \text{ GeV}$



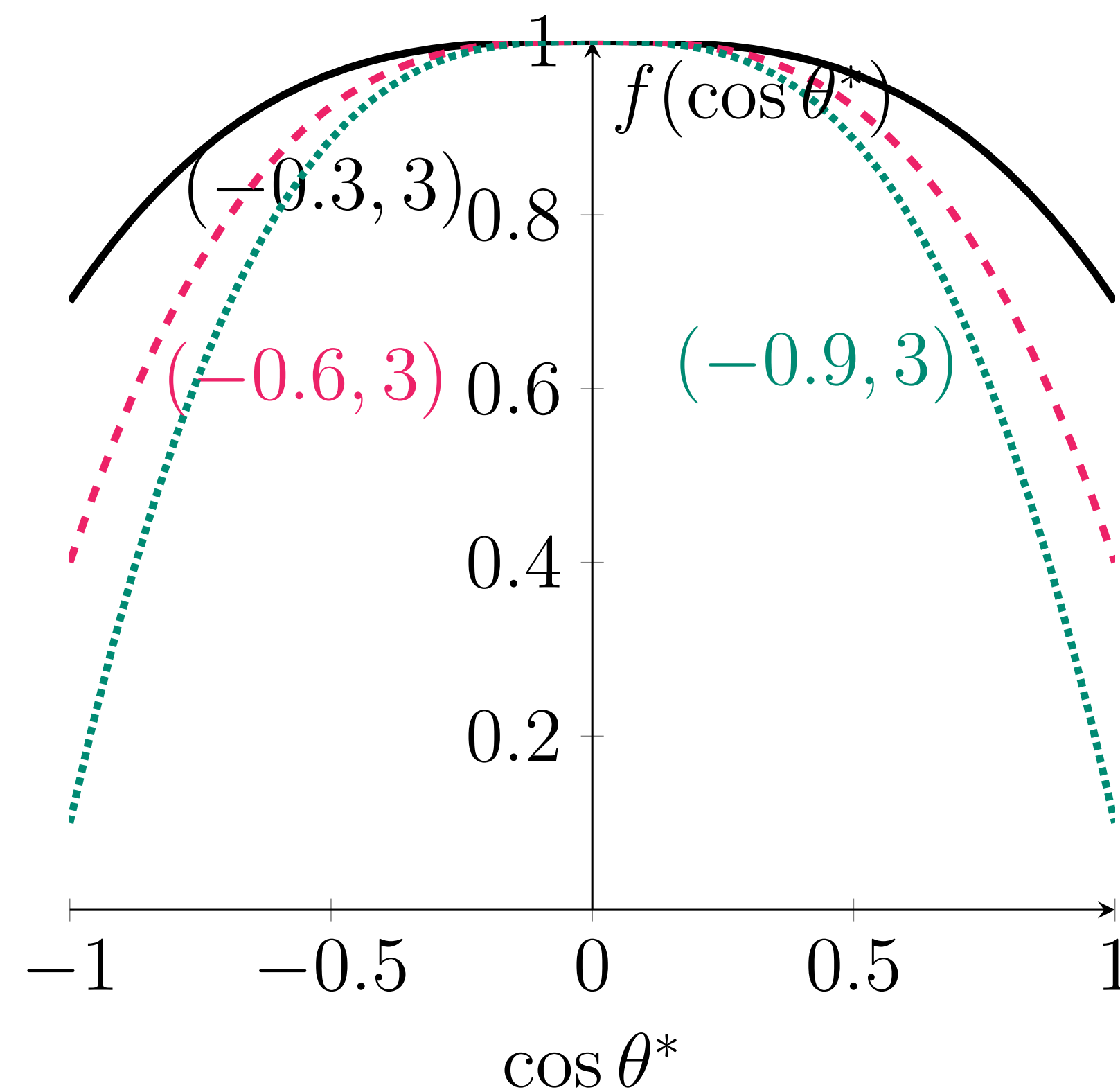
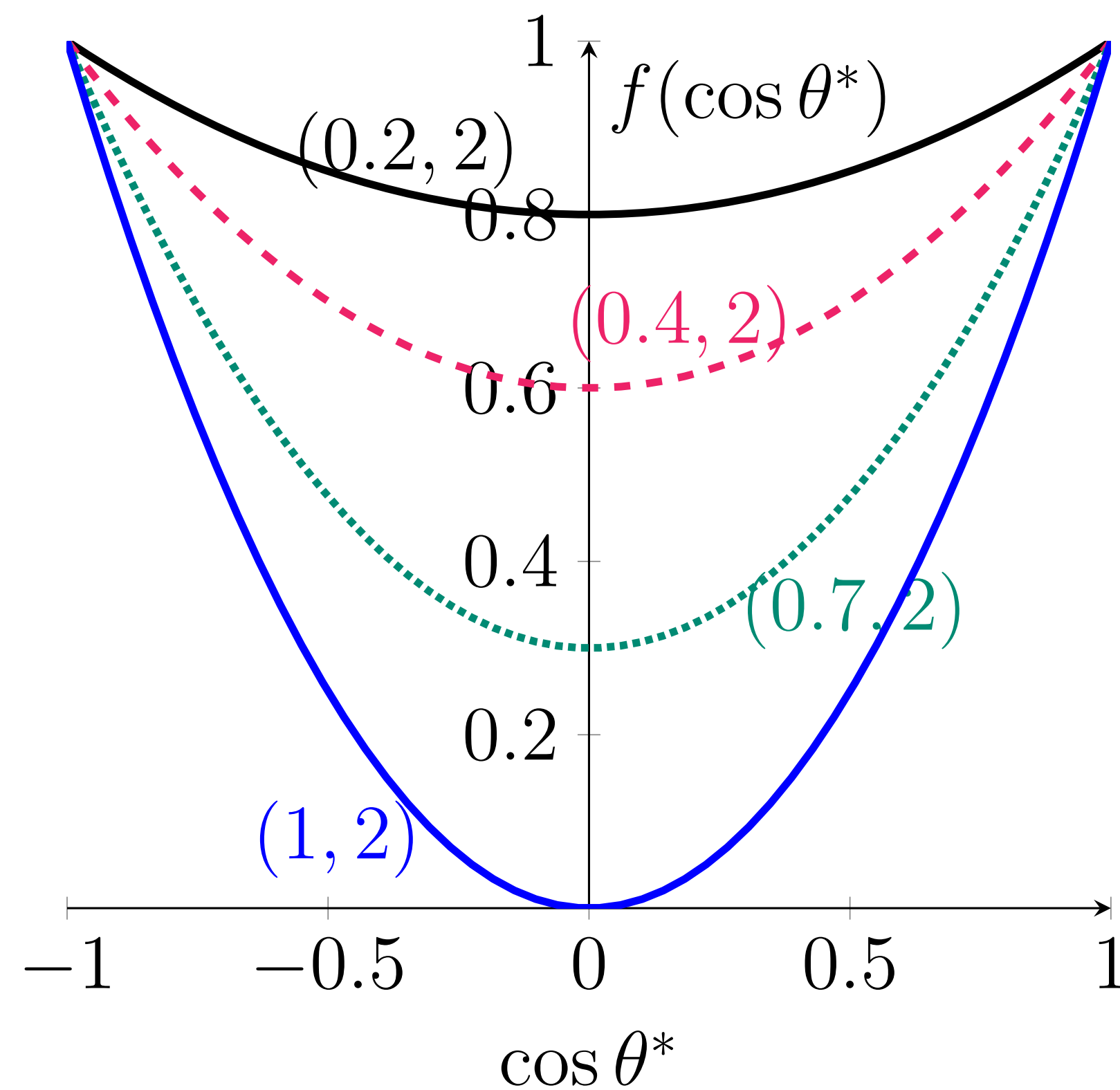
Modelling Correlations Between Nucleons

Neutron and a proton in the final state



Nucleon sampling function

Shape



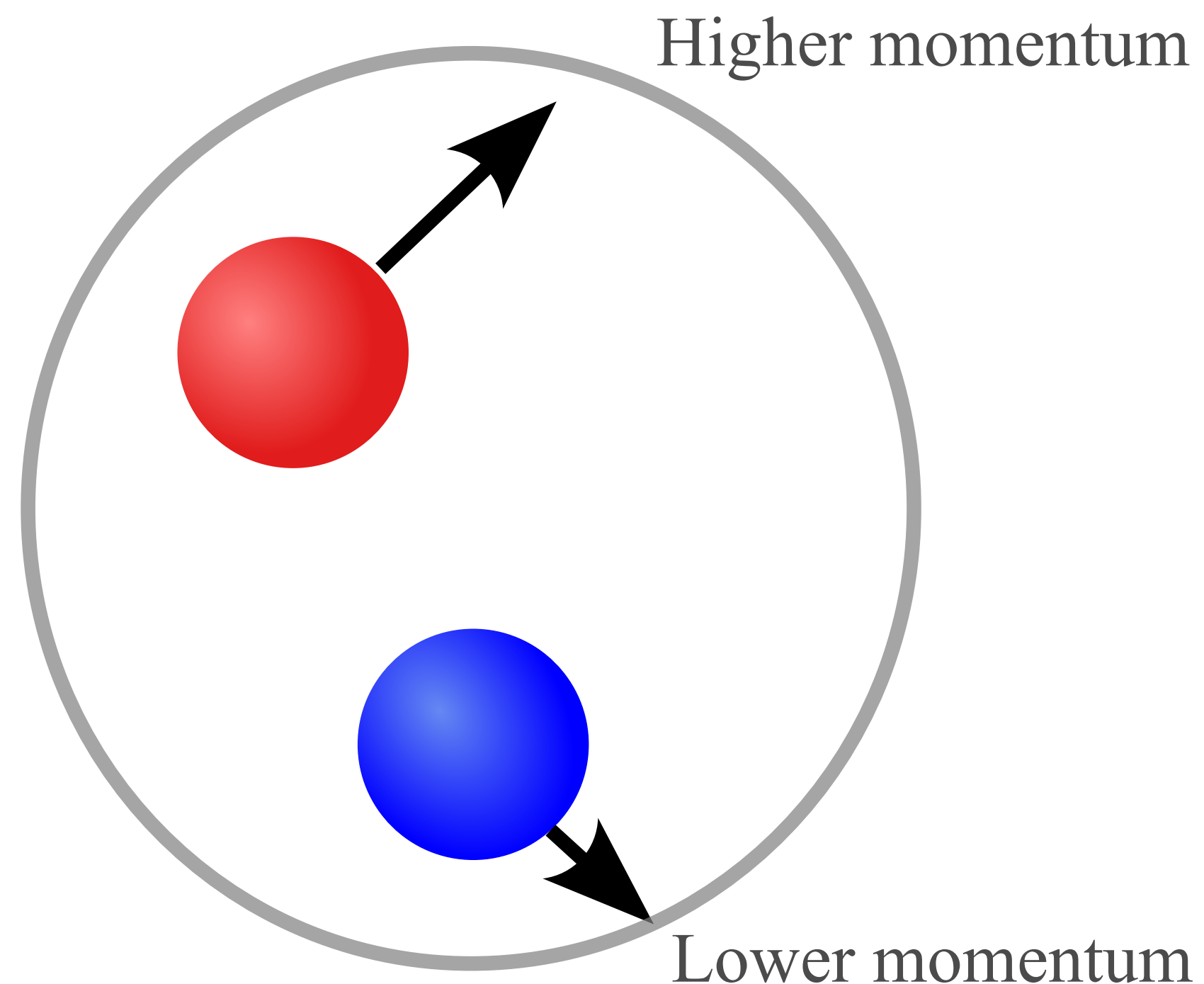
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Modelling Correlations Between Nucleons

Motivation (for 2p2h only)

Modelling Correlations Between Nucleons

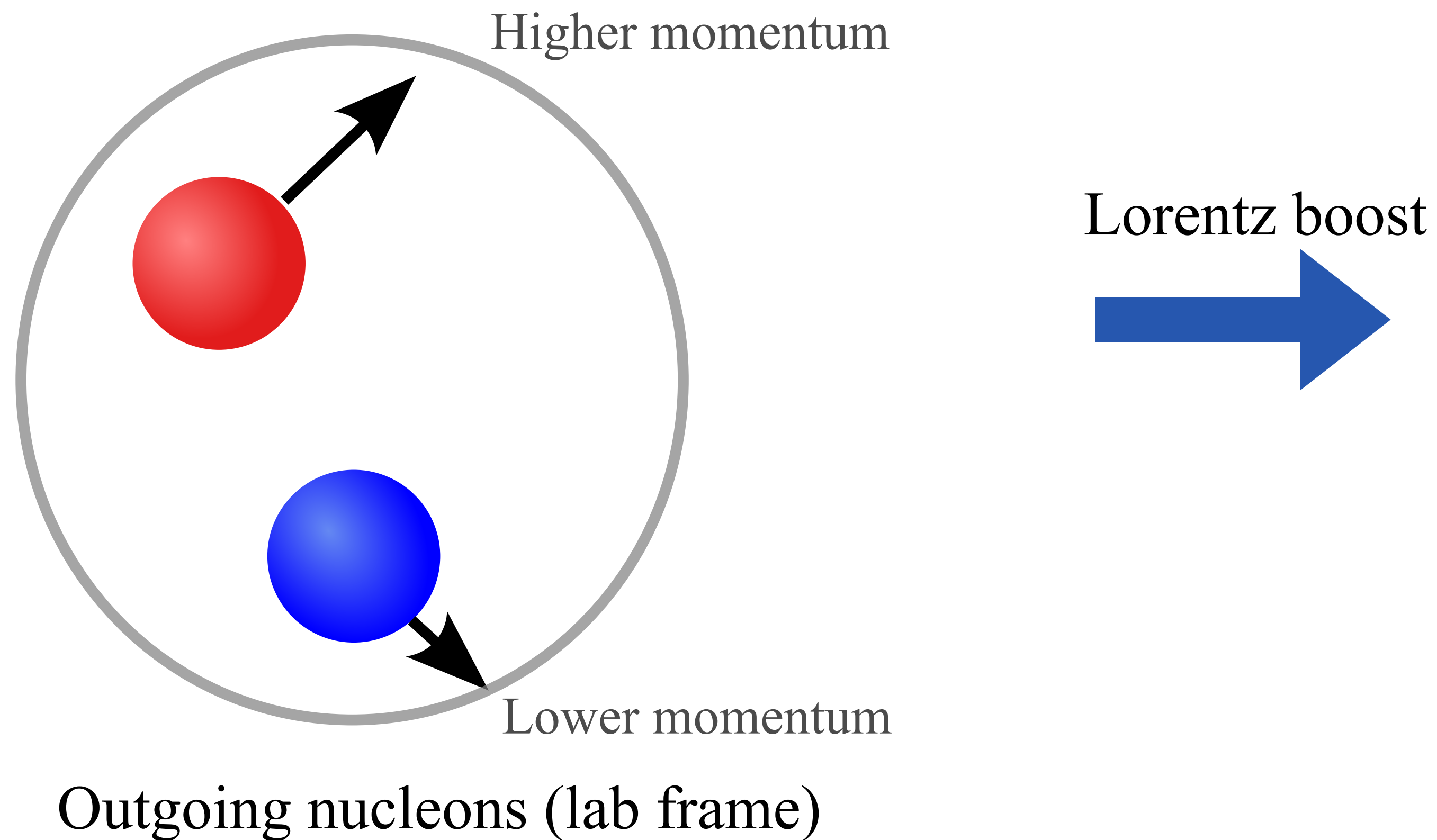
Motivation (for 2p2h only)



Outgoing nucleons (lab frame)

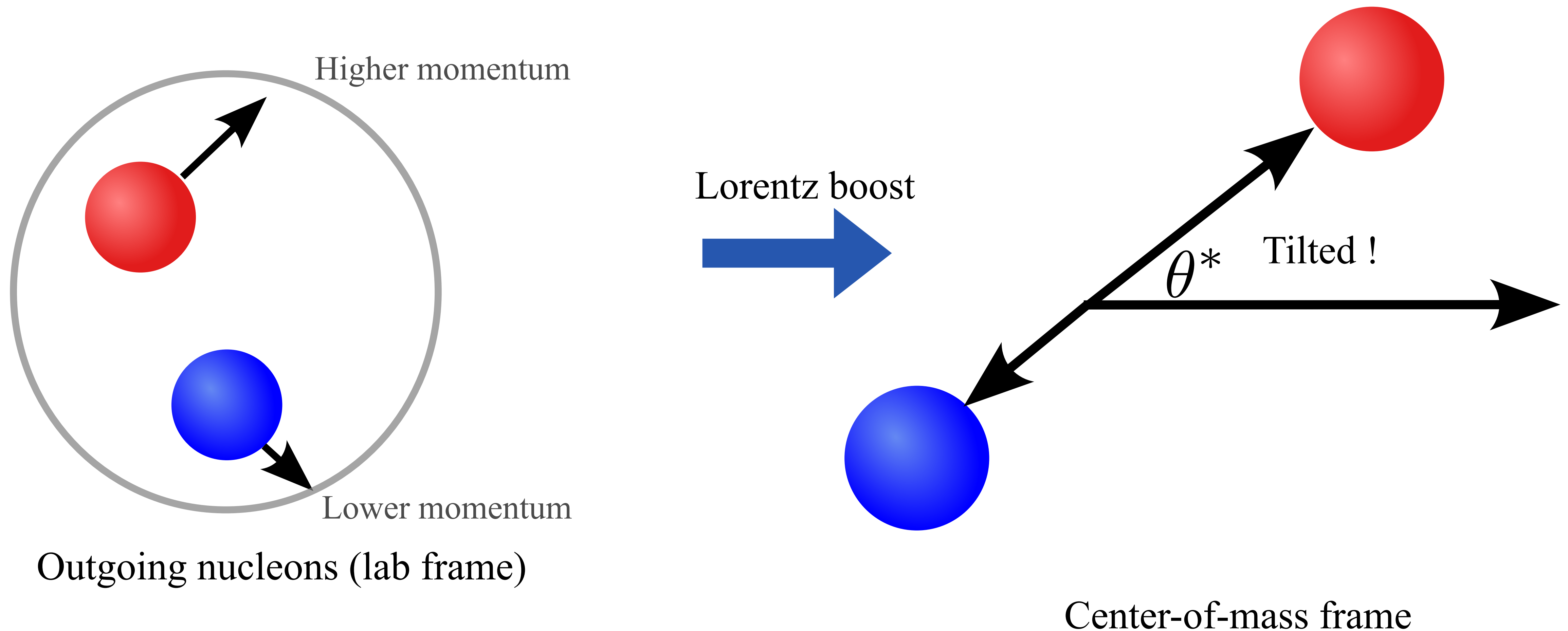
Modelling Correlations Between Nucleons

Motivation (for 2p2h only)



Modelling Correlations Between Nucleons

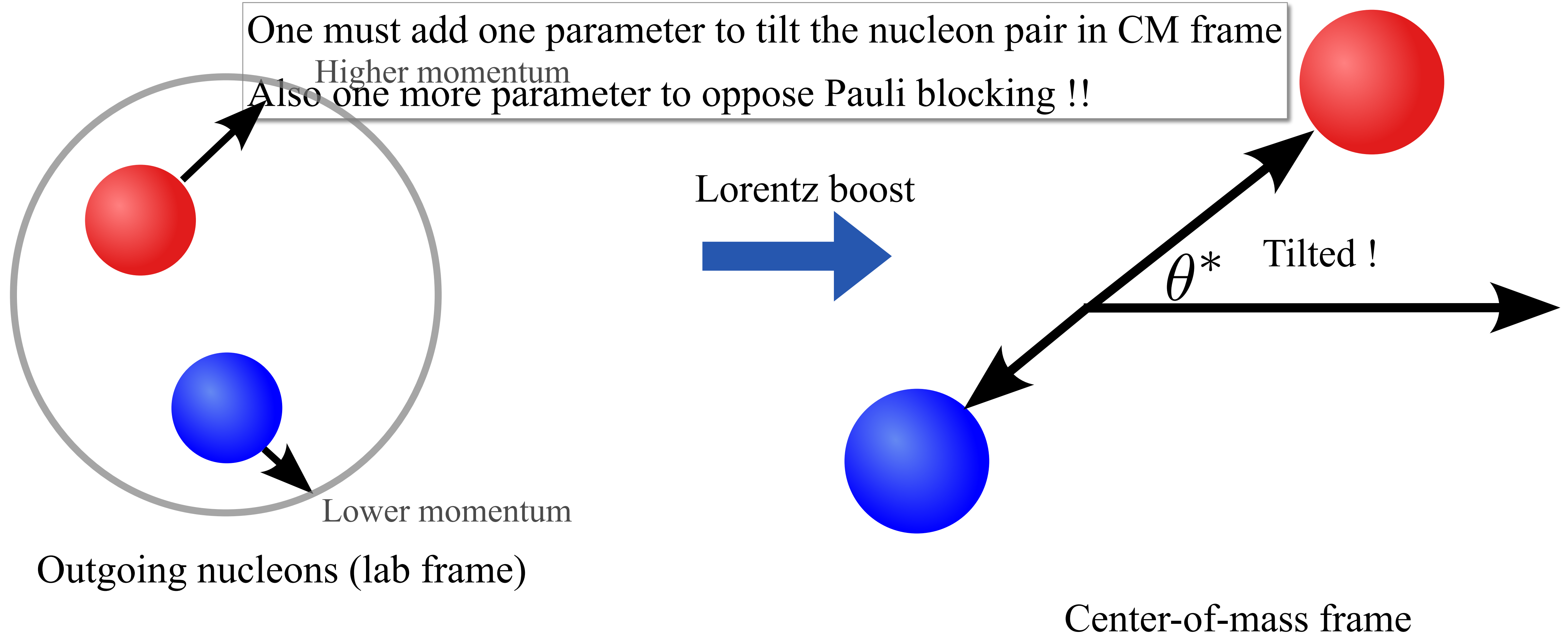
Motivation (for 2p2h only)



Modelling Correlations Between Nucleons

Motivation (for 2p2h only)

One must add one parameter to tilt the nucleon pair in CM frame
Higher momentum
Also one more parameter to oppose Pauli blocking !!

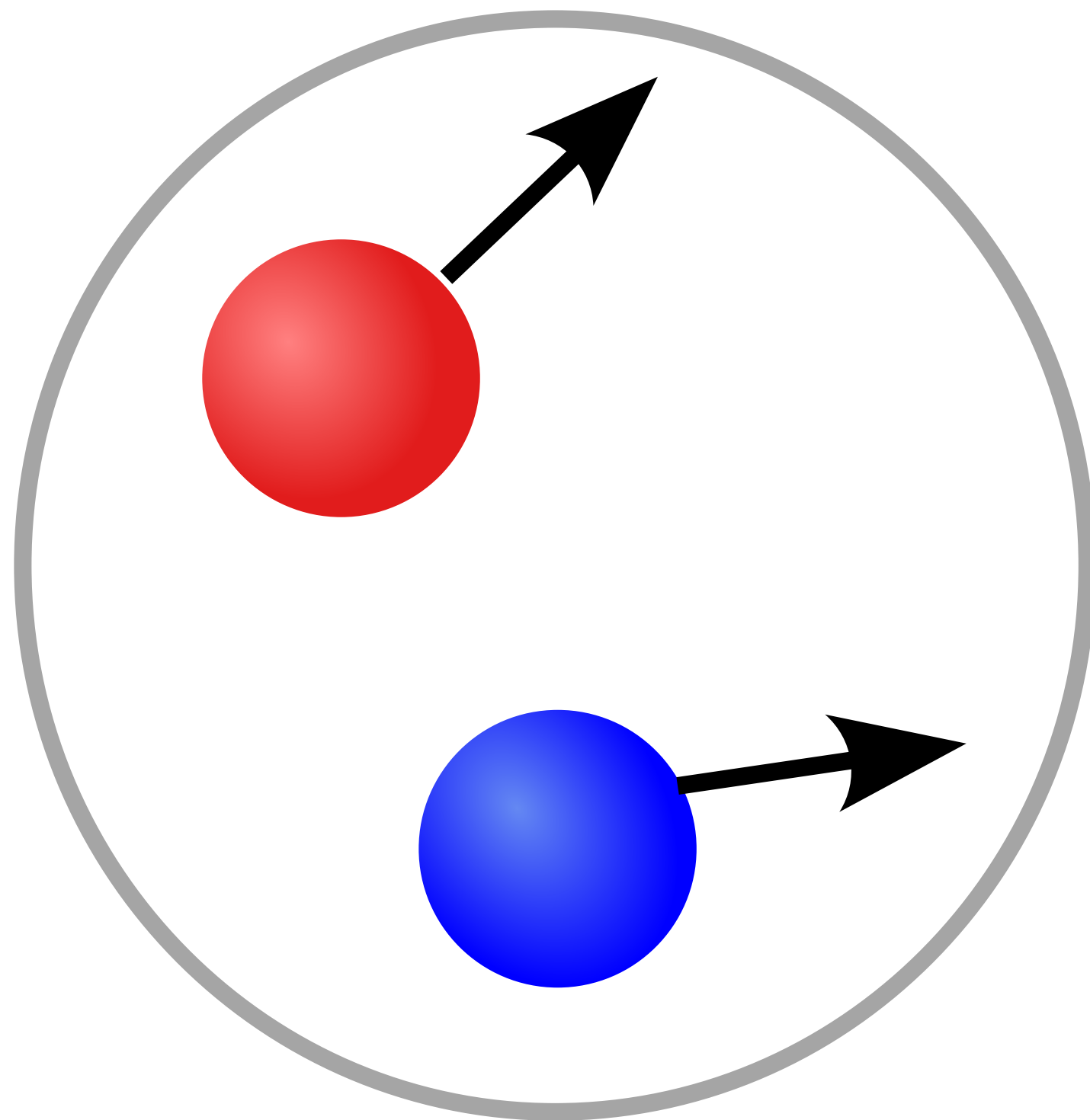


Modelling Correlations Between Nucleons

Motivation (for 2p2h only)

Modelling Correlations Between Nucleons

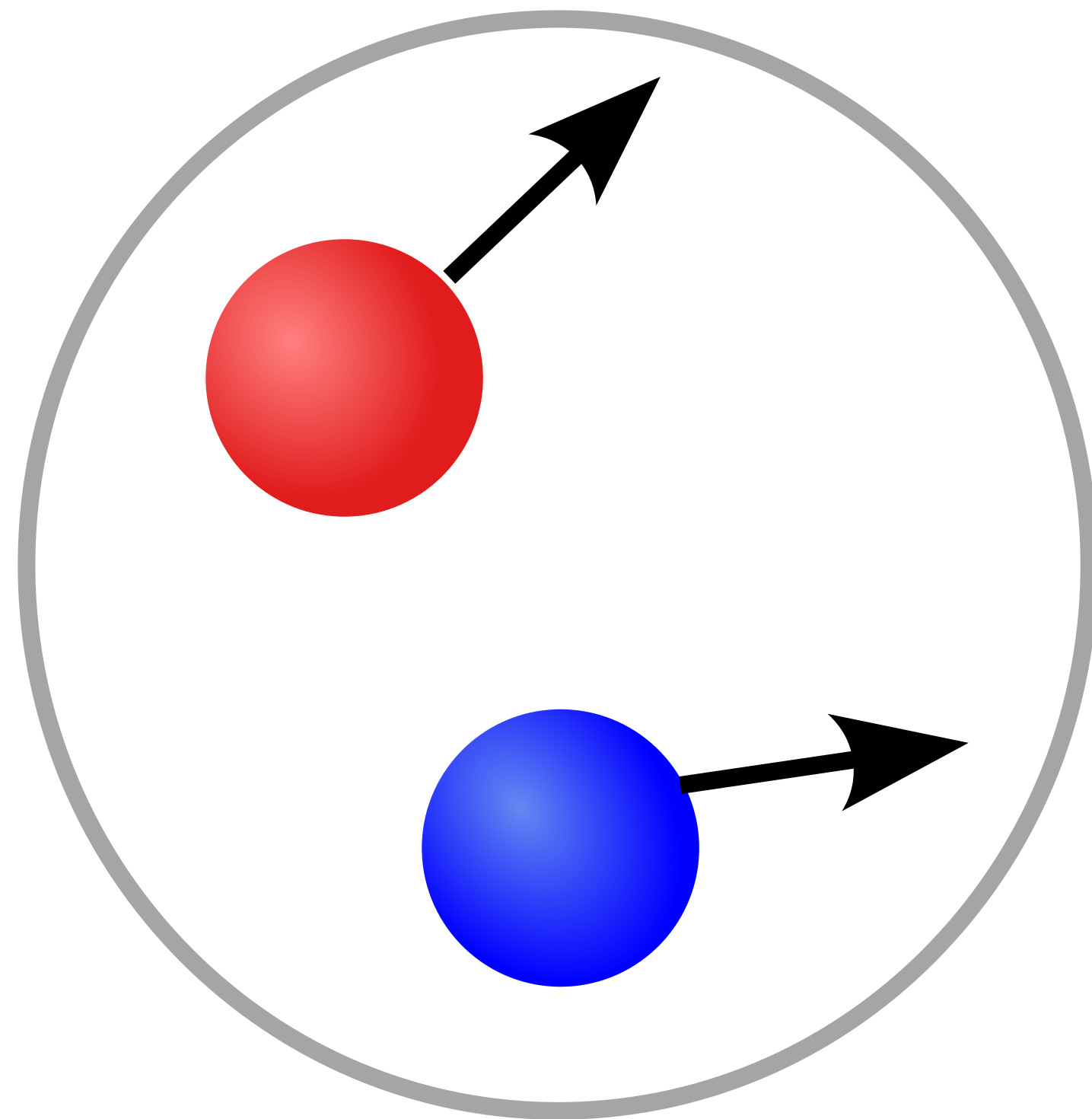
Motivation (for 2p2h only)



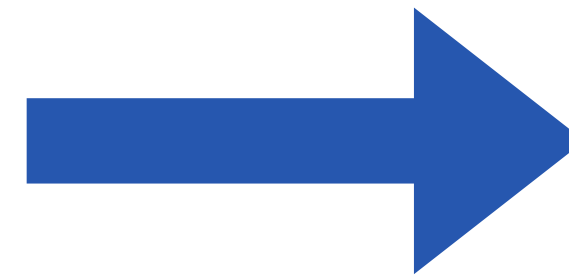
Outgoing nucleons (lab frame)

Modelling Correlations Between Nucleons

Motivation (for 2p2h only)



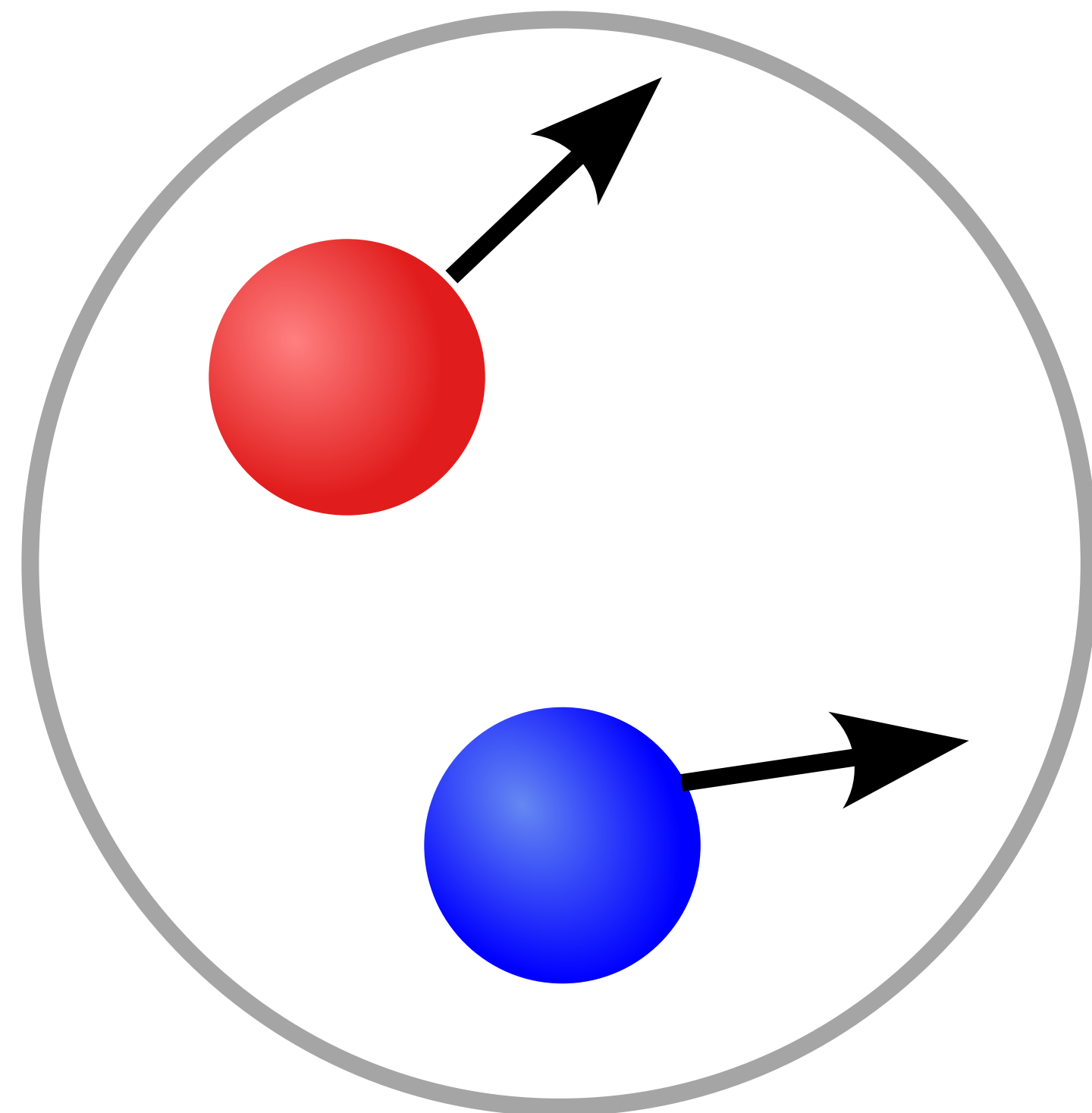
Lorentz boost



Outgoing nucleons (lab frame)

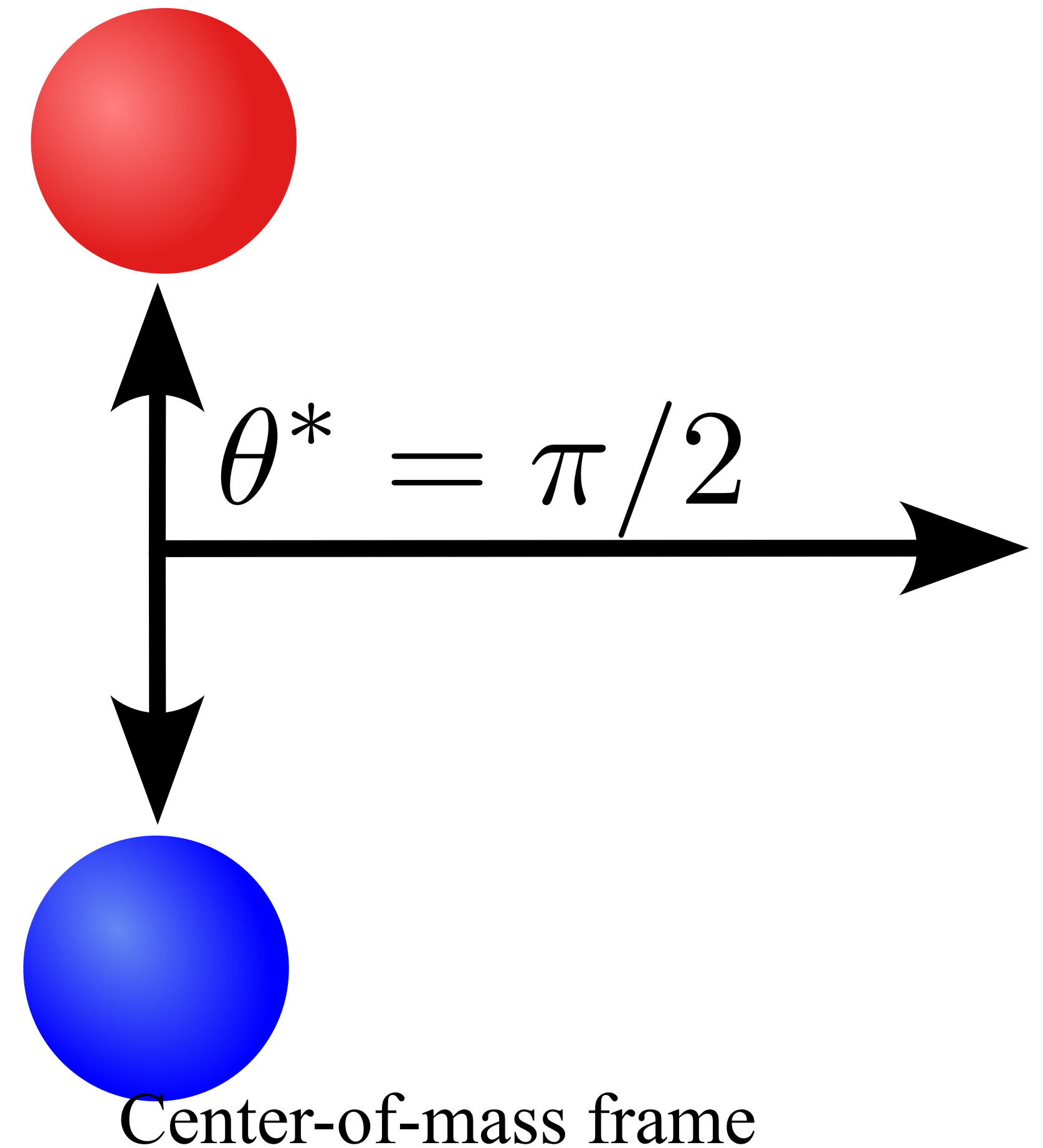
Modelling Correlations Between Nucleons

Motivation (for 2p2h only)



Outgoing nucleons (lab frame)

Lorentz boost



Center-of-mass frame

Modelling Correlations Between Nucleons

Nucleons Sampling Function (for 2p2h only)

$$f(x; P, l) = \begin{cases} N(\kappa) \left(1 - P + P \left| \frac{x}{\kappa} \right|^l \right) \\ N'(\kappa) \left(1 + P \left| \frac{x}{\kappa} \right|^l \right) \end{cases}$$

$$l \in \{1, 2, 3, \dots\}$$

$$P \in [0, 1]$$

$$P \in [-1, 0]$$

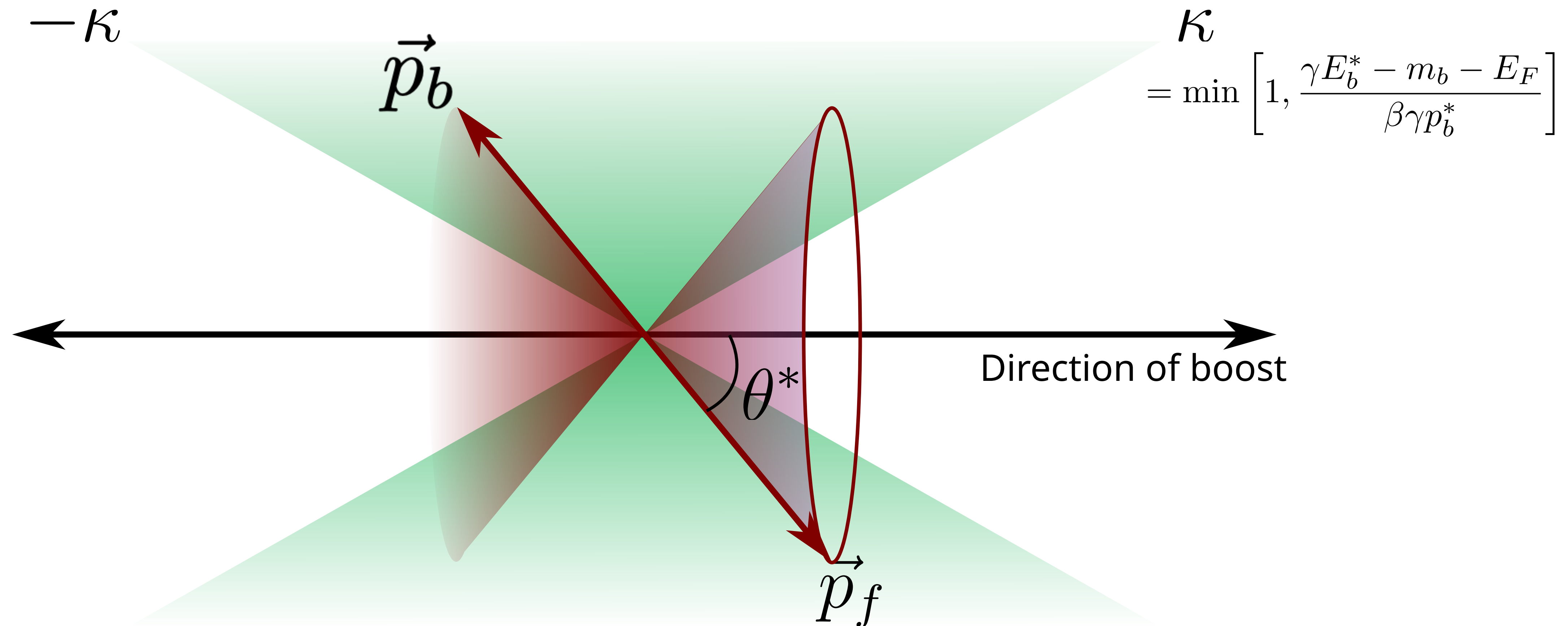
$$l \in \{1, 2, 3, \dots\}$$

Normalisation factor

$$= \min \left[1, \frac{\gamma E_b^* - m_b - E_F}{\beta \gamma p_b^*} \right]$$

Pauli blocking in center-of-mass frame

Schematics

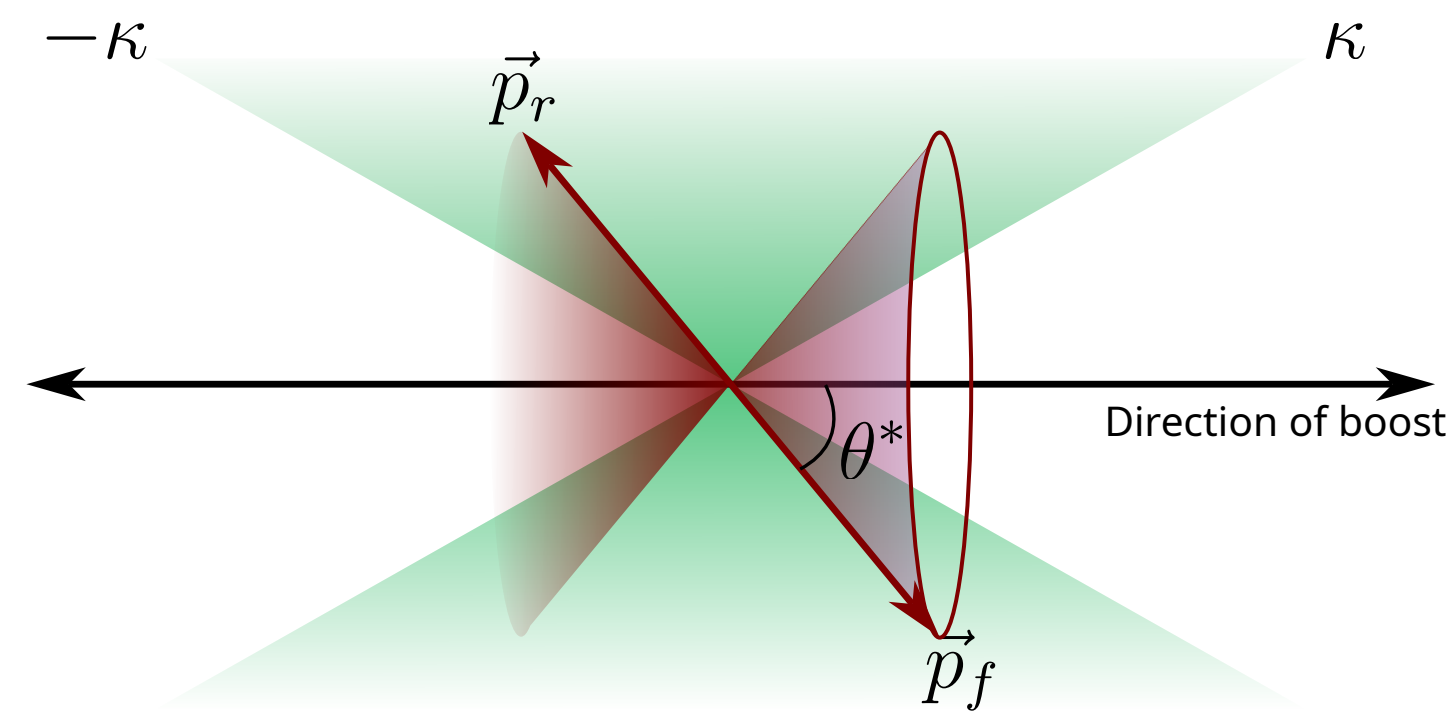


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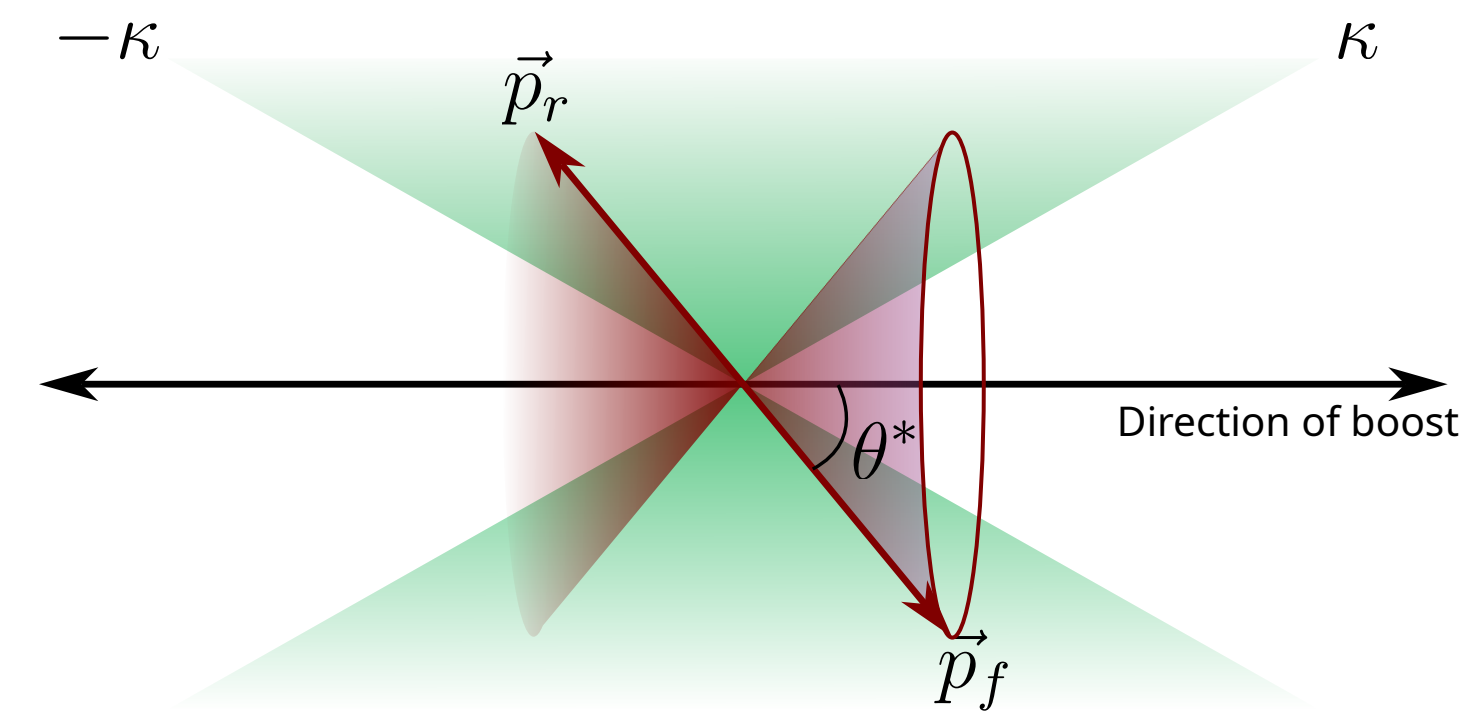
Fitting adjustable parameters

Optimal parameters obtained after fit

- We first assume the following:
 - In np (pn) configuration neutron (proton) is a forward nucleon.



Forward nucleon $\rightarrow$ neutron



Forward nucleon $\rightarrow$ proton

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Fitting adjustable parameters

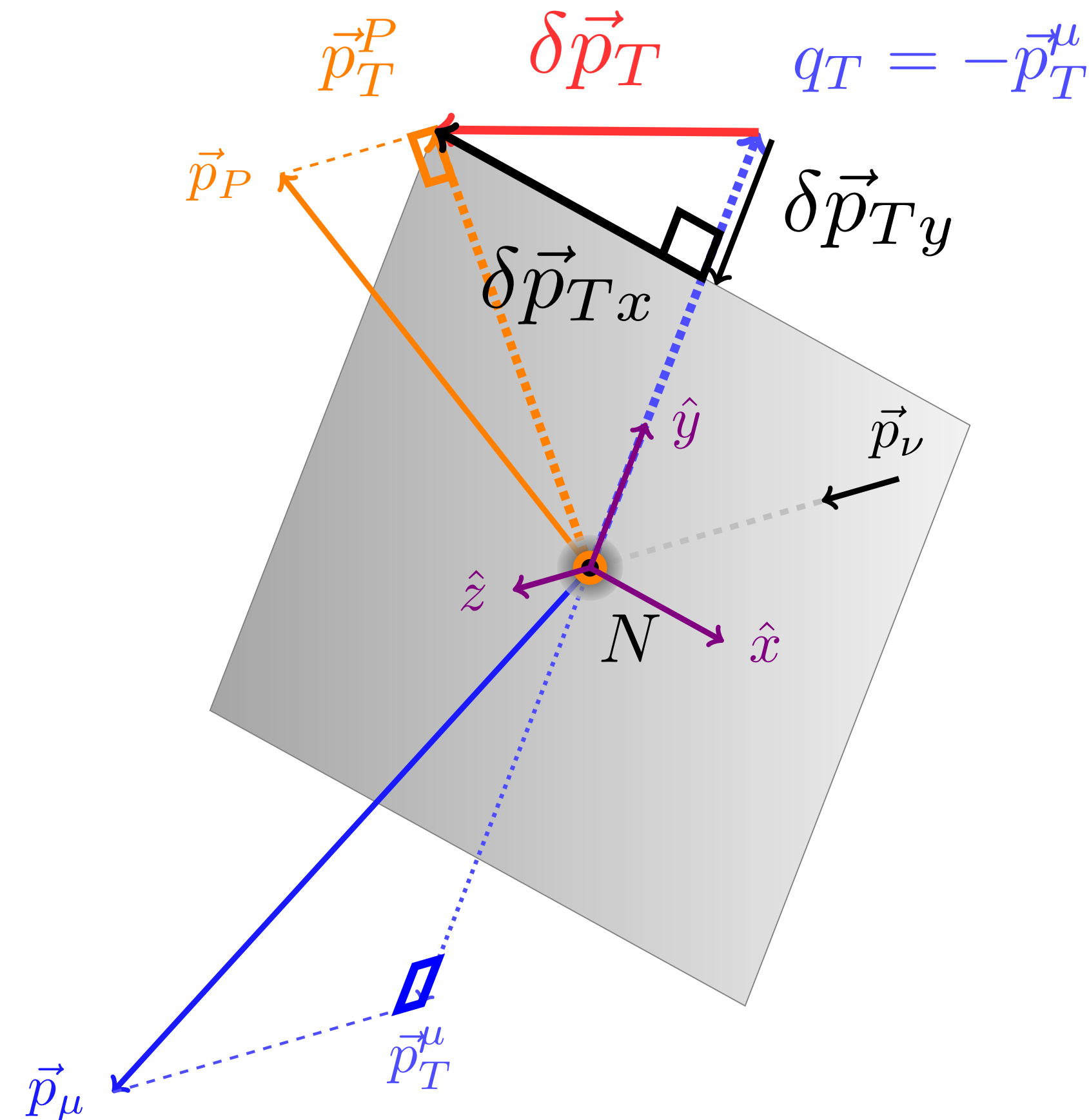
Optimal parameters obtained after fit

- Each outgoing nucleon pair is provided with a separate parameter set .
- The momentum distributions of the outgoing nucleons, and , are fit separately.
 - has 2 parameters to fit.
 - has 4 parameters to fit.
- Optimal values $(\hat{P}, \hat{l})_{\{pp,np,pn\}}$ are found after **minimisation procedure**.
- These optimal values work sufficiently good for all neutrino energies. Hence they are called “global fit values”

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Transverse kinematic imbalance

Impact of nuclear effects



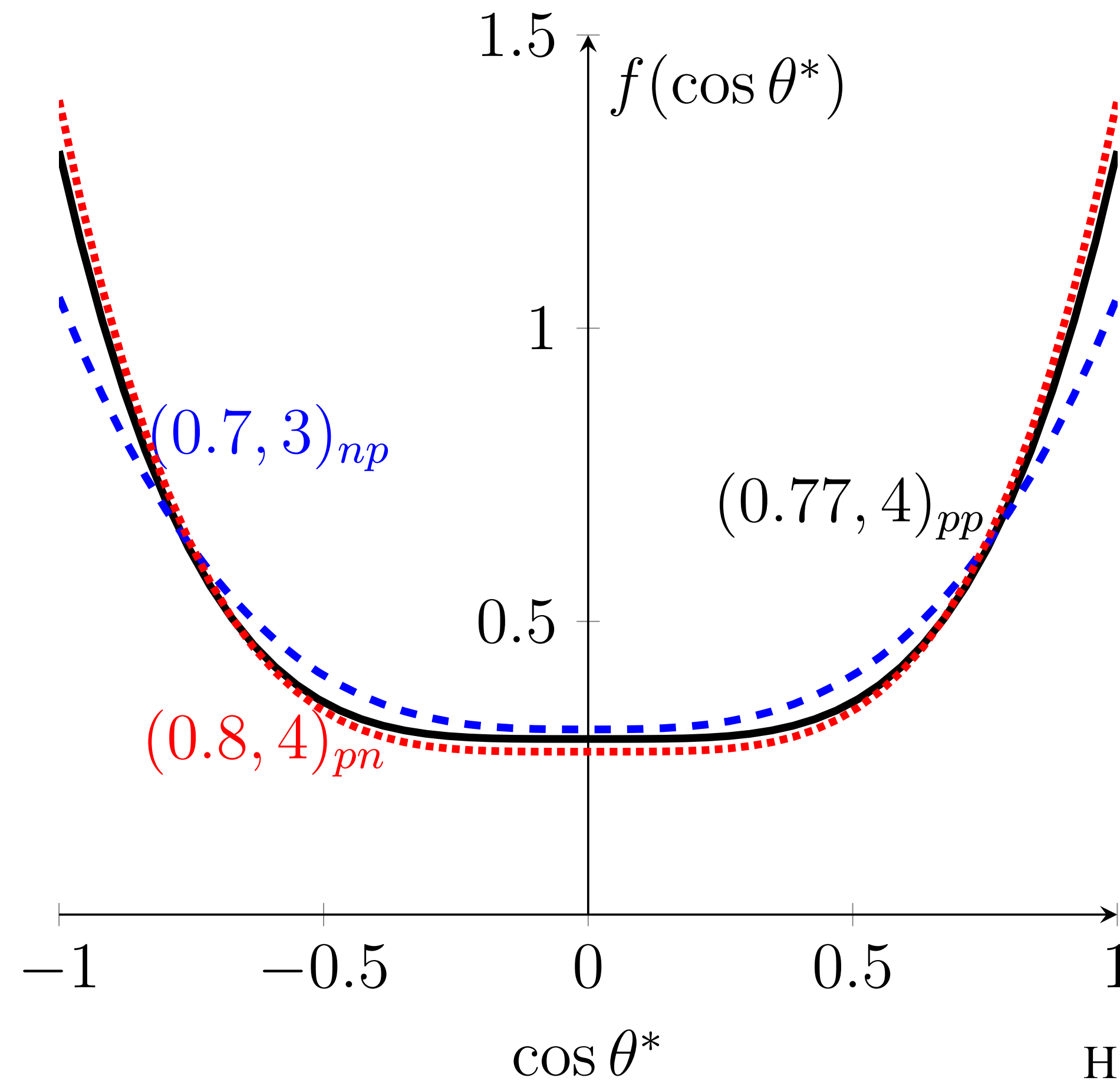
$\delta \mathbf{p}_T = 0$
For static and free nucleon !!

$\delta\alpha_T$
Uniform for Fermi motion !!

X.-G. Lu *et al.* Phys. Rev. D 101, (2020) 092001T. Cai *et al.* Phys. Rev. Lett. 121, (2018) 022504

Fitting adjustable parameters

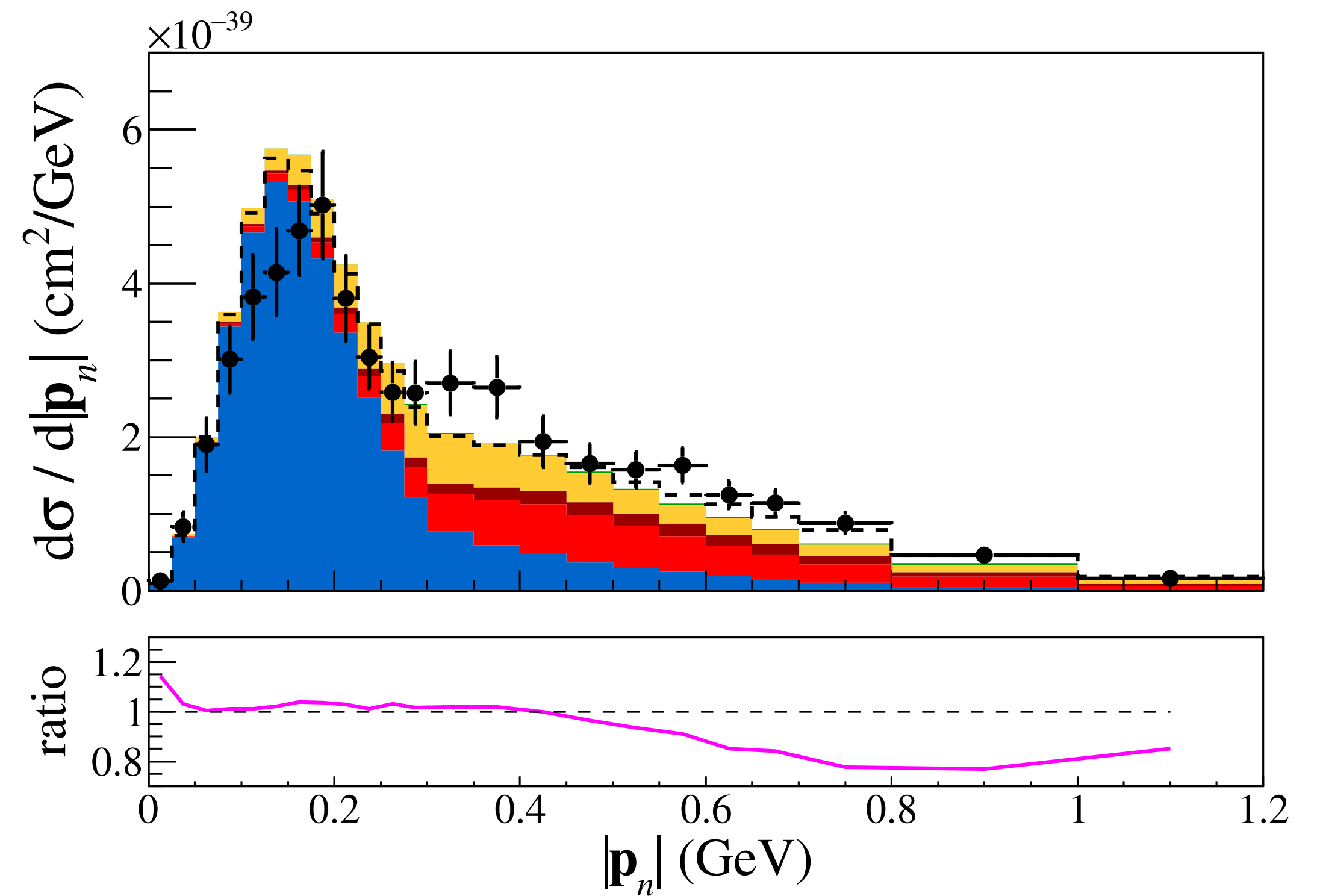
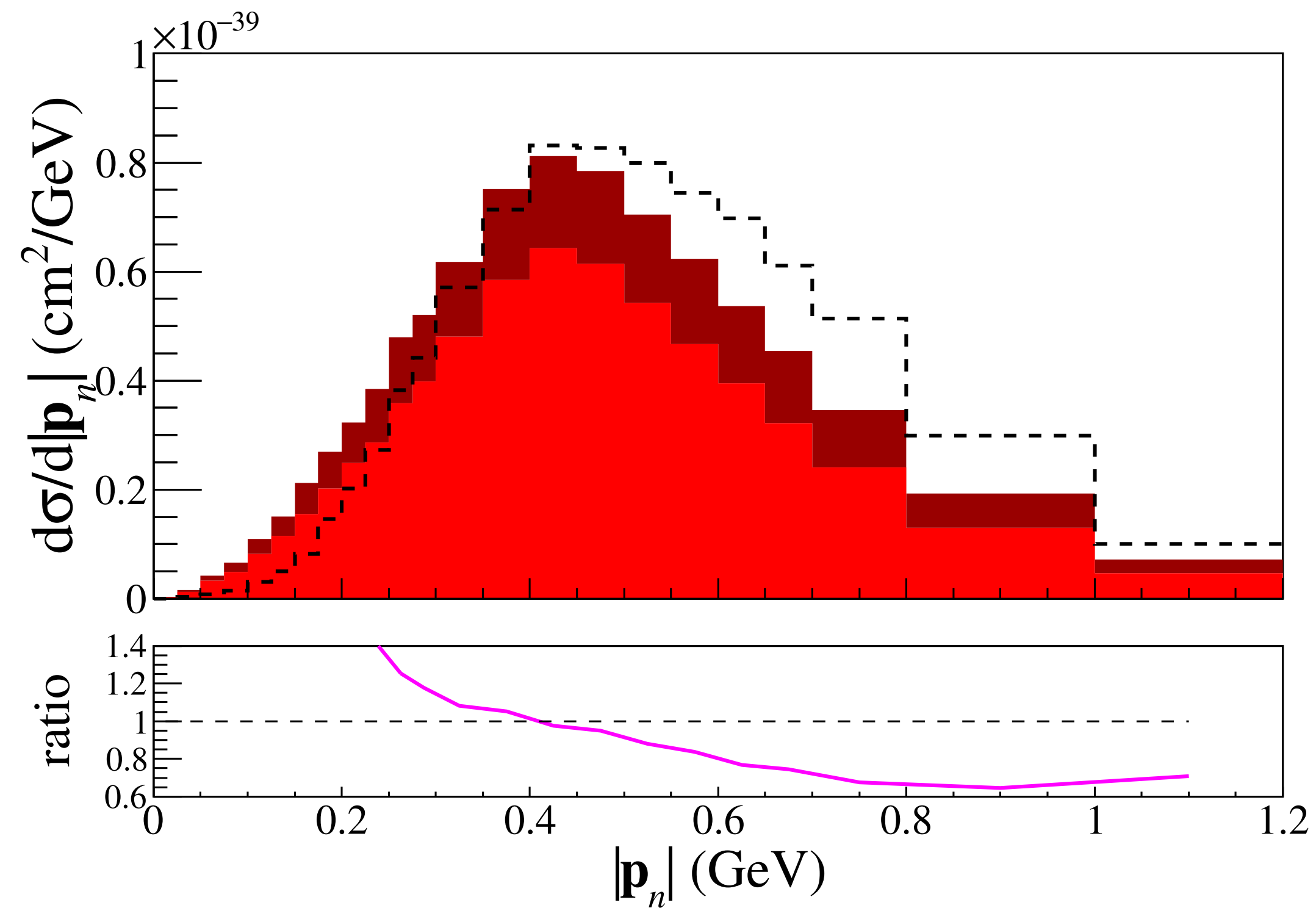
Optimal parameters obtained after fit



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MINER ν A CC1p0 π data*

Comparison : old vs new MEC model



H. Prasad *et al.* Phys. Rev. D 111 (2025) 036032

X.-G. Lu *et al.* Phys. Rev. D 101, (2020) 092001

T. Cai *et al.* Phys. Rev. Lett. 121, (2018) 022504

MINER ν A CC1p0 π data*

$\chi^2/\text{d.o.f}$ comparison : old vs new MEC model

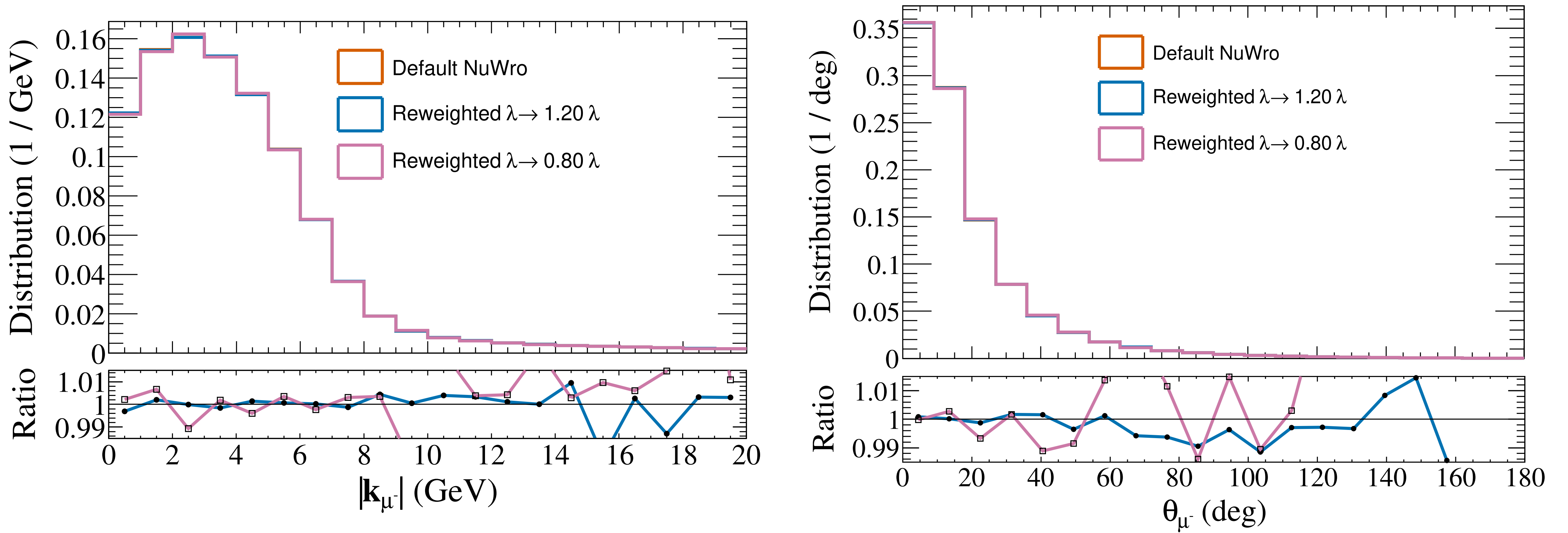
	Old MEC model	New MEC model
$d\sigma/d \mathbf{p}_p $	0.99 (24.76/25)	0.79 (19.72/25)
$d\sigma/d\theta_p$	1.73 (44.88/26)	1.62 (42.08/26)
$d\sigma/d\delta\Phi_T$	2.69 (63.78/23)	1.95 (44.82/23)
$d\sigma/d\delta \mathbf{p}_T $	3.05 (73.10/24)	2.82 (67.68/24)
$d\sigma/d\delta\alpha_T$	1.76 (21.13/12)	1.63 (19.52/12)
$d\sigma/d \mathbf{p}_n $	2.70 (64.85/24)	3.21 (77.09/24)

	Old MEC model		New MEC model	
	-0.2 - 0.2 (GeV)	-0.7 - 0.7 (GeV)	-0.2 - 0.2 (GeV)	-0.7 - 0.7 (GeV)
$\delta\mathbf{p}_{T_x}$	2.21 (17.69/8)	2.29 (64.25/28)	2.85 (22.77/8)	1.95 (54.49/28)
$\delta\mathbf{p}_{T_y}$	0.44 (3.51/8)	3.23 (90.44/28)	0.58 (4.63/8)	2.05 (57.55/28)

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Exact-reweighting scheme

Invariance in lepton kinematics



$\nu_{\mu}(\sim 6 \text{ GeV})$ on Fe_{56}

Selection cuts (Nature Vol 599, Page 569)

Electron data selection

The experiment measured electron scattering from ^4He , ^{12}C and ^{56}Fe nuclei at beam energies of 1.159, 2.257 and 4.453 GeV, and detected the scattered electron and knocked-out particles over a wide range of angles and momenta in the CEBAF Large Acceptance Spectrometer (CLAS)²⁰ at the Thomas Jefferson National Accelerator Facility (Jefferson Lab). We detected electrons with energy $E_e \geq 0.4$, 0.55 and 1.1 GeV for $E_{\text{beam}} = 1.159, 2.257$ and 4.453 GeV, respectively, and angles $15^\circ \leq \theta_e \leq 45^\circ$; hadrons with momenta above 150 to 300 MeV/c and $10-20^\circ \leq \theta_h \leq 140^\circ$; and photons with energy $E_\gamma \geq 300$ MeV. These hadron detection thresh-

Target: carbon

protons with momenta greater than 600 MeV/c.

We detected protons with momenta $p_p \geq 300$ MeV/c and angles $\theta_p \geq 10^\circ$, charged pions with momenta $p_\pi \geq 150$ MeV/c and angles $\theta_{\pi^+} \geq 10^\circ$ and $\theta_{\pi^-} \geq 22^\circ$, and photons with energy $E_\gamma \geq 300$ MeV and $8^\circ \leq \theta_\gamma \leq 45^\circ$. We applied separate fiducial cuts for electrons, π^- , positive particles and photons to select momentum-dependent regions of CLAS where the detection efficiency was constant and close to one. We also determined the minimum electron angle (as a function of electron momentum p) for each beam energy as:

$$\theta_e^{1.1} \geq 17^\circ + \frac{7^\circ}{p [\text{GeV}]}, \quad (12)$$

$$\theta_e^{2.2} \geq 16^\circ + \frac{10.5^\circ}{p [\text{GeV}]}, \quad (13)$$

$$\theta_e^{4.4} \geq 13.5^\circ + \frac{15^\circ}{p [\text{GeV}]}, \quad (14)$$

and the minimum π^- angle as

$$\theta_{\pi^-}^{1.1} \geq 17^\circ + \frac{4^\circ}{p [\text{GeV}]},$$

and

$$\theta_{\pi^-}^{2.2,4.4} \geq 25^\circ + \frac{7^\circ}{p [\text{GeV}]},$$

for $p_{\pi^-} < 0.35$ GeV/c and

$$\theta_{\pi^-}^{2.2,4.4} \geq 16^\circ + \frac{10^\circ}{p [\text{GeV}]}$$

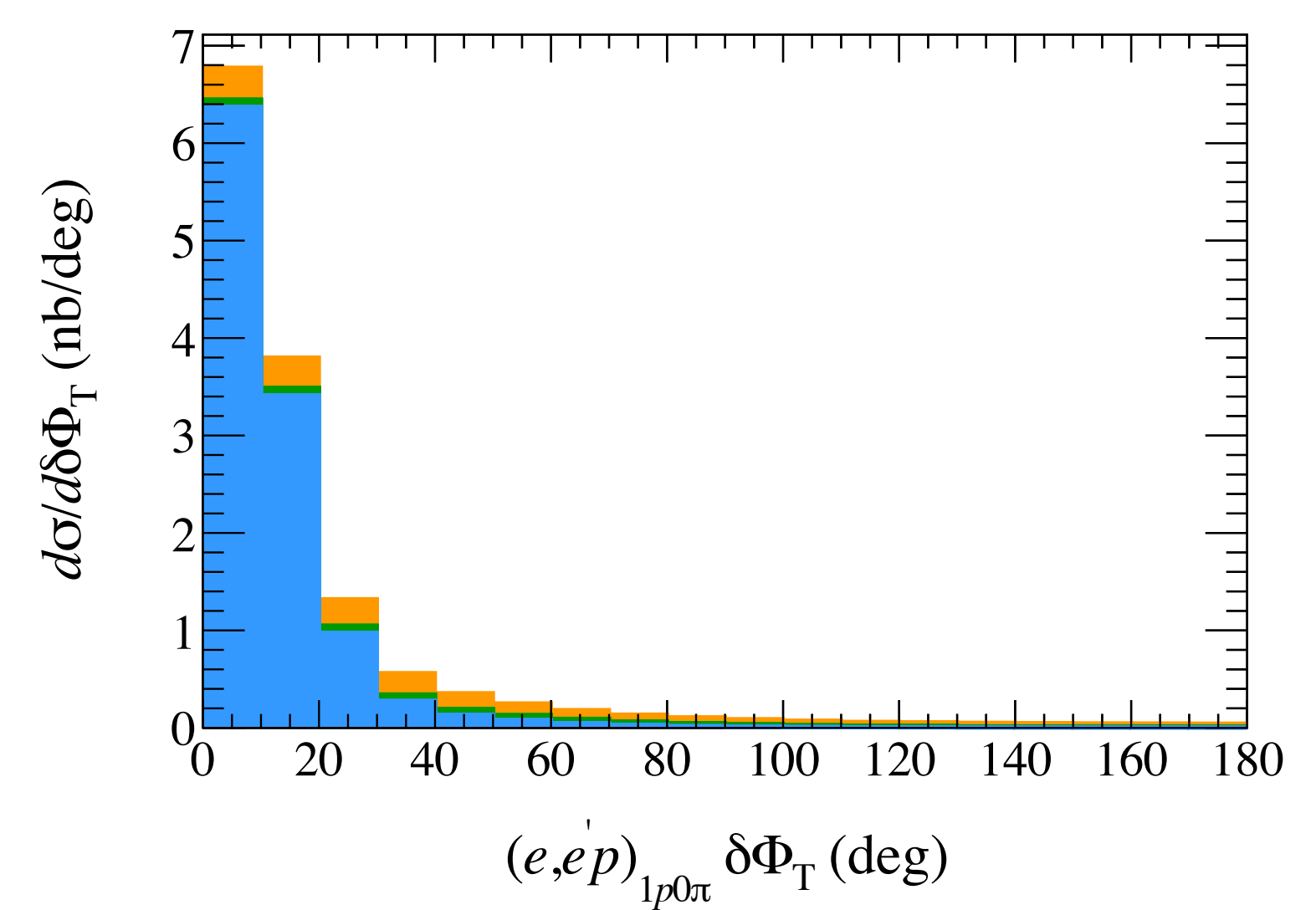
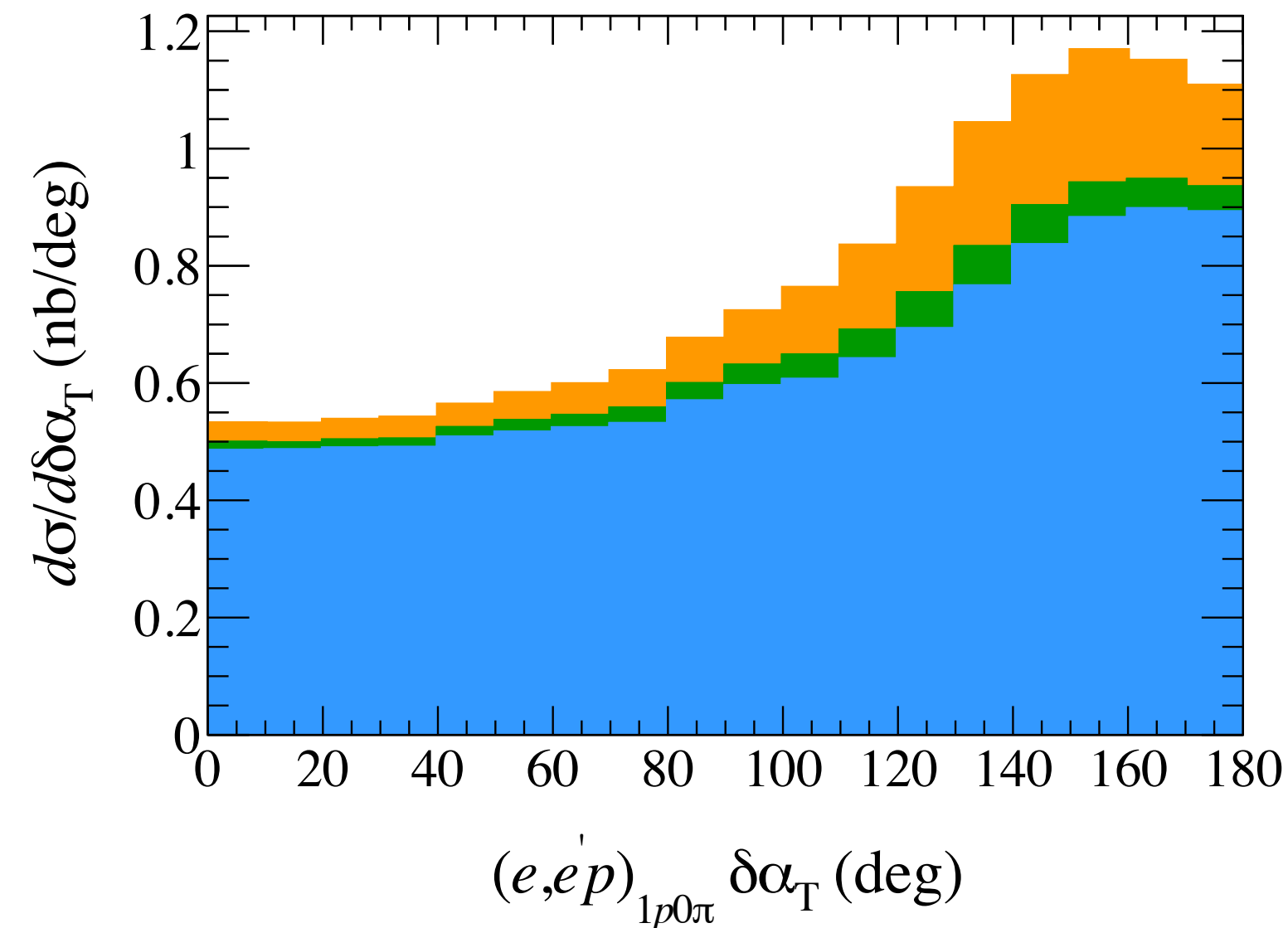
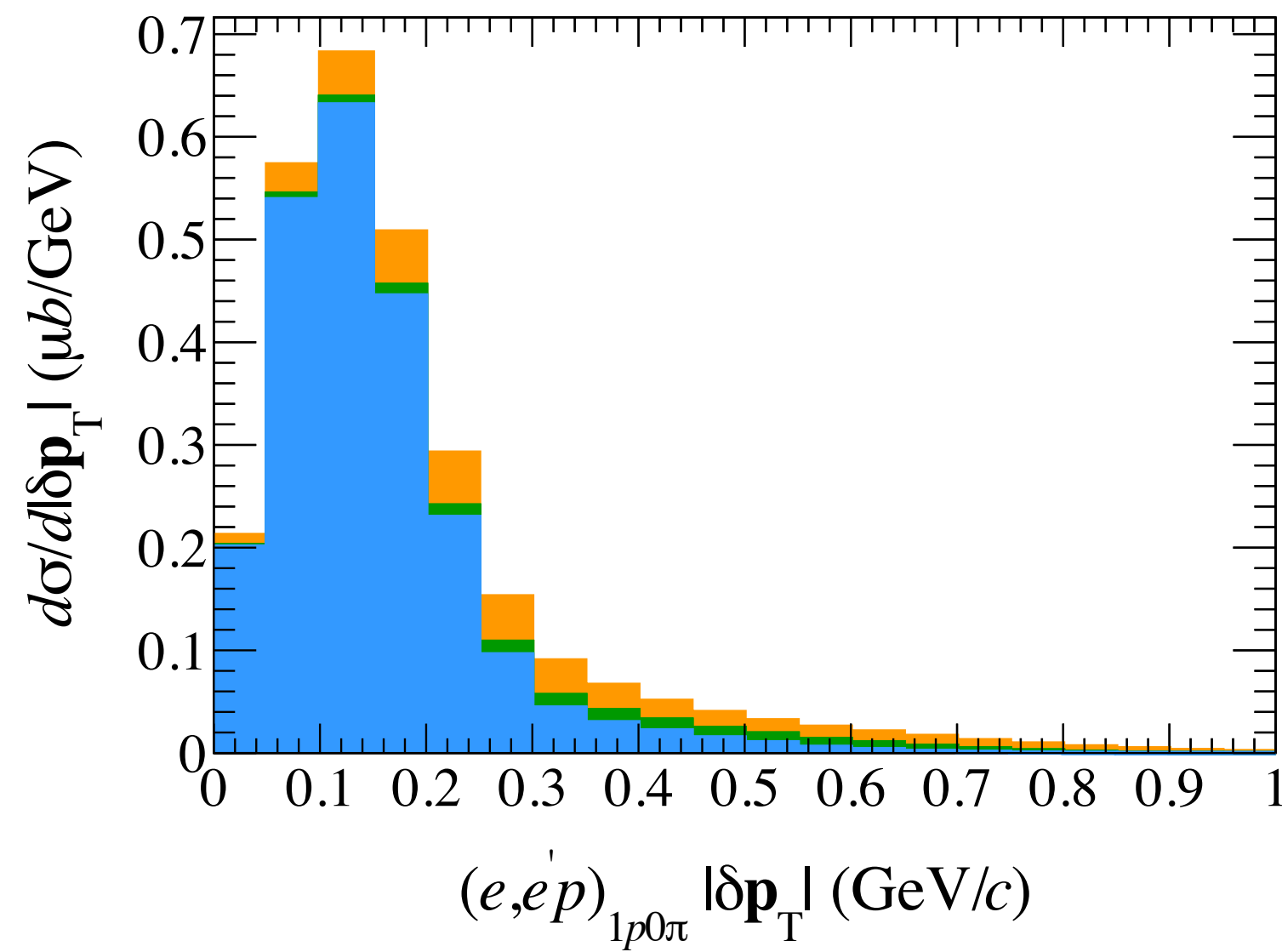
for $p_{\pi^-} \geq 0.35$ GeV/c. The minimum π^+ and proton angle was $\theta > 12^\circ$ for all datasets and momenta.

eWroNet

Preliminary results on *exclusive* predictions

- This tool can also be used to produce exclusive observables (such as transverse-kinematic-imbalance, TKIs) to compare nuclear models **without worrying about the overall normalization.**

$E=1.159$ GeV; $15^\circ \leq \theta \leq 45^\circ$; Carbon

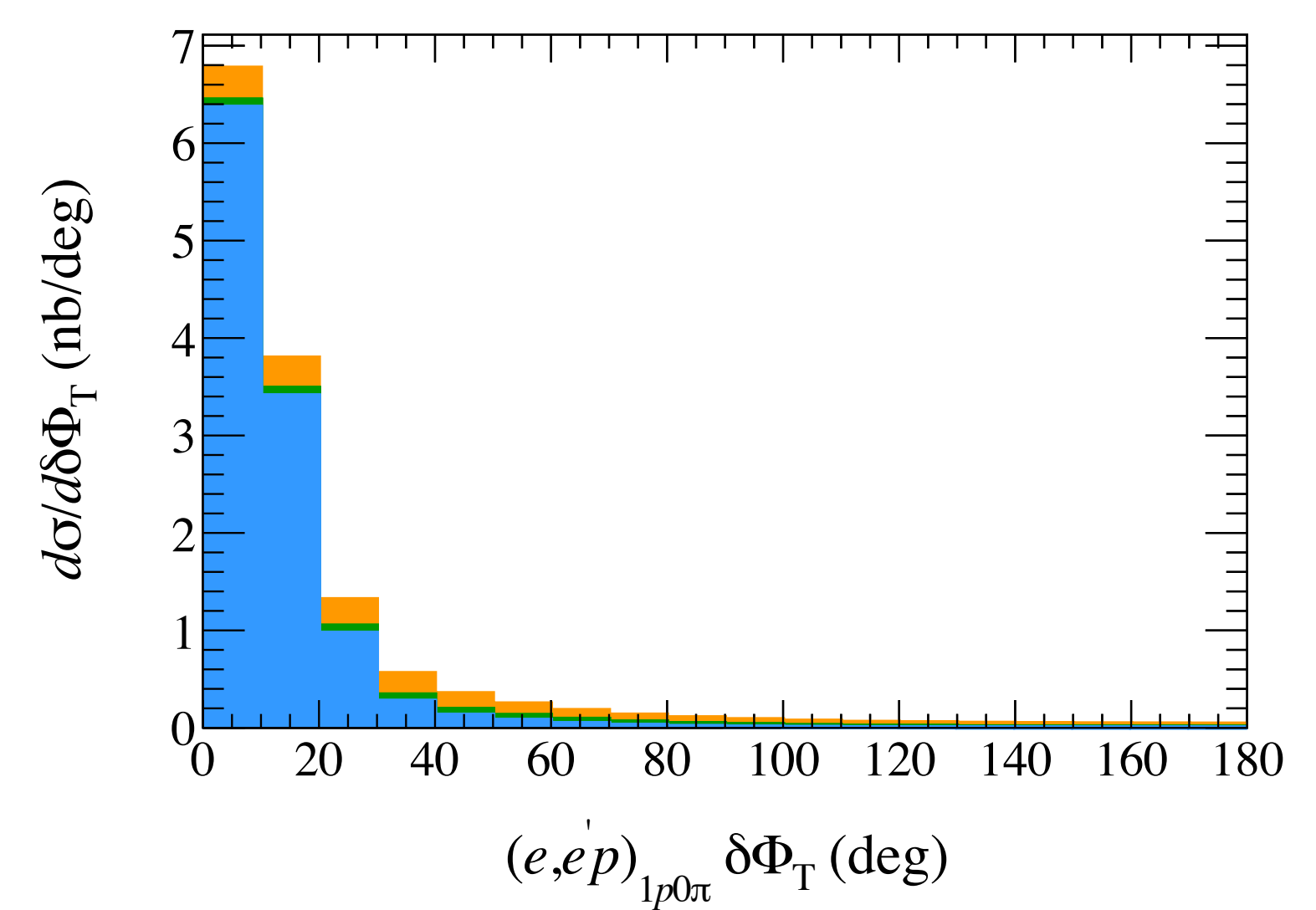
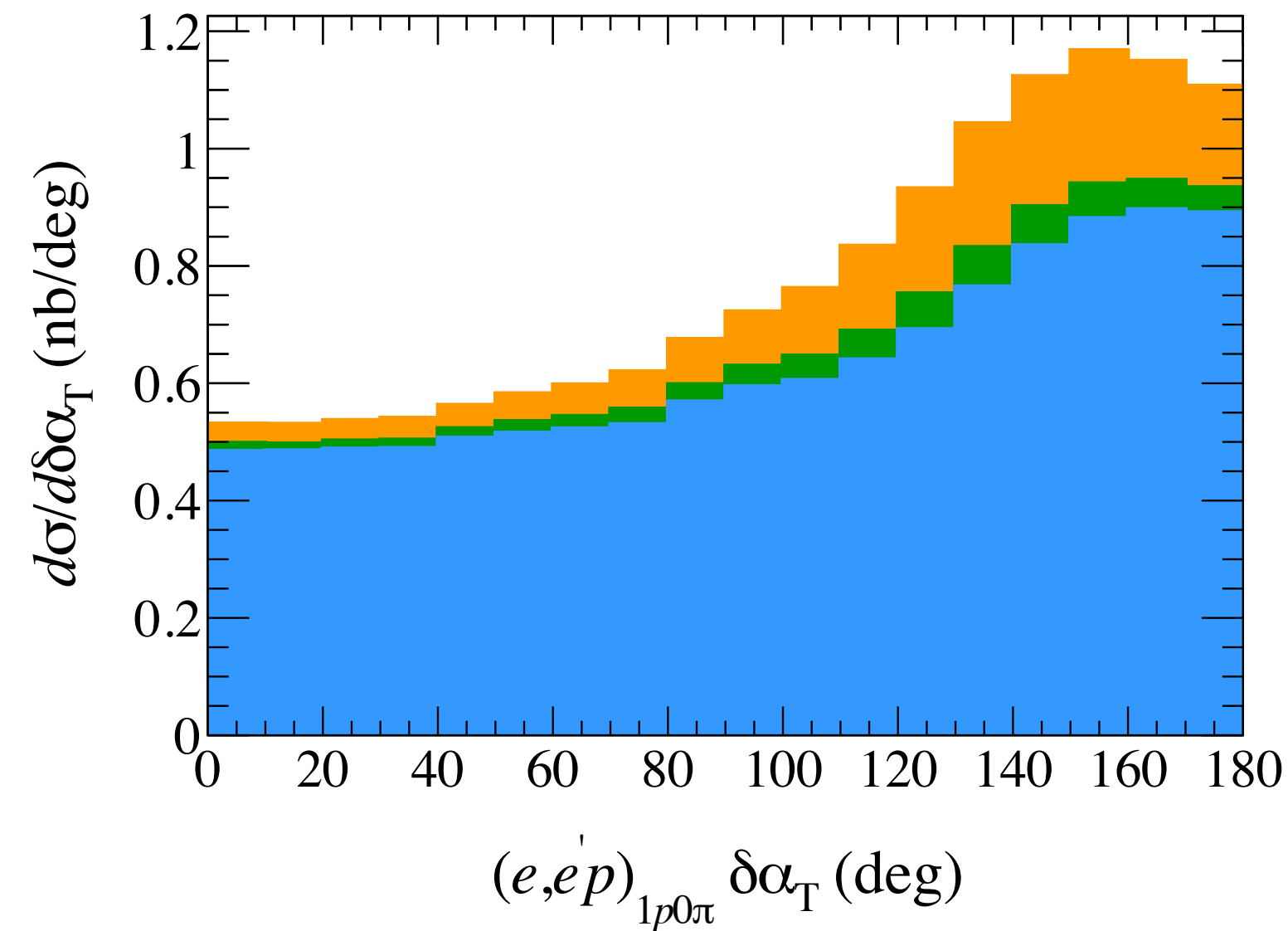
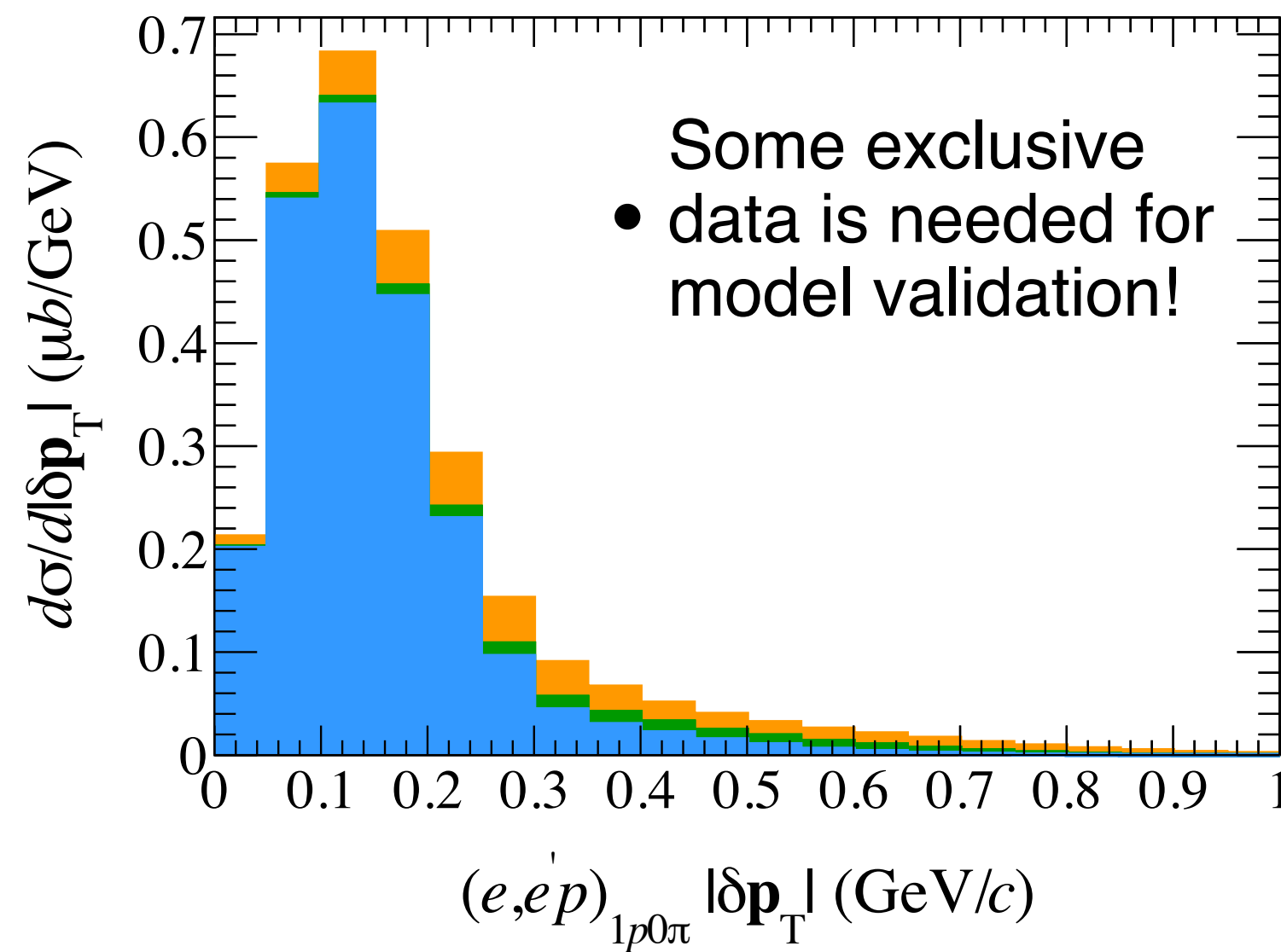


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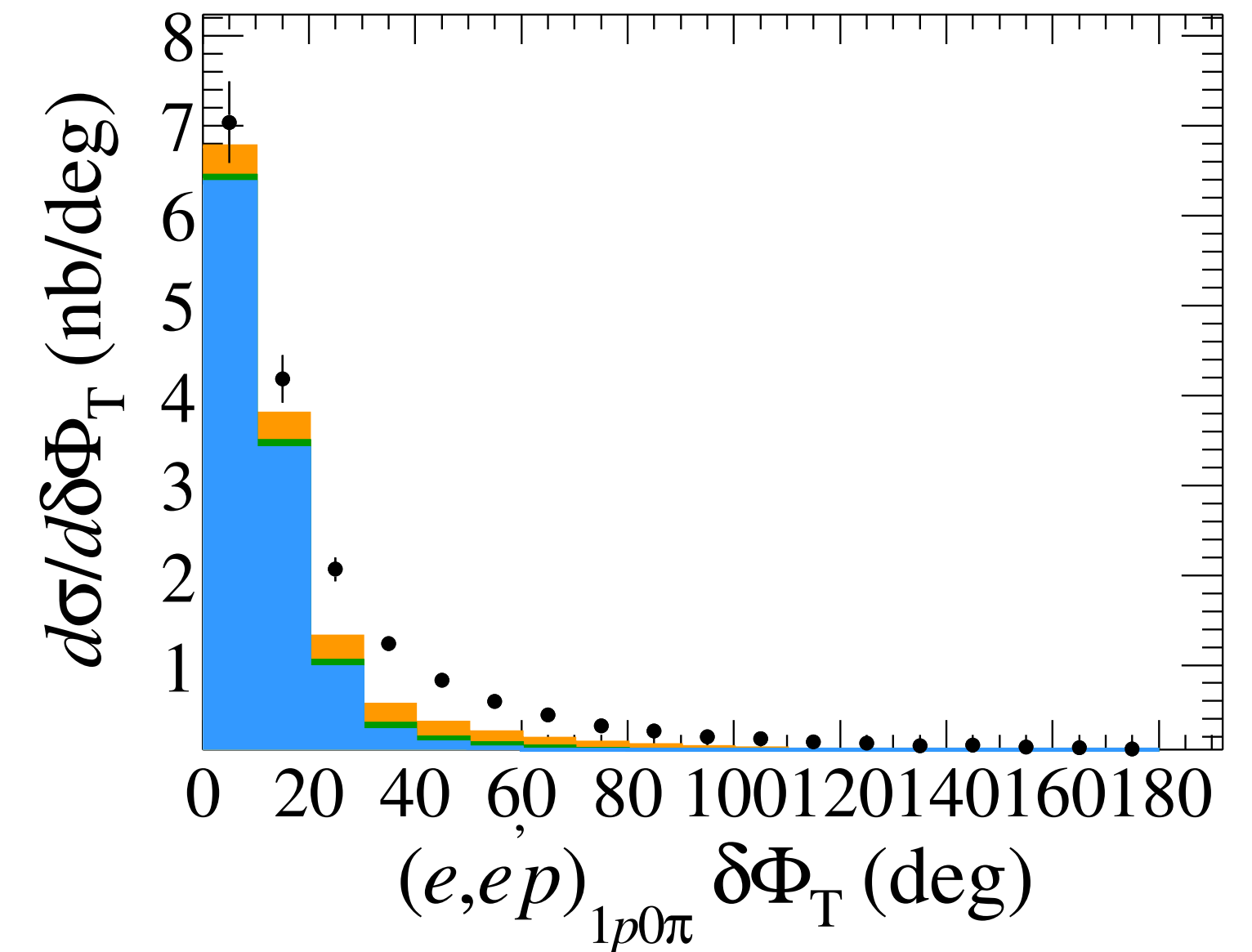
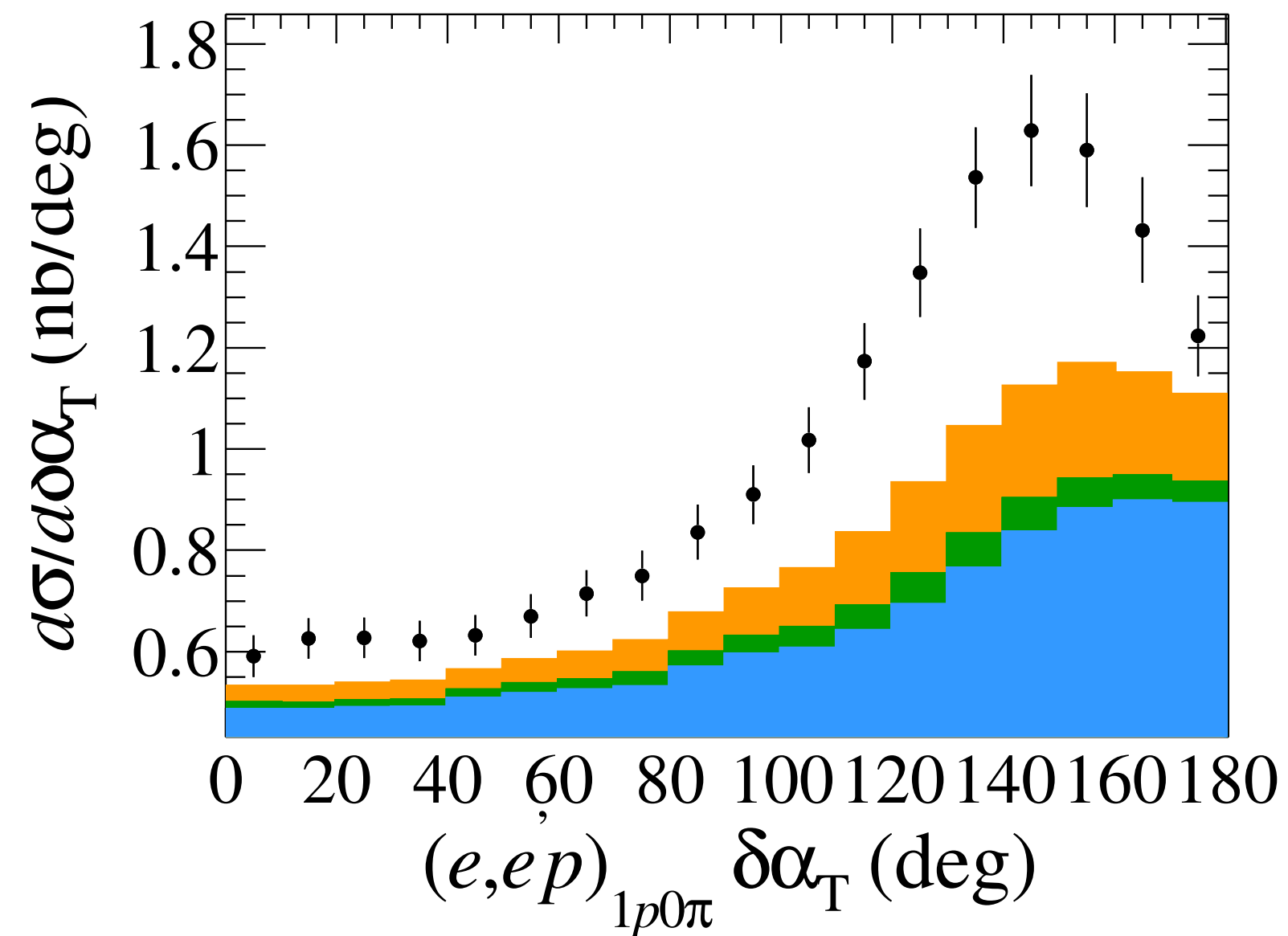
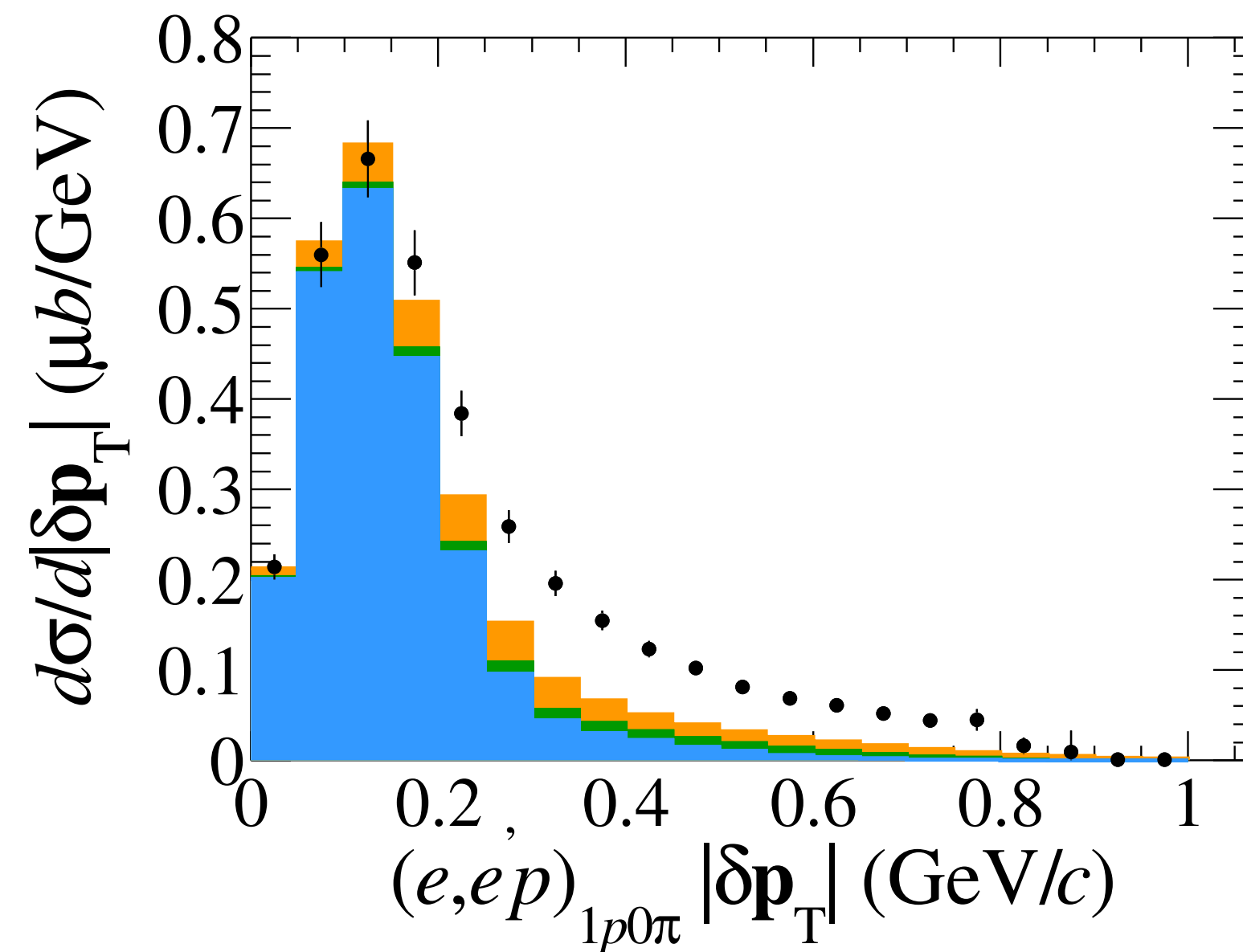


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